STRUCTURAL ECONOMICS

The General Case

Solutions Manual

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About this manual

This manual gives detailed solutions to every exercise in *Structural Economics: The General Case*. Numbered references (Proposition 6.4, equation (7.12)) point to the book.

Numbers. Every numerical answer is produced by a script in `code/solutions/solNN.py` (Python, `numpy`), run from the repository root: `python3 code/solutions/sol06.py`. The scripts print each value used here, so an instructor can change a parameter and regenerate the answers.

Data exercises. Exercises that ask students to work with real data (INE, OECD, CEPAL, the Atlas of Economic Complexity) are answered in two parts. First comes the method, step by step. Then come the results, wherever the data could be fetched into the repository's `data/` directory, with the source and retrieval date. Where the data could not be fetched, the manual says so and gives the method and the expected qualitative result only.

Open-ended questions. For interpretive questions, the manual gives a model answer: the argument a strong student should make, the formal object it rests on, and common mistakes.

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Chapter 1

Economics in Matrix Notation

All numbers below are printed by `python3 code/solutions/sol01.py`.

Exercise 1.1 Reading the allocation matrix

(a) Agent 3's bundle is row 3 of X: $x_3 = (1, 1, 5)'$, one unit of Food, one of Clothing and five of Housing.

What she is "good at" cannot be read off X. The allocation matrix records *consumption* (Definition 1.1); production and endowments are in Ω. After trade, consumption reflects preferences and wealth, not what an agent brings to the market: in the Cobb–Douglas example of Section 1.2, Ana (a farmer) and Boris (a tailor) end up with identical bundles. The inference "she consumes housing, so she produces housing" is valid only under autarky ($W = I$, $X = \Omega$). With that caveat, a strong answer says: if there were no trade, agent 3 would be a housing producer (a builder or landlord); with trade, her bundle says only that she spends about $5/7 = 0.7143$ of her consumption on housing, much more than anyone else.

Common mistake: treating a row of X as a row of Ω.

(b) Total housing is the third entry of $\mathbf{1}'X$: $1 + 2 + 5 + 2 = 10$. Agent 3 consumes the most housing (5 units, half the total).

(c) $\mathbf{1}'X = (5+2+1+4,\ 3+4+1+2,\ 1+2+5+2) = (12, 10, 10) = \mathbf{1}'\Omega$. The allocation is feasible with equality (no waste), Definition 1.2.

(d) The rank of X is $3 = \min(N, L)$, the largest possible. Rows 1–3 alone are linearly independent: their determinant is

$$\det \begin{pmatrix} 5 & 3 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 5 \end{pmatrix} = 64 \neq 0.$$

So yes: in the sense of the Key Insight of Section 1.2, the economy is structurally differentiated. Rows are not proportional, so agents differ in *what* they consume, not only in how much. The budget shares by row are $(0.5556, 0.3333, 0.1111)$, $(0.25, 0.5, 0.25)$, $(0.1429, 0.1429, 0.7143)$ and $(0.5, 0.25, 0.25)$. Under identical Cobb–Douglas preferences all four rows would be equal and X would have rank one (equation (1.13)).

Rank is a coarse measure: almost every real matrix has full rank. The singular values are a finer one. Here they are $(9.3515, 4.2242, 2.1694)$, so the best rank-one approximation captures $9.3515^2 / \sum_k \sigma_k^2 = 0.7950$ of $\|X\|_F^2$. About 80% of the matrix is "same bundle, different scale"; the rest is differentiation.

Exercise 1.2 Cobb-Douglas equilibrium

(a) Agent i maximizes $\frac{1}{3}\ln x_{i1} + \frac{2}{3}\ln x_{i2}$ (a monotone transform of U_i) subject to $p_1 x_{i1} + p_2 x_{i2} = w_i$, $w_i = p'\omega_i$; the budget binds because U_i is strictly increasing. The Lagrangian conditions are $\frac{1}{3x_{i1}} = \mu p_1$ and $\frac{2}{3x_{i2}} = \mu p_2$, so $p_1 x_{i1} = \frac{1}{3\mu}$ and $p_2 x_{i2} = \frac{2}{3\mu}$. Adding them, $w_i = 1/\mu$, hence

$$x_{i1}^* = \frac{w_i}{3p_1}, \qquad x_{i2}^* = \frac{2w_i}{3p_2}.$$

Each agent spends one third of her wealth on good 1 and two thirds on good 2. The solution is interior (log utility goes to $-\infty$ at the boundary), and the objective is strictly concave, so this is the unique maximizer.

(b) Totals: $\mathbf{1}'\Omega = (4+1+3,\ 1+3+2) = (8, 6)$. With $p_1 = 1$, $w_i = \omega_{i1} + p_2\omega_{i2}$ and $\sum_i w_i = 8 + 6p_2$. Market clearing for good 1:

$$\sum_i \frac{w_i}{3} = 8 \iff \frac{8+6p_2}{3} = 8 \iff p_2^* = \frac{16}{6} = \frac{8}{3} \approx 2.6667.$$

By Walras' law, $p'z(p) = 0$ for every p, where z is aggregate excess demand (the script checks it at $p_2 = 0.5, 1, 4$), so good 2 clears too. Directly: $\frac{2(8+16)}{3\cdot 8/3} = 6$.

(c) Wealth at $p^* = (1, 8/3)$: $w = (4 + \frac{8}{3},\ 1+8,\ 3+\frac{16}{3}) = (6.6667, 9, 8.3333)$. Then $x_{i1} = w_i/3$ and $x_{i2} = 2w_i/(3\cdot 8/3) = w_i/4$:

$$X^* = \begin{pmatrix} 2.2222 & 1.6667 \\ 3.0000 & 2.2500 \\ 2.7778 & 2.0833 \end{pmatrix}, \qquad \mathbf{1}'X^* = (8, 6) = \mathbf{1}'\Omega.$$

Feasible with equality. As in the chapter, $X^* = \mathbf{w}\,(1/3,\ 1/4)$ has rank one: everyone spends the same budget shares $(1/3, 2/3)$, and agents differ only in wealth. Agent 2, whose endowment is rich in the expensive good, is now the richest.

Exercise 1.3 Welfare loss computation

By Proposition 1.5, $L = \frac{\kappa N}{2}\,\mathrm{Var}(\omega) = 0.25\ \mathrm{Var}(\omega)$ here, with $\mathrm{Var}(\omega) = \frac{1}{N}\sum_i \omega_i^2 - 1$ since $\Omega/N = 1$. First-best welfare is $\mathcal{W}^{FB} = 5\,U(1) = 4.75$. The largest endowment in all three cases is 5, and $\kappa \cdot 5 = 0.5 < 1$, so every U is increasing on the relevant range.

(a) $\boldsymbol{\omega} = \mathbf{1}$: $\mathrm{Var} = 0$ and $L = 0$.

(b) $\frac{1}{5}(9 + 1 + 0.25 + 0.0625 + 0.0625) = 2.075$, so $\mathrm{Var} = 1.075$ and $L = 0.25 \times 1.075 = 0.26875$. Directly, $\mathcal{W}^{FB} - \sum_i U(\omega_i) = 0.26875$.

(c) $\frac{1}{5} \cdot 25 = 5$, so $\mathrm{Var} = 4$ and $L = 1.0$: about 21% of first-best welfare is lost when one person holds everything.

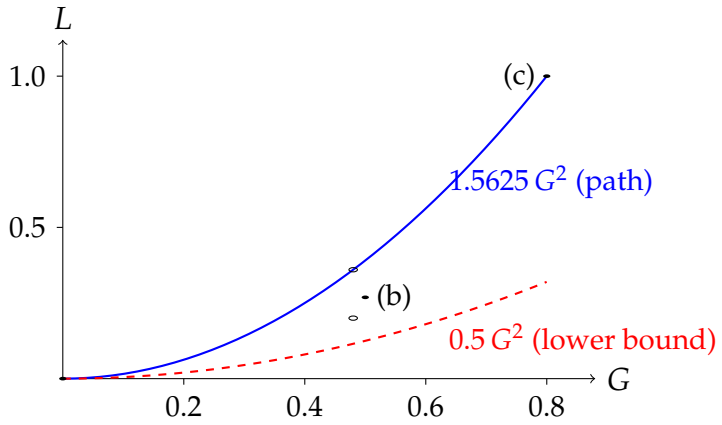
(d) L is *not* a function of the Gini coefficient. We use $G(\boldsymbol{\omega}) = \frac{1}{2N^2\mu}\sum_{i,j}|\omega_i - \omega_j|$, whose maximum is $(N-1)/N = 0.8$. The three cases have $G = 0$, 0.5 and 0.8.

- **Along a path.** On the family $\boldsymbol{\omega}(t) = (1-t)\mathbf{1} + t \cdot 5e_1$, both measures are explicit: $G = 0.8t$ and $\mathrm{Var} = 4t^2$, so $L = t^2 = 1.5625\,G^2$, a convex curve through $(0.8, 1)$.

- **Same Gini, different loss.** $(2, 2, 1, 0, 0)$ and $(3.4, 0.4, 0.4, 0.4, 0.4)$ both have $G = 0.48$, but $L = 0.2$ and $L = 0.36$. Gini weights absolute differences; L weights squared deviations, so it penalizes one very rich agent more.

- **A bound that always holds.** By Cauchy–Schwarz, $\frac{1}{N^2}\sum_{i,j}|\omega_i - \omega_j| \le \left(\frac{1}{N^2}\sum_{i,j}(\omega_i - \omega_j)^2\right)^{1/2} = \sqrt{2\,\mathrm{Var}}$. Hence $2\mu G \le \sqrt{2\,\mathrm{Var}}$, i.e. $\mathrm{Var} \ge 2\mu^2 G^2$ and
$$L \ \ge\ \kappa N \mu^2 G^2 = 0.5\,G^2.$$

- **A sample.** The script draws 20,000 random endowment vectors (Dirichlet, scaled to $\Omega = 5$). All satisfy the bound. For $G \in [0.39, 0.41)$, L ranges over $[0.1200, 0.2315]$; for $G \in [0.59, 0.61)$, over $[0.3142, 0.5366]$.



Model answer. The welfare cost of inequality with quadratic utility is a variance, not a Gini. Two economies with the same Gini can have different losses, and L grows at least quadratically in G. *Common mistake:* plotting one point per Gini value, as if $L(G)$ were a function.

Exercise 1.4 Bargaining with different power

The surplus is $\Omega - \sum_j d_j = 2 - 0.8 = 1.2$, and $U(x) = x - 0.25x^2$.

(a) The utilitarian planner equalizes marginal utilities $1 - \kappa x_i$ (Example 1.4), so $x^{FB} = (1,1)$, $U = (0.75, 0.75)$ and $\mathcal{W}^{FB} = 1.5$. Both agents get more than their disagreement payoffs, so the first-best is also individually rational.

(b) By Proposition 1.9, $x_i = d_i + \lambda_i \cdot 1.2$. With $\boldsymbol{\lambda} = (0.5, 0.5)$: $x = (1.2, 0.8)$, $U = (0.84, 0.64)$ and $\mathcal{W} = 1.48$. The loss relative to the first-best is 0.02. Because U is quadratic and $\sum_i (x_i - 1) = 0$, the loss is exactly $\frac{\kappa}{2} \sum_i (x_i - 1)^2 = 0.25(0.04 + 0.04) = 0.02$, the same computation as in the proof of Proposition 1.23. Equal power does not give equal division, because outside options differ.

(c) With $\boldsymbol{\lambda} = (0.8, 0.2)$: $x = (0.6 + 0.96,\ 0.2 + 0.24) = (1.56, 0.44)$, $U = (0.9516, 0.3916)$, $\mathcal{W} = 1.3432$, and the loss is $0.1568 = 0.25(0.3136 + 0.3136)$. Agent 1 benefits: she gains 0.1116 in utility over equal power, and agent 2 loses 0.2484. Both still beat disagreement ($U(d) = (0.51, 0.19)$), and $\kappa x = (0.78, 0.22) < 1$, so the allocation is Pareto efficient (Proposition 1.9). The loss is distributional, not an inefficiency.

(d) Solve $d_i + \lambda_i \cdot 1.2 = 1$: $\boldsymbol{\lambda}^* = \left(\frac{0.4}{1.2}, \frac{0.8}{1.2}\right) = (0.3333, 0.6667)$. It sums to one because $\sum_i (1 - d_i) = 2 - 0.8 = 1.2$. As in Example 1.10, reaching the first-best requires giving *more* power to the agent with the worse outside option.

Exercise 1.5 The cascade in action

(a) $A' = \begin{pmatrix} 0.2 & 0.3 \\ 0.1 & 0.1 \end{pmatrix}$, and

$$(A')^2 = \begin{pmatrix} 0.07 & 0.09 \\ 0.03 & 0.04 \end{pmatrix}, \quad (A')^3 = \begin{pmatrix} 0.023 & 0.030 \\ 0.010 & 0.013 \end{pmatrix}, \quad (A')^4 = \begin{pmatrix} 0.0076 & 0.0099 \\ 0.0033 & 0.0043 \end{pmatrix}.$$

For example, $(A')^2_{11} = 0.2 \cdot 0.2 + 0.3 \cdot 0.1 = 0.07$. Each round is roughly a third of the previous one, the rate $\rho(A)$ of part (c).

(b) $S_4 = \sum_{k=0}^{4} (A')^k = \begin{pmatrix} 1.3006 & 0.4299 \\ 0.1433 & 1.1573 \end{pmatrix}$. The exact inverse is $\det(I - A') = 0.8 \cdot 0.9 - 0.3 \cdot 0.1 = 0.69$ and

$$(I - A')^{-1} = \frac{1}{0.69} \begin{pmatrix} 0.9 & 0.3 \\ 0.1 & 0.8 \end{pmatrix} = \begin{pmatrix} 1.3043 & 0.4348 \\ 0.1449 & 1.1594 \end{pmatrix}.$$

The error is $L - S_4 = \begin{pmatrix} 0.0037 & 0.0049 \\ 0.0016 & 0.0021 \end{pmatrix}$, i.e. relative errors from 0.18% to 1.12%; the off-diagonal entries converge slowest because they need at least one indirect round. The error has an exact form: telescoping (Appendix 1.9) gives $L - S_K = (A')^{K+1} L$, which the script confirms for $K = 4$. The error is nonnegative and shrinks geometrically.

(c) Appendix 1.9: trace 0.3 and determinant $0.02 - 0.03 = -0.01$, so $\lambda = (0.3 \pm \sqrt{0.13})/2 = (0.3 \pm 0.36056)/2$, giving $\lambda_1 = 0.33028$ and $\lambda_2 = -0.03028$. Then $\rho(A) = 0.33028 < 1$. Sanity check: $\rho(A)$ is at most the largest row sum, 0.4.

(d) The output multipliers are the column sums of L: $(1.4493, 1.5942)$. Sector 2 has the stronger backward linkages. The reason is in the recipes: sector 2 needs $0.3 + 0.1 = 0.4$ of inputs per unit, sector 1 only 0.3, and sector 2's main input (good 1, $a_{21} = 0.3$) is itself input-intensive. The forward linkages (row sums of L) are $(1.7391, 1.3043)$: good 1 is the more widely used input. Both vectors have the same mean, 1.5217, as they must.

Exercise 1.6 Properties of W^{CE}

(a) $\mathbf{1}'\mathbf{1} = N$, so $(W^{CE})^2 = \frac{1}{N^2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = \frac{N}{N^2}\mathbf{1}\mathbf{1}' = W^{CE}$. W^{CE} is the orthogonal projection onto $\text{span}(\mathbf{1})$.

(b) $(W^{CE}\mathbf{x})_i = \sum_j \frac{1}{N} x_j = \bar{x}$ for every i, so $W^{CE}\mathbf{x} = \bar{x}\mathbf{1}$. *Interpretation:* one round of anonymous interaction replaces every individual value (a belief, a price signal, an endowment-weighted demand) by the population mean. Individual identity is erased in a single step. This is the "anonymous" property (a) of Section 1.6.

(c) $W^{CE}\mathbf{1} = \mathbf{1}$ by (b), since $\bar{1} = 1$. If $\mathbf{1}'\mathbf{x} = 0$ then $W^{CE}\mathbf{x} = \bar{x}\mathbf{1} = 0$, so the subspace $\mathbf{1}^{\perp}$, of dimension $N-1$, consists of eigenvectors with eigenvalue 0. $\mathbb{R}^N = \text{span}(\mathbf{1}) \oplus \mathbf{1}^{\perp}$, so the geometric multiplicities add to N and there are no other eigenvalues. Since W^{CE} is symmetric, the algebraic and geometric multiplicities agree; equivalently, its characteristic polynomial is $\lambda^{N-1}(\lambda - 1)$. For $N = 3$, a basis of the eigenspace for 0 is $(1,-1,0)'$, $(1,0,-1)'$; an orthonormal one is $(1,-1,0)'/\sqrt{2}$, $(1,1,-2)'/\sqrt{6}$.

(d) $W = \begin{pmatrix} 0.3333 & 0.3333 & 0.3333 \\ 0.6667 & 0.1667 & 0.1667 \\ 0 & 0.5 & 0.5 \end{pmatrix}$. Row sums $(1,1,1)$; column sums $\frac{1}{3}(3,3,3) = (1,1,1)$. So *yes*, W is doubly stochastic, and by Proposition 1.21 $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$.

With $\mathbf{x} = (3,6,9)'$: $W\mathbf{x} = (6, 4.5, 7.5)'$, while $W^{CE}\mathbf{x} = (6,6,6)'$. What is different:

- **Non-anonymity.** W preserves the mean ($\mathbf{1}'W\mathbf{x} = 18 = \mathbf{1}'\mathbf{x}$, because columns sum to one) but not the identity of who listens to whom. Agent 2 puts 2/3 of her weight on agent 1 and moves to 4.5; agent 3 ignores agent 1 and moves to 7.5.
- **Same coordinates, different matrix.** W has rank 2 (columns 2 and 3 are equal), and its eigenvalues are $\{1,0,0\}$: trace 1 and determinant 0 force the other two to sum to 0 and multiply to 0. So $\Gamma(W) = 1$, as for W^{CE}. But 0 has algebraic multiplicity 2 and geometric multiplicity 1, so $W - W^{CE}$ is a nonzero nilpotent matrix with $(W - W^{CE})^2 = 0$. Hence $W^2 = W^{CE}$ exactly (use $WW^{CE} = W^{CE}W = W^{CE}$), and $W^2\mathbf{x} = (6,6,6)'$.

Lesson: the two coordinates of Section 1.7 (power $\boldsymbol{\pi}$ and integration Γ) are identical for W and W^{CE}, yet the matrices differ in what happens along the way. Uniform power does not imply anonymity.

Exercise 1.7 Bolivia's input-output table (data exercise)

Data (real results). INE Bolivia, supply-use tables, base 2017, year 2019 (Instituto Nacional de Estadística (INE), Bolivia, 2026), retrieved 2026-11-05. The repository

converts them to an industry-by-industry table of *domestic* flows at basic prices (`data/compute/compute_bolivia_io.py`: purchasers' to basic prices, import proportionality, fixed product-sales structure) and aggregates it to the book's 12 sectors (`data/processed/bolivia_Z/Z_2017_2019_book12.csv`, millions of current Bs). Gross output comes from `data/processed/bolivia_io_sector_facts.csv`. We aggregate further to six sectors: Hydrocarbons, Mining, Agriculture, Manufacturing (which includes refining), Construction, and Services (the remaining seven sectors). Gross output, in million Bs: 25,863; 25,009; 46,473; 114,506; 51,045; 287,812.

(a) Published flows have rows = sellers: $Z_{ss'}$ is the sales of s to s'. In the book's convention, $a_{ij} = Z_{ji}/x_i$ (the purchases of i from j per unit of i's output), i.e. $A = (Z\,\mathrm{diag}(\mathbf{x})^{-1})'$:

$$A = \begin{pmatrix} 0.0056 & 0.0017 & 0.0009 & 0.0131 & 0.0015 & 0.2019 \\ 0.0043 & 0.0009 & 0.0042 & 0.0567 & 0.0017 & 0.1700 \\ 0.0047 & 0.0013 & 0.1171 & 0.0641 & 0.0067 & 0.1601 \\ 0.0357 & 0.1121 & 0.1922 & 0.1091 & 0.0027 & 0.1059 \\ 0.0005 & 0.0313 & 0.0010 & 0.2045 & 0.0380 & 0.0632 \\ 0.0093 & 0.0010 & 0.0109 & 0.0646 & 0.0076 & 0.1811 \end{pmatrix}$$

(rows and columns in the order above). The row sums are the domestic intermediate inputs per unit of output: $(0.2246, 0.2378, 0.3540, 0.5576, 0.3385, 0.2746)$. They match the INE-derived figures quoted in Chapter 3 (hydrocarbons 0.225, manufacturing 0.558). $\rho(A) = 0.3473 < 1$, so the economy is productive.

(b) $L = (I - A')^{-1}$:

$$L = \begin{pmatrix} 1.0094 & 0.0096 & 0.0116 & 0.0460 & 0.0116 & 0.0154 \\ 0.0061 & 1.0107 & 0.0138 & 0.1320 & 0.0618 & 0.0125 \\ 0.0120 & 0.0259 & 1.1586 & 0.2582 & 0.0594 & 0.0365 \\ 0.0356 & 0.0838 & 0.1046 & 1.1693 & 0.2578 & 0.0965 \\ 0.0038 & 0.0040 & 0.0104 & 0.0073 & 1.0419 & 0.0105 \\ 0.2574 & 0.2284 & 0.2466 & 0.2410 & 0.1410 & 1.2480 \end{pmatrix}.$$

(c) The output multipliers (column sums of L) are: Hydrocarbons 1.324, Mining 1.363, Agriculture 1.546, Manufacturing 1.854, Construction 1.573, Services 1.419. With Mining and Gas merged into one sector, the five-sector multipliers are: Mining/Gas 1.343, Agriculture 1.546, Manufacturing 1.852, Services 1.419 (Construction 1.572). Sanity check: mean backward linkage equals mean forward linkage

(1.5132).

(d) The highest multiplier is Manufacturing (1.854); the lowest is Hydrocarbons (1.324), with Mining close behind. On Pinto's question, the measured table does *not* show a block-diagonal A. Take the extractive block {Hydrocarbons, Mining} against the rest. The cross-block mass is $\chi(A) = 0.3250$ (Definition 3.5), and the six-sector $\Gamma(W^P) = 0.7467$, with $N_{\text{eff}} = 2.843$ of 6 and $\pi_{\text{Services}} = 0.527$. The extractive sectors buy 97.29% of their domestic intermediate inputs from the rest of the economy (mostly services), and the rest buys 12.34% of its domestic inputs from them. Bolivia's heterogeneity lies elsewhere: in how *little* the extractive sectors buy domestically per unit of output (0.22–0.24, against 0.56 in manufacturing) and in where their output goes (part (e)). This matches the measured values in Chapter 3, Section 3.4.

(e) *Challenge.* The gas sector is an enclave in two precise senses, but not in the sense of zero linkages.

- **Backward.** One unit of final demand for hydrocarbons generates $Le_{\text{hyd}} = (1.0094, 0.0061, 0.0120, 0.0356, 0.0038, 0.2574)'$. That is 0.315 of output in other sectors, almost all of it in services: the lowest spillover of the six sectors (manufacturing: 0.685).

- **Forward.** Its forward linkage is 1.104, the second lowest, because 70.1% of its output is exported rather than sold to domestic users.

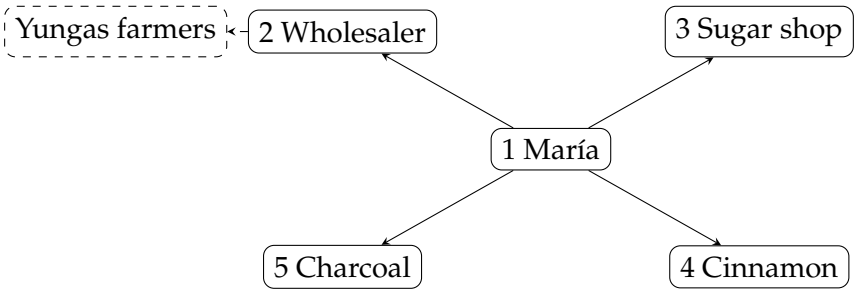
- **Income.** Compensation of employees is 5.95% of its value added, so the rents reach households through the fiscal system rather than through wages or input purchases (Chapter 3, Section 3.2).

In the production reading, its Perron weight is $\pi_{\text{hyd}} = 0.0357$ at six sectors (against $1/6$ if uniform). Answer: an export enclave with thin domestic backward and forward linkages, connected to the rest of the economy mainly through services purchases and taxes.

Caveats. (i) Domestic flows depend on the import-proportionality assumption, because the SUT does not split domestic from imported use. (ii) Multipliers depend on aggregation: the 12-sector values in `data/processed/bolivia_io_linkages.csv` differ slightly. (iii) *Common mistake:* computing multipliers from W^P. For any row-stochastic W, $(I - \gamma W)^{-1}\mathbf{1} = \mathbf{1}/(1-\gamma)$, so every row-sum linkage is identical (Section 1.7).

Exercise 1.8 From story to matrix

(a) The agents are 1 María, 2 Wholesaler, 3 Sugar Shop, 4 Cinnamon Distributor, 5 Charcoal Supplier. Following the book's convention (Appendix 5.9), draw $i \to j$ when i uses j's output; influence and demand impulses flow against the arrows.



The Yungas farmers are outside the five-agent system (dashed).

(b) With $G_{ij} = 1$ if i uses j's output (written G to avoid a clash with the coefficient matrix; Appendix 5.9):

$$G = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The in-degrees (column sums) are $(0, 1, 1, 1, 1)$ and the out-degrees $(4, 0, 0, 0, 0)$.

(c) G is not symmetric. The asymmetry is the *direction of dependence*: María depends on four suppliers, and none of them depends on her output. A demand shock to api travels to all four, and a shock to any supplier reaches only María. A symmetric matrix would mean mutual dependence: each supplier would buy api as an input.

(d) After the wholesaler's row and column are set to zero, only María is *directly* affected: $G_{12} = 1$ was the only entry in column 2. For the cascade, note that $G^2 = 0$ (nobody buys from María), so the Neumann series stops after one term: with any coefficient c per link, $(I - cG')^{-1} = I + cG'$. In the demand-driven Leontief model the cascade therefore has depth one. Removing the wholesaler changes the output of agents 3–5 not at all: with $c = 0.25$ and api demand $\mathbf{d} = e_1$, $\mathbf{y}$ goes from $(1, 0.25, 0.25, 0.25, 0.25)$ to $(1, 0, 0.25, 0.25, 0.25)$.

That conclusion is wrong economically, and the reason is instructive. The disappearance of a supplier is a *supply* shock, which travels downstream to users, not upstream like the demand impulses the quantity model $\mathbf{y} = (I - A')^{-1}\mathbf{d}$ handles

(Chapter 3, Section 3.2). With fixed proportions and no substitute source of corn, María cannot produce api. Her purchases of sugar, cinnamon and charcoal then fall to zero, so agents 3–5 are *indirectly* affected through her. The Yungas farmers, outside the system, lose their buyer.

(e) W^{CE} for $N = 5$ has every entry equal to $1/5$: everyone depends equally on everyone, including herself. G has 4 nonzero entries out of 25. It contains exactly the information W^{CE} destroys: *who* depends on *whom* (support), in which *direction* (asymmetry), the hub (María, the only buyer), and the absence of any cycle (nilpotency, hence no feedback). As a weight matrix, G cannot even be row-normalized, because rows 2–5 are zero (the suppliers' own inputs are outside the system). It is reducible, so a Perron vector is not defined until the system is closed. *Common mistake:* treating the binary adjacency matrix as a technical-coefficient matrix with entries 1, for which $\rho(A) < 1$ would fail in any graph with a cycle.

Exercise 1.9 From sectors to countries

(a) $\rho(A) = 0.1564$. $L = (I - A')^{-1} = \begin{pmatrix} 1.0601 & 0.0118 & 0.0118 \\ 0.3534 & 1.0595 & 0.0595 \\ 0.3534 & 0.0595 & 1.0595 \end{pmatrix}$. The output multipliers (column sums) are $(1.7668, 1.1307, 1.1307)$. Country 1 has by far the strongest backward linkages: its recipe draws 0.30 from each of the others.

(b) $Le_2 = (0.0118, 1.0595, 0.0595)'$: a unit of final demand in Country 2 generates 1.0595 in Country 2, 0.0595 in Country 3 and only 0.0118 in Country 1, a total of 1.1307. By contrast, a unit in Country 1 generates $(1.0601, 0.3534, 0.3534)$, a total of 1.7668.

(c) The answer depends on which half of Prebisch's description we look at, and a strong answer says so.

- **The demand-pull half matches.** Countries 2 and 3 sell inputs to Country 1 ($a_{12} = a_{13} = 0.30$). Expansion in Country 1 pulls their output ($\ell_{21} = \ell_{31} = 0.3534$), and their own expansion barely reaches Country 1 ($\ell_{12} = 0.0118$). Countries 2 and 3 export primary inputs, and their activity is conditioned by Country 1's demand.

- **The import half is reversed.** The periphery imports almost nothing from Country 1 as an input ($a_{21} = a_{31} = 0.01$), whereas in Prebisch the periphery imports manufactures *from* the center.

The modification that makes Countries 2 and 3 depend on Country 1's output is

$$A_{\text{mod}} = \begin{pmatrix} 0.05 & 0.01 & 0.01 \\ 0.30 & 0.05 & 0.05 \\ 0.30 & 0.05 & 0.05 \end{pmatrix} = A'.$$

Then $L_{\text{mod}} = L'$, and the multipliers become $(1.0836, 1.4723, 1.4723)$: now the periphery has the larger backward linkages (it pulls inputs from the center), and Country 1 has the largest *forward* linkage (1.7668).

In the production reading of $\boldsymbol{\pi}$ (row-normalized W^P), Country 1's Perron weight rises from 0.0897 to 0.7241. Who is the "center" is thus not read from backward linkages alone. The center is the economy whose output others depend on: column sums of W^P, $(0.2587, 1.3706, 1.3706)$ before and $(2.2143, 0.3929, 0.3929)$ after.

(d) *Claim: with a symmetric A there is no center–periphery structure in the linkages.* If $A = A'$, then $L' = ((I - A')^{-1})' = (I - A)^{-1} = L$, so every country's backward linkage (column sum) equals its forward linkage (row sum), and $\ell_{ij} = \ell_{ji}$: an impulse from i to j is as strong as from j to i. For example, the symmetric part $(A + A')/2$ gives BL $=$ FL $= (1.4995, 1.3694, 1.3694)$. Center–periphery is a statement about *direction* (who conditions whom), so it lives in the antisymmetric part $(A - A')/2$. In the original A, BL $-$ FL $= (0.6832, -0.3416, -0.3416)$. Asymmetry is necessary but not sufficient: a periphery also needs low weight in others' recipes and a low π. The rigorous version is Chapter 16's.

Chapter 2

Welfare Theorems as Matrix Properties

All numbers below are printed by `python3 code/solutions/sol02.py`. Throughout, $\mathrm{Var}(\pi_i) = \frac{1}{N}\sum_i(\pi_i - \frac{1}{N})^2$ and $\Gamma = 1 - |\lambda_2|$ (Chapter 1, Section 1.7).

Exercise 2.1 Doubly stochastic check

All four matrices are row-stochastic and irreducible. For such W, $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ if and only if every column sums to one (Proposition 1.21), so the test is the column sums.

(a) Column sums $(1, 1, 1)$: doubly stochastic. $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$, $\mathrm{Var}(\pi_i) = 0$. (Also $\Gamma = 1$: the matrix is W^{CE}.)

(b) Column sums $(0.6 + 0.1 + 0.3,\ 0.3 + 0.6 + 0.1,\ 0.1 + 0.3 + 0.6) = (1, 1, 1)$. The matrix is a circulant, so it is doubly stochastic: $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$, $\mathrm{Var}(\pi_i) = 0$. Its other eigenvalues are $0.6 + 0.3\omega + 0.1\omega^2$ with $\omega = e^{\pm 2\pi i/3}$, i.e. $0.4 \pm 0.1732i$, of modulus 0.4359, so $\Gamma = 0.5641$. The matrix is asymmetric and non-anonymous, yet power is equal.

(c) Column sums $(1.3, 0.7)$: not doubly stochastic. With $W = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$, $a = 0.2$, $b = 0.5$, Chapter 3, Proposition 3.7(2) gives $\boldsymbol{\pi} = (b, a)/(a + b) = (5/7, 2/7) = (0.7143, 0.2857)$. Check: $\frac{5}{7} \cdot 0.8 + \frac{2}{7} \cdot 0.5 = \frac{4+1}{7} = \frac{5}{7}$. Then $\mathrm{Var}(\pi_i) = \frac{1}{2}[(\frac{5}{7} - \frac{1}{2})^2 + (\frac{2}{7} - \frac{1}{2})^2] = (\frac{3}{14})^2 = 0.04592$, and $N_{\mathrm{eff}} = 1.6897$. The other eigenvalue is $1 - a - b = 0.3$, so $\Gamma = 0.7$.

(d) The matrix is symmetric with unit row sums, hence doubly stochastic: $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$, $\mathrm{Var}(\pi_i) = 0$. Its eigenvalues are 1 and 0.1 (twice), so $\Gamma = 0.9$.

Lesson. Matrices (b) and (d) have equal power and different integration ($\Gamma = 0.5641$ and 0.9); (c) has unequal power and $\Gamma = 0.7$. Power and the gap are sep-

arate coordinates. *Common mistake:* concluding from asymmetry, as in (b), that power is unequal.

Exercise 2.2 Is there a clawback?

The setting is Example 2.15: $N = 4$, $\Omega = 20$, $\bar{d} = 3$, $\Phi = 8$, $\kappa = 0.1$ and $\boldsymbol{\pi} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6})$.

(a) A transfer τ of disagreement payoff from Agent 2 to Agent 3 gives $\mathbf{d}' = (3, 3-\tau, 3+\tau, 3)$. The total $\sum_j d'_j = 12$ is unchanged, so the surplus is still $\Phi = 8$ and $x'_i = d'_i + 8\pi_i$ (Proposition 1.9):

τ	x'_1	x'_2	x'_3	x'_4
0.5	5.6667	5.1667	4.8333	4.3333
1	5.6667	4.6667	5.3333	4.3333
2	5.6667	3.6667	6.3333	4.3333
3	5.6667	2.6667	7.3333	4.3333

Agent 3 receives exactly τ more than her baseline 4.3333 in every case, and Agent 2 exactly τ less: no clawback, as in Proposition 2.14(1). Checks: every row sums to 20, and $\max_i \kappa x'_i \leq 0.733 < 1$. The power terms $\pi_i \Phi = (2.667, 2.667, 1.333, 1.333)$ are untouched (part (2) of the proposition).

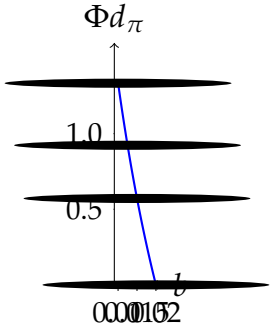
(b) With the gas block's cross-block entry equal to b (two entries per row) and the traditional block's fixed at 0.02, each gas agent puts total weight $\varepsilon_1 = 2b$ on the other block, and each traditional agent $\varepsilon_2 = 0.04$. Within-block weights are uniform, so the partition is exactly lumpable, and Chapter 6, Proposition 6.12, gives the gas block's mass

$$\pi_1 + \pi_2 = \frac{\varepsilon_2}{\varepsilon_1 + \varepsilon_2} = \frac{0.04}{2b + 0.04}.$$

Power is split equally within each block (by symmetry). With $\bar{d}$ equal, $d_\pi = (\pi_1 + \pi_2) - \frac{1}{2}$, and the per-round transfer that restores equal division costs

$$\Phi\, d_\pi = 8\Big(\frac{0.04}{2b + 0.04} - \frac{1}{2}\Big) = \frac{4(0.02 - b)}{b + 0.02}.$$

b	$\boldsymbol{\pi}$	$\mathbf{x}^*$	Φd_π	L_{power}
0.0100	(0.3333, 0.3333, 0.1667, 0.1667)	(5.6667, 5.6667, 4.3333, 4.3333)	1.3333	0.0889
0.0125	(0.3077, 0.3077, 0.1923, 0.1923)	(5.4615, 5.4615, 4.5385, 4.5385)	0.9231	0.0426
0.0150	(0.2857, 0.2857, 0.2143, 0.2143)	(5.2857, 5.2857, 4.7143, 4.7143)	0.5714	0.0163
0.0200	(0.25, 0.25, 0.25, 0.25)	(5, 5, 5, 5)	0	0



The curve is decreasing and convex. Rebalancing the *direction* of dependence ($b \to 0.02$) removes the recurring transfer altogether: at $b = 0.02$ the matrix is doubly stochastic.

(c) Multiplying all cross-block entries by 5 gives $\varepsilon_1 = 10b$ and $\varepsilon_2 = 0.2$. The block mass $\varepsilon_2/(\varepsilon_1 + \varepsilon_2) = 0.2/(10b + 0.2)$ is unchanged, so $\boldsymbol{\pi}$, $\mathbf{x}^*$, Φd_π and L_{power} are the same as in (b), for every b. What changes is the spectral gap. The lumped 2×2 chain has second eigenvalue $1 - \varepsilon_1 - \varepsilon_2$, and the within-block deviation vectors ($x_1 = -x_2$ or $x_3 = -x_4$) have eigenvalue 0, because within-block weights are equal. So the spectrum is $\{1, 1 - \varepsilon_1 - \varepsilon_2, 0, 0\}$ and $\Gamma = \varepsilon_1 + \varepsilon_2$:

b	0.0100	0.0125	0.0150	0.0200
Γ (original)	0.060	0.065	0.070	0.080
Γ (cross entries $\times 5$)	0.300	0.325	0.350	0.400

Why: power depends on the *ratio* of the cross-block flows, integration on their *level* (Chapter 3, Proposition 3.7). Scaling every cross flow by the same factor is "integration only" (Policy B1 of the example): it speeds up mixing and does nothing for the distribution.

Exercise 2.3 Welfare decomposition

Parameters: $m = \Omega/N = 1$, $1 - \kappa m = 0.5$, $\gamma = 0.5$, $\Phi = 1$. We use Theorem 2.16 with $c_j = \sum_{i \neq j} W_{ij}$ (off-diagonal column sums; the diagonal never enters (2.2)), $\alpha_j = 1 + \gamma c_j$, $H = N / \sum_j \alpha_j^{-1}$, and the exact formula (2.8).

(a) For $W = \frac{1}{3}\mathbf{1}\mathbf{1}'$: $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$, so $\mathrm{Var}(\pi_i) = 0$ and $L_{\text{power}} = 0$. Every $c_j = \frac{2}{3}$, so all $\alpha_j = \frac{4}{3} = H$ and $L_{\text{externality}} = \frac{(1-\kappa m)^2}{2\kappa}\sum_j (\alpha_j - H)^2/\alpha_j = 0$. Both terms of $L_{\text{interaction}}$ contain $\pi_j - \frac{1}{3}$ or $\mathrm{Cov}(\pi_j, c_j)$, so it vanishes too. The social optimum and the bargaining allocation are both $(1, 1, 1)$.

(b) The Perron vector is $\boldsymbol{\pi} = (0.2340, 0.4043, 0.3617)$, with $\mathrm{Var}(\pi_i) = 0.005231$ and $N_{\text{eff}} = 2.8651$. (The full column sums are $(0.8, 1.2, 1.0)$, so W is not doubly stochastic.) The off-diagonal column sums are

$$\mathbf{c} = (0.2 + 0.1,\ 0.4 + 0.3,\ 0.1 + 0.3) = (0.3, 0.7, 0.4),$$
$$\mathrm{Var}(c_j) = 0.028889, \qquad \mathrm{Cov}(\pi_j, c_j) = 0.010402,$$

so $\boldsymbol{\alpha} = (1.15, 1.35, 1.20)$ and $H = 1.227677$. The three terms:

$$\begin{aligned} L_{\text{externality}} &= \tfrac{0.25}{1}\textstyle\sum_j (\alpha_j - H)^2/\alpha_j = 0.004242, \\ L_{\text{power}} &= \tfrac{\kappa N}{2}\Phi^2\,\mathrm{Var}(\pi_i) = 0.75 \times 0.005231 = 0.003923, \\ L_{\text{interaction}} &= \underbrace{\tfrac{\kappa\gamma\Phi^2}{2}\textstyle\sum_j c_j(\pi_j - \frac{1}{3})^2}_{0.000850}\ \underbrace{-\gamma\Phi N(1-\kappa m)\,\mathrm{Cov}(\pi_j, c_j)}_{-0.007801} \\ &= -0.006951. \end{aligned}$$

The second-order approximation to $L_{\text{externality}}$ in (2.8) gives 0.005417; $\gamma = 0.5$ is not small. The total is $L = 0.001214$. The script confirms it directly as $\sum_j \alpha_j U(x_j^{\text{soc}}) - \sum_j \alpha_j U(x_j^{\text{barg}})$, with social optimum $\mathbf{x}^{\text{soc}} = (0.9325, 1.0906, 0.9769)$ (Proposition 2.9) and bargaining allocation $\mathbf{x}^{\text{barg}} = (0.9007, 1.0709, 1.0284)$. Both satisfy $\kappa x_j \leq 0.545 < 1$, so the interior hypothesis of the theorem holds.

(c) $L_{\text{interaction}} = -0.006951 < 0$. It is negative because $\mathrm{Cov}(\pi_j, c_j) > 0$: agent 2 is both the most powerful ($\pi_2 = 0.4043$) and the most cared about ($c_2 = 0.7$). The bargaining tilt toward her is in the same direction as the planner's tilt toward the agent whose consumption others value, so the two distortions partly cancel. The tilt even overshoots the social optimum for agent 3 ($1.0284 > 0.9769$), which is why $L_{\text{interaction}}$ cannot cancel everything. If power sat with agents nobody cares about (Cov < 0), the interaction term would be positive and the losses would compound. *Common mistake:* reading $L_{\text{interaction}} < 0$ as "power is good". It says only that, *under a social welfare function that counts the caring*, part of the power tilt is endorsed; whether such caring should count is normative (Chapter 2, Section 2.4).

Exercise 2.4 Spectral gap across topologies

The constructions, for $N = 8$:

- (a) $W^{CE} = \frac{1}{8}\mathbf{1}\mathbf{1}'$.
- (b) The undirected ring: $W_{i,i\pm1} = \frac{1}{2}$ (indices mod 8), no self-loops.
- (c) Two blocks: $W = \mathrm{diag}(\frac{1}{4}\mathbf{1}_4\mathbf{1}_4', \frac{1}{4}\mathbf{1}_4\mathbf{1}_4')$.
- (d) The star: the spokes put weight 1 on the hub, and the hub spreads $\frac{1}{7}$ over the spokes ($a = 1$, $s = 0$ in Chapter 6, Proposition 6.11).

	W			lazy $\frac{1}{2}(I+W)$		
	Γ	N_{eff}	$\mathrm{Var}(\pi)$	Γ	N_{eff}	$\mathrm{Var}(\pi)$
(a) W^{CE}	1	8	0	0.5	8	0
(b) ring	0	8	0	0.1464	8	0
(c) two blocks	0	—	—	0	—	—
(d) star	0	3.5	0.02009	0.5	3.5	0.02009

Details.

- (b) The ring is a circulant with eigenvalues $\cos(2\pi k/8)$, $k = 0, \ldots, 7$: $1, 0.7071, 0, -0.7071, -1, \ldots$. It is doubly stochastic, so $\boldsymbol{\pi}$ is uniform.
- (d) Power concentrates on the hub: $\pi_h = a/(a+1-s) = \frac{1}{2}$, each spoke has $\frac{1}{14} = 0.0714$, and $N_{\text{eff}} = 1/(\frac{1}{4} + 7 \cdot \frac{1}{196}) = 3.5$.
- Lazy versions: the eigenvalues map as $\lambda \mapsto (1+\lambda)/2$, so the lazy ring has $\lambda_2 = (1 + \cos\frac{\pi}{4})/2 = 0.8536$, and the lazy star has spectrum $\{1, 0.5^{(6)}, 0\}$, matching Proposition 6.11 with $a = s = \frac{1}{2}$. Laziness never changes $\boldsymbol{\pi}$, since $\boldsymbol{\pi}'\frac{1}{2}(I+W) = \boldsymbol{\pi}'$.

Why the ring and the star have $\Gamma = 0$ although they are connected. Both are irreducible but *periodic*: their graphs are bipartite. For the ring, colour the nodes alternately ±1; $\mathbf{x} = (1, -1, 1, -1, \ldots)'$ satisfies $(W\mathbf{x})_i = \frac{1}{2}(x_{i-1} + x_{i+1}) = -x_i$. For the star, $\mathbf{x}$ with $x_h = 1$ and $x_{\text{spoke}} = -1$ gives $(W\mathbf{x})_h = 7 \cdot \frac{1}{7}(-1) = -x_h$ and $(W\mathbf{x})_{\text{spoke}} = x_h = -x_{\text{spoke}}$. In both cases -1 is an eigenvalue, $|\lambda_2| = 1$, and $\Gamma = 0$: deviations oscillate forever instead of decaying (Chapter 6, Theorem 6.6, item 5). The directed ring $i \to i+1$ is periodic with period 8 and also has $\Gamma = 0$. A self-loop breaks the periodicity, which is what laziness does.

Why the two blocks have no unique $\boldsymbol{\pi}$. The matrix is reducible, and each block is a closed class with its own stationary vector $\frac{1}{4}\mathbf{1}_4$. Hence $(\frac{1}{4}\mathbf{1}_4', 0)$, $(0, \frac{1}{4}\mathbf{1}_4')$ and every convex combination $(t \cdot \frac{1}{4}\mathbf{1}_4', (1-t) \cdot \frac{1}{4}\mathbf{1}_4')$ are stationary; the eigenvalue 1 has multiplicity 2 (Chapter 3, Proposition 3.7(1)). N_{eff} ranges from 4 to 8 depending on t, so "power" is undefined until the blocks are coupled.

Do Γ and N_{eff} move together? No. The lazy W^{CE} and the lazy star have the same $\Gamma = 0.5$ with $N_{\text{eff}} = 8$ and 3.5. The ring and W^{CE} have the same $N_{\text{eff}} = 8$ with $\Gamma = 0$ and 1. Making a matrix lazy halves its distance from 1 in the spectrum (it lowers Γ of W^{CE} from 1 to 0.5 and raises that of the star from 0 to 0.5) without touching $\boldsymbol{\pi}$. This is Chapter 6, Proposition 6.16 in action.

Exercise 2.5 Block disconnection with data

Data (real results, with a stated limitation).

- **Value added.** Gross value added by sector, 2019, from INE's supply-use tables, base 2017, aggregated to the book's 12 sectors (Instituto Nacional de Estadística (INE), Bolivia, 2026) (`data/processed/bolivia_io_sector_facts.csv`, million current Bs).
- **Employment.** Employment by ISIC Rev. 4 section, 2019, from ILOSTAT (International Labour Organization, 2026), which processes INE's Encuesta Continua de Empleo microdata (`data/processed/ilostat_bolivia_emp_eco_sex.csv`, thousands of persons).

Both were retrieved 2026-11-05.

Method.

1. *Map sectors to ISIC sections.* Hydrocarbons and Mining together form section B. The other matches are: Agriculture A, Manufacturing C, Construction F, Commerce G, Transport H, Financial services K, Personal services I+R+S, Domestic services T. INE puts non-market education and health in Public administration, so Public administration and Other services are merged and matched to D+E+J+L+M+N+O+P+Q+U. The mapped employment adds to the ILOSTAT total of 5,622.1 thousand.
2. *Value added per worker* $y_g = \text{VA}_g / n_g$, in Bs per worker per year.
3. *Two blocks:* B (Hydrocarbons + Mining) and everything else. Compute the employment-weighted between-group variance $\sum_b \frac{n_b}{n}(\bar{y}_b - \bar{y})^2$ and the

within-group component of Appendix 2.7, using the nine sector means inside block 2.

Results.

	VA (million Bs)	workers (thousand)	VA per worker (Bs)
Hydrocarbons + Mining	34,923.3	75.2	464,610
Everything else		5,546.9	49,585
Economy		5,622.1	55,134

Value added per worker in the extractive block is 9.37 times that of the rest, and the block employs 1.34% of workers. Variance of VA per worker (Bs^2):

- between blocks: 2.2721×10^9;
- within block 2, across sectors: 1.9458×10^9 (driven by financial services, 378,118 Bs per worker, and public administration plus other services, 122,495);
- total: 4.2180×10^9.

The between-group share is 0.5387 and the ratio between/within is 1.1677. By Proposition 2.11, $L_{\text{block}} = \frac{\kappa}{2} n \cdot (\text{between variance})$, so the ratio of L_{block} to the within-group loss is the same 1.1677 for every κ. For a scale: with $\kappa = 10^{-6}$ per Bs ($\kappa y_g \leq 0.465 < 1$ for every sector), L_{block} is 1,136 utility units per worker, against 973 within.

Interpretation.

- *Levels versus logs.* The between-block term is large in levels because a tiny, extremely capital-intensive block sits far above the mean. In logs, which weight proportional gaps, the between share falls to 0.1477 (between 0.0828, within 0.4776). The "two-block" diagnosis is sensitive to the metric.
- *VA is not income.* Compensation of employees is only 5.95% of hydrocarbons' value added (29.13% in mining), and compensation per worker in the block is 76,662 Bs a year. Most of the block's VA per worker is rent and capital income, which reaches households through the state (Chapter 3), not through the workers.
- *Lower bound.* The within component here is across sectors only. True within-group dispersion across individuals is larger and needs household-survey microdata (INE's Encuesta de Hogares), which were not fetched. The between share reported is therefore an upper bound on the share in an individual-level decomposition.

Exercise 2.6 El Alto vs. Zona Sur

[ILLUSTRATIVE] The figures are the exercise's, not survey estimates.

(a) $\bar{\omega} = (5 \cdot 800 + 3 \cdot 4500)/8 = 2187.5$ Bs. By Proposition 2.11,

$$\begin{aligned} L_{\text{block}} &= \frac{0.0001}{2}\left[5(800 - 2187.5)^2 + 3(4500 - 2187.5)^2\right] \\ &= 0.00005\,(9{,}625{,}781.25 + 16{,}042{,}968.75) = 1283.4375. \end{aligned}$$

Satiation check: $\kappa\bar{\omega}_{\text{Alto}} = 0.08$ and $\kappa\bar{\omega}_{\text{Sur}} = 0.45$, both < 1.

(b) Each El Alto agent puts total weight $\varepsilon_1 = 3\varepsilon$ on Zona Sur and $(1 - 3\varepsilon)/5$ on each El Alto agent. Each Zona Sur agent puts $\varepsilon_2 = 5\varepsilon$ on El Alto and $(1 - 5\varepsilon)/3$ on each Zona Sur agent. The partition is exactly lumpable, so $1 - \varepsilon_1 - \varepsilon_2 = 1 - 8\varepsilon$ is an eigenvalue (Proposition 6.12). Any vector supported on one block and summing to zero there is mapped to zero, because within-block weights are equal. The spectrum is therefore $\{1, 1 - 8\varepsilon, 0^{(6)}\}$ and $\Gamma = 1 - |1 - 8\varepsilon|$:

- $\varepsilon = 0.02$: $\lambda_2 = 0.84$, $\Gamma = 0.16$; cross-block deviations halve in 3.976 periods.
- $\varepsilon = 0.15$: $\lambda_2 = -0.2$, $\Gamma = 0.80$; they halve in 0.431 periods, and the sign alternates.

L_{block} is 1283.4375 in both cases. Γ is a property of how fast an averaging process on W (beliefs, prices, norms) mixes across the blocks. L_{block} is a property of where *resources* are: it compares within-block equal division with full equal division and contains no W at all. A faster-mixing information network does not move a single boliviano from Zona Sur to El Alto. Integration lowers L_{block} only through a mechanism that makes resources cross: labour mobility, cross-block trade at better terms, fiscal transfers.

Two more facts. Column sums are $5 \cdot \frac{1-3\varepsilon}{5} + 3\varepsilon = 1$ for El Alto and $5\varepsilon + 3 \cdot \frac{1-5\varepsilon}{3} = 1$ for Zona Sur, so W is doubly stochastic and $\boldsymbol{\pi} = \frac{1}{8}\mathbf{1}$ ($N_{\text{eff}} = 8$) for every ε. Equivalently, the block mass $\varepsilon_2/(\varepsilon_1 + \varepsilon_2) = 5/8$ equals El Alto's population share. This economy has segregation and slow integration without any power concentration.

(c) ε is the weight each agent puts on the other block: in information terms, how much people in El Alto hear, see and rely on people in Zona Sur, and vice versa. Interventions that raise it create repeated cross-block contact:

- urban transport that cuts travel time between the blocks, such as the Mi Teleférico cable-car network linking El Alto and La Paz (opened 2014);

- socially mixed schools and universities;
- hiring and procurement that cross the blocks (public employment, supplier programmes);
- shared media and public spaces.

A strong answer distinguishes interventions that raise ε (contact, integration) from those that close the gap in block means (transfers, labour-market integration). Only the second lower L_{block} directly, though the first may help the second happen.

Exercise 2.7 Furtado's thesis formalized

(a) A two-block economy with self-reproducing disconnection. Aggregate the economy into two blocks, Modern (M) and Traditional (T), with

$$W_t = \begin{pmatrix} 1-\varepsilon_t & \varepsilon_t \\ \varepsilon_t & 1-\varepsilon_t \end{pmatrix}, \qquad \mathbf{x}^*(W_t) = W_t\boldsymbol{\omega},$$

so the traditional block's income is $x_T(\varepsilon) = \omega_T + \varepsilon(\omega_M - \omega_T)$: coupling lets part of the modern block's resources reach the traditional one. Linkages are built out of the traditional block's surplus over subsistence $\bar{c}$, with a cap $I_{\max}$ on what one period's surplus can finance, and depreciate at rate δ toward a floor ε_0 (minimal trade):

$$\varepsilon_{t+1} = f(\varepsilon_t) = (1-\delta)\varepsilon_t + \delta\varepsilon_0 + \sigma \min\{\max\{x_T(\varepsilon_t) - \bar{c}, 0\},\ I_{\max}\}.$$

This is $W_{t+1} = f(W_t, \mathbf{x}^*(W_t))$. [ILLUSTRATIVE] Parameters: $\omega_M = 8$ and $\omega_T = 3.2$ (Example 2.12), $\bar{c} = 3.5$, $\delta = 0.1$, $\sigma = 0.05$, $\varepsilon_0 = 0.02$, $I_{\max} = 0.5$.

The block-diagonal fixed point. The traditional block starts investing only above $\varepsilon_{\text{th}} = (\bar{c} - \omega_T)/(\omega_M - \omega_T) = 0.0625$. Below it, $f(\varepsilon) = (1-\delta)\varepsilon + \delta\varepsilon_0$, whose fixed point is $\varepsilon^* = \varepsilon_0 = 0.02$. There, traditional income is $3.2960 < \bar{c} = 3.5$: the structure produces an allocation too poor to finance the linkages that would change the structure. This W^* has weak coupling and, with the block means of Example 2.12, $L_{\text{block}} = 2.16 > 0$ (Proposition 2.11).

The map has two more fixed points. Between the threshold and saturation ($\varepsilon \le 0.1667$), $\delta(\varepsilon - \varepsilon_0) = \sigma(x_T(\varepsilon) - \bar{c})$ gives the middle one,

$$\hat{\varepsilon} = \frac{\sigma(\bar{c} - \omega_T) - \delta\varepsilon_0}{\sigma(\omega_M - \omega_T) - \delta} = 0.0929.$$

Above saturation, $\varepsilon_{\text{high}} = \varepsilon_0 + \sigma I_{\max}/\delta = 0.27$. The script verifies $f(\varepsilon) = \varepsilon$ at all three.

(b) Stability. f is piecewise linear with slope $1 - \delta = 0.9$ below the threshold and above saturation, and slope $1 - \delta + \sigma(\omega_M - \omega_T) = 1.14$ in between. A fixed point of a one-dimensional map is locally stable when $|f'(\varepsilon^*)| < 1$. So:

- $\varepsilon^* = 0.02$ is stable, with basin $[0, 0.0929)$;
- $\hat{\varepsilon} = 0.0929$ is unstable (the threshold between the two basins);
- $\varepsilon_{\text{high}} = 0.27$ is stable, with basin $(0.0929, 1]$.

Simulations: starting at 0.05 or 0.09 the coupling returns to 0.02; starting at 0.095 or 0.15 it converges to 0.27. In general, the underdevelopment fixed point W^* is stable when two conditions hold:

1. At W^* the allocation leaves the traditional block below its investment threshold: $\varepsilon_0(\omega_M - \omega_T) < \bar{c} - \omega_T$.
2. Small increases in coupling do not raise investment faster than linkages depreciate: $|1 - \delta + \sigma\, \partial x_T/\partial\varepsilon| < 1$ at W^*. Here $\partial(\text{investment})/\partial\varepsilon = 0$ below the threshold, so this reduces to $\delta > 0$.

For a matrix-valued map, the condition is that the Jacobian of $W \mapsto f(W, \mathbf{x}^*(W))$, restricted to perturbations E with $E\mathbf{1} = 0$ (which keep rows stochastic), has spectral radius below one at W^*. The case study's smooth map $F(\varepsilon) = \bar{\varepsilon}\varepsilon^2/(\varepsilon^2 + h^2)$ has the same three-fixed-point structure.

(c) Breaking out means leaving the basin of W^*. There are two ways.

1. *A big push in the state.* Raise ε past the unstable threshold, a one-off jump of at least $\hat{\varepsilon} - \varepsilon_0 = 0.0729$ in coupling. A one-off income transfer cannot do this here. It raises next period's coupling by at most $\sigma I_{\max} = 0.025$, and the script confirms that one-off transfers of 0.5 to 2 all return to $\varepsilon = 0.02$. This is the dynamic counterpart of Proposition 2.14: transfers move $\mathbf{x}$, not W.
2. *A change in parameters that destroys the low fixed point* (a bifurcation). A permanent rise of 0.2040 in ω_T, a lower subsistence threshold, or a higher floor ε_0 (a public linkage programme) all make condition 1 fail, so that only the high equilibrium remains.

In Furtado's terms, underdevelopment is not a stage but a basin of attraction, and development is a structural change large enough to cross its boundary. *Common*

mistake: identifying "breaking out" with growth of ω_M. Modern-sector growth reaches the traditional block only through the floor coupling: it raises $x_T(\varepsilon_0) = \omega_T + \varepsilon_0(\omega_M - \omega_T)$, and here it removes the trap only if $\omega_M > 18.20$, more than doubling the modern endowment. With $\varepsilon_0 = 0$, no growth of ω_M helps at all.

Chapter 3

Growth and Distribution in Matrix Form

All numbers below are printed by `python3 code/solutions/sol03.py`. Leontief inverses use the book's convention $L = (I - \gamma A')^{-1}$ with $\gamma = 1$ unless stated, and $W^P = D^{-1}A$ (Chapter 1, Section 1.7).

Exercise 3.1 Solow with two groups

(a) A young agent with wage w chooses saving s to maximize $\ln(w - s) + \theta \ln((1 + r)s)$ (Example 3.1). The first-order condition is $\frac{1}{w-s} = \frac{\theta}{s}$, so $s = \frac{\theta}{1+\theta} w$ and the savings rate is $\sigma(\theta) = \theta/(1 + \theta)$, equation (3.3). It does not depend on r, because log utility makes the income and substitution effects cancel. Then

$$\sigma(\theta^M) = \tfrac{2}{3} = 0.6667, \qquad \sigma(\theta^F) = \tfrac{0.5}{1.5} = 0.3333.$$

(b) $\bar{s} = \frac{1}{2}(0.6667 + 0.3333) = 0.5$.

(c) From (3.4), $K_{t+1} = \bar{s}(1 - \alpha)K_t^\alpha$. The map $K \mapsto \bar{s}(1 - \alpha)K^\alpha$ is increasing and strictly concave, with slope ∞ at 0 and slope $\to 0$ at ∞, so it has a unique positive fixed point, which is globally stable on $(0, \infty)$:

$$K^* = \left[\bar{s}(1 - \alpha)\right]^{1/(1-\alpha)} = (0.35)^{1/0.7} = 0.2232.$$

(d) With $\theta = 2$ for both groups, $\bar{s} = 2/3$ and $K^* = (0.4667)^{1/0.7} = 0.3366$. The ratio is $(0.5/0.6667)^{1/0.7} = 0.6630$: steady-state capital is 33.70% lower. Output $Y = K^\alpha$ is $1 - 0.6630^{0.3} = 11.60\%$ lower (0.6377 against 0.7213), and the wage $(1 - \alpha)K^\alpha$ falls from 0.5049 to 0.4464.

A caution a strong answer adds. "Lower K^*" is not "lower welfare". In this economy (no population growth, full depreciation), the golden-rule capital stock solves $\alpha K^{\alpha-1} = 1$, i.e. $K_{gr} = 0.3^{1/0.7} = 0.1791$. Both steady states lie *above* it: their gross marginal products are 0.8571 and 0.6429, both below 1, so both are dynamically inefficient. Steady-state aggregate consumption $C = K^{\alpha} - K$ is 0.4145 with the gender gap and 0.3847 without it. The "macroeconomic cost of the formation mechanism" is a cost in capital and output. Whether it is a welfare cost depends on the calibration, and on who bears it: women's lower old-age consumption, if θ^F is produced by the structure (Chapter 10), is the distributional object.

Exercise 3.2 Propagation in a 3-sector economy

(a) The row sums of A are $(0.35, 0.40, 0.25) < 1$, so the Neumann series converges. The inverse is

$$L = (I - A')^{-1} = \begin{pmatrix} 1.1623 & 0.2083 & 0.0877 \\ 0.2705 & 1.1806 & 0.1462 \\ 0.1096 & 0.2083 & 1.1404 \end{pmatrix}.$$

(b) $\Delta \mathbf{Y} = Le_2 = (0.2083, 1.1806, 0.2083)'$: sector 2 rises by 1.1806 (the impulse plus own-use feedback), and sectors 1 and 3 by 0.2083 each. The total is 1.5972.

(c) The multipliers (column sums of L) are $(1.5424, 1.5972, 1.3743)$, so sector 2 has the highest. It has the most input-intensive recipe (row sum 0.40), and its largest input is good 1 ($a_{21} = 0.15$), which itself buys heavily from sector 2 ($a_{12} = 0.20$). That two-sector loop $2 \to 1 \to 2$ recirculates demand. Sector 3 has the lightest recipe (0.25) and the lowest multiplier.

(d) $\rho(A) = 0.3386$. Row-normalizing,

$$W^P = \begin{pmatrix} 0.2857 & 0.5714 & 0.1429 \\ 0.375 & 0.25 & 0.375 \\ 0.2 & 0.4 & 0.4 \end{pmatrix},$$

with eigenvalues $1, -0.2416, 0.1774$, so $\Gamma(W^P) = 0.7584$. Its Perron vector is $\boldsymbol{\pi} = (0.2937, 0.3916, 0.3147)$, with $N_{\text{eff}} = 2.9529$ of 3. The cross-block mass of $\{1,2\}$ against $\{3\}$ is $\chi(A) = 0.35$. This economy is *connected*: impulses mix fast, and every sector puts at least 14% of its inputs on each other sector. As a check on Proposition 3.4: the mean off-diagonal Leontief entry is 0.1718, below the norm bound $\|A\|_2/(1 - \|A\|_2) = 0.5198$ ($\|A\|_2 = 0.3420$).

Exercise 3.3 The enclave

(a) With $a_{31} = a_{32} = a_{13} = a_{23} = 0.01$:

$$L_{\text{enc}} = \begin{pmatrix} 1.1541 & 0.1925 & 0.0150 \\ 0.2566 & 1.1541 & 0.0157 \\ 0.0157 & 0.0150 & 1.1115 \end{pmatrix}.$$

A unit shock to sector 3 gives $L_{\text{enc}} e_3 = (0.0150, 0.0157, 1.1115)'$: 0.0306 in total reaches sectors 1 and 2.

(b) In the connected case, $Le_3 = (0.0877, 0.1462, 1.1404)'$, i.e. 0.2339 reaches sectors 1 and 2. Propagation falls by a factor of 5.86 to sector 1, 9.33 to sector 2, and 7.64 in total. The reverse direction collapses too: shocks to sectors 1 and 2 now reach sector 3 by 0.0157 and 0.0150, against 0.1096 and 0.2083. The multipliers fall to $(1.4264, 1.3615, 1.1421)$.

(c) The new production network is

$$W^P_{\text{enc}} = \begin{pmatrix} 0.3226 & 0.6452 & 0.0323 \\ 0.5769 & 0.3846 & 0.0385 \\ 0.0833 & 0.0833 & 0.8333 \end{pmatrix},$$

with eigenvalues $1, 0.7978, -0.2573$. So $\Gamma = 0.2022$, down from 0.7584, and the second eigenvalue is now close to $+1$, the signature of weak coupling (Chapter 3, Proposition 3.7). The cross-block mass falls from 0.35 to $\chi = 0.0580$. Sector 3 now buys 16.67% of its inputs from the other block (before: 60%), and sectors 1 and 2 buy 3.23% and 3.85% from it (before: 14.29% and 37.5%). Power moves toward the core: the $\{1, 2\}$ block's Perron mass rises from 0.6853 to 0.8243, because sector 3 depends on it more (in share terms) than it depends on sector 3. Both indicators move here, but for different reasons. χ falls because the cross entries of A fell. Γ falls because, after row-normalization, the cross *shares* fell.

(d) This is the stylized version of Bolivia's gas sector as an enclave (Section 3.5): a sector with few domestic purchases and few domestic sales, whose growth barely reaches the rest of the economy. The measured economy is more nuanced (Exercise 1.7 of this manual):

- In INE's 2019 table, hydrocarbons buy only 0.225 of domestic inputs per unit of output and export 70% of their output, so cases 1 and 3 of the trickle-down intuition hold.

- But those few inputs come mostly from services, and the six-sector $\Gamma(W^P)$ is high (0.7467).

The enclave shows up in the size of the hydrocarbons column of L and in its low Perron weight, not in a small aggregate Γ (Section 3.4). *Common mistake:* inferring an enclave from Γ alone.

Exercise 3.4 Transfer vs. structural policy

Set-up ([ILLUSTRATIVE]; the exercise does not fix technology, so we take it from Example 3.2). Technology is $f_i = Z_i K_i^{\alpha_i} L_i^{1-\alpha_i}$ with $\alpha_M = 0.4$, $\alpha_T = 0.3$, $Z = (2, 1)$, $L_M = L_T = 1$. Saving and depreciation are $s = 0.2$, $\delta = 0.05$, and $A = A_{\text{dual}}$ of Example 3.6. Output is $\mathbf{Y} = (I - A')^{-1}\mathbf{f}(\mathbf{K})$ and capital accumulates out of each sector's *income*: $K_{i,t+1} = sI_i + (1-\delta)K_i$. Both policies start from the no-policy steady state $\mathbf{K}^* = (42.4315, 12.1740)$, with $\mathbf{Y}^* = (10.6079, 3.0435)$ and $Y_M/Y_T = 3.4854$. The productivity gap is measured by gross output per worker, Y_M/Y_T; value-added ratios are reported too.

(a) *Transfer:* $I_M = 0.9Y_M$, $I_T = Y_T + 0.1Y_M$. On impact, traditional income rises from 3.0435 to 4.1043 (by $1.0608 = 0.1Y_M$), and modern income falls to 9.5471. The productivity gap does not move on impact: neither A nor $\mathbf{K}$ has changed. Over time, higher saving out of the transfer raises K_T and Y_T (items 1–2 of Conjecture 3.9). After 10 periods $Y_M/Y_T = 3.3128$, and in the long run 2.9971. The income ratio falls at once to 2.3261.

(b) *Linkages:* the cross coefficients rise from $(0.03, 0.02)$ to 0.10, and the cost is the same $0.1Y_M$ each period, so $I_M = 0.9Y_M$ and $I_T = Y_T$.

$$L_{\text{dual}} = \begin{pmatrix} 1.1775 & 0.0294 \\ 0.0442 & 1.2511 \end{pmatrix} \longrightarrow L_{\text{link}} = \begin{pmatrix} 1.1940 & 0.1493 \\ 0.1493 & 1.2687 \end{pmatrix}.$$

Propagation from Modern to Traditional rises from 0.0442 to 0.1493 (a factor of 3.38), and the multipliers from $(1.2217, 1.2805)$ to $(1.3433, 1.4179)$. Because the new A applies to the existing $\mathbf{f}(\mathbf{K})$, the productivity ratio drops at once to 2.7374.

(c) After 10 periods:

Policy	Y_M/Y_T	VA_M/VA_T	I_M/I_T	total income	K_T
None	3.4854	3.6642	3.4854	13.6514	12.1740
Transfer	3.3128	3.4826	2.2396	13.5722	13.9504
Linkages	2.6514	2.8408	2.3863	13.9200	13.8206

Linkage policy closes the *productivity* gap: it is 2.6514 against 3.3128, and in the long run (cost continuing) 2.4941 against 2.9971. It also raises total income. The transfer closes the *income* gap slightly more (2.2396 against 2.3863) and lowers total income below the no-policy level. It shifts accumulation toward the traditional sector, where the marginal product of capital at the baseline steady state is lower (0.0522 against 0.0844 in **f**). A strong answer reports both rankings and names the objective. Transfers act on ω (who has what), and linkages act on A (who is connected to whom), as the chapter's intuition box says.

Caveats.

- The reduced form is demand-pull. A genuine cost-reducing technology shock would travel downstream through the price dual (Section 3.2).
- Gross output double-counts intermediate sales, which is why the value-added ratio is also shown.
- The model has no labour mobility, so items 3–5 of Conjecture 3.9 cannot be assessed here.

Exercise 3.5 Prebisch in the global I-O table (data exercise)

What was and was not fetched. The table the exercise asks for, a multi-region input-output table with Bolivia as a separate economy (Eora, or CEPAL's regional input-output matrices for Latin America), was *not* fetched. Eora requires registration, and Bolivia is in the rest-of-world block of the OECD ICIO tables (`data/README.md`, §9). Parts (b) and (c) therefore get the method and the expected result only. Parts (a), (d) and (e) need only a row-stochastic country network. For them we use a *trade-based proxy* for W^P, built from real data: Growth Lab bilateral goods trade, HS92, 2019 (The Growth Lab at Harvard University, 2026), retrieved 2026-11-05 (`data/processed/atlas_bilateral_selected_years.csv.gz`). Row i gives the shares of importer i's goods imports by exporter. This is the construction of `data/compute/compute_global_wp.py`, aggregated to six nodes (USA, Germany, China, Bolivia, Nigeria, rest of world). It is *not* an input-output table: it mixes intermediate, capital and final goods and has no gross output.

Method for the full exercise.

1. Take the MRIO table for one year: the flows $Z_{(c,s),(c',s')}$ (sales of sector s in country c to sector s' in c') and the gross outputs $x_{(c,s)}$.
2. Aggregate sectors within each country, and all other countries into "rest of world", by summing flows and outputs (`iolib.aggregate`).

3. Form A in the book's convention, $a_{ij} = Z_{ji}/x_i$; $L = (I - A')^{-1}$; and $W^P = D^{-1}A$.

4. Multipliers are the column sums of L, computed from A, never from W^P.

5. For a Bolivian productivity shock, use the price dual $\mathbf{p} = (I - A)^{-1}\mathbf{v}$: a cost reduction travels downstream to the users of Bolivian output (Section 3.2).

Results from the trade-based proxy (2019). Rows are importers and columns exporters:

	USA	DEU	CHN	BOL	NGA	ROW
USA	0	0.0566	0.1661	0.0002	0.0016	0.7754
DEU	0.0588	0	0.0898	0.0001	0.0018	0.8495
CHN	0.0623	0.0643	0	0.0002	0.0013	0.8719
BOL	0.0754	0.0201	0.1258	0	0.0000	0.7787
NGA	0.0672	0.0240	0.3475	0.0000	0	0.5612
ROW	0.1145	0.0943	0.1608	0.0006	0.0036	0.6262

(a) Column sums (how much the others rely on each country's goods): USA 0.3783, DEU 0.2594, CHN 0.8901, BOL 0.0011, NGA 0.0083 (ROW 4.4629). China and the United States have the highest. Bolivia's column is the smallest by two orders of magnitude. The full 230-economy matrix (`data/processed/global_wp_pi.csv`) gives the same ranking: China 30.75, USA 24.17, Germany 11.13, Nigeria 1.08, Bolivia 0.10.

(b) *Method only* (no gross output in the proxy). Expected result: in an MRIO the large, diversified economies (USA, Germany, China) have the highest output multipliers, because a larger share of their intermediate inputs is sourced domestically. Small commodity exporters (Bolivia, Nigeria) have low ones, because their inputs leak abroad. The measured domestic multipliers of Exercise 1.7 (Bolivia's hydrocarbons 1.324) are consistent with this.

(c) *Method only* for the propagation itself. A first-order exposure measure is available from the proxy: Bolivia's goods are 0.000201 of US imports and 0.000004 of Nigeria's (US imports from Bolivia were 462 million USD in 2019; Nigeria's, 0.19 million). Expected result: a Bolivian agricultural productivity shock propagates to both by orders of magnitude less than to Bolivia's own sectors, and about 50 times more to the USA than to Nigeria ($0.000201/0.000004 = 50.4$), in proportion to these exposures.

(d) Take center {USA, DEU, CHN} against periphery {BOL, NGA}, in the five-country system (rows renormalized over the five). Only 0.0340 of the flows cross the center–periphery boundary, and they run one way. Bolivia and Nigeria source 100% of their within-system imports from the center, while the center sources 0.82%, 1.28% and 1.15% (USA, DEU, CHN) of its within-system imports from the periphery. The six-node matrix has $\Gamma = 0.8004$: it mixes fast, because every row puts most of its weight on the rest of world. The aggregation matters: the 230-economy matrix has $\Gamma = 0.452$ (`data/README.md`, §8). So the network is *integrated* (large Γ) and *asymmetric* (one-way dependence). It is not block-diagonal.

(e) In the six-node proxy the Perron weights are USA 0.0923, DEU 0.0791, CHN 0.1346, BOL 0.00046, NGA 0.0029 (ROW 0.6906). The full 230-economy matrix gives USA 0.0889, DEU 0.0824, CHN 0.1302, BOL 0.000526, NGA 0.00286, close to world export shares (0.0905, 0.0802, 0.1422, 0.000463, 0.00296). This is consistent with a center–periphery structure: three economies hold most of the production-reading power, and the US weight is 169 times Bolivia's in the full matrix (202 in the six-node proxy). Two qualifications a strong answer adds:

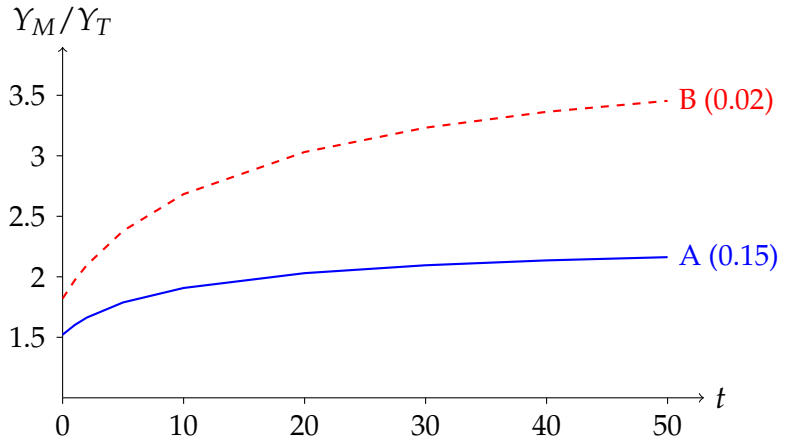
- In a trade-share network, π largely reproduces export shares (a gravity effect), so it measures size as much as position.
- The production reading of π (Chapter 1, Section 1.7) is a statement about long-run influence in the network, not about the terms-of-trade mechanism Prebisch emphasized (Chapter 16).

Exercise 3.6 Kuznets revisited

Set-up ([ILLUSTRATIVE]). $N = 2$, $f_i = Z_i K_i^{\alpha_i}$ with $L_i = 1$, $Z = (2, 1)$, $\alpha = (0.4, 0.3)$, $s = 0.2$, $\delta = 0.05$, $K_0 = (1, 1)$, and $\mathbf{Y}_t = (I - A')^{-1}\mathbf{f}(\mathbf{K}_t)$, $\mathbf{K}_{t+1} = s\mathbf{Y}_t + (1-\delta)\mathbf{K}_t$. The diagonal of A is $(0.15, 0.20)$, as in Example 3.6.

(a, b) The ratio Y_M/Y_T for Case A (off-diagonal 0.15) and Case B (off-diagonal 0.02):

t	0	1	2	5	10	20	30	40	50
A	1.5217	1.6015	1.6631	1.7883	1.9077	2.0302	2.0954	2.1357	2.1624
B	1.8202	1.9709	2.0967	2.3795	2.6834	3.0313	3.2324	3.3636	3.4543



Both ratios rise monotonically toward their long-run values, 2.2292 (A) and 3.7033 (B). The modern sector's advantage in Z and α makes it accumulate faster. Linkages pull the traditional sector up with it, so the long-run ratio in A is 0.6019 times that in B.

(c) With each sector's output assigned to its own workers ($L_M = L_T = 1$, labour shares $\frac{1}{2}$), the between-sector variance of log output per worker is $\frac{1}{4}(\ln y_M - \ln y_T)^2$. In Case A it rises monotonically from 0.0441 to 0.1487; in Case B from 0.0897 to 0.3842. Neither traces an inverted U: with fixed labour shares the index is a monotone function of the ratio, and the ratio is monotone. The Kuznets mechanism is missing from the model. In Kuznets the inverted U comes from the *reallocation of workers*, not from the sectoral gap. The between-group variance with modern-sector employment share w is $V_B = w(1-w)(\Delta \ln y)^2$. Holding the long-run Case-B log gap 1.3092 fixed, $V_B = 0.1543, 0.36, 0.4285, 0.36, 0.1543$ at $w = 0.1, 0.3, 0.5, 0.7, 0.9$: an inverted U in w. To get Kuznets, add labour mobility (a Lewis or Harris–Todaro block). *Common mistake:* reading a rising and then stable Y_M/Y_T as an inverted U.

(d) Long-run Y_M/Y_T falls monotonically with the cross-linkage strength c:

c	0	0.02	0.05	0.10	0.15	0.20	0.30	0.40
$\rho(A)$	0.2000	0.2070	0.2309	0.2781	0.3271	0.3766	0.4760	0.5758
Y_M/Y_T	4.2098	3.7033	3.1688	2.5964	2.2292	1.9718	1.6325	1.4171

Claim: with diminishing returns ($\alpha_i < 1$) and $\gamma\rho(A) < 1$, the ratio converges for every linkage strength, including $c = 0$.

Proof sketch. The growth map $G(\mathbf{K}) = SL\mathbf{f}(\mathbf{K}) + (I - \Delta)\mathbf{K}$ of (3.10) has the following properties:

- it is order-preserving, because $L \geq 0$ and f_i is increasing;

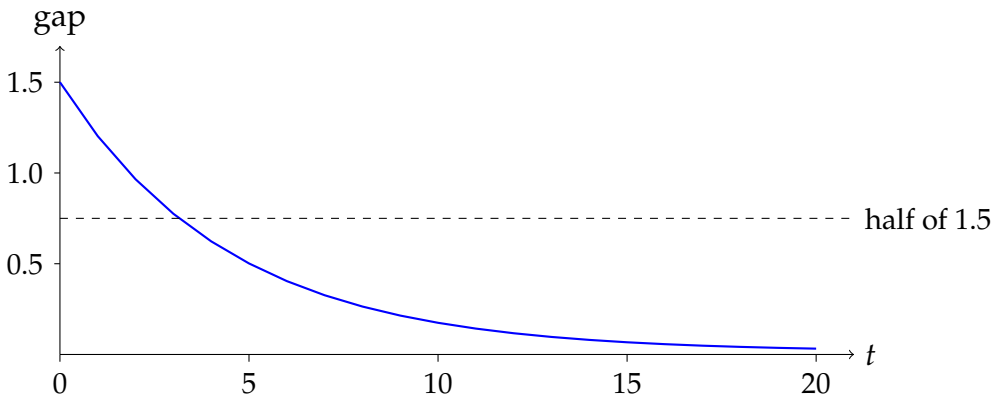
- it is strictly subhomogeneous, $G(t\mathbf{K}) > tG(\mathbf{K})$ for $t \in (0,1)$, because $t^{\alpha_i} > t$;
- it satisfies $G(\mathbf{K}) \ll \mathbf{K}$ for large $\mathbf{K}$, because $f_i/K_i \to 0$, and $G(\mathbf{K}) \gg \mathbf{K}$ for small $\mathbf{K} \gg 0$, because $f_i/K_i \to \infty$.

An order-preserving, strictly subhomogeneous map of the positive cone has at most one positive fixed point, and its orbits converge to it (Krause's theorem on concave monotone maps; the argument is a contraction in Thompson's part metric). Hence $\mathbf{K}_t \to \mathbf{K}^* \gg 0$, and $Y_M/Y_T \to$ a finite positive limit. Divergence needs constant returns. In the AK case ($\alpha_i = 1$, $Z = (0.3, 0.2)$), the decoupled economy ($c = 0$) has growth factors 1.0206 and 1, and the ratio grows without bound (83.15 at $t = 200$). Any $c > 0$ makes the growth matrix primitive, so the ratio converges to its Perron-direction value (11.72 for $c = 0.02$), though slowly.

Exercise 3.7 The formation connection

The law of motion is $\theta^F_{t+1} = 0.8\theta^F_t + 0.4 + 0.1(0.1\lambda - 0.1)$. With $\lambda = 0.7$ the signal term is -0.003, so the fixed point is $\theta^* = (0.4 - 0.003)/0.2 = 1.9850$, and the long-run gap is $\theta^M - \theta^* = 0.0150$. This is the formation multiplier $\mathcal{M} = \mu/(1-\rho) = 0.5$ (Definition 10.3) times the perceived-return shortfall $(1-\lambda)r = 0.03$. Deviations from θ^* decay as 0.8^t, so $\text{gap}_t = 0.015 + 1.485 \cdot 0.8^t$.

(a, b) The simulated gap $\theta^M - \theta^F_t$ for $t = 0, 1, \ldots, 11$ is 1.5, 1.203, 0.9654, 0.7753, 0.6233, 0.5016, 0.4043, 0.3264, 0.2641, 0.2143, 0.1745, 0.1426, falling to 0.0364 and 0.0321 at $t = 19, 20$. Meanwhile θ^F_t rises from 0.5 to 1.9679 at $t = 20$.



(c) The gap first falls below half its initial value (0.75) in generation 4 (0.6233; generation 3 is 0.7753). The continuous solution of $0.015 + 1.485 \cdot 0.8^t = 0.75$ is $t = 3.152$.

(d) With $\lambda = 1$ from $t = 10$ on, the fixed point becomes $\theta^* = 2 = \theta^M$, so the gap decays to zero as $\text{gap}_{10} \cdot 0.8^k$, starting from $\text{gap}_{10} = 0.1745$. It falls to

$0.1396, 0.1116, 0.0893, 0.0715$, so it first halves after 4 more generations. The half-life is $\ln 2/(-\ln 0.8) = 3.106$ generations, the same as in part (c): it depends only on the persistence $\rho = 0.8$, not on where the gap starts or on the signal correction. This is the persistence result of Chapter 10 (Theorem 10.7): with $J = \rho$ in this scalar case, deviations decay at rate ρ, so removing a distortion has its full effect only over several generations.

What a strong answer notices. Most of the gap closes even with $\lambda = 0.7$, because the anchor $\bar{\theta} = 2 = \theta^M$ pulls women toward men's patience. The distortion sustains only a gap of 0.015. The exercise therefore illustrates persistence, not the size of the distortion. With an anchor $\bar{\theta}$ itself formed by the structure (Chapter 10), the permanent gap would be much larger. *Common mistake:* crediting the signal correction at $t = 10$ with the closing of the gap.

Chapter 4

What the Standard Model Assumes

All numbers below are printed by `python3 code/solutions/sol04.py`. A1–A6 are the assumptions of Definition 4.1.

Exercise 4.1 Classifying failures

For each scenario, the answer names the violated assumptions, the formal object, and the one that matters most.

(a) *Landlord and three tenants, no outside labour market.*

- Violated: A4 (no anonymous market; terms are set by bilateral bargaining) and A5 (the landlord's position gives him a large π_i).
- Formal object: the surplus is divided by $x_i = d_i + \pi_i(\Omega - \sum_j d_j)$ (Proposition 1.9). With no outside labour market, the tenants' disagreement payoffs d_i are also low.
- Most important: A4/A5. Note that the loss is distributional, not a Pareto inefficiency: weighted Nash bargaining is efficient (Proposition 4.5). A1 may also fail (patron–client obligations), but it is not the defining feature.

(b) *Algorithmic content shifts purchasing behaviour.*

- Violated: A3 (a platform-shaped W^I that is star-like, with the platform at the hub; Section 4.4) and, if purchases move because tastes move rather than beliefs about products, A6.
- Most important: A6 if the shift persists after the content stops (preferences changed), A3 if it reverses (beliefs changed). A strong answer names this test: persistence after the signal is removed is the empirical signature of formation (Chapter 11).

(c) *Gas revenues redistributed through transfers, the gas sector an enclave.*

- Violated: A2. The production structure is treated as given, and policy acts on $\boldsymbol{\omega}$, not on A (Chapter 3, Section 3.5). The rents also give the state a central position in the fiscal network.
- Most important: A2. The transfers can equalize consumption without changing A, so the enclave persists.

(d) *A girl in Potosí develops new aspirations.*

- Violated: A6, through a change in A3: W^I changed (the phone), and θ_i responds via the formation equation (4.6).
- Most important: A6, the deepest assumption (Remark 4.8).

(e) *Two ethnic groups trade but do not intermarry, share schools or attend the same churches.*

- A4 roughly holds in goods markets (anonymous trade).
- A1 fails with a block-diagonal W^U (caring stays within groups), and A3 fails with a block-diagonal W^I (separate schools and churches, hence separate information ecosystems).
- Most important: A3. Block-diagonal W^I means the two groups converge to different consensuses, which is the formation-level mechanism of reproduction (Chapter 5, Section 5.5). Whether power is unequal depends on the cross-block flows, not on the segregation itself (Proposition 6.12).

Exercise 4.2 Measuring departure

(a) Column sums: $(0.5+0.1+0.2,\ 0.3+0.7+0.1,\ 0.2+0.2+0.7) = (0.8, 1.1, 1.1)$. W is row-stochastic but not doubly stochastic.

(b) $\delta_{\text{col}}(W) = \sqrt{(-0.2)^2 + 0.1^2 + 0.1^2} = \sqrt{0.06} = 0.2449$.

(c) Solving $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$, $\mathbf{1}'\boldsymbol{\pi} = 1$ gives $\boldsymbol{\pi} = (7, 11, 12)/30 = (0.2333, 0.3667, 0.4)$. Check: $0.5 \cdot \frac{7}{30} + 0.1 \cdot \frac{11}{30} + 0.2 \cdot \frac{12}{30} = \frac{3.5+1.1+2.4}{30} = \frac{7}{30}$. Then $\text{Var}(\pi_i) = 0.005185$ (the maximum for $N = 3$ is $2/9 = 0.2222$), $N_{\text{eff}} = 2.8662$, $N_{\text{eff}}/N = 0.9554$, and $d_\pi = 0.1$. The eigenvalues are $1, 0.5, 0.4$, so $\Gamma = 0.5$.

(d) With only the composite in hand, three departures can be measured:

- **A5 (power).** $\boldsymbol{\pi} \neq \frac{1}{3}\mathbf{1}$, but mildly: $N_{\text{eff}}/N = 0.9554$, $d_\pi = 0.1$.
- **A4 (anonymity).** The rows differ. The distance from the anonymous matrix with the same power is $\|W - \mathbf{1}\boldsymbol{\pi}'\|_F = 0.6683$, comparable to the entries themselves. The strong self-weights (0.5, 0.7, 0.7) say each sector interacts mostly with itself.
- **Integration.** $\Gamma = 0.5$, half the benchmark's.

The most severely violated assumption is therefore A4: interaction is far from anonymous, while power is close to equal. Two cautions belong in the answer. First, the composite does not identify *which* layer (W^U, W^P or W^I) causes the departure, so A1–A3 cannot be assessed from W alone (Proposition 4.6 needs the layers). Second, a "Bolivia" composite with three sectors would mix node sets: utility and information networks live on people, production on sectors, and a composite requires the same node set (Chapter 1, Section 1.7).

Exercise 4.3 The hierarchy in action

Set-up. $N = 4$, $\Omega = 4$ ($m = 1$), $\kappa = 0.5$, $\Phi = 1$, $\gamma_U = 0.5$. The utility network is zero-diagonal and row-stochastic, with agent 1 a patron whom everybody cares about:

$$W^U = \begin{pmatrix} 0 & 0.5 & 0.25 & 0.25 \\ 0.6 & 0 & 0.2 & 0.2 \\ 0.6 & 0.2 & 0 & 0.2 \\ 0.6 & 0.2 & 0.2 & 0 \end{pmatrix}, \qquad \mathbf{c} = (1.8, 0.9, 0.65, 0.65),\ \mathrm{Var}(c_j) = 0.2237.$$

The market layer is W^{CE} when A4 holds. When A4 fails, it is a market-power star with hub h: spokes put $a = 0.8$ on the hub and 0.2 on themselves, and the hub keeps $s = 0.8$. Its Perron weight is $\pi_h = a/(a + 1 - s) = 0.8$ (Proposition 6.11). The composite is $W = 0.2\,W^U + 0.8\,W^{\text{mkt}}$ (Proposition 4.6), and losses follow Theorem 2.16. Every allocation below satisfies $\kappa x_i < 1$.

(a) *A1 fails, A4 holds:* $W = 0.2W^U + 0.8W^{CE}$. Its column sums are $(1.16, 0.98, 0.93, 0.93)$, so $\boldsymbol{\pi} = (0.2857, 0.2473, 0.2335, 0.2335)$ is *not* uniform ($N_{\text{eff}} = 3.971$). Utility externalities enter power through the composite, even though the market is symmetric. The losses are $L_{\text{ext}} = 0.03233$, $L_{\text{power}} = 0.00046$ and $L_{\text{int}} = -0.00976$, so $L = 0.02302$. Uniformity is restored exactly when W^U is doubly stochastic: with the "ring of care" (each agent cares 0.5 about two neighbours), $\boldsymbol{\pi} = \frac{1}{4}\mathbf{1}$ (Proposition 1.21).

(b) *A4 fails, A1 holds* ($W^U = 0$, so the composite is the star alone, hub agent 4): $\boldsymbol{\pi} = (0.0667, 0.0667, 0.0667, 0.8)$, $N_{\text{eff}} = 1.531$. Power is not uniform, and the loss is purely distributional: $L = L_{\text{power}} = 0.10083$.

(c) *Both fail.* The answer depends on *who* holds market power, and that is the content of the interaction term.

	$\boldsymbol{\pi}$	L_{ext}	L_{power}	L_{int}	L
(c1) hub = agent 1 (the patron)	(0.6786, 0.1328, 0.0943, 0.0943)	0.03233	0.06147	−0.06908	0.02472
(c2) hub = agent 4	(0.1614, 0.0962, 0.0870, 0.6554)	0.03233	0.05561	+0.05502	0.14296

By symmetry, the single-failure losses are the same whichever agent is the hub: $L(a) + L(b) = 0.02302 + 0.10083 = 0.12385$.

- In (c1), market power sits with the agent everybody cares about: $\text{Cov}(\pi, c) = 0.11589 > 0$, $L_{\text{int}} < 0$, and the joint loss (0.02472) is far *smaller* than the sum.
- In (c2), it sits with an agent nobody cares about especially: $\text{Cov}(\pi, c) = -0.03508 < 0$, $L_{\text{int}} > 0$, and the joint loss (0.14296) *exceeds* the sum by 0.01910.

Answer: superadditivity is not a general property. Its sign is the sign of L_{int}, led by $-\gamma\Phi N(1 - \kappa m)\,\text{Cov}(\pi_j, c_j)$.

Caution. $L(c) - [L(a) + L(b)]$ is not exactly L_{int}: the composite changes $\boldsymbol{\pi}$ when layers are combined, so L_{power} in (c) differs from both (a) and (b). The clean test of interaction is the three-term decomposition at the same $\boldsymbol{\pi}$.

Exercise 4.4 Bolivia's six assumptions (data exercise)

Starting point. The stylized dual economy has $W^P = \begin{pmatrix} 0.833 & 0.167 \\ 0.091 & 0.909 \end{pmatrix}$, so $\Gamma = 0.2576$ and $\boldsymbol{\pi} = (0.3529, 0.6471)$, with $N_{\text{eff}}/N = 0.9204$. Chapter 3, Section 3.4, shows that the measured economy differs on the integration coordinate.

(a) *Assessment.*

- **A1 fails clearly.** Compadrazgo, kinship and community obligations are central to Bolivian economic life (Example 5.2 is a stylization), so $W^U \neq 0$.
- **A2 fails.** A changed substantially over time and was a policy object: the 2006 hydrocarbons nationalization, and industrialization plans for gas and lithium.

- **A3 fails.** Information ecosystems are segmented by language and region (Chapter 5).
- **A4 holds approximately** in some retail markets (urban food markets with many sellers) and **fails** in hydrocarbons, where YPFB is the dominant buyer and seller, and in thin rural markets.
- **A5 fails** in the production layer (below) and in wealth shares. Income inequality is high: INE's Gini of per-capita household income was 0.6 in 2005 and 0.426 in 2019.
- **A6** cannot be assessed without cohort data. The case for failure is the speed of change in aspirations and female participation, which Chapter 11 asks how to measure.

(b) *Departure measures* (Table 4.1). Two of the six can be computed from data in the repository; the rest were not fetched.

Computed (real results). Source: INE supply-use tables and input-output matrices (Instituto Nacional de Estadística (INE), Bolivia, 2026), as processed by `data/compute/compute_bolivia_io.py`, retrieved 2026-11-05. Domestic flows, book12 aggregation.

- **A2**, $\|\Delta A\|_F / \|A\|_F$ per decade, computed from A in the book's convention ($a_{ij} = Z_{ji}/x_i$), 12 sectors:
  - 0.3154 for 1990–2000;
  - 0.4152 per decade for 2000–2009 (0.3737 over nine years);
  - 0.3014 for 2003–2013;
  - 0.3300 per decade for 2017–2024 (0.2310 over seven years, on base 2017; 2023–2024 are preliminary).

  Never compare across the change of base (1990 to 2017). These are 30–42% per decade, *above* the chapter's illustrative 5–20% range. Part of this is measurement: the base-1990 make matrices are diagonal in most years, and aggregation changes the norm. But it confirms that A is not a constant.
- **A5 (production layer only)**, $N_{\text{eff}}(\boldsymbol{\pi})/N$ for 2019 W^P, input shares: 6.4795 of $N = 11$, i.e. $N_{\text{eff}}/N = 0.5890$. The power Gini is 0.4446, $d_\pi = 0.3228$, and $\pi_{\max} = 0.2702$ (other services). The domestic W^P also has $\Gamma = 0.4674$. These are measured values for one layer, not for the composite.

Not fetched (method only).

- *A4*, $d_\pi(W^{\text{mkt}})$: market shares by firm within markets (for example, from INE's industrial surveys or the national tax service), with $W^{\text{mkt}} = \mathbf{1}\mathbf{s}'$ built per market. Expected result: near the maximum $1 - 1/N$ in hydrocarbons distribution and electricity, low in urban food retail.
- *A3*, $N_{\text{eff}}(\boldsymbol{\pi}^I)/N$: audience shares by outlet (Kantar IBOPE ratings are paywalled; Latinobarómetro requires registration; `data/README.md`, §9).
- *A1*, $\gamma_U \|W^U\|_F$: survey modules on transfers and support networks (EH microdata; not fetched).
- *A6*, Ψ: cohort-by-year panels (Chapter 11).

(c) *Ranking.* The measures are in different units, so any ranking is a judgment. It should be stated as such and defended. On the available evidence:

1. **A2**: measured, large (≈ 0.3 per decade) and policy-driven.
2. **A5 in the production layer**: measured, moderate ($N_{\text{eff}}/N = 0.589$). Lower concentration than in the OECD comparators (`data/README.md`, §3.1).
3. **A3 and A1**: likely large, but unmeasured here.
4. **A4**: severe in specific markets, mild in others.
5. **A6**: unmeasured.

The most severe *documented* deviation is the instability and policy-dependence of the production structure (A2), together with the enclave position of hydrocarbons inside it ($\pi = 0.039$, 70% of output exported; Chapter 3). The stylized two-sector picture of a small $\Gamma(W^P)$ is *not* borne out by the 2019 table (0.467, not 0.26). *Common mistake:* ranking by comparing numbers in different units, for example a Γ against an N_{eff}/N.

Exercise 4.5 Sunkel's five processes

Model answer. Put the domestic agents and the foreign center on one node set: center C, domestic modern elite E, and traditional groups T_1 (urban informal) and T_2 (rural). The composite is W, and power $\boldsymbol{\pi}(W)$ divides surpluses by the allocation rule $x_i = \frac{\Omega}{N} + (\pi_i - \frac{1}{N})\Phi$ (Proposition 1.23). Each process is then a property of W:

1. *Development* (of the modern pole): a rising π_E and a rising x_E.
2. *Underdevelopment*: a falling block power $\pi_{T_1} + \pi_{T_2}$, and with it a falling x_T.
3. *Dependency*: domestic attention and inputs pointing abroad, $\sum_{i \neq C} \pi_i W_{iC} / \sum_{i \neq C} \pi_i$ (Dos Santos's "conditioned by another economy").
4. *Marginalization*: vanishing in-weights of T from the modern pole ($W_{ET} \to 0$), so $\pi_T \to 0$. Nobody depends on, cares about or listens to the marginalized.
5. *Spatial inequality*: a widening gap between block allocations, $x_E - \bar{x}_T$, i.e. a growing between-group term L_{block} (Proposition 2.11).

The single process is one rewiring path $W(t) = W_0 + tE$ with $E\mathbf{1} = 0$. It moves weight from the domestic cross-block entries (W_{ET}, W_{TE}) to the transnational ones (W_{EC}, W_{CE}): "transnational integration and national disintegration". By the perturbation lemma (Lemma 6.19), all five observables move together, with first-order effect $d\boldsymbol{\pi}'/dt = \boldsymbol{\pi}' E Z$, $Z = (I - W_0 + \mathbf{1}\boldsymbol{\pi}')^{-1}$.

A numerical example ([ILLUSTRATIVE]). Order (C, E, T_1, T_2):

$$W_0 = \begin{pmatrix} 0.90 & 0.10 & 0 & 0 \\ 0.20 & 0.40 & 0.20 & 0.20 \\ 0.05 & 0.35 & 0.40 & 0.20 \\ 0.05 & 0.25 & 0.20 & 0.50 \end{pmatrix}, \quad W_1 = \begin{pmatrix} 0.80 & 0.20 & 0 & 0 \\ 0.55 & 0.40 & 0.025 & 0.025 \\ 0.05 & 0.10 & 0.65 & 0.20 \\ 0.05 & 0.05 & 0.20 & 0.70 \end{pmatrix},$$

with $E = W_1 - W_0$, $\Omega = 4$ and $\Phi = 2$:

t	$\boldsymbol{\pi}$	dependency	$W_{CE} + W_{EC}$	$E \leftrightarrow T$ weight	π_T	$x_E - \bar{x}_T$	Γ(domestic)
0	(0.5447, 0.2114, 0.1138, 0.1301)	0.1196	0.300	1.000	0.2439	0.1789	0.7105
0.5	(0.5903, 0.2094, 0.0942, 0.1060)	0.2161	0.525	0.600	0.2003	0.2186	0.5380
1	(0.6701, 0.2350, 0.0452, 0.0497)	0.4062	0.750	0.200	0.0949	0.3751	0.1878

One parameter produces all five observables:

- the elite's power and allocation rise (π_E from 0.2114 to 0.2350; x_E from 0.9228 to 0.9701);
- dependency more than triples;
- the traditional blocks are marginalized (π_T falls from 0.2439 to 0.0949);
- the elite–traditional allocation gap doubles;
- the domestic network disintegrates (the gap of the renormalized domestic block falls from 0.7105 to 0.1878).

At $t = 0$ the lemma gives $d\boldsymbol{\pi}/dt = (0.1068, -0.0233, -0.0374, -0.0462)$, which matches finite differences. That is the formal content of "one process, five manifestations". The case study's caution applies: this is a formalization of Sunkel's thesis, not evidence for it. The empirical test is whether measured W (global I-O plus domestic I-O) moved along such a direction (Chapter 23).

Exercise 4.6 A6 is the deepest

Set-up ([ILLUSTRATIVE]). There are two groups of equal size: the modern group with income $y_E = 10$, and the traditional group, whose income depends on its aspiration/patience parameter θ through human-capital investment: $y_T = 4 + 2\theta + \tau$, where τ is a transfer from E to T. Welfare is utilitarian with $U(x) = x - 0.025x^2$ ($\kappa = 0.05$; every income is below satiation at 20), evaluated on consumption with a fixed criterion (see the caution below). The transfer is conditional on remaining in the traditional sector. Recipients therefore attend more to traditional-sector peers: the weight on them in W^I is $w(\tau) = 0.5 + 0.2\tau$, and perceived returns are $\hat{r} = wr_T + (1-w)r_M$, with $r_T = 0.05$, $r_M = 0.25$, $r^* = 0.15$. Preferences form by (4.6) with $\rho = 0.9$ and $\mu = 0.5$, so the steady state is $\theta^*(\tau) = \bar{\theta} + \mathcal{M}(\hat{r}(\tau) - r^*)$, with $\bar{\theta} = 1$ and $\mathcal{M} = \mu/(1-\rho) = 5$ (Definition 10.3).

(a) *Economist A* holds θ at $\theta^*(0) = 1$, so $y_T = 6 + \tau$. She equates marginal utilities, $1 - 0.05(10 - \tau) = 1 - 0.05(6 + \tau)$, so $T^* = \tau_A = 2$. Her welfare rises from 12.6 to 12.8.

(b) *Economist B* lets θ respond: $d\theta^*/d\tau = -\mathcal{M} \cdot 0.2 \cdot (r_M - r_T) = -0.2$, so in the long run $dy_T/d\tau = 1 - 2 \cdot 0.2 = 0.6$. At $\tau_A = 2$, $w = 0.9$, $\theta^* = 0.6$ and $y_T = 7.2$. The policy has changed W^I, which changed θ, which changed the mapping from resources to welfare.

(c) Under B's model, A's policy lowers long-run welfare from 12.6 to 12.304. The marginal effect of a first transfer is $dW_B/d\tau|_{0^+} = -U'(10) + 0.6\,U'(6) = -0.5 + 0.42 = -0.08 < 0$, so over $\tau \in [0, 3]$ B's optimum is $\tau = 0$. The welfare path under A's policy, starting from $\theta_0 = 1$, is 12.8, 12.7518, 12.7082 at $t = 0, 1, 2$; then 12.6008 at $t = 5$, 12.4806 at $t = 10$, 12.366 at $t = 20$ and 12.3257 at $t = 30$. It falls below the no-transfer level in generation 6. Short-run evaluations (one generation, θ fixed) confirm A's policy; long-run ones reverse it.

The mechanism is the conditionality, not the transfer as such. The same transfer unconditionally (w unaffected) keeps $\theta = 1$ and yields 12.8. Conditioned on education, so that it lowers the weight on traditional peers ($\omega = -0.1$), it raises θ to 1.2 and yields 13.036. A6 matters because, once preferences form on W^I, the *de-*

sign of a policy enters welfare through its effect on the network, not only through its size.

Caution. We evaluated welfare on consumption. Evaluating with each period's own preferences (patience θ_t) raises the endogenous-preference welfare problem of Chapter 12: the policy can make people prefer the outcome it produces. A rigorous answer states which criterion is used. *Common mistake:* "proving" the reversal by comparing utilities computed with different θ.

(d) The "growth first, redistribute later" consensus relies on the separation of efficiency and equity of the Second Welfare Theorem (Theorem 2.13): redistribution is a later, separable act. Without A6, three things change.

- *Timing matters*: the structure in place during growth forms preferences (ρ is persistence), and "later" redistribution meets a population whose θ was formed by the structure. With $\rho = 0.9$ the half-life of a formation effect is $\ln 2/(-\ln 0.9) = 6.579$ generations.
- *Form matters*: transfers that change W^I change the welfare they are meant to improve.
- *Structural policy is not separable*: building linkages and information access (changing A and W^I) is part of distribution, not something to do after it.

The consensus is right that transfers can move consumption one-for-one (Proposition 2.14). It is wrong to assume they leave the network, and hence future preferences and incomes, untouched.

Exercise 4.7 The spectral gap of the assumptions

Construction ([ILLUSTRATIVE]). $W^{(0)} = \frac{1}{8}\mathbf{1}\mathbf{1}'$, and $W^{(k)} = \frac{1}{k+1}(W^{(0)} + F_1 + \cdots + F_k)$ adds one row-stochastic feature layer at a time:

- F_1, a W^U cluster: agents 1–3 attend only to each other, and the others are anonymous;
- F_2, a W^P block structure $\{1..4\}, \{5..8\}$ with cross-block weight 0.05;
- F_3, a W^I hub: weight 0.5 on agent 1 plus a uniform remainder;
- F_4, a market hub: weight 0.4 on agent 2;
- F_5, a core–periphery layer: the core $\{1, 2\}$ keeps 0.9 in the core, and the periphery puts 0.7 on the core.

The allocation is $x_i = 1 + (\pi_i - \frac{1}{8})$ ($\Omega = 8$, $\Phi = 1$), and $\kappa x_i \le 0.5744 < 1$ throughout. In the misaligned variant the hubs and the core are agents 8, 6 and $\{7, 8\}$ instead.

		aligned (hubs on 1, 2)			misaligned	
k	added	Γ	N_{eff}	L_{power}	N_{eff}	L_{power}
0	—	1.0000	8.0000	0	8.0000	0
1	W^U cluster	0.6875	7.1176	0.00387	7.1176	0.00387
2	W^P blocks	0.5468	7.4235	0.00243	7.4235	0.00243
3	W^I hub	0.6601	6.3965	0.00783	7.2641	0.00317
4	market hub	0.7281	6.4787	0.00734	7.3523	0.00275
5	core–periphery	0.7567	5.7342	0.01235	6.9633	0.00465

(Γ in the misaligned variant is identical up to $k = 4$ and is 0.7620 at $k = 5$; the reason is given below.)

Do Γ and N_{eff} move together? No. The block layer ($k = 2$) *lowers* Γ (from 0.6875 to 0.5468) and *raises* N_{eff}, because it dilutes the cluster. The hub layers ($k = 3, 4$) *raise* Γ while they lower N_{eff}. Each hub layer has identical rows, $F = \mathbf{1}\mathbf{v}'$. For any row-stochastic W, a left eigenvector $\mathbf{u}$ with eigenvalue $\lambda \neq 1$ satisfies $\mathbf{u}'\mathbf{1} = 0$, so $\mathbf{u}'(aW + (1-a)\mathbf{1}\mathbf{v}') = a\lambda\mathbf{u}'$. Adding the layer therefore shrinks every non-unit eigenvalue by the factor $k/(k+1)$, *wherever* the hub is: $\Gamma(W^{(3)}) = 1 - 0.75(1 - 0.5468) = 0.6601$. This is why the aligned and misaligned variants share Γ until the core–periphery layer, which is not of this form. This is Chapter 6, Proposition 6.16 again.

Shape of the loss. The loss is neither linear, nor convex, nor concave in k. Its increments in the aligned case are $(0.0039, -0.0014, 0.0054, -0.0005, 0.005)$, with second differences of alternating sign. Three reasons:

1. $L_{\text{power}} = \frac{\kappa N}{2}\Phi^2 \operatorname{Var}(\pi)$ is quadratic in $\boldsymbol{\pi}$, so features that concentrate power on the *same* agents compound. Aligned features end at $L = 0.01235$, about 2.654 times the misaligned 0.00465.

2. Features that *dilute* (blocks, misaligned hubs) can lower the loss.

3. In this construction each new layer also lowers the weight of the earlier ones (equal weights $\frac{1}{k+1}$).

The number of structural features is not a sufficient statistic. *Common mistake:* attributing a fixed loss to each "feature" and adding them up.

Chapter 5

The Three Networks

All numbers below are printed by `python3 code/solutions/sol05.py`. Raw intensities K are row-normalized to $W = D^{-1}K$, and an all-zero row becomes a self-loop (Remark 5.3).

Exercise 5.1 Utility externalities: keeping up and falling behind

(a) With $U_i(x) = x - \frac{1}{2}x^2$ and $\mathbf{x} = (0.6, 0.4, 0.8)$: $U = (0.42, 0.32, 0.48)$. By (5.1),

$$V_2 = 0.32 + 0.5(0.25 \cdot 0.42 + 0.05 \cdot 0.48) = 0.32 + 0.5(0.105 + 0.024) = 0.3845,$$
$$V_3 = 0.48 + 0.5(0.02 \cdot 0.42 + 0.05 \cdot 0.32) = 0.48 + 0.5(0.0084 + 0.016) = 0.4922.$$

Check: $V_1 = 0.4728$, as in Example 5.2. Rosa's externality term is the largest (0.0645): she cares most about María, who has the second-highest private utility.

(b) Social welfare is $\sum_i V_i = \sum_j \alpha_j U(x_j)$, with $\alpha_j = 1 + \gamma_U c_j$ (Chapter 2, equation (2.4)). The column sums are $\mathbf{c} = (0.25 + 0.02,\ 0.30 + 0.05,\ 0.02 + 0.05) = (0.27, 0.35, 0.07)$, so $\boldsymbol{\alpha} = (1.135, 1.175, 1.035)$. The first-order conditions $\alpha_j(1 - x_j) = \nu$ and $\sum_j x_j = 1.8$ give $\nu = 1.2/\sum_j \alpha_j^{-1} = 0.44472$ and

$$\mathbf{x}^V = (0.6082,\ 0.6215,\ 0.5703), \qquad \mathbf{x}^U = (0.6, 0.6, 0.6).$$

Rosa, the most cared about ($c_2 = 0.35$), gets the most; Carlos, whom nobody much cares about, gets less than equal division. The gain in $\sum V$ over equal division is 0.000766, exactly the formula of Proposition 2.9 with $m = 0.6$, $\kappa = 1$. The cost in $\sum U$ is small: 1.259295 against 1.26. Satiation is not an issue ($x_j < 1$).

(c) As $\gamma_U \to 1$, the weights become $\boldsymbol{\alpha} = (1.27, 1.35, 1.07)$ and the tilt grows: $\mathbf{x}^V = (0.6163, 0.6391, 0.5446)$, with a welfare gain of 0.002841. The loss of equal

division grows roughly with $\gamma_U^2 \operatorname{Var}(c_j)$, as in (2.8). As $\gamma_U \to 0$, $\alpha_j \to 1$ and the two allocations coincide: at $\gamma_U = 0.1$, $\mathbf{x}^V = (0.6016, 0.6047, 0.5937)$, and at $\gamma_U = 0$, $\mathbf{x}^V = (0.6, 0.6, 0.6)$. This is the benchmark A1 of Chapter 4. *Common mistake:* using row sums (how much i cares) instead of column sums (how much j is cared about). Only the latter enter the planner's weights.

Exercise 5.2 Three networks for a factory town

(a) [ILLUSTRATIVE] Agents are Owner F, Worker W, Shopkeeper S and Mayor M. A plausible set of raw intensities (rows: who cares about, depends on, or listens to whom):

$$K^U = \begin{pmatrix} 0 & .1 & .1 & .3 \\ .1 & 0 & .4 & .1 \\ .1 & .3 & 0 & .1 \\ .3 & .2 & .2 & 0 \end{pmatrix}, \; K^P = \begin{pmatrix} .1 & .5 & .05 & .1 \\ .6 & .2 & .1 & .1 \\ .3 & 0 & .3 & .1 \\ .5 & .1 & .2 & .2 \end{pmatrix}, \; K^I = \begin{pmatrix} .5 & .1 & .1 & .3 \\ .3 & .2 & .3 & .2 \\ .2 & .3 & .3 & .2 \\ .4 & .1 & .2 & .3 \end{pmatrix}.$$

The owner cares about the mayor (politics). The worker and the shopkeeper are neighbours. The owner's production depends on labour; the worker's on the factory (capital and employment); the shop on the factory's products; and the municipality on the factory's taxes. Row-normalizing gives, for example, $W^P_{F\cdot} = (0.1333, 0.6667, 0.0667, 0.1333)$ and $W^U_{W\cdot} = (0.1667, 0, 0.6667, 0.1667)$. The layer Perron vectors are:

- $\boldsymbol{\pi}(W^U) = (0.202, 0.2801, 0.2917, 0.2262)$;
- $\boldsymbol{\pi}(W^P) = (0.3826, 0.3354, 0.1497, 0.1324)$;
- $\boldsymbol{\pi}(W^I) = (0.3835, 0.1548, 0.1968, 0.2648)$.

(b) The composite with $\alpha = \frac{1}{3}$ each is

$$W = \begin{pmatrix} 0.2111 & 0.3222 & 0.1222 & 0.3444 \\ 0.3556 & 0.1333 & 0.3556 & 0.1556 \\ 0.2762 & 0.3 & 0.2429 & 0.181 \\ 0.4429 & 0.1619 & 0.2286 & 0.1667 \end{pmatrix}.$$

Its column sums are $(1.2857, 0.9175, 0.9492, 0.8476)$.

(c) $\boldsymbol{\pi} = (0.3118, 0.2367, 0.2287, 0.2227)$, with $N_{\text{eff}} = 3.9186$ and $\Gamma = 0.7236$. The owner has the most power. By Proposition 5.11, his power is the power directed

at him through all three channels: everyone depends on his output (W^P column F: 0.6, 0.4286, 0.5), the mayor listens to him (0.4), and the mayor cares about him (0.4286). The mayor has the smallest column sum in the composite (0.8476), so he comes last.

(d) When the factory closes, column F of K^P goes to zero (nobody can depend on output that no longer exists), and so does row F (the factory buys no inputs). The owner's production row becomes a self-loop, and the other rows are renormalized: $W^P_{W\cdot} = (0, 0.5, 0.25, 0.25)$, $W^P_{S\cdot} = (0, 0, 0.75, 0.25)$, $W^P_{M\cdot} = (0, 0.2, 0.4, 0.4)$. The new Perron vector is $\boldsymbol{\pi} = (0.2721, 0.2065, 0.2802, 0.2411)$. The owner loses 0.0397; the worker loses 0.0302 (the factory was the main user of his labour); the shopkeeper gains 0.0515 and becomes the most powerful agent, because the production weight shifts onto her shop. Power becomes *less* concentrated ($N_{\text{eff}} = 3.9468$), and the town integrates more slowly ($\Gamma = 0.6103$). The owner keeps a residual weight through W^U and W^I: losing the factory does not erase his social and informational position at once.

Exercise 5.3 DeGroot learning

(a) Row sums: $0.5 + 0.3 + 0.2 = 0.1 + 0.6 + 0.3 = 0.2 + 0.2 + 0.6 = 1$, so W^I is row-stochastic.

(b) The powers are

$$(W^I)^2 = \begin{pmatrix} 0.32 & 0.37 & 0.31 \\ 0.17 & 0.45 & 0.38 \\ 0.24 & 0.30 & 0.46 \end{pmatrix}, \quad (W^I)^5 = \begin{pmatrix} 0.2341 & 0.3751 & 0.3908 \\ 0.2287 & 0.3740 & 0.3973 \\ 0.2353 & 0.3685 & 0.3962 \end{pmatrix}.$$

The rows become more similar: the largest within-column spread falls from 0.15 at $t = 2$ to 0.0066 at $t = 5$. The rows converge to $\boldsymbol{\pi}'$ (Proposition 5.7).

(c) Solving $\boldsymbol{\pi}' W^I = \boldsymbol{\pi}'$ gives $\boldsymbol{\pi} = (10, 16, 17)/43 = (0.2326, 0.3721, 0.3953)$. Agent 3 has the most epistemic power, as in Appendix 5.9. Power iteration from $\frac{1}{3}\mathbf{1}$ gives $(0.267, 0.367, 0.367)$, $(0.243, 0.373, 0.383)$ and $(0.233, 0.373, 0.395)$ at $t = 1, 2, 5$. The other eigenvalues are $0.35 \pm 0.0866i$, of modulus 0.3606, so $\Gamma = 0.6394$.

(d) With $\mathbf{b}^{(0)} = (0, 0.5, 1)'$, $\mathbf{b}^{(t)} = (W^I)^t \mathbf{b}^{(0)}$:

t	$\mathbf{b}^{(t)}$	$\max_i \lvert b_i^{(t)} - b^* \rvert$
1	(0.35, 0.6, 0.7)	2.31×10^{-1}
2	(0.495, 0.605, 0.61)	8.64×10^{-2}
5	(0.5783, 0.5843, 0.5804)	3.08×10^{-3}
10	(0.5814, 0.5814, 0.5814)	1.96×10^{-5}
50	(0.5814, 0.5814, 0.5814)	7.77×10^{-16}

The limit is $\boldsymbol{\pi}'\mathbf{b}^{(0)} = \frac{16 \cdot 0.5 + 17}{43} = 0.5814$, as in (5.10). The errors shrink roughly by $|\lambda_2|^t = 0.3606^t$ per round, which is the geometric rate of Appendix 5.9.

Exercise 5.4 The phone as structural change

(a) [ILLUSTRATIVE] The agents are Mother, Friend, Cousin, TikTok, Teacher. The pre-phone network has the Mother row the exercise gives. Cousin and TikTok are unreachable from Potosí: nobody there listens to them, and the cousin hears nobody in Potosí (TikTok did not exist; it is coded as a self-loop).

$$W^I_{\text{pre}} = \begin{pmatrix} .6 & .2 & 0 & 0 & .2 \\ .3 & .5 & 0 & 0 & .2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ .2 & .1 & 0 & 0 & .7 \end{pmatrix}, \qquad W^I_{\text{post}} = \begin{pmatrix} .55 & .15 & .05 & .05 & .2 \\ .2 & .4 & .05 & .25 & .1 \\ .1 & 0 & .8 & .1 & 0 \\ 0 & .05 & .05 & .85 & .05 \\ .2 & .1 & .05 & .05 & .6 \end{pmatrix}.$$

Post-phone, the mother and the teacher glance at the cousin and TikTok, the friend watches TikTok heavily, the cousin stays in touch by WhatsApp, and the feed responds weakly to local creators. Luisa's own rows are those of Example 5.8: $(0.5, 0.3, 0, 0, 0.2)$ pre-phone and $(0.35, 0.25, 0.15, 0.15, 0.10)$ post-phone. Nobody in the five-agent network listens to Luisa, so she is a *listener* outside the matrix.

(b) *Pre-phone*, W^I_{pre} is reducible: the eigenvalue 1 has multiplicity 3, one for each closed class {Mother, Friend, Teacher}, {Cousin}, {TikTok}. The Perron vector is not unique (Chapter 3, Proposition 3.7(1)). Within the Potosí class it is $(0.3714, 0.2286, 0.4)$ on (Mother, Friend, Teacher): the teacher, whom both others heed, has the most epistemic power. *Post-phone*, the network is irreducible and aperiodic, with

$$\boldsymbol{\pi} = (0.1548,\ 0.097,\ 0.2,\ 0.397,\ 0.1513), \qquad N_{\text{eff}} = 3.9395, \quad \Gamma = 0.2315.$$

Epistemic power shifts massively to TikTok (0.397) and the cousin (0.2). They are

listened to a little by everyone and listen mostly to themselves: the self-reliance effect of Example 5.12.

(c) With $\mathbf{b}^{(0)} = (0.3, 0.3, 0.8, 0.6, 0.7)$:

- *Pre-phone.* There is no single consensus. The Potosí class converges to $(0.3714, 0.2286, 0.4) \cdot (0.3, 0.3, 0.7) = 0.46$, while the cousin stays at 0.8 and TikTok at 0.6. Luisa listens only to Potosí, so her long-run belief is 0.46.
- *Post-phone.* Everyone converges to $\boldsymbol{\pi}'\mathbf{b}^{(0)} = 0.5796$, and so does Luisa.

(d) The phone raises Luisa's long-run belief about returns to education from 0.46 to 0.5796. The decomposition is the instructive part.

- If only *Luisa* got a phone (her post-phone row, everyone else's pre-phone rows), her long-run belief would be 0.532: 0.70×0.46 from Potosí plus $0.15 \times 0.8 + 0.15 \times 0.6$ from the cousin and TikTok, which keep their own beliefs.
- Once the *network* has changed, Luisa's own row is irrelevant to her long-run belief: with the pre-phone row she still converges to 0.5796. A listener whom nobody listens to has zero Perron weight (adding Luisa as a sixth agent gives her $\pi = 0$) and inherits the consensus.

The structural change that matters is in other people's rows. Whether the higher belief changes Luisa's θ is the formation question of Chapter 10.

Exercise 5.5 When networks align

(a) [ILLUSTRATIVE] A village with

$$W = \begin{pmatrix} 0.4 & 0.3 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.1 & 0.2 \\ 0.2 & 0.1 & 0.4 & 0.3 \\ 0.1 & 0.2 & 0.3 & 0.4 \end{pmatrix} = W^U = W^P = W^I.$$

The composite is $\alpha_U W + \alpha_P W + \alpha_I W = W$ for every $\boldsymbol{\alpha}$, because the weights sum to one. W is symmetric with unit row sums, hence doubly stochastic: $\boldsymbol{\pi} = \frac{1}{4}\mathbf{1}$, and $\Gamma = 0.6$. In this village no agent has more structural power than another, through any channel. (Using $W^U = W$ with a nonzero diagonal is a simplification; Remark 5.3.)

(b) Agent 1 starts importing: half of its purchases from agent 2 (0.15 of 0.3) are replaced by imports. Imports are not a domestic node, so the domestic production row loses mass, and after renormalization (Definition 5.10) it is $W^P_{1\cdot} = (0.4706, 0.1765, 0.2353, 0.1176)$. Only W^P changes. With equal weights the composite's first row becomes $(0.4235, 0.2588, 0.2118, 0.1059)$, and

$$\boldsymbol{\pi} = (0.2556,\ 0.2368,\ 0.2556,\ 0.2519) \quad (N_{\text{eff}} = 3.9962,\ \Gamma = 0.6101).$$

Agent 2, the displaced supplier, loses power (-0.0132), and the others gain. With a heavier production weight, $\boldsymbol{\alpha} = (0.2, 0.5, 0.3)$, agent 2 loses 0.02. In Leontief terms, take domestic technical coefficients $A = 0.5\,K$ (scaled so that $\rho(A) < 1$). Agent 1's output multiplier falls from 2 to 1.8032, because imported inputs leak. Agent 2's forward linkage falls from 2 to 1.8032. The other multipliers fall slightly too (1.9582, 1.9684, 1.9765), because the cascade now leaks.

(c) Entering the global supply chain breaks the alignment that made the village simple. The production network now points partly outside the village, while the utility and information networks still point inside. Three things follow:

- a leakage of demand (lower multipliers);
- a redistribution of local power away from the agents whose sales are displaced;
- a divergence between the channels: agent 2 is still cared about and listened to, but less depended upon.

This is the first step of Dos Santos's third regime (case study): bridging links outward that do not connect inward. *Common mistake:* representing the import as a new column in W^P alone. The composite then mixes node sets (Chapter 1, Section 1.7). The outside world must be added as a node in every layer, or kept out of all of them.

Exercise 5.6 Prebisch in W^I

Construction ([ILLUSTRATIVE]). Agents 1–2 are the center and 3–6 the periphery. Each center agent puts 0.45 on each center agent and 0.025 on each periphery agent, so total weight $\varepsilon_1 = 0.1$ on the periphery. Each periphery agent puts 0.35 on each center agent and 0.075 on each periphery agent, so $\varepsilon_2 = 0.7$ on the center. Within-block weights are uniform, so the partition is exactly lumpable.

(a) By Proposition 6.12, the center's mass is $\varepsilon_2/(\varepsilon_1+\varepsilon_2) = 0.7/0.8 = 0.875$: $\boldsymbol{\pi} = (0.4375, 0.4375, 0.0313, 0.0313, 0.0313, 0.0313)$, with $N_{\text{eff}} = 2.5859$ and $\Gamma = 0.8$. The two center countries hold 87.5% of epistemic power, against $1/3$ under uniformity.

(b) With $\mathbf{b}^{(0)} = (0, 0, 1, 1, 1, 1)$, the consensus is $\boldsymbol{\pi}'\mathbf{b}^{(0)} = 0.125$. After 30 rounds every agent holds 0.125.

(c) The center determines 87.5% of the consensus, although it is a third of the agents. This is *consistent with* a DeGroot reading of the Washington Consensus (Chapter 5's Prebisch box): the periphery's long-run view converges close to the center's. But the model does not show that the actual consensus arose this way. That would require measuring W^I and the initial beliefs, and the chapter's case study is explicit that it offers no such evidence.

(d) The periphery's belief carries weight $\varepsilon_1/(\varepsilon_1+\varepsilon_2)$ in the consensus. It "survives" (dominates the consensus) if and only if $\varepsilon_1 > \varepsilon_2$: the center must listen to the periphery more than the periphery listens to the center.

ε_1	ε_2	consensus	Γ
0.10	0.70	0.1250	0.8
0.10	0.30	0.2500	0.4
0.40	0.30	0.5714	0.7
0.70	0.10	0.8750	0.8

Two routes lead there: the periphery listens less to the center (lower ε_2), or the center listens more to the periphery (higher ε_1). Only the *ratio* matters: rows 1 and 4 have the same Γ and opposite consensuses. In the limit $\varepsilon_2 \to 0$ the periphery becomes a closed class and keeps its own belief, but there is then no common consensus. Integration (Γ) is not what protects the periphery's view; the direction of attention is.

Exercise 5.7 Composite power in the La Paz model

(a) Hold $\alpha_U/\alpha_I = 0.2/0.3$, so $\alpha_U = 0.4(1-\alpha_P)$ and $\alpha_I = 0.6(1-\alpha_P)$. With W^U, W^P and W^I from Example 5.12:

α_P	0	0.2	0.4	0.5	0.6	0.8	0.9	0.99	1
π_3	0.4539	0.4870	0.5323	0.5625	0.6005	0.7198	0.8239	0.9770	1.0000
Γ	0.3323	0.2813	0.2299	0.2040	0.1781	0.1260	0.0998	0.0762	0.0735

π_3 increases monotonically in α_P (on a grid of 1,001 points). Its minimum is 0.4539 at $\alpha_P = 0$, and its maximum 1 at $\alpha_P = 1$. At $\alpha_P = 0.5$ we recover the example's $(0.2108, 0.2267, 0.5625)$. At $\alpha_P = 1$ the composite is W^P, in which the vendor is an *absorbing* state: her row is $(0, 0, 1)$ and the Minister's row leaks $1/6$ to her. So $\boldsymbol{\pi} = e_3$, the unique stationary vector of a chain with one closed class.

(b) The vendor matters most in an economy organized around production dependence (high α_P). The reason is not that others depend heavily on her; it is that weight flows into her and none flows out. She is least prominent when information and care dominate, but even then she leads (0.4539), because her column of W^I is the heaviest. "Invisible" does not happen in this example. It would require that nobody depend on, care about or listen to her, i.e. a zero column in every layer, which gives $\pi_3 = 0$ (marginalization; Exercise 4.5 of this manual).

(c) [ILLUSTRATIVE] Luisa (student) is agent 4, with raw rows:

- utility: she cares about the vendor (her mother) 0.5, and the vendor now cares about her 0.6;
- production: she helps at the stall, so the vendor's row gains 0.1 on Luisa, and Luisa's own production row is empty (a self-loop);
- information: Luisa listens to Minister 0.1, Executive 0.2, Vendor 0.3 and herself 0.4, and the vendor listens to her 0.1.

Other agents' rows are unchanged. The composite $W_4 = 0.2W^U + 0.5W^P + 0.3W^I$ has

$$\boldsymbol{\pi} = (0.2058,\ 0.2332,\ 0.29,\ 0.2709), \qquad N_{\text{eff}} = 3.9327.$$

The vendor's power falls from 0.5625 to 0.29: she now sends weight to Luisa through all three channels, so weight no longer accumulates in her node. The household {Vendor, Luisa} holds 0.561, almost exactly the vendor's former share. Including a dependent redistributes power within the household rather than adding to it.

(d) Replacing the vendor's self-loop in the normalized W^U by $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ gives $\boldsymbol{\pi} = (0.3125, 0.3471, 0.3404)$: π_3 falls by 0.222, from 0.5625. If the production self-loop is replaced too, π_3 falls to 0.1713. The vendor's top rank in the example is largely an artifact of how all-zero rows are coded. The self-loop convention makes a person who cares about nobody count as fully self-reliant, and the Perron vector rewards self-reliance. A strong answer reports the measure's sensitivity to this convention and treats "power" attributed to autonomous agents with caution, as the example's last paragraph warns.

Exercise 5.8 From story to matrix: the chain from gas well to school (data)

Data (real results). INE Bolivia, supply-use tables, base 2017, year 2019, 39 industries × 42 products (Instituto Nacional de Estadística (INE), Bolivia, 2026), retrieved 2026-11-05. They are converted to industry-by-industry flows at basic prices by the repository's own code (`data/compute/compute_bolivia_io.py`: `load_b17`, `sut_to_iot`). Z^{tot} includes imported inputs; Z^{dom} removes them by import proportionality. Coefficients follow the book's convention $a_{ij} = Z_{ji}/x_i$.

(a) *The links as entries of A and W^P.* In the table, a is the input per unit of the buyer's output, and W^P is the domestic share of the buyer's inputs:

link	entry	a (total)	a (domestic)	imported share	W^P
gas well → processing (refining)	$a_{17,05}$	0.3616	0.3616	0.0000	0.5855
gas → thermoelectric plant	$a_{23,05}$	0.0503	0.0493	0.0201	0.1069
refined fuel → power plant	$a_{23,17}$	0.0178	0.0078	0.5598	0.0170
electricity → flour mill	$a_{09,23}$	0.0067	0.0066	0.0096	0.0119
flour → bakery (same industry)	$a_{09,09}$	0.1328	0.1147	0.1363	0.2074

The remaining links are not entries of A at all.

- *Bread → miner* is household final consumption: 8,596.9 million Bs of milling and bakery products go to households, against 4,688.8 to intermediate use. Mining's purchases from the milling industry are $a_{06,09} = 0.0003$.
- *Tin → exports* is final demand: mining exports 39.14% of its output.
- *Export revenue → school* is a fiscal flow (royalties and taxes) that finances non-market education. It is not an intermediate purchase.

The input–output table sees only the first half of the story.

(b) *Testing the story's import claims.* Each claim is a hypothesis.

- *The processing plant's equipment is imported.* This cannot be tested on intermediate inputs: equipment is capital formation. At product level, imports are 95.77% of the total supply of machinery and equipment, so the claim is very plausible. The intermediate inputs of the processing step are domestic: refining buys its gas domestically (imported share 0), though 24.43% of its total intermediate inputs are imported.

- *The power plant was built by a foreign firm.* This is ownership or FDI, not an I-O fact, and not testable here. The plant's fuel is domestic gas (imported share 2.01%); refined fuels for power are 55.98% imported, but they are a small input (0.0178).

- *The mill buys imported wheat.* Wheat is not a separate product in the 42-product classification; it is inside "agricultural products". Under import proportionality the mill's agricultural inputs are 3.60% imported ($a_{09,01} = 0.2464$, of which 0.2375 domestic), and imports are only 3.50% of agricultural supply. That figure is an *assumption* (the same import share in every use), not an observation. Testing the wheat claim needs product-by-industry import matrices or customs data by HS code (wheat, HS 1001; flour, HS 1101), which were not fetched. Imports are 11.77% of the supply of milling and bakery products themselves, so imported *flour*, not wheat, is the more visible leak.

(c) *Two blocks.* We use the 2019 12-sector domestic table, without domestic services, which have no intermediate flows. With the gas block = {Hydrocarbons}:

- $\varepsilon_1 = 0.9751$: the share of hydrocarbons' domestic inputs bought from the rest;

- $\varepsilon_2 = 0.0390$: the share of the rest's inputs bought from hydrocarbons.

The two-block formula of Proposition 6.12 predicts block power $\varepsilon_2/(\varepsilon_1 + \varepsilon_2) = 0.0384$, against an actual $\pi_{\text{hyd}} = 0.0392$ in the 11-sector W^P. The prediction is close, although the hypothesis of equal ε for every agent fails: the rest's sectors put between 0.0014 and 0.1050 on hydrocarbons. With the block {Hydrocarbons, Mining}: $\varepsilon_1 = 0.9729$, $\varepsilon_2 = 0.1234$, prediction 0.1125, actual 0.0767. Here the heterogeneity (from 0.0068 to 0.2649) matters more.

Is the chain "mostly imaginary"? Partly.

- *What exists.* The first link is real and domestic: refining's largest input is domestic gas. The gas-to-power link exists (ε-sized).

- *What is thin.* Power-to-mill is tiny ($a = 0.0067$).

- *What is not a production link.* The second half of the chain (bread, tin, school) runs through final demand and the budget, not through A.

The gas block buys almost all its domestic inputs from the rest, but the rest depends on it very little ($\varepsilon_2 = 0.039$), and it exports 70% of its output. That is Pinto's

heterogeneity in its *one-way* form: weak forward linkages, not an absence of backward ones. *Common mistake:* reading a story's sequence of transactions as a chain of technical coefficients.

Chapter 6

Centrality, Power, and the Perron Vector

Throughout, W is row-stochastic, $\boldsymbol{\pi}$ is its left Perron vector normalized to sum one (Definition 6.4), $\Gamma = 1 - |\lambda_2|$ (Definition 6.14) and $N_{\text{eff}} = 1/\sum_i \pi_i^2$ (Definition 6.15). Every number below is printed by the script `code/solutions/sol06.py` (run from the repository root).

Exercise 6.1 Computing centralities

(a) The matrix is (6.1); its rows sum to one, so $\rho(W) = 1$ and the admissible parameter ranges are $\delta \in (0, 1)$ and $|\beta| < 1$. The four measures are

- in-degree $\mathbf{d}^{\text{in}} = W'\mathbf{1}$ (Definition 6.1): the column sums;
- Katz, $\mathbf{c}^{\text{Katz}} = [(I - 0.5\,W')^{-1} - I]\mathbf{1}$ (equation (6.2));
- Bonacich, $\mathbf{c}^{\text{Bon}} = (I - 0.3\,W')^{-1}W'\mathbf{1}$ (Definition 6.3);
- the eigenvector: the solution of $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$ with $\mathbf{1}'\boldsymbol{\pi} = 1$. The script computes it with numpy's eigenvalue routine applied to W'; power iteration as in Appendix 6A gives the same vector.

	Agent 1	Agent 2	Agent 3	Agent 4	Agent 5
In-degree	0.80	1.10	1.10	1.10	0.90
Katz ($\delta = 0.5$)	0.7735	1.0808	1.1343	1.1119	0.8995
Bonacich ($\beta = 0.3$)	1.1228	1.5567	1.5992	1.5808	1.2833
Eigenvector π_i	0.1435	0.2083	0.2361	0.2269	0.1852

The table reproduces Example 6.5; $N_{\text{eff}} = 4.866$ and $\Gamma = 0.400$.

Sanity checks. (i) Because $W\mathbf{1} = \mathbf{1}$, $\mathbf{1}'(W')^k\mathbf{1} = N$ for every k, so $\mathbf{1}'\mathbf{c}^{\text{Katz}} = \sum_{k\geq 1}\delta^k N = N\delta/(1-\delta) = 5.000$, and $\mathbf{1}'\mathbf{c}^{\text{Bon}} = \sum_{k\geq 0}\beta^k N = N/(1-\beta) = 7.1429$; both sums match the table. (ii) $\sum_i \pi_i = 1$. (iii) By (6.4), $\mathbf{c}^{\text{Bon}}(0.3) = \mathbf{c}^{\text{Katz}}(0.3)/0.3$; the script computes $\mathbf{c}^{\text{Katz}}(0.3) = (0.3368, 0.4670, 0.4798, 0.4742, 0.3850)$, which is 0.3 times the Bonacich row.

(b) Katz, Bonacich and the eigenvector give the same strict ranking $3 \succ 4 \succ 2 \succ 5 \succ 1$. In-degree agrees on the bottom ($5 \succ 1$) but ties Agents 2, 3 and 4 at 1.10. The tie is broken by *who* supplies the weight. Write $\pi_i = \sum_j \pi_j W_{ji}$ (6.6). The inflow terms $\pi_j W_{j2}$ are $(0.0574, 0.0625, 0.0472, 0.0227, 0.0185)$, and Agent 2's largest single inflow comes from the weakest agent ($\pi_1 W_{12} = 0.0574$). The terms $\pi_j W_{j3}$ are $(0.0144, 0.0417, 0.0708, 0.0907, 0.0185)$, and Agent 3's largest comes from the strong Agent 4 ($\pi_4 W_{43} = 0.0907$).

The measures disagree on *magnitudes*, not on order. In-degree puts the weakest agent at 73% of the strongest (0.80/1.10). The eigenvector puts her at 61% (0.1435/0.2361). The normalized Katz vector $\mathbf{1}'(I-\delta W)^{-1}/\mathbf{1}'(I-\delta W)^{-1}\mathbf{1}$ moves monotonically towards $\boldsymbol{\pi}$ as $\delta \to 1$:

$$\begin{aligned}
\delta = 0.5: &\quad (0.1774,\ 0.2081,\ 0.2134,\ 0.2112,\ 0.1899),\\
\delta = 0.9: &\quad (0.1521,\ 0.2095,\ 0.2306,\ 0.2230,\ 0.1848),\\
\delta = 0.999: &\quad (0.1436,\ 0.2084,\ 0.2361,\ 0.2268,\ 0.1852).
\end{aligned}$$

Short paths flatten the distribution. Only the parameter-free limit counts paths of every length without discount.

Agreement of the rankings is a feature of this matrix, not a theorem. At small δ, $\mathbf{c}^{\text{Katz}} = \delta\mathbf{d}^{\text{in}} + \delta^2 W'\mathbf{d}^{\text{in}} + O(\delta^3)$. Katz therefore ranks by in-degree first and only breaks ties by the second-order term. The eigenvector can reverse an in-degree ranking whenever a high-degree agent is supplied by weak agents. *Common mistake:* computing column sums of $(I-\delta W)^{-1}$ but calling them "multipliers". For a row-stochastic W every row sum of $(I-\delta W)^{-1}$ equals $1/(1-\delta)$; multipliers come from the technical coefficients A (Chapter 1), never from W.

(c) Row 3 becomes $(0.2, 0.2, 0.2, 0.2, 0.2)$: W_{31} rises from 0.1 to 0.2 and W_{33} falls from 0.3 to 0.2. The new Perron vector is

$$\begin{aligned}
\boldsymbol{\pi}^{\text{new}} &= (0.1788,\ 0.2182,\ 0.2092,\ 0.2215,\ 0.1723),\\
\Delta\boldsymbol{\pi} &= (+0.0353,\ +0.0099,\ -0.0269,\ -0.0054,\ -0.0129).
\end{aligned}$$

Agent 1's power rises by 0.0353, i.e. by 24.6%, and N_{eff} rises from 4.866 to 4.948.

Agent 3, who redirected her dependence, loses most (-0.0269). The ranking becomes $4 \succ 2 \succ 3 \succ 1 \succ 5$: Agent 1 is no longer last. Agent 2 gains too ($+0.0099$), because Agent 1, who now has more power, places 0.4 on Agent 2.

Check against Proposition 6.10. The move is "in row $j = 3$, move $t = 0.1$ from $k = 3$ to $i = 1$". The proposition allows $k = j$; it only needs $i \neq k$. Since $j \neq i$, its proof applies directly. Delete row and column 1 to get Q; the move lowers Q_{33} and touches nothing else in Q. The expected hitting times of agent 1 from agents 2–5 therefore fall, from $(8.624, 11.763, 13.419, 14.129)$ to $(6.789, 8.364, 10.398, 11.269)$. The mean return time $m_1 = 1 + \sum_{l \neq 1} W_{1l} h_l$ falls from 6.9677 to 5.5917, and Kac's formula gives $\pi_1 = 1/m_1$: $0.1435 \to 0.1788$, as computed.

To first order, Lemma 6.19 with $E = \mathbf{e}_3(\mathbf{e}_1 - \mathbf{e}_3)'$ and $t = 0.1$ predicts $\Delta\boldsymbol{\pi}' \approx t\,\boldsymbol{\pi}' E Z = (0.0399, 0.0111, -0.0304, -0.0060, -0.0146)$. The signs are right. The magnitudes overstate the exact change by about 13%, because $t = 0.1$ is not small relative to the entries of row 3.

Exercise 6.2 Doubly stochastic or not

All four matrices are row-stochastic and irreducible, so by Proposition 6.8: $\boldsymbol{\pi} = \frac{1}{N}\mathbf{1}$ if and only if every column sums to one. For a 2×2 row-stochastic matrix $\begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ we use $\boldsymbol{\pi} = (b, a)/(a + b)$ (the cut balance $\pi_1 a = \pi_2 b$) and $\lambda_2 = \operatorname{tr} W - 1 = 1 - a - b$.

(a) $W = \frac{1}{3}\mathbf{1}\mathbf{1}' = W^{\mathrm{CE}}$. Its column sums are $(1, 1, 1)$, so it is doubly stochastic and $\boldsymbol{\pi} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. It has rank one, so its spectrum is $\{1, 0, 0\}$ and $\Gamma = 1$.

(b) The column sums are $(1.3, 0.7)$, so W is not doubly stochastic. With $a = 0.5$ and $b = 0.8$, $\boldsymbol{\pi} = (0.8, 0.5)/1.3 = (8/13, 5/13) = (0.6154, 0.3846)$. Check: $0.5 \cdot \frac{8}{13} + 0.8 \cdot \frac{5}{13} = \frac{8}{13}$. Agent 1 has more power because agent 2 depends on her heavily ($W_{21} = 0.8$). Also $\lambda_2 = 1 - 1.3 = -0.3$, so $\Gamma = 0.7$.

(c) The column sums are $(0.5 + 0.5,\ 0.5 + 0.5,\ 0.5 + 0.5) = (1, 1, 1)$, so W is doubly stochastic and $\boldsymbol{\pi} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, the same as in (a). But Γ differs. $W = \frac{1}{2}(I + P)$ with P the cyclic permutation, so its eigenvalues are $\frac{1}{2}(1 + \omega^k)$, $\omega = e^{2\pi \mathrm{i}/3}$: 1 and $0.25 \pm 0.433\mathrm{i}$, of modulus $\frac{1}{2}|1 + \omega| = 0.5$. Hence $\Gamma = 0.5$, against 1 in (a). For example W^3 has entries 0.25 on the diagonal and 0.375 off it: it is still far from $\frac{1}{3}\mathbf{1}\mathbf{1}'$ after three rounds, while the matrix in (a) is already there after one.

The pair (a)/(c) is the content of Proposition 6.16 in miniature. The two matrices have the same power distribution but different speeds of integration. The cycle is slower because influence travels one way round the ring.

(d) $W = 0.1\,I + 0.3\,\mathbf{1}\mathbf{1}'$ is symmetric with unit row sums, hence doubly stochastic, so $\boldsymbol{\pi} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Its spectrum is $\{0.1 + 0.9, 0.1, 0.1\} = \{1, 0.1, 0.1\}$, so $\Gamma = 0.9$. *Common mistake:* reading a large diagonal ("self-reliance") as a source of power. In (d) the diagonal exceeds the off-diagonal entries, yet power is exactly equal, because power depends only on the column sums.

Exercise 6.3 The spectral gap and convergence

Construction ($N = 8$, as in Example 6.17 and `code/ch06.py`):

- complete: $\frac{1}{8}\mathbf{1}\mathbf{1}'$;
- lazy star: $\frac{1}{2}(I + W^{\star})$, where in $W^{\star}$ the hub (agent 1) spreads 1 equally over the spokes and each spoke puts 1 on the hub; this is $a = s = \frac{1}{2}$ in Proposition 6.11;
- lazy ring: $\frac{1}{2}I + \frac{1}{4}(S + S')$, with S the cyclic shift;
- weak blocks: two blocks of four, with weight $(1 - \varepsilon)/4$ within and $\varepsilon/4$ across, $\varepsilon = 0.01$.

(a)

Topology	$\lvert\lambda_2\rvert$	Γ	N_{eff}	spectrum
Complete	0	1.000	8.00	$\{1, 0^{(7)}\}$
Star (lazy)	0.500	0.500	3.50	$\{1, 0.5^{(6)}, 0\}$
Ring (lazy)	0.8536	0.1464	8.00	$\{\frac{1}{2} + \frac{1}{2}\cos\frac{2\pi k}{8}\}_{k=0}^{7}$
Weak blocks	0.980	0.020	8.00	$\{1, 0.98, 0^{(6)}\}$

The star's values follow from (6.8): $\pi_h = a/(a + 1 - s) = \frac{1}{2}$, each spoke has $\frac{1}{14}$, so $N_{\text{eff}} = 1/(\frac{1}{4} + 7 \cdot \frac{1}{196}) = 3.5$, and the spectrum is $\{1, s - a, 1 - a\} = \{1, 0, \frac{1}{2}\}$. The ring is a symmetric circulant, so its eigenvalues are $\frac{1}{2} + \frac{1}{2}\cos(2\pi k/8)$ and $\lvert\lambda_2\rvert = \frac{1}{2} + \frac{\sqrt{2}}{4} = 0.8536$. The weak-block matrix is doubly stochastic and lumpable, and its only nontrivial eigenvalue is $1 - 2\varepsilon = 0.98$ (Proposition 6.12 with $\varepsilon_1 = \varepsilon_2 = 0.01$). Three of the four networks have perfectly equal power ($N_{\text{eff}} = 8$) and gaps that range from 0.02 to 1.

Without self-loops. The ring $\frac{1}{2}(S + S')$ with $N = 8$ and the pure star $W^{\star}$ both have spectrum containing $\{1, -1\}$, so $\Gamma = 0$ (script). Both are irreducible but *bipartite*: the ring splits into odd and even sites, the star into hub and spokes. Every closed walk then has even length, and the graph has period 2. By Theorem 6.6

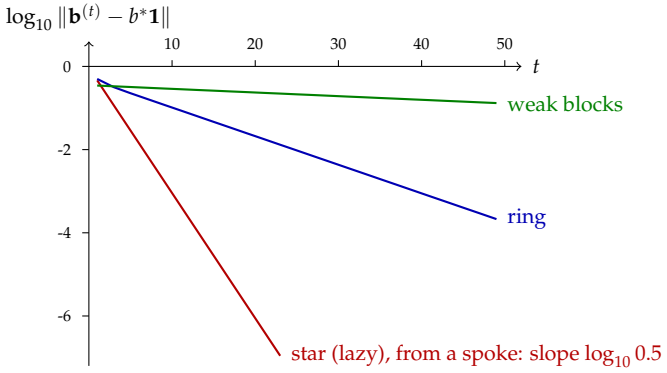
(item 5) and Step 9 of Appendix 6B, an irreducible matrix of period d has the d-th roots of unity in its spectrum; here -1 is an eigenvalue, with eigenvector $x_i = (-1)^{r(i)}$. The dynamics oscillate rather than fail to integrate. Starting at the hub, the pure star gives $\mathbf{b}^{(1)} = (0, 1, \ldots, 1)'$, $\mathbf{b}^{(2)} = (1, 0, \ldots, 0)' = \mathbf{b}^{(0)}$, and so on for ever. Power is unaffected: the Perron vectors are $(\frac{1}{2}, \frac{1}{14}, \ldots)$ and $\frac{1}{8}\mathbf{1}$, the same as for the lazy versions. Laziness changes the eigenvalues ($\lambda \mapsto \frac{1}{2}(1+\lambda)$) but not $\boldsymbol{\pi}$.

(b) Here $b^* = \boldsymbol{\pi}'\mathbf{b}^{(0)} = \pi_1$. The distances $\|\mathbf{b}^{(t)} - b^*\mathbf{1}\|_2$ (initial distance 0.9354 for the complete graph, the lazy ring and the weak blocks) are:

t	1	2	5	10	20	50
Complete	0	0	0	0	0	0
Star, from the hub	0	0	0	0	0	0
Star, from a spoke	$4.63 \cdot 10^{-1}$	$2.31 \cdot 10^{-1}$	$2.89 \cdot 10^{-2}$	$9.04 \cdot 10^{-4}$	$8.83 \cdot 10^{-7}$	0
Ring (lazy)	$5.00 \cdot 10^{-1}$	$3.85 \cdot 10^{-1}$	$2.27 \cdot 10^{-1}$	$1.03 \cdot 10^{-1}$	$2.11 \cdot 10^{-2}$	$1.82 \cdot 10^{-4}$
Weak blocks	$3.47 \cdot 10^{-1}$	$3.40 \cdot 10^{-1}$	$3.20 \cdot 10^{-1}$	$2.89 \cdot 10^{-1}$	$2.36 \cdot 10^{-1}$	$1.29 \cdot 10^{-1}$

The starting vector $\mathbf{e}_1$ puts the belief on the hub, and for the lazy star it converges in *one* step: $\mathbf{b}^{(1)} = W\mathbf{e}_1$ is the hub column, $(\frac{1}{2}, \ldots, \frac{1}{2})'$, which is already $b^*\mathbf{1}$. The deviation $\mathbf{e}_1 - \frac{1}{2}\mathbf{1}$ lies in the eigenspace of the eigenvalue $s - a = 0$ (the lumped hub/spoke mode in the proof of Proposition 6.11), so the slow modes $\frac{1}{2}$ are never excited. Starting at a spoke (row "Star, from a spoke"; $b^* = \frac{1}{14}$) excites them. The distance then halves every round, and it is below machine precision from $t = 44$.

The figure plots $\log_{10}$ of the distance (every second t; complete graph and star-from-hub omitted, they are 0 from $t = 1$).



(c) On a log scale each curve becomes a straight line of slope $\log|\lambda_2|$. For the ring, the ratio d_{t+1}/d_t is 0.7706 at $t = 1$ (faster modes are still present), 0.8522 at $t = 5$, and 0.8536 from $t = 20$ on. A least-squares fit of $\ln d_t$ on t over $t = 11, \ldots, 50$ gives $e^{\text{slope}} = 0.8536 = |\lambda_2|$. For the weak blocks the ratio is 0.98 at every t, because

$\mathbf{e}_1 - b^*\mathbf{1}$ excites only the eigenvalues 0.98 and 0: the fitted rate is 0.9800, and the half-life is $\ln 2/(-\ln 0.98) = 34.3$ rounds. For the star from a spoke the fitted rate over $t = 11, \dots, 30$ is 0.5000.

The reason is the proof of Proposition 8.3. $\mathbf{b}^{(t)} - b^*\mathbf{1} = (W - \mathbf{1}\boldsymbol{\pi}')^t(\mathbf{b}^{(0)} - b^*\mathbf{1})$, and $W - \mathbf{1}\boldsymbol{\pi}'$ has spectral radius $|\lambda_2|$. All four matrices are symmetric or reversible and so diagonalizable, so $\|\mathbf{b}^{(t)} - b^*\mathbf{1}\| \le C|\lambda_2|^t$ with equality in the limit whenever the initial deviation has a component on the λ_2 eigenspace. *Common mistake:* expecting every starting vector to converge at rate $|\lambda_2|$. The rate is an upper bound; the hub start of the star shows that a deviation orthogonal to the slow modes converges faster.

Exercise 6.4 Bonacich with negative β

(a) The matrix is symmetric with all row and column sums equal to one, so $\mathbf{d}^{\text{in}} = W'\mathbf{1} = \mathbf{1}$. Then $W'\mathbf{1} = \mathbf{1}$ gives $(I - \beta W')\mathbf{1} = (1 - \beta)\mathbf{1}$, hence

$$\mathbf{c}^{\text{Bon}}(\beta) = (I - \beta W')^{-1}\mathbf{1} = \frac{1}{1-\beta}\mathbf{1},$$

so $\mathbf{c}^{\text{Bon}}(0.3) = 1.4286\,\mathbf{1}$ and $\mathbf{c}^{\text{Bon}}(-0.3) = 0.7692\,\mathbf{1}$. The inverse exists because the spectrum of W is $\{1, -1, 0, 0\}$ and $|\beta| < 1$.

(b) *As posed, the question has no non-trivial answer.* In this network every agent has the same centrality for every β. The graph is the complete bipartite graph $K_{2,2}$ with equal weights, and it is vertex-transitive: any agent can be mapped to any other by a permutation P with $PWP' = W$, and Bonacich centrality is invariant under such permutations. Negative β changes the level (from 1.4286 to 0.7692) but cannot separate agents who are structurally identical.

To see what negative β does, break the symmetry. Buyer 1 trades only with seller 3, buyer 2 with sellers 3 and 4, and seller 4 only with buyer 2. Each row is uniform over the agent's partners, so W has rows $(0, 0, 1, 0)$, $(0, 0, \tfrac{1}{2}, \tfrac{1}{2})$, $(\tfrac{1}{2}, \tfrac{1}{2}, 0, 0)$, $(0, 1, 0, 0)$ and in-degrees $(0.5, 1.5, 1.5, 0.5)$. Then

$$\begin{aligned}\mathbf{c}^{\text{Bon}}(0.3) &= (0.8075,\ 2.0497,\ 2.0497,\ 0.8075),\\ \mathbf{c}^{\text{Bon}}(-0.3) &= (0.3167,\ 1.2217,\ 1.2217,\ 0.3167).\end{aligned}$$

The two well-connected agents (2 and 3) lead in both cases, but their lead widens under bargaining: $c_2/c_1 = 2.54$ at $\beta = 0.3$ and 3.86 at $\beta = -0.3$. With $\beta < 0$ the series $\mathbf{c} = \sum_k \beta^k (W')^k \mathbf{d}^{\text{in}}$ subtracts the centrality of your partners at odd path

lengths. Agents 2 and 3 are partnered with agents (1 and 4) who have a single partner each, hence no outside option. That weakness is a source of bargaining strength for 2 and 3. This is Bonacich's reading of $\beta < 0$, and in the symmetric network of the exercise it has nothing to act on.

(c) *Eigenvector.* W is doubly stochastic, so by Proposition 6.8 $\boldsymbol{\pi} = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$. This agrees with the result the exercise attributes to Manea (2011): equal bargaining power in a symmetric bipartite network. The agreement is a consequence of symmetry, not of a general correspondence: Assumption 7.A is a modelling postulate, and strategic bargaining payoffs are not eigenvector centralities in general.

Gap. $W = \frac{1}{2}\begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}$ with $J = \mathbf{1}_2\mathbf{1}_2'$. Its eigenvalues are 1 (eigenvector $\mathbf{1}$), -1 (eigenvector $(1,1,-1,-1)'$) and $0, 0$. So $|\lambda_2| = 1$ and $\Gamma = 0$. The value is zero because the network is bipartite, i.e. periodic with period 2: beliefs or influence alternate between the two sides, W^t does not converge, and $(1,1,-1,-1)'$ flips sign every period. It does *not* reflect fragmentation: the network is irreducible and the two sides are as connected as they can be. The lazy version $\frac{1}{2}(I + W)$ has $\Gamma = 0.5$ and the same $\boldsymbol{\pi}$. *Common mistake:* reading $\Gamma = 0$ here as "disconnected" or "unequal". It is neither.

Exercise 6.5 Power iteration by hand

(a) $W' = \begin{pmatrix} 0.6 & 0.2 \\ 0.4 & 0.8 \end{pmatrix}$. Starting from $\mathbf{v}^{(0)} = (1,1)'$:

t	$W'\mathbf{v}^{(t-1)}$	$\mathbf{v}^{(t)}$ (normalized)	$\max_i \lvert v_i^{(t)} - \pi_i \rvert$
1	(0.8, 1.2)	(0.4, 0.6)	0.06667
2	(0.36, 0.64)	(0.36, 0.64)	0.02667
3	(0.344, 0.656)	(0.344, 0.656)	0.01067
4	(0.3376, 0.6624)	(0.3376, 0.6624)	0.00427
5	(0.33504, 0.66496)	(0.33504, 0.66496)	0.00171

For instance, $t = 2$: $0.6(0.4) + 0.2(0.6) = 0.36$ and $0.4(0.4) + 0.8(0.6) = 0.64$. After the first step no normalization is needed: $\mathbf{1}'W'\mathbf{v} = (W\mathbf{1})'\mathbf{v} = \mathbf{1}'\mathbf{v}$, so W' preserves the sum of a vector. The initial $\mathbf{v}^{(0)} = (1,1)'$ sums to 2, and only that first normalization matters.

(b) The question depends on what "converged to two decimal places" means. The error first falls below 0.005 at $t = 4$. The rounded iterate first equals the

rounded limit $(0.33, 0.67)$ only at $t = 6$ ($\mathbf{v}^{(6)} = (0.334, 0.666)$). The iterates at $t = 3, 4, 5$ round to $(0.34, 0.66)$, because $\mathbf{v}^{(5)} = (0.33504, \ldots)$ is still just above the rounding boundary 0.335. The error criterion is the right one; rounding agreement depends on how close the limit lies to a rounding boundary.

(c) $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$ gives, from the first column, $0.6\pi_1 + 0.2\pi_2 = \pi_1$, i.e. $0.4\pi_1 = 0.2\pi_2$, so $\pi_2 = 2\pi_1$. With $\pi_1 + \pi_2 = 1$, $\boldsymbol{\pi} = (\frac{1}{3}, \frac{2}{3})$. (The second column gives the same equation, as it must, since $W - I$ is singular.)

(d) The iterates approach $(\frac{1}{3}, \frac{2}{3})$ monotonically from above in the first coordinate. The spectrum is $\{1, \operatorname{tr} W - 1\} = \{1, 0.4\}$, so $\Gamma = 0.6$. The error ratios are exactly 0.4 at every step: $0.02667/0.06667 = 0.01067/0.02667 = \cdots = 0.4 = |\lambda_2|$. In two dimensions this is exact, not asymptotic. Write $\mathbf{v}^{(t)} = \boldsymbol{\pi} + c_t\mathbf{u}$ with $\mathbf{u} = (1, -1)'$, a left eigenvector of W for 0.4 ($\mathbf{u}'W = (0.4, -0.4)$). Then $W'(\boldsymbol{\pi} + c\mathbf{u}) = \boldsymbol{\pi} + 0.4c\,\mathbf{u}$, and the sum stays 1, so $c_t = 0.4^t c_0$. Each additional correct decimal costs $\ln 10 / \ln(1/0.4) = 2.5$ iterations (Appendix 6A).

Exercise 6.6 Structural power vs. formal power

A matrix (agents ordered President, General, Businesswoman, Farmer; row i is whom i depends on):

$$W = \begin{pmatrix} 0.30 & 0.20 & 0.20 & 0.30 \\ 0.30 & 0.30 & 0.10 & 0.30 \\ 0.10 & 0.10 & 0.40 & 0.40 \\ 0.05 & 0.05 & 0.20 & 0.70 \end{pmatrix}. \text{[ILLUSTRATIVE]}$$

Reading. The President depends on the army (0.2), on business tax revenue (0.2) and on food for the cities and the palace (0.3). The General depends on the President's orders and budget (0.3), on his own corps (0.3) and on food rations (0.3). The Businesswoman's agro-processing and retail depend on farm output (0.4). The Farmer depends mostly on her own land and family labor (0.7), sells through the Businesswoman (0.2), and needs little from the state (0.05 each). Every row sums to one and every entry is positive, so W is primitive.

Result. $\boldsymbol{\pi} = (0.1178,\ 0.1060,\ 0.2367,\ 0.5395)$, $N_{\text{eff}} = 2.687$, $\Gamma = 0.576$. The Farmer has the highest eigenvector centrality, more than four times the President's.

Verification. The balance equation (6.6) for the Farmer is

$$\textstyle\sum_j \pi_j W_{j4} = 0.30(0.1178) + 0.30(0.1060) + 0.40(0.2367) + 0.70(0.5395) = 0.5395.$$

Her column sum is 1.7, the largest; but the ranking is not just in-degree. The Businesswoman (column sum 0.9) ranks above the President (0.75) because the powerful Farmer places 0.2 on her: indirect dependence ("directly or indirectly") at work.

How robust is it? If the President, General and Businesswoman each cut their food weight to f (adding the difference to their own diagonal), the Farmer stays on top at $f = 0.30$ ($\pi_4 = 0.500$) and at $f = 0.20$ ($\pi_4 = 0.400$). At $f = 0.10$ she falls to 0.250, behind the Businesswoman (0.364). Structural power from food dependence requires that the dependence be large *relative* to the other links in each row, not merely present.

Interpretation. Formal rank is a property of an organizational chart. Structural power is a property of the dependence matrix (Section 6.5). It is the Perron weight that enters bargaining in Chapter 7 (by Assumption 7.A). *Common mistakes:* (i) writing the Farmer's dependence on the others into her *row* and expecting that to make her powerful. Power comes from the *column*, from being depended upon (Proposition 6.10). (ii) Forgetting to make rows sum to one, which breaks the reading of π as power (Remark after Proposition 6.8).

Exercise 6.7 Power in a stylized dual economy

(a) Example 3.6 has $A_{\text{dual}} = \begin{pmatrix} 0.15 & 0.03 \\ 0.02 & 0.20 \end{pmatrix}$ (rows Modern, Traditional), with row sums $(0.18, 0.22)$. Following the convention $W^P = D^{-1}A$ (Chapter 1, Section 1.7),

$$W^P = \begin{pmatrix} 0.15/0.18 & 0.03/0.18 \\ 0.02/0.22 & 0.20/0.22 \end{pmatrix} = \begin{pmatrix} 0.8333 & 0.1667 \\ 0.0909 & 0.9091 \end{pmatrix}.$$

This is a two-block matrix with leakages $\varepsilon_M = 0.1667$ and $\varepsilon_T = 0.0909$, so by Proposition 6.12, exactly,

$$\pi_M = \frac{\varepsilon_T}{\varepsilon_M + \varepsilon_T} = \frac{0.0909}{0.2576} = 0.3529, \qquad \pi_T = 0.6471,$$

with $\Gamma = \varepsilon_M + \varepsilon_T = 0.2576$ and $N_{\text{eff}} = 1.841$. Check: $0.3529 \times 0.8333 + 0.6471 \times 0.0909 = 0.3529$. *The Traditional sector has the most structural power.* The answer surprises many students, but it follows from the definition. In W^P, being powerful means that others spend a large share of their input costs on you. The Modern sector spends 16.7% of its (domestic intermediate) input costs on traditional inputs, while the Traditional sector spends only 9.1% on modern inputs. Note that the

absolute coefficients (0.03 vs 0.02) and the input-cost shares point the same way here. The reading of π in force is the production reading of `docs/conventions.md`, §2: π is the $\gamma \to 1$ limit of the normalized influence vector, *not* bargaining power, unless Assumption 7.A is adopted.

(b) No. $\boldsymbol{\pi}$ is a function of W^P alone, and $W^P = D^{-1}A$ depends only on the technical coefficients. A transfer changes incomes and endowments $\boldsymbol{\omega}$, but not the recipe of either sector. So, holding A fixed, $\boldsymbol{\pi}$ is unchanged (Section 6.5, "Power vs. Wealth"). The transfer would move $\boldsymbol{\pi}$ only if it changed A, for instance by changing what the traditional sector buys. That is an indirect, behavioral channel, which a fixed-W model rules out by construction.

(c) With $a_{TM} = 0.15$, the Traditional row of A becomes $(0.15, 0.20)$, with sum 0.35:

$$W^P_{\text{new}} = \begin{pmatrix} 0.8333 & 0.1667 \\ 0.4286 & 0.5714 \end{pmatrix}, \qquad \boldsymbol{\pi}_{\text{new}} = (0.7200,\ 0.2800),$$

with $\Gamma = 0.5952$ and $N_{\text{eff}} = 1.676$. *The Traditional sector's power falls*, from 0.6471 to 0.2800 (-0.3671). The plant makes the Traditional sector depend on the Modern sector: weight in the Traditional row moves from T to M. By Proposition 6.10 (with $j = T$, $i = M$, $k = T$) that can only raise π_M. Integration rises: Γ goes from 0.258 to 0.595, as the bound $\Gamma \leq \varepsilon_M + \varepsilon_T$ of Proposition 6.12 allows. Concentration of power also rises (N_{eff} falls from 1.841 to 1.676), now in favor of the Modern sector.

The output side moves the other way, and it must be computed from A, not from W^P. The Leontief inverse $(I - A')^{-1}$ has column sums (output multipliers) $(1.2217, 1.2805)$ before and $(1.2287, 1.4804)$ after: a unit of traditional final demand now generates 0.2221 units of modern output instead of 0.0294. By contrast, a link in the other direction, a_{MT} from 0.03 to 0.15 (the modern sector buying more from traditional suppliers), gives $\boldsymbol{\pi} = (0.1538, 0.8462)$: it raises the Traditional sector's power.

(d) Only the plant changes the power structure: it changes A, hence W^P, hence $\boldsymbol{\pi}$. The transfer leaves $\boldsymbol{\pi}$ unchanged. But the plant, as specified, changes it *against* the Traditional sector, because buying from someone confers power on the seller, not the buyer. A structural policy meant to raise the traditional sector's Perron weight must make *others* depend on it: purchases by the modern sector, by government procurement, or by processing plants that use traditional output as an input. *Common mistake:* assuming that any new cross-block link "integrates" and therefore "empowers" the periphery. The plant does raise Γ and the multipliers,

but it lowers π_T. Integration and the distribution of power are separate coordinates (Proposition 6.16).

(e) *Data: real results.*

- *Dataset.* INE Bolivia, *Cuadros de Oferta y Utilización*, base 2017, year 2019, current prices (Instituto Nacional de Estadística (INE), Bolivia, 2026). The table was converted in `data/compute/compute_bolivia_io.py` to a domestic industry-by-industry flow matrix Z at basic prices (import proportionality, product-sales-structure model; `data/README.md` §2), aggregated to the book's 12 sectors (`book12`). File: `data/processed/bolivia_Z/Z_2017_2019_book12.csv`; retrieved 2026-11-05. Domestic services has no intermediate flows and is dropped, leaving 11 sectors. (INE's older *Matriz Insumo-Producto*, base 1990, ends in 2014; the base-2017 tables are the current ones.)

- *Check.* The 11-sector $W^P[s, s'] = Z[s', s] / \sum_k Z[k, s]$ (input-cost shares, rows = buyers) reproduces the values quoted in the exercise: $\Gamma = 0.4674$ and $N_{\text{eff}} = 6.4795$ of 11 (Chapter 3, Section 3.4). The largest Perron weights are other services (0.270), financial services (0.162) and manufacturing (0.159); hydrocarbons has 0.039.

- *Aggregation.* Choose a partition into Modern and Traditional, let M be the 11×2 membership matrix, set $Z_2 = M'ZM$ (sum the flows), and form W_2^P from Z_2 in the same way. The 12-sector classification does not separate large from small firms, so no partition matches Example 3.6 exactly; we report three.

Modern block	ε_M	ε_T	π_M	N_{eff} (of 2)	λ_2	Γ
Hydrocarbons, Mining, Financial	0.7620	0.1904	0.2000	1.471	0.0476	0.952
+ Manufacturing	0.5435	0.4283	0.4407	1.972	0.0283	0.972
+ Publ. adm., Transport, Oth. serv.	0.3222	0.7023	0.6855	1.758	−0.0246	0.975
Stylized (Example 3.6)	0.1667	0.0909	0.3529	1.841	0.7424	0.258

Here ε_M is the share of the modern block's domestic intermediate input costs spent on the traditional block, and vice versa. For a 2×2 row-stochastic matrix $\lambda_2 = 1 - \varepsilon_M - \varepsilon_T$, so $\Gamma = 1 - |1 - \varepsilon_M - \varepsilon_T|$.

Comparison and answer. Every two-sector aggregation makes Bolivia look *far more* integrated ($\Gamma \approx 0.95$–0.98) than the 11-sector table (0.467), and far more than the stylized model (0.258). Two reasons:

1. *The data do not have the stylized enclave's small cross-block shares.* In the narrow definition the extractive and financial block spends 76% of its domestic input

costs on the rest of the economy. The stylized model assumes 17%. On the integration coordinate, Bolivia's measured W^P is not block-diagonal, as the Prebisch box of Section 6.4 reports.

2. *Aggregation removes slow modes.* If a partition is exactly lumpable ($WM = M\bar{W}$), every eigenvalue of $\bar{W}$ is an eigenvalue of W (if $\bar{W}\bar{\mathbf{x}} = \lambda\bar{\mathbf{x}}$ then $W(M\bar{\mathbf{x}}) = \lambda M\bar{\mathbf{x}}$), so $\Gamma(\bar{W}) \geq \Gamma(W)$. The slow mode $\lambda_2 = 0.533$ of the 11-sector matrix runs *within* what the partitions lump together, and the two-block matrix cannot represent it.

Power is a separate coordinate, and it depends on the partition: π_M ranges from 0.20 to 0.69 as the modern block grows. Compare concentration as N_{eff}/N (Chapter 6 conventions): 0.735–0.986 for the two-block versions and $6.48/11 = 0.589$ at 11 sectors. Like Γ, a single number from an arbitrary partition is not informative; only like-for-like comparisons are.

Common mistakes: comparing Γ or N_{eff} across aggregation levels or node sets; computing W^P by normalizing the rows of the published table (those are *sales* shares, rows = suppliers); and including imported inputs, which mixes the domestic structure with the import structure.

Chapter 7

The Bargaining Equilibrium

All solutions use the transferable-utility rule of Proposition 7.3,

$$x_i^* = d_i + \pi_i\Big(\Omega - \sum_j d_j\Big),$$

and, with equal outside options, $x_i^* = \Omega/N + (\pi_i - 1/N)\Phi$. The weights are $\boldsymbol{\pi}(W)$ by Assumption 7.A, a modelling postulate. Allocations are evaluated with $U(x) = x - \frac{\kappa}{2}x^2$ below satiation. Every number is printed by the script `code/solutions/sol07.py`.

Exercise 7.1 Landlord–Farmer revisited

(a) With $K = 10$, $L = 2$ and $Y = AK^{0.3}L^{0.7} = 20$, $A = 20/(10^{0.3}2^{0.7}) = 6.1703$. Under Cobb–Douglas each factor's marginal-product payment is its exponent times Y: $MPK = 0.3 \cdot 20/10 = 0.60$ and $MPL = 0.7 \cdot 20/2 = 7.00$, so the Landlord is paid $0.6 \times 10 = 6$ and the Farmer $7 \times 2 = 14$ (they exhaust Y by Euler's theorem). Relative to the original example, $(10, 10)$, the higher labor share benefits the *Farmer* $(+4)$ at the Landlord's expense (-4).

(b) The bargaining allocation does not depend on the technology, only on Y and the outside options: $S = 20 - 10 - 2 = 8$ and, by (7.1), $x_L = 10 + 0.7 \cdot 8 = 15.6$, $x_F = 2 + 0.3 \cdot 8 = 4.4$. Compared with the competitive allocation $(6, 14)$, the Landlord gets 9.6 more and the Farmer 9.6 less. The Farmer's share of the surplus is 30% under bargaining and $(14 - 2)/8 = 150\%$ under marginal-product pricing.

(c) A bargaining allocation gives the Farmer $x_F = 2 + 8\pi_F$. Reproducing $x_F = 14$ requires $\pi_F = 12/8 = 1.5 > 1$, so no $\pi_F \in [0, 1]$ works. The general reason: every TU bargaining allocation is individually rational, so $x_L \geq d_L = 10$ and $x_F \geq d_F = 2$. The bargaining set is the segment $x_F \in [2, 10]$. The competitive

payment to the Landlord, 6, is *below* his outside option 10. That allocation violates his participation constraint, and no bargaining weight can deliver it: the Landlord would rather use his land elsewhere. More generally, the competitive allocation is reproducible iff the capital share a satisfies $aY \geq d_L$ and $(1-a)Y \geq d_F$, i.e. $a \in [0.5, 0.9]$ here. Marginal-product pricing ignores the outside options; the bargaining model is built on them.

In the original example ($a = 0.5$) the competitive allocation $(10, 10)$ requires $10 = 2 + 8\pi_F$, i.e. $\boldsymbol{\pi} = (\pi_L, \pi_F) = (0, 1)$: the Landlord is held exactly to his outside option and the Farmer takes the whole surplus. This is the boundary of the simplex. A Perron vector of an irreducible W is strictly positive (Theorem 6.6), so it can approach $(0, 1)$ but never equal it. Under Assumption 7.A, the competitive outcome of that example is the limit of a network in which the Landlord's power vanishes.

Exercise 7.2 The FOC in practice

(a) With one good, endowments $\boldsymbol{\omega} = (1.5, 1.0, 0.5)$ summing to $\Omega = 3$, and $d_i = U(\omega_i)$ (or $d_i = \omega_i$), a strictly positive surplus for every agent requires $x_i > \omega_i$ for all i, because U is increasing below satiation. Summing gives $\sum_i x_i > \sum_i \omega_i = 3$, which contradicts feasibility. The only individually rational allocation is $\mathbf{x} = \boldsymbol{\omega}$, where every $S_i = 0$ and the Nash product $\sum_i \pi_i \ln S_i$ is $-\infty$ (Appendix 7A). There are no gains from cooperation, so hypothesis (iv) of Theorem 7.5 fails. In the exercise as posed, cooperation produces $\Omega = 3$ while the outside options sum to 1.2, so there is a surplus of 1.8 to divide.

(b) With $S_i = x_i - d_i$, $\partial S_i / \partial x_i = 1$, and the Lagrangian of (7.4) gives the FOC (7.6):

$$\frac{0.5}{x_1 - 0.6} = \mu, \qquad \frac{0.3}{x_2 - 0.4} = \mu, \qquad \frac{0.2}{x_3 - 0.2} = \mu, \qquad x_1 + x_2 + x_3 = 3.$$

That is four equations in (x_1, x_2, x_3, μ). Solving the first three for $x_i = d_i + \pi_i/\mu$ and substituting into the fourth gives $1/\mu = 3 - 1.2 = 1.8$.

(c) A student with `scipy` would write

```
minimize(lambda x: -np.sum(pi*np.log(x-d)),
         x0=np.ones(3), method="SLSQP",
         constraints=[{"type": "eq",
                       "fun": lambda x: x.sum()-3}],
         bounds=[(di+1e-9, None) for di in d])
```

The manual's script keeps to `numpy` and instead applies Newton's method to the FOC system of (b), with Jacobian $\mathrm{diag}(-\pi_i/S_i^2)$ bordered by the constraint. It converges to

$$\mathbf{x}^* = (1.5000,\ 0.9400,\ 0.5600), \qquad \mu = 0.5556 = 1/1.8,$$

which is the closed form (7.7): $x_1 = 0.6 + 0.5(1.8)$, $x_2 = 0.4 + 0.3(1.8)$, $x_3 = 0.2 + 0.2(1.8)$. The objective is strictly concave, so this stationary point is the global maximum. As a check, none of 5000 random feasible allocations attains a higher weighted Nash product. Satiation: $\kappa\mathbf{x}^* = (0.375, 0.235, 0.140)$, all below 1. The utilities are $U(\mathbf{x}^*) = (1.2188, 0.8296, 0.5208)$.

(d) With $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$, $\mathbf{x} = \boldsymbol{d} + \frac{1}{3}(1.8)\mathbf{1} = (1.2, 1.0, 0.8)$. The difference is $\mathbf{x}(\boldsymbol{\pi}) - \mathbf{x}(\frac{1}{3}\mathbf{1}) = (\boldsymbol{\pi} - \frac{1}{3}\mathbf{1}) \cdot 1.8 = (+0.30, -0.06, -0.24)$. Agent 1 ($\pi_1 = 0.5 > \frac{1}{3}$) gains, and agents 2 and 3 ($\pi < \frac{1}{3}$) lose, in proportion to their power deficit. Equal power does not mean equal consumption here, because the outside options differ: it means equal division of the *surplus*. Utilitarian welfare is $\sum_i U(x_i) = 2.5691$ with unequal power, 2.6150 with equal power, and 2.6250 at equal division $(1, 1, 1)$. Both bargaining outcomes are Pareto efficient (Proposition 7.7); the welfare differences are distributional.

Remark (a common confusion). The FOC (7.6) is written for surpluses in utility, $S_i = U_i(x_i) - d_i$. If the agents bargained over utility, with $S_i = U(x_i) - U(d_i)$, the FOC would be $\pi_i(1 - \kappa x_i)/S_i = \mu$. Newton's method gives $\mathbf{x} = (1.4627, 0.9550, 0.5823)$. Concavity of U shifts consumption slightly away from the powerful agent, because her marginal utility falls as she consumes more. The exercise's TU specification, $S_i = x_i - d_i$, is the one in which the book's allocation rule holds exactly.

Exercise 7.3 Welfare decomposition

Both economies have $N = 4$, $\Omega = 20$, $\bar{d} = 3$, $\Phi = 8$ and $\kappa = 0.1$ (Section 7.6).

(a) By (7.9), $L_{\text{power}} = \frac{\kappa N}{2}\Phi^2 \,\mathrm{Var}(\boldsymbol{\pi}) = 12.8\ \mathrm{Var}(\boldsymbol{\pi})$.

	$\boldsymbol{\pi}$	$\mathrm{Var}(\boldsymbol{\pi})$	L_{power}
A (village)	$(0.2564, 0.2821, 0.2179, 0.2436)$	0.000534	0.0068
B (enclave)	$(0.3342, 0.4158, 0.1250, 0.1250)$	0.016456	0.2106

Economy B's loss is 30.8 times Economy A's (the chapter's "about 31"). The loss is

computed both from the formula and as $\sum_i [U(5) - U(x_i)]$, with

$$\mathbf{x}_A = (5.0513,\ 5.2564,\ 4.7436,\ 4.9487), \qquad \mathbf{x}_B = (5.6739,\ 6.3261,\ 4.0000,\ 4.0000);$$

the two agree to machine precision.

(b) *Set-up.* $W^U = W^P - \mathrm{diag}(W^P)$, $\gamma_U = 0.3$. The caring weights are the column sums $c_j = \sum_{i \neq j} W^U_{ij}$:

- A: $\mathbf{c} = (0.4, 0.6, 0.5, 0.4)$;
- B: $\mathbf{c} = (0.46, 0.56, 0.32, 0.32)$.

(W^U is substochastic here, with row sums 0.4–0.6. The theorem only needs $W^U \geq 0$.) Social welfare is $\mathcal{S}(\mathbf{x}) = \sum_j (1 + 0.3 c_j) U(x_j)$. Its maximizer on $\sum_j x_j = 20$ solves $(1 + \gamma_U c_j)(1 - \kappa x_j) = \mu$. That gives $x_j = (1 - \mu / w_j)/\kappa$ with $w_j = 1 + \gamma_U c_j$ and $\mu = (N - \kappa\Omega) / \sum_j w_j^{-1}$:

- A: $\mathbf{x}^{FB} = (4.9019, 5.1612, 5.0349, 4.9019)$;
- B: $\mathbf{x}^{FB} = (5.0629, 5.1897, 4.8737, 4.8737)$.

The three terms of Theorem 7.8 are:

	L_{power}	$L_{\text{externality}}$	$L_{\text{interaction}}$	(linear, quadratic)	L
A	0.006838	0.002680	−0.002730	(−0.003846, 0.001116)	0.006787
B	0.210633	0.004075	−0.034386	(−0.061891, 0.027505)	0.180322

Check: the components add up to $L = \mathcal{S}(\mathbf{x}^{FB}) - \mathcal{S}(\mathbf{x}^B)$. For B, $\mathcal{S}(\mathbf{x}^{FB}) = 16.8716$ and $\mathcal{S}(\mathbf{x}^B) = 16.6913$. The interaction term equals the sum of its linear and quadratic parts in the theorem. *The power component is the largest in both economies*: by a factor of about 2.5 in A and 50 in B over the externality term. The externality term is small because the c_j vary little (weights $1 + \gamma_U c_j$ between 1.10 and 1.18). In A, $L_{\text{externality}}$ and $L_{\text{interaction}}$ almost cancel, so $L \approx L_{\text{power}}$.

(c) The interaction term is *negative* in both economies. Its leading term is

$$-\gamma_U N\, U'(\Omega/N)\, \Phi\ \mathrm{Cov}(c_j, \pi_j), \qquad U'(5) = 0.5 > 0.$$

Here $\mathrm{Cov}(c_j, \pi_j) = 0.000801$ in A and 0.012894 in B (correlations 0.418 and 0.992). The positive quadratic term ($\gamma_U \frac{\kappa}{2} \Phi^2 \sum_j c_j (\pi_j - \frac{1}{4})^2$) does not overturn the sign. The covariance is positive by construction. π_j is large when column j of W^P receives a lot of weight, and c_j is the off-diagonal part of the same column. "Your

production partners are your friends" therefore makes the agents others care about the same agents who hold power.

Interpretation. A welfare function that counts interdependent utilities, $V_i = U_i + \gamma_U \sum_j W^U_{ij} U_j$, gives extra weight to the consumption of the cared-about. When the cared-about are also the powerful, the extra consumption that bargaining gives them is partly counted as a gain, and the measured loss falls: in B from $0.2106 + 0.0041$ to 0.1803. This is not evidence that elite cohesion is benign. It is a property of the evaluation, and the chapter recommends evaluating structural outcomes with private utilities U. *Common mistake:* reading $L < L_{\text{power}}$ as "externalities reduce the cost of power". The externalities raise the first-best benchmark ($L_{\text{externality}} > 0$); the reduction comes entirely from re-weighting the same allocation.

Exercise 7.4 Structural policy vs. transfer

Baseline (Economy B):

$$\boldsymbol{\pi} = (0.3342,\ 0.4158,\ 0.125,\ 0.125), \qquad \mathbf{x} = (5.6739,\ 6.3261,\ 4.0000,\ 4.0000).$$

Utilitarian welfare is $\sum_i U(x_i) = 14.7894$ and the loss relative to equal division is 0.2106. The Gini coefficient of consumption is 0.1082 and max / min = 1.582.

(a) *Transfer.* $\boldsymbol{d}' = (2, 3, 3, 4)$ still sums to 12, so $\Omega - \sum_j d'_j = 8$ and

$$\mathbf{x}' = \boldsymbol{d}' + 8\boldsymbol{\pi} = (4.6739,\ 6.3261,\ 4.0000,\ 5.0000).$$

The transfer passes through one-for-one: $\Delta\mathbf{x} = \Delta\boldsymbol{d}$, since $\partial x_i / \partial d_i = 1 - \pi_i$ and $\partial x_i / \partial d_j = -\pi_i$ cancel when $\sum_j d_j$ is fixed (Proposition 7.9). Welfare is $\sum_i U(x'_i) = 14.8568$, a gain of 0.0674. The loss relative to equal division falls to 0.1432, the Gini to 0.0913; max / min stays at 1.582, because the Minister is untouched.

(b) *Structural policy.* Raising every cross-block entry to 0.05 (from 0.01 in the enclave rows and 0.03 in the traditional rows), with the difference taken from the diagonal, gives

$$W^P_{\text{new}} = \begin{pmatrix} 0.40 & 0.50 & 0.05 & 0.05 \\ 0.40 & 0.50 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.60 & 0.30 \\ 0.05 & 0.05 & 0.30 & 0.60 \end{pmatrix}.$$

Now both blocks leak $\varepsilon_1 = \varepsilon_2 = 0.10$, so by Proposition 6.12 each block holds exactly half of the power:

- $\boldsymbol{\pi}_{\text{new}} = (0.225, 0.275, 0.250, 0.250)$, with N_{eff} rising from 3.166 to 3.980;
- the spectrum is $\{1, 0.8, 0.3, 0\}$, so Γ rises from 0.08 to 0.20;
- $\mathbf{x}_{\text{new}} = (4.80, 5.20, 5.00, 5.00)$, and $\sum_i U = 14.9960$, a gain of 0.2066;
- the loss relative to equal division is 0.0040, the Gini 0.015 and max / min $=$ 1.083.

(c)

	welfare gain	Gini($\mathbf{x}$)	Farmer's x_4	$\boldsymbol{\pi}$ changed?
Transfer	0.0674	0.0913	5.00	no
Structural	0.2066	0.0150	5.00	yes

The structural policy produces three times the welfare gain and reduces inequality much more. The transfer helps only the Farmer, and only by the amount transferred. The structural policy equalizes the two *blocks*, lifting the Vendor as well, and compresses the gap within the enclave. Only the structural policy changes the power structure, because only it changes W. The Farmer's consumption happens to be 5.00 under both. Note also why the structural policy works. It does not merely add cross-block links; it makes the enclave's dependence on the traditional block (ε_1: 0.02 $\to$ 0.10) rise more than the reverse (ε_2: 0.06 $\to$ 0.10), and the block masses depend only on the ratio $\varepsilon_2 : \varepsilon_1$. The two policies are complements: applying both gives $\mathbf{x} = (3.8, 5.2, 5.0, 6.0)$. *Caveat:* the comparison holds the transfer's size fixed at one unit; a large enough transfer can always beat a given structural change on welfare. The difference is that a transfer must be repeated, while a change in W persists.

Exercise 7.5 The escape threshold

(a) In the notation of Proposition 6.11, $N = 5$, $a = 0.9$ and $s = 0.5$:

$$\pi_h = \frac{a}{a + 1 - s} = \frac{0.9}{1.4} = 0.6429, \qquad \pi_{\text{spoke}} = \frac{1 - \pi_h}{4} = 0.0893,$$

so $\boldsymbol{\pi} = (0.6429, 0.0893, 0.0893, 0.0893, 0.0893)$ and $N_{\text{eff}} = 2.246$. The spectrum is $\{1, s - a, 1 - a\ (\times 3)\} = \{1, -0.4, 0.1, 0.1, 0.1\}$, so $\Gamma = 1 - \max(0.4, 0.1) = 0.6$. Integration is high and power is concentrated.

(b) The spokes are symmetric, so "the most peripheral agent" is any spoke. Raising her power by 50% means $\pi_{\text{spoke}} = 1.5 \times 0.0893 = 0.1339$, hence $\pi_h = 1 -$

$4(0.1339) = 0.4643$. Inverting $\pi_h = a/(a+1-s)$ gives $a = \pi_h(1-s)/(1-\pi_h) = 0.4333$. Every spoke's dependence on the hub must fall from 0.9 to 0.433, by 0.467, or 51.9%. (The new $\Gamma = 0.433$, with spectrum $\{1, 0.567^{(3)}, 0.067\}$.) The response is smooth: $d\pi_{\text{spoke}}/da = -(1-s)/((N-1)(a+1-s)^2) = -0.0638$ at $a = 0.9$. The linear approximation is poor over this range: it predicts a gain of only $0.0638 \times 0.467 = 0.030$ instead of the required 0.045, because $\pi_{\text{spoke}} = (1-s)/((N-1)(a+1-s))$ is convex in a.

If a *single* spoke lowers her own dependence, moving the weight onto herself while the others keep $a = 0.9$, she reaches +50% at $a_1 = 0.571$. Her power rises because she retains more of her own weight (Proposition 6.10 with $j = i$), and the other spokes lose a little (e.g. at $a_1 = 0.6$: $\pi_1 = 0.1282$, the others 0.0855, the hub 0.6154). Whether that "self-reliance" power translates into bargaining power is precisely what Assumption 7.A takes for granted; Proposition 7.11 treats the cost of such moves as a lower bound only.

(c) In each case a is the share of the agent's dependence that runs through a single hub, and the computation says that it must roughly halve before the periphery's position improves by half.

- *Monopsony.* The spoke is a worker in a company town, the hub the employer. Lowering a means alternative employers, a second job, or a union that the employer depends on. It is costly and gradual: each worker's outside options must change, and one worker's exit raises her own weight only modestly unless others move too.

- *Export diversification.* The spoke is a commodity exporter whose earnings depend on one buyer or one commodity market. Lowering a is diversification of markets and products. It moves the country's Perron weight only when the new partners also come to depend on it, i.e. when the change appears in their rows as well.

- *Media.* The spoke is a viewer of state TV. Switching to independent sources lowers a in W^I. Since π_h falls only when *many* spokes switch, the epistemic dominance of a broadcaster is robust to individual exit and fragile to coordinated exit.

In all three cases, Proposition 7.11 says the spectral obstacle is mild in a well-mixed star ($\Gamma = 0.6$, $\|Z\|_\infty$ small); the real obstacle is the cost of creating the alternatives, which the hub can influence. *Common mistake:* expecting the hub's power to fall in proportion to N. By (6.8) it does not depend on N at all.

Exercise 7.6 Bolivia: the full model

This is an open construction; any internally consistent calibration is acceptable if it is stated and its matrices are row-stochastic on the same four nodes (`docs/conventions.md`, §1). The following is one model answer [ILLUSTRATIVE]. Nodes, in order: Gas (G), Government (Gov), Urban informal (U), Rural traditional (R).

$$W^P = \begin{pmatrix} .50 & .30 & .15 & .05 \\ .30 & .40 & .20 & .10 \\ .15 & .10 & .50 & .25 \\ .10 & .05 & .25 & .60 \end{pmatrix}, \; W^U = \begin{pmatrix} .60 & .30 & .05 & .05 \\ .30 & .50 & .10 & .10 \\ .10 & .10 & .60 & .20 \\ .05 & .05 & .20 & .70 \end{pmatrix}, \; W^I = \begin{pmatrix} .60 & .30 & .05 & .05 \\ .30 & .60 & .05 & .05 \\ .20 & .30 & .40 & .10 \\ .10 & .40 & .10 & .40 \end{pmatrix}.$$

Reading. In W^P (input-cost shares) the sectors buy mostly within their own block, and rural producers buy few modern inputs. In W^U caring is concentrated within classes. In W^I urban and rural agents listen to the government (state media) and to gas-sector news; the elites listen almost only to each other. Composite weights $\boldsymbol{\alpha} = (0.4, 0.3, 0.3)$. Outside options $\boldsymbol{d} = (4, 3, 2, 1)$, joint output $\Omega = 20$ (surplus 10), and $\kappa = 0.05$.

(a) $W = 0.4W^P + 0.3W^U + 0.3W^I$ is row-stochastic with rows $(0.56, 0.30, 0.09, 0.05)$, $(0.30, 0.49, 0.125, 0.085)$, $(0.15, 0.16, 0.50, 0.19)$ and $(0.085, 0.155, 0.19, 0.57)$. Perron vectors:

	G	Gov	U	R	N_{eff}	Γ
W^P	0.2563	0.1975	0.2860	0.2602	3.934	0.423
W^U	0.2656	0.2324	0.2282	0.2739	3.975	0.285
W^I	0.3929	0.4405	0.0833	0.0833	2.761	0.600
Composite	0.3107	0.3024	0.2018	0.1851	3.803	0.430

Power in this calibration comes from the information layer: production and caring are nearly balanced, but in W^I the elites hold 83%. Note that $\boldsymbol{\pi}(W) \neq \sum_k \alpha_k \boldsymbol{\pi}(W^k)$ in general; the composite must be computed.

(b) $\mathbf{x}^* = \boldsymbol{d} + 10\boldsymbol{\pi} = (7.1072, 6.0235, 4.0185, 2.8509)$. The largest value of κx_i is $0.355 < 1$, and $\sum_i U(x_i) = 17.2233$. Rural Traditional gets the least: it has the lowest outside option and the lowest power.

(c) Three policies, each defined so that rows stay stochastic:

(i) *Transfer:* one unit of outside option from Gas to Rural, $\boldsymbol{d} = (3, 3, 2, 2)$.

(ii) *Gas-to-fertilizer plant:* Rural's W^P row moves 0.10 from itself to Gas (it now buys fertilizer), and fertilizer raises joint output by 2 ($\Omega = 22$), an assumed productivity effect.

(iii) *Rural internet:* in W^I, Rural moves 0.10 of attention from itself to Urban (it listens more), and, because rural voices become visible online, Urban moves 0.10 and Government 0.05 of their own weight onto Rural (Rural is heard more).

Policy	π	x_R	Δx_R	$\sum_i U$
Baseline	$(0.3107, 0.3024, 0.2018, 0.1851)$	2.8509	—	17.2233
(i) Transfer	unchanged	3.8509	$+1.0000$	17.3861
(ii) Plant, network effect only	$(0.3247, 0.3054, 0.1995, 0.1704)$	2.7043	-0.1465	17.1891
(ii) Plant, with $\Omega = 22$	same	3.0452	$+0.1943$	18.6164
(ii′) Plant buys rural output, $\Omega = 22$	$(0.2872, 0.2973, 0.2057, 0.2098)$	3.5174	$+0.6665$	18.7269
(iii) Internet, two-way	$(0.3078, 0.2951, 0.2026, 0.1945)$	2.9450	$+0.0942$	17.2401
(iii′) Internet, Rural only listens	$(0.3118, 0.3026, 0.2093, 0.1763)$	2.7627	-0.0881	17.2159

Answer. In this calibration the transfer has the largest effect on Rural consumption ($+1.00$). It passes through one-for-one (Exercise 7.4). A structural policy beats it only if it raises π_R by more than $1/(\Omega - \sum_j d_j) = 0.10$, a large change in a four-node network.

The plant as specified *lowers* Rural's power ($0.1851 \to 0.1704$). Buying fertilizer makes Rural depend on Gas, which raises the seller's Perron weight (Proposition 6.10; compare Exercise 6.7(c)). Rural gains only through the larger pie: $\partial x_R / \partial \Omega = \pi_R$ (Proposition 7.9). A plant that instead *buys* rural output (row (ii′): Gas's W^P row moves 0.10 onto Rural) raises π_R to 0.2098 and gives $+0.67$. The internet works only to the extent that it makes Rural *heard*. One-way access, where Rural listens more, lowers π^I_R and her consumption.

What the comparison leaves out. (iii) is primarily a *formation* policy: by changing whose returns Rural agents observe, it shifts $\boldsymbol{\theta}$ over generations (Chapters 9–10). Those effects, the Ψ ratio of Definition 9.13, are not in this one-period bargaining calculation and could reverse the ranking at long horizons. Likewise, the transfer is a flow that must be financed every period, while (ii) and (iii) change W permanently. *Common mistakes:* composing layers defined on different node sets (sectors in W^P, persons in W^I), and assuming that "more connection" raises the connected agent's power whatever the direction of the new links.

Exercise 7.7 Data: labor shares as bargaining outcomes

Status of the data. CEPALSTAT labor-share series were *not* fetched into the repository (`data/README.md` lists what was fetched), so no CEPALSTAT values are reported here. For Bolivia, the same concept can be computed from INE's supply-

use tables, which are in `data/processed`. Those results are reported below. For Chile and Mexico only the method is given.

Method.

1. *Series.* CEPALSTAT, national accounts, "Gross domestic product by income component": compensation of employees (D.1) and gross value added or GDP. Take annual values, 2010–latest, for BOL, CHL and MEX. Use the same denominator for all three (GVA at basic prices is preferable to GDP at market prices, because net taxes on products are not factor income).

2. *Adjustment for self-employment.* Compensation of employees omits the labor income of the self-employed, which sits in *mixed income* (B.3). Self-employment is 68.1% of employment in Bolivia, 24.4% in Chile and 30.3% in Mexico (2019; WDI series SL.EMP.SELF.ZS, ILO modelled estimates, in `data/processed/wdi_long.csv`, retrieved 2026-11-05). The unadjusted share is therefore not comparable across these countries. Report an adjusted share that attributes to labor the fraction $D1/(D1+B2)$ of mixed income (the economy-wide employee share of the non-mixed income).

3. *Two-agent TU model.* Normalize $\Omega = \text{GVA} = 1$, with labor and capital as the agents. Assume outside options (d_L, d_K) as shares of GVA, satisfying $d_L \le s_L$ and $d_K \le 1 - s_L$ (individual rationality). By (7.7), $s_L = d_L + \pi_L(1 - d_L - d_K)$, so
$$\pi_L = \frac{s_L - d_L}{1 - d_L - d_K}.$$

Results for Bolivia (INE, *Cuadros de Oferta y Utilización*, base 2017, use table, value-added block summed over the 39 industries; Instituto Nacional de Estadística (INE), Bolivia, 2026; `data/processed/ine_sut_long.csv`, retrieved 2026-11-05). The accounting identity $B1b = D1 + D2 - D3 + B2b + B3b$ holds to 10^{-6} in every year (sanity check).

	2017	2018	2019	2020	2021	2022	2023	2024
$s_L = D1/B1b$	0.409	0.414	0.411	0.433	0.401	0.386	0.393	0.410
s_L adjusted	0.525	0.531	0.531	0.557	0.518	0.497	0.513	0.532
mixed income / GVA	0.221	0.219	0.224	0.220	0.225	0.223	0.233	0.228

(2023 and 2024 are preliminary in INE's release.)

(a) With 2019 values ($s_L = 0.4114$; adjusted 0.5307), the implied labor weight under five outside-option assumptions is:

(d_L, d_K)	$(0,0)$	$(0.2,0.2)$	$(0.3,0.1)$	$(0.1,0.3)$	$(0.35,0.2)$
π_L, unadjusted s_L	0.411	0.352	0.186	0.519	0.136
π_L, adjusted s_L	0.531	0.551	0.384	0.718	0.402

A natural baseline is d_L equal to subsistence or informal-sector earnings and d_K equal to the return capital could earn abroad. With no outside options, $\pi_L = s_L$: the labor share *is* the bargaining weight. For Chile and Mexico the computation is identical once the CEPALSTAT series are downloaded. Given the much lower self-employment rates there, one expects the adjustment to matter less than in Bolivia. That is a qualitative expectation, not a result.

(b) The implied weight is very sensitive to the assumed outside options. Differentiating,

$$\frac{\partial \pi_L}{\partial d_L} = -\frac{(1 - s_L) - d_K}{(1 - d_L - d_K)^2} < 0, \qquad \frac{\partial \pi_L}{\partial d_K} = \frac{s_L - d_L}{(1 - d_L - d_K)^2} > 0;$$

at $s_L = 0.4114$ and $\boldsymbol{d} = (0.2, 0.2)$ these are -1.080 and $+0.587$. Moving 0.1 of GVA between the two outside options changes π_L from 0.186 to 0.519, a factor of nearly three, at the same observed labor share. As d_L ranges over $[0, s_L)$ and d_K over $[0, 1 - s_L)$, π_L ranges over the whole interval $(0, 1)$.

Implication for identification. One observed share, s_L, is one equation in three unknowns (π_L, d_L, d_K). Distributional data alone identify $\boldsymbol{\pi}$ only up to the outside options, and not at all without restrictions on them. Identification needs independent evidence on outside options (unemployment or informal-sector earnings, capital mobility), variation that shifts $\boldsymbol{d}$ but not $\boldsymbol{\pi}$ (or the reverse), or an independent measurement of W, from which $\boldsymbol{\pi}(W)$ can be computed and the allocation rule tested rather than inverted. This is the same lesson as Chapter 11's for Ψ: an outcome is not a structural parameter. *Common mistakes:* using unadjusted compensation shares in economies with large self-employment; comparing shares with different denominators (GDP vs GVA); and treating the inverted π_L as a measurement of power rather than as a residual that absorbs every misspecified outside option.

Chapter 8

Belief Manipulation and Epistemic Power

The reading of $\boldsymbol{\pi}$ in force is the information reading: $\boldsymbol{\pi}^I$ is the vector of DeGroot consensus weights, $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)}$ (Theorem 8.1). Every number is printed by the script `code/solutions/sol08.py`.

Exercise 8.1 DeGroot by hand

(a) $\mathbf{b}^{(t)} = W^I\mathbf{b}^{(t-1)}$, row by row:

$$b_1^{(1)} = 0.5(1) + 0.3(0) + 0.2(0.5) = 0.600, \quad b_2^{(1)} = 0.1(1) + 0.6(0) + 0.3(0.5) = 0.250,$$
$$b_3^{(1)} = 0.3(1) + 0.2(0) + 0.5(0.5) = 0.550;$$
$$b_1^{(2)} = 0.300 + 0.075 + 0.110 = 0.485, \quad b_2^{(2)} = 0.060 + 0.150 + 0.165 = 0.375,$$
$$b_3^{(2)} = 0.180 + 0.050 + 0.275 = 0.505;$$
$$\mathbf{b}^{(3)} = (0.456,\ 0.425,\ 0.473),$$

the last from $b_1^{(3)} = 0.2425 + 0.1125 + 0.1010$, $b_2^{(3)} = 0.0485 + 0.2250 + 0.1515$ and $b_3^{(3)} = 0.1455 + 0.0750 + 0.2525$. The spread $\max_i b_i - \min_i b_i$ falls from 1 to 0.35, 0.13 and 0.048. Each round averages, so every $b_i^{(t)}$ lies in $[\min_j b_j^{(t-1)}, \max_j b_j^{(t-1)}]$, a useful check.

(b) The column sums are $(0.9, 1.1, 1.0)$, so W^I is not doubly stochastic and $\boldsymbol{\pi}^I$ is not uniform (Proposition 6.8). Solving $\boldsymbol{\pi}'W^I = \boldsymbol{\pi}'$ with $\mathbf{1}'\boldsymbol{\pi} = 1$ (e.g. replace one balance equation by the normalization) gives

$$\boldsymbol{\pi}^I = \left(\tfrac{14}{50},\ \tfrac{19}{50},\ \tfrac{17}{50}\right) = (0.28,\ 0.38,\ 0.34).$$

Check, first column: $0.28(0.5) + 0.38(0.1) + 0.34(0.3) = 0.140 + 0.038 + 0.102 = 0.28$.

(c) $b^* = (\boldsymbol{\pi}^I)'\mathbf{b}^{(0)} = 0.28(1) + 0.38(0) + 0.34(0.5) = 0.45$. The iterates confirm it: $\max_i |b_i^{(t)} - 0.45| = 2.68 \cdot 10^{-3}$ at $t = 5$, $5.37 \cdot 10^{-6}$ at $t = 10$ and $5.43 \cdot 10^{-11}$ at $t = 20$. $W^I > 0$ is primitive, so Theorem 8.1 applies. Agent 2, the most epistemically powerful agent ($\pi_2 = 0.38$), starts at 0, which is why the consensus lies below the simple mean of the priors (0.5).

(d) The eigenvalues other than 1 satisfy $\lambda_2 + \lambda_3 = \operatorname{tr} W - 1 = 0.6$ and $\lambda_2\lambda_3 = \det W = 0.1$, so $\lambda_{2,3} = 0.3 \pm 0.1\mathrm{i}$. Then $|\lambda_2| = \sqrt{0.1} = 0.3162$ and $\Gamma^I = 0.6838$. By Proposition 8.3 the half-life of disagreement is

$$t_{1/2} = \frac{\ln 2}{\ln(1/0.3162)} = 0.602 \text{ rounds.}$$

Disagreement falls by more than half every round. With complex λ_2 the convergence oscillates slightly; the modulus governs the rate. *Common mistake:* using $\operatorname{Re}\lambda_2 = 0.3$ instead of $|\lambda_2|$.

Exercise 8.2 Effective sample size

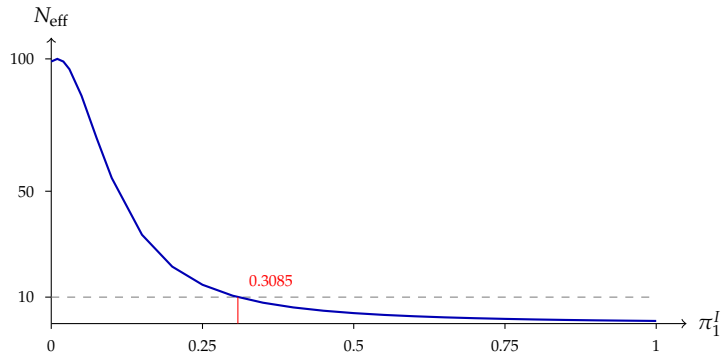
With $\pi_1 = p$ and the rest $(1-p)/99$ each, $N_{\text{eff}}(p) = 1/(p^2 + (1-p)^2/99)$ (Remark 8.6).

(a) $\boldsymbol{\pi} = \frac{1}{100}\mathbf{1}$: $N_{\text{eff}} = 100$, the maximum (Definition 6.15).

(b) $p = 0.5$: $N_{\text{eff}} = 1/(0.25 + 0.25/99) = 3.96$.

(c) $p = 0.9$: $N_{\text{eff}} = 1/(0.81 + 0.01/99) = 1.234$.

(d) $N_{\text{eff}}(p)$ rises from 99 at $p = 0$ to its maximum 100 at $p = 0.01$ (uniform power) and then falls monotonically to 1 at $p = 1$.



$N_{\text{eff}} = 10$ solves $p^2 + (1-p)^2/99 = 0.1$, i.e. $100p^2 - 2p - 8.9 = 0$, whose positive root is

$$p^* = \frac{2 + \sqrt{4 + 3560}}{200} = 0.3085.$$

N_{eff} drops below 10 once a single voice holds more than 30.85% of epistemic weight. With equally precise independent signals, the consensus error variance is then more than 10 times the $\sigma^2/100$ that equal weighting would achieve (Proposition 8.5). *Common mistake:* assuming N_{eff} is monotone on all of $[0, 1]$; it is not, because $p < 1/N$ concentrates weight on the other 99 agents.

Exercise 8.3 Manipulation cost

(a) The current consensus is $b^* = 0.45$, so the target requires $\Delta b = 0.35$. Any target in $[\min_j b_j^{(0)}, \max_j b_j^{(0)}] = [0, 1]$ is reachable (Definition 8.8), so 0.8 is. A shift $\Delta\boldsymbol{\pi}$ must satisfy $\Delta\boldsymbol{\pi}'\mathbf{b}^{(0)} = 0.35$ and $\mathbf{1}'\Delta\boldsymbol{\pi} = 0$, and it must keep $\boldsymbol{\pi} + \Delta\boldsymbol{\pi} \geq 0$. The shift of smallest Euclidean norm is proportional to the demeaned beliefs $\mathbf{b}^{(0)} - \bar{b}\mathbf{1} = (0.5, -0.5, 0)$:

$$\Delta\boldsymbol{\pi} = 0.35\,\frac{(0.5, -0.5, 0)}{0.5} = (+0.35,\ -0.35,\ 0), \qquad \boldsymbol{\pi}^{\text{new}} = (0.63,\ 0.03,\ 0.34).$$

Check: $0.63(1) + 0.03(0) + 0.34(0.5) = 0.80$. Epistemic weight must move from agent 2 (belief 0) to agent 1 (belief 1): agent 2's weight must fall by 92%. This is also the cheapest shift in ℓ_1, since it moves mass between the two extreme beliefs.

(b) By Proposition 8.9, an elementary move $E = \mathbf{e}_j(\mathbf{e}_i - \mathbf{e}_k)'$ (row j moves weight from k to i) changes the consensus at rate $db^*/dt = \pi_j(z_i - z_k)$, with $\mathbf{z} = Z(\mathbf{b}^{(0)} - b^*\mathbf{1})$. Here

$$Z = (I - W^I + \mathbf{1}\boldsymbol{\pi}')^{-1} = \begin{pmatrix} 1.296 & -0.084 & -0.212 \\ -0.304 & 1.316 & -0.012 \\ 0.096 & -0.284 & 1.188 \end{pmatrix}, \qquad \mathbf{z} = (0.74,\ -0.76,\ 0.24).$$

($Z\mathbf{1} = \mathbf{1}$ is a check.) The rate is largest for $i = 1$ (largest z), $k = 2$ (smallest z) and j with the largest π_j. The ranking:

Edge change	db^*/dt	capacity W_{jk}
Agent 2 moves weight from herself to agent 1 ($W_{22} \to W_{21}$)	0.570	0.6
Agent 3 moves weight from agent 2 to agent 1 ($W_{32} \to W_{31}$)	0.510	0.2
Agent 1 moves weight from agent 2 to herself ($W_{12} \to W_{11}$)	0.420	0.3

The cheapest edge makes the most influential agent (agent 2) listen to the agent whose belief is the target (agent 1) instead of to herself. To first order the target needs $t = 0.35/0.57 = 0.614$, more than the capacity 0.6. Exactly, moving all of W_{22} only reaches $b^* = 0.636$, so no single edge suffices. Combining edges, the projected steepest-ascent path in W (rows kept stochastic and nonnegative) reaches $b^* = 0.80$ with $\|\Delta W^I\|_F = 0.484$ (path length 0.506), which is an upper bound on $C(0.8)$. The first-order estimate ignoring sign constraints is $0.35/\|P(G)\|_F = 0.557$, where $G_{jk} = \pi_j z_k$ is the gradient and P removes row means. The resulting network has $\boldsymbol{\pi} = (0.6216, 0.0199, 0.3586)$, with agent 3 no longer listening to agent 2 at all ($W_{32} = 0$).

(c) $\|Z\|_\infty = 1.632$ and $\max_j |b_j^{(0)} - b^*| = 0.55$, so the bound (8.10) gives

$$t\,\|E\|_\infty \;\geq\; \frac{0.35}{0.55 \times 1.632} = 0.390.$$

Also $\|Z\|_\infty = 1.632 \geq \max_k 1/|1-\lambda_k| = 1/|0.7 - 0.1\mathrm{i}| = 1.414$, as the proposition requires.

Lazy networks. $W_\alpha = \alpha W^I + (1-\alpha)I$ has the same $\boldsymbol{\pi}$, and $I - W_\alpha = \alpha(I - W^I)$. On the invariant complement $\{\mathbf{x} : \boldsymbol{\pi}'\mathbf{x} = 0\}$ the inverse therefore scales by $1/\alpha$, which gives the exact identity

$$Z_\alpha = \mathbf{1}\boldsymbol{\pi}' + \frac{1}{\alpha}\left(Z - \mathbf{1}\boldsymbol{\pi}'\right)$$

(verified numerically). Since $\boldsymbol{\pi}'(\mathbf{b}^{(0)} - b^*\mathbf{1}) = 0$, $\mathbf{z}_\alpha = \mathbf{z}/\alpha$, and every edge's effect on the consensus scales by $1/\alpha$:

α	$\Gamma(W_\alpha)$	$\|Z_\alpha\|_\infty$	bound on $t\|E\|_\infty$	best edge db^*/dt
1	0.6838	1.632	0.3899	0.570
0.5	0.3481	3.624	0.1756	1.140
0.25	0.1746	7.688	0.0828	2.280
0.1	0.0699	19.880	0.0320	5.700

As α falls, Z grows like $1/\alpha$, the lower bound on the cost falls in proportion to α, and manipulation becomes cheaper. Two readings, both correct:

1. The eigenvalues $1 - \alpha(1-\lambda_k)$ approach $+1$, which is the case Lemma 6.19 identifies as sensitive.

2. In a sticky network agents give little weight to others, so a perturbation of given size is large relative to the existing off-diagonal weights. Redirecting even a little of a large self-weight changes who matters.

Common mistake: concluding that small Γ protects a society from manipulation because "beliefs move slowly". Slow mixing and a manipulable consensus go together when the small gap comes from eigenvalues near $+1$.

Exercise 8.4 The algorithm as W^I designer

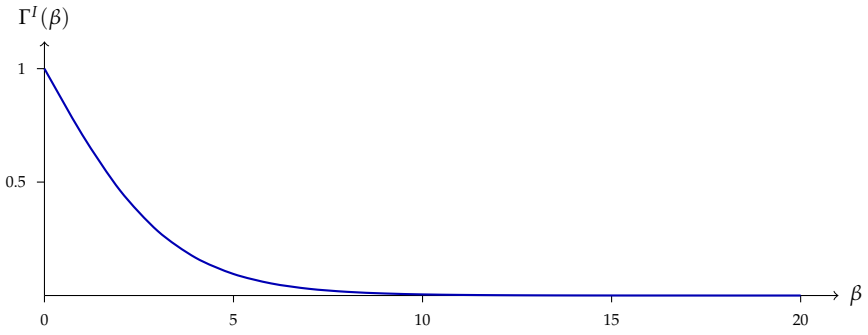
Agents 1–4 have $b^{(0)} = 0.8$ and agents 5–8 have $b^{(0)} = 0.2$.

(a) $W^I = \frac{1}{8}\mathbf{1}\mathbf{1}'$: $\boldsymbol{\pi}^I = \frac{1}{8}\mathbf{1}$, so $b^* = 0.5$. The matrix has rank one, so $\Gamma^I = 1$ and consensus is reached in one round.

(b) Block-diagonal with $\frac{1}{4}$ on every within-block entry. Each block averages to its own prior in one round, so the limits are 0.8 for liberals and 0.2 for conservatives. The eigenvalue 1 has multiplicity two (spectrum $\{1, 1, 0^{(6)}\}$), so $\Gamma^I = 0$. W^I is reducible and $\boldsymbol{\pi}^I$ is not unique; there is no global consensus.

(c) Let $q = e^{-0.6\beta}$, since $|b_i - b_j| \in \{0, 0.6\}$. Every row has four entries 1 and four entries q before normalization, so within-group weights are $1/(4(1+q))$ and cross-group weights $q/(4(1+q))$:

β	within W_{ij}	across W_{ij}	$\Gamma^I = 1 - \tanh(0.3\beta)$	half-life (rounds)
0	0.125000	$1.250 \cdot 10^{-1}$	1.0000	0
1	0.161414	$8.859 \cdot 10^{-2}$	0.7087	0.562
5	0.238144	$1.186 \cdot 10^{-2}$	0.0949	6.96
20	0.249998	$1.536 \cdot 10^{-6}$	$1.23 \cdot 10^{-5}$	$5.64 \cdot 10^{4}$



(d) *Claim.* For every finite β, W^I is symmetric and doubly stochastic, so $b^* = 0.5$.

Proof. The kernel $K_{ij} = e^{-\beta|b_i - b_j|}$ is symmetric. Its row sums are all equal: each row contains four entries 1 (own group, including i) and four entries q, so $K\mathbf{1} = 4(1+q)\mathbf{1}$. Hence $W^I = K/(4(1+q))$ is symmetric with unit row sums, and therefore unit column sums. For $\beta < \infty$, $q > 0$ and $W^I > 0$ is primitive. By

Proposition 6.8, $\boldsymbol{\pi}^I = \frac{1}{8}\mathbf{1}$, and by Theorem 8.1, $b^* = \frac{1}{8}(4(0.8) + 4(0.2)) = 0.5$. The society always reaches *one* consensus, and it is the democratic one.

Spectrum. $W^I = \frac{1}{4(1+q)}\begin{pmatrix} J & qJ \\ qJ & J \end{pmatrix}$ with $J = \mathbf{1}_4\mathbf{1}_4'$. The partition into the two groups is exactly lumpable, with $\bar{W} = \frac{1}{1+q}\begin{pmatrix} 1 & q \\ q & 1 \end{pmatrix}$. Its eigenvalues are 1 and $(1-q)/(1+q)$. The other six eigenvalues are 0 (vectors summing to zero within each group). Hence

$$|\lambda_2| = \frac{1 - e^{-0.6\beta}}{1 + e^{-0.6\beta}} = \tanh(0.3\beta), \qquad \Gamma^I = 1 - \tanh(0.3\beta).$$

What diverges is the time to consensus. The half-life is $t_{1/2}(\beta) = \ln 2 / \ln(\coth(0.3\beta))$. Since $1 - \tanh x \approx 2e^{-2x}$, $t_{1/2} \approx \frac{\ln 2}{2} e^{0.6\beta}$ for large β: 6.96 rounds at $\beta = 5$, about 140 at $\beta = 10$, and $5.64 \cdot 10^4$ at $\beta = 20$. The algorithm does not change *what* the society eventually believes. It changes *when*: for any realistic horizon, at large β the society behaves as if it were split into two echo chambers. The limit $\beta = \infty$ is the reducible matrix of (b). *Common mistake:* inferring polarization of the *limit* from polarization at every observed date. Here the limit is unique and equal-weighted; only the speed is affected. Concentration and integration are different coordinates (Proposition 6.16).

Exercise 8.5 Bolivia's epistemic structure

A model answer [ILLUSTRATIVE]. No audience data are used in (a)–(d); the matrices are stylized and stated so that they can be replaced by data (part (e)).

(a) Each city has three nodes: the state outlet (S), the private outlet (P) and citizens (C). The within-city matrices (rows S, P, C; row i is whom i listens to) are

$$B_{\text{LP}} = \begin{pmatrix} .8 & 0 & .2 \\ .1 & .6 & .3 \\ .5 & .2 & .3 \end{pmatrix}, \quad B_{\text{SC}} = \begin{pmatrix} .6 & .1 & .3 \\ 0 & .8 & .2 \\ .15 & .55 & .3 \end{pmatrix}, \quad B_{\text{CB}} = \begin{pmatrix} .7 & .1 & .2 \\ .1 & .7 & .2 \\ .35 & .35 & .3 \end{pmatrix}.$$

In La Paz citizens put 0.5 on state media; in Santa Cruz 0.55 on private media; Cochabamba is mixed. Outlets mostly reproduce their own line and listen somewhat to citizens. Alone, each city's Perron vector over (S, P, C) is $(0.647, 0.118, 0.235)$ in La Paz, $(0.087, 0.681, 0.232)$ in Santa Cruz and $(0.389, 0.389, 0.222)$ in Cochabamba.

(b) *National composite.* Every node keeps $1 - \eta = 0.9$ of its city row and gives $\eta = 0.1$ to the node of the same type in the other two cities (0.05 each): state

outlets follow the national state network, private outlets each other, citizens other citizens. All nine rows sum to one. The national Perron vector is

	State	Private	Citizens
La Paz	0.1851	0.0672	0.0811
Santa Cruz	0.0576	0.1956	0.0802
Cochabamba	0.1282	0.1304	0.0748
total by type	0.3709	0.3931	0.2361

$N_{\text{eff}} = 7.56$ of 9 and $\Gamma^I = 0.15$. Each city holds exactly $\frac{1}{3}$. The city partition is lumpable with the symmetric chain $(1-\eta)I + \frac{\eta}{2}(J-I)$, whose stationary vector is uniform (Proposition 6.12 generalized to three blocks). Its non-unit eigenvalue $1 - \frac{3}{2}\eta = 0.85$ (double) sets Γ^I: the cities mix slowly. Citizens, the majority of the population, hold only 24% of epistemic weight. The outlets hold 76%, split almost evenly between state and private media.

(c) $b^* = \sum_j \pi_j^I b_j^{(0)}$ is linear in the priors, so "the consensus favors nationalization" ($b^* > 0.5$, with $b = 1$ full nationalization) is a half-space of initial beliefs:

$$0.3709\,\bar{b}_S + 0.3931\,\bar{b}_P + 0.2361\,\bar{b}_C > 0.5,$$

where $\bar{b}_S$ is the π-weighted average belief of the state outlets, and so on. For example, with state outlets at 0.9 and private outlets at 0.2, the consensus favors nationalization iff the citizens' (common) prior exceeds 0.371: $b^* = 0.483$ at a citizen prior of 0.3, 0.530 at 0.5 and 0.578 at 0.7. Liberalization wins in the complementary half-space. Because state and private outlets carry similar total weight, the outcome is decided by the citizens' priors and by the small asymmetry between the media blocks, not by population.

(d) Calibrate the citizens' prior so that $b^* = 0.6$ (state 0.9, private 0.2, citizens 0.7947). The cost of reaching $b_T = 0.4$:

- *Lower bound* (Proposition 8.9): $\|Z\|_\infty = 9.11$, so $t\|E\|_\infty \geq 0.2/(0.4 \cdot 9.11) = 0.055$.
- *First-order Frobenius estimate*: $0.2/\|P(G)\|_F = 0.133$.
- *Achieved* by the projected steepest-descent path (rows stochastic and nonnegative): $\|\Delta W^I\|_F = 0.212$ (path length 0.229), an upper bound on $C(0.4)$. The optimal path spreads small changes over all nine rows (row norms 0.035–0.095) and makes the Santa Cruz private outlet the dominant voice ($\pi^I = 0.551$).

Natural targeted policies cost more:

- every citizen moving 0.1 of attention from her state outlet to her private outlet costs $\|\Delta W\|_F = 0.245$ and reaches only $b^* = 0.551$ (larger moves are infeasible in Santa Cruz, whose citizens give state media only 0.135);
- state outlets moving 0.2 of their own weight onto the city's private outlet costs 0.490 and reaches 0.504.

Shifting the consensus by 0.2 thus costs a Frobenius perturbation of about 0.2 when chosen optimally, far less than the targeted policies. It is cheap because inter-city coupling is weak ($\lambda_2 = 0.85$, $\|Z\|_\infty = 9.11$).

(e) *Data: method only.* Latinobarómetro microdata require registration and were *not* fetched (`data/README.md` §9); audience ratings (Kantar IBOPE) are paywalled. No values are reported. Method:

1. *Survey.* Latinobarómetro, Bolivia sample, the most recent wave that asks how respondents obtain political information (sources: television, radio, newspapers, social networks, family and friends) and records city or department. Check the codebook of the chosen wave for the exact item.
2. *Citizens' rows.* For each city, take the weighted shares of respondents naming each source. Map "family and friends" to the citizens' own node C. Split television and radio into state and private using channel-level audience shares (needed from a second source, or from a wave that asks which channel). Normalize to sum to $1 - \eta$, and calibrate the cross-city weight η from the share of national (non-local) media.
3. *Outlets' rows.* These are not in the survey. Use content data, such as the share of an outlet's political stories sourced from government, from other outlets or from citizens, or keep the stylized rows and report the sensitivity of the results to them.
4. *Compute* $\boldsymbol{\pi}^I$, N_{eff} (and $N_{\text{eff}}/9$), the city and type masses, and Γ^I, with bootstrap intervals from the survey weights.

Expected result (qualitative, not a finding). Since citizens' rows place most weight on outlets and outlets place little on citizens, $\boldsymbol{\pi}^I$ should concentrate on the outlets, as in the stylized model. Whether state or private media hold more weight nationally depends on the measured city rows and on η; this exercise cannot settle it without the data.

Exercise 8.6 The belief-action-signal loop

(a) With $W^I_{\text{block}} = \text{diag}(W_{11}, W_{22})$ and each block row-stochastic, an agent in sector s observes only returns from her own sector. Since all of them are r_s, $\hat{r}_i = \sum_j W^I_{ij} r_j = r_s$ (rows sum to one). Modern agents believe $\hat{r}_M = 0.15$ and Traditional agents $\hat{r}_T = 0.05$. Both beliefs are correct about the returns each sector observes (step 1 of Definition 8.12), and neither sector learns the other's return.

(b) $a = \theta\hat{r}$ rises from 0.05θ to 0.10θ: by 0.05θ, i.e. it doubles (+100%). The proportional effect does not depend on θ.

(c) Let 10% of the Traditional sector (say 10 of 100 agents, next to 50 Modern agents; uniform averaging within each block) put 0.5 on the Modern block. *Beliefs, one step* (the loop as defined, where agents observe returns): the treated agents believe 0.10, the rest 0.05, so the sector average rises from 0.05 to $0.9(0.05) + 0.1(0.10) = 0.055$ (+10%).

Spectral gap. Before, W^I is block-diagonal with two closed classes, so $\Gamma^I = 0$. After, the Modern block is the only closed class and the Traditional block is transient. The Traditional sub-block is rank one, $Q = \mathbf{q}\mathbf{1}'/n_T$ with $q_i \in \{1, 0.5\}$, so its only non-zero eigenvalue is $\mathbf{1}'\mathbf{q}/n_T = 1 - f/2 = 0.95$. Hence

$$|\lambda_2| = 0.95, \qquad \Gamma^I = f/2 = 0.05.$$

If agents also average each other's beliefs (iterated DeGroot), there is now a single consensus equal to the Modern return. The Traditional sector's mean belief is 0.055 after one round, 0.101 after 14 and 0.149 after 100, with half-life $\ln 2/\ln(1/0.95) = 13.5$ rounds. A few smartphones change the long-run belief of the whole sector, slowly. Which reading applies depends on whether agents transmit beliefs or only observe outcomes, and that is an empirical question.

(d) *The loop.* Generation t:

$$\hat{\mathbf{r}}_t = W^I \mathbf{r}_{t-1} \;\longrightarrow\; a_{i,t} = \theta_i \hat{r}_{i,t} \;\longrightarrow\; r_{i,t} = g_i(a_{i,t}) \;\longrightarrow\; \text{next signal } \mathbf{r}_t.$$

With block-diagonal W^I and linear returns $g_T(a) = r_0 + \gamma a$, the Traditional sector's return follows $r_{T,t} = r_0 + \gamma\theta\, r_{T,t-1}$. Its fixed point is $r^*_T = r_0/(1 - \gamma\theta)$.

- *Dynamic stability.* The fixed point is stable iff $|\gamma\theta| < 1$. In general, a fixed point is locally stable iff the spectral radius of $W^I \,\text{diag}(g_i'(a_i^*)\theta_i)$ is below one. Example: $\theta = 1$, $\gamma = 0.4$, $r_0 = 0.03$ gives $r^*_T = 0.05$, stable. With the smartphone share $f = 0.1$ (sector signal $(1 - f/2)r_T + (f/2)r_M$), the fixed

point becomes $r_T^* = (r_0 + \gamma\theta\frac{f}{2}r_M)/(1 - \gamma\theta(1 - \frac{f}{2})) = 0.0532$. The simulated loop converges there.

- *A trap.* For the low-return equilibrium to be a *trap*, i.e. stable and coexisting with a high-return one, g_T must be nonlinear: S-shaped, with slope $\gamma\theta > 1$ in a middle range. Then the low fixed point is stable if $\theta g_T'$ at it is below one *and* W^I gives the sector no weight on modern returns. With linear g there is a unique steady state and no trap (compare Example 9.10).

The block structure of W^I is what makes the low equilibrium self-confirming: beliefs are correct about the returns observed, and actions reproduce those returns (keyinsight after Definition 8.12). Part III replaces $\hat{r}$ by the preference parameter θ (Definition 9.7).

Exercise 8.7 Quijano in matrix form

(a) A minimal model [ILLUSTRATIVE]: $N = 10$, with 2 European-descended agents (E) and 8 Indigenous agents (I).

- E rows: 0.45 on each E agent, so $\varepsilon_{EI} = 0.1$ on I, spread as 0.0125 each.
- I rows: 0.3 on each E agent, so $\varepsilon_{IE} = 0.6$ on E; 0.2 on oneself and $0.2/7 = 0.0286$ on each other I agent.

Every row of a block places the same total weight on the other block, so Proposition 6.12 gives the E block's mass exactly: $\varepsilon_{IE}/(\varepsilon_{EI} + \varepsilon_{IE}) = 0.6/0.7 = 0.857$. Hence $\pi_E^I = 0.4286$ each (4.3 times $1/N$) and $\pi_I^I = 0.0179$ each (0.18 times $1/N$), with $N_{\text{eff}} = 2.70$ of 10 and $\Gamma^I = 0.7$. The network is well integrated and very unequal, which is the combination Proposition 6.16 allows. The mechanism is *deference*: Indigenous agents listen to Europeans, who do not listen back.

(b) Let b_i be the belief that Western science or formal schooling is the more valuable form of knowledge. Suppose $b = 0.9$ for E and $b = 0.3$ for I. The consensus is $b^* = 0.857(0.9) + 0.143(0.3) = 0.814$, against a population mean of priors of 0.42. Society converges on the colonial valuation although 80% of its members start from the opposite one. If the priors are noisy signals of the true value, the consensus error variance is $\sigma^2/N_{\text{eff}} = 0.370\sigma^2$, against $0.1\sigma^2$ with equal voice (Proposition 8.5). The network discards most of the information held by Indigenous agents. Their priors are neither less accurate by assumption nor ignored by anyone's choice; they simply carry little weight in $\boldsymbol{\pi}^I$ (Definition 8.2).

(c) Through the belief-action-signal loop (Definition 8.12) and the formation equation (9.1), the consensus valuation becomes a signal about the returns to Spanish-language formal schooling, migration to the city and wage work, relative to Indigenous languages, traditional medicine and artisanal skill. Agents form θ accordingly: aspirations toward the valued forms, and away from the devalued ones even where the latter have real returns.

For illustration, take Example 9.10's $\mu/(1-\rho) = 1.5$, anchor 1 and reference 0.5. Then $\theta^* = 1 + 1.5(0.814 - 0.5) = 1.47$ under the colonial consensus and $1 + 1.5(0.42 - 0.5) = 0.88$ under equal voice. The point is qualitative: preferences inherit the epistemic hierarchy. They then look like "free choices" (Section 9.1.4), and they reproduce the asymmetry in W^{UI} and W^{P} that Quijano emphasizes.

(d) *Decoloniality as a matrix perturbation.* Equal voice is $\boldsymbol{\pi}^I = \frac{1}{10}\mathbf{1}$, i.e. $W^I + \Delta W^I$ doubly stochastic (Proposition 6.8). Within the two-parameter family above it requires the E block's mass to equal 2/10, i.e. $\varepsilon_{IE}/(\varepsilon_{EI} + \varepsilon_{IE}) = 0.2$, or $\varepsilon_{EI} = 4\,\varepsilon_{IE}$. Write $\Delta W^I = \Delta^E + \Delta^I$, where

- Δ^E changes the E rows: Europeans listen to Indigenous agents, moving weight from E columns to I columns;
- Δ^I changes the I rows: Indigenous agents stop deferring, moving weight from E columns to I columns.

Both terms have zero row sums. Two facts follow:

1. *Giving voice alone is impossible.* With $\varepsilon_{IE} = 0.6$ unchanged, equal power needs $\varepsilon_{EI} = 2.4 > 1$. Deference must fall too: $\varepsilon_{IE} \leq 0.25$.
2. *The cheapest equalizing perturbation in the family* has $\varepsilon_{IE} = 0.1625$ and $\varepsilon_{EI} = 0.650$, with $\|\Delta W^I\|_F = 1.258$. Europeans must place two-thirds of their attention on Indigenous voices, and Indigenous agents must cut their deference from 0.6 to 0.16. The consensus becomes the population mean, 0.42, and $N_{\text{eff}} = 10$; $\Gamma^I = 0.641$ stays high.

Other equalizing choices, such as $(0.8, 0.2)$ or $(1.0, 0.25)$, cost 1.272 and 1.335. A partial reform, Europeans listening ($0.1 \to 0.8$) with deference unchanged, costs 0.783 but leaves the E mass at 0.429.

Interpretation. By Proposition 6.10, power is conferred by *others'* attention, so decoloniality is a change in the rows of the powerful (who listens) as much as in the rows of the subordinated (who defers). The perturbation is large ($\|\Delta W^I\|_F$ above 1 in a 10×10 row-stochastic matrix), consistent with the chapter's point

that the obstacle is structural, not informational. Note what it does not claim: equal voice maximizes accuracy only if traditions carry independent and comparably precise information (Proposition 8.5). *Common mistake:* proposing to "raise π^I of Indigenous agents" by changing their own rows toward themselves. Self-weight does raise one's Perron weight, but it isolates rather than integrates, and it typically lowers Γ^I.

Chapter 9

Where Preferences Come From

All exercises use the formation equation (9.1),

$$\theta_{i,t+1} = \rho\,\theta_{i,t} + (1-\rho)\bar{\theta} + \mu\Big(\sum_j W^I_{ij} r_{j,t} - r^*\Big),$$

with t in generations and $\kappa := \mu/(1-\rho)$ the formation multiplier. With a constant signal s the solution is $\theta_t = \theta^* - (\theta^* - \theta_0)\rho^t$, with $\theta^* = \bar{\theta} + \kappa(s - r^*)$; this closed form is used repeatedly. Every number is printed by the script `code/solutions/sol09.py`.

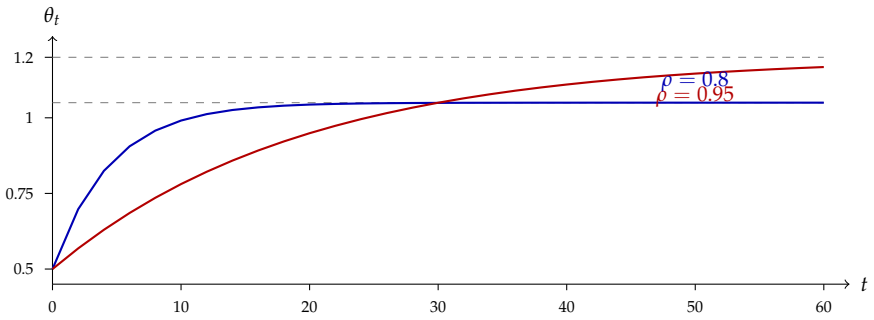
Exercise 9.1 The formation equation by hand

(a) With $\rho = 0.8$, $\bar{\theta} = 1$, $\mu = 0.2$, $r = 0.15$ and $r^* = 0.1$, the map is $\theta_{t+1} = 0.8\,\theta_t + 0.2 + 0.2(0.05) = 0.8\,\theta_t + 0.21$. From $\theta_0 = 0.5$, $\theta_1 = 0.8(0.5) + 0.21 = 0.61$, $\theta_2 = 0.8(0.61) + 0.21 = 0.698$, and so on:

t	1	2	3	4	5	6	7	8	9	10
θ_t	0.6100	0.6980	0.7684	0.8247	0.8698	0.9058	0.9347	0.9577	0.9762	0.9909

(b) $\theta^* = 0.21/(1-0.8) = 1.05$, i.e. $\bar{\theta} + \kappa(r - r^*) = 1 + 1(0.05)$ (Proposition 9.9, eq. (9.2)). The deviation obeys $\theta_{t+1} - \theta^* = 0.8(\theta_t - \theta^*)$, so $\theta_t = 1.05 - 0.55(0.8)^t$, which reproduces the table (check: $1.05 - 0.55(0.8)^{10} = 0.9909$). Since $|0.8| < 1$ the map is a contraction and $\theta_t \to 1.05$ from any θ_0. The return is constant, so there is no feedback and the local rate of Proposition 9.11 is $\lambda = \rho$.

(c) Trajectories (both parts; every second generation):



"90% of the steady state" has two readings; state which one you use.

- *Level:* $\theta_t \geq 0.9\,\theta^* = 0.945$ requires $0.55(0.8)^t \leq 0.105$, i.e. $t \geq 7.42$. It is first reached at $t = 8$ ($\theta_8 = 0.9577$).
- *Gap:* 90% of the initial gap closed requires $0.8^t \leq 0.1$, i.e. $t \geq 10.32$, first reached at $t = 11$ (θ_{10} has closed 89.3%).

The half-life is $\ln 2 / \ln 1.25 = 3.11$ generations (eq. (9.5)).

(d) With $\rho = 0.95$, $\theta_{t+1} = 0.95\,\theta_t + 0.05 + 0.01 = 0.95\,\theta_t + 0.06$, and $\theta^* = 1.20$. The steady state changes too, because $\kappa = 0.2/0.05 = 4$ instead of 1. $\theta_1, \ldots, \theta_{10} = 0.535, 0.5682, 0.5998, 0.6298, 0.6584, 0.6854, 0.7112, 0.7356, 0.7588, 0.7809$. The level criterion is met at $t = 35$ (exact 34.38) and the gap criterion at $t = 45$ (exact 44.89); the half-life is 13.5 generations.

Convergence is about 4.3 times slower, the ratio $\ln 0.8 / \ln 0.95$. Raising ρ with μ fixed changes *two* things: the speed $(1 - \rho)$ and the long-run sensitivity (κ). The slow–fast scaling of Appendix 9B separates them by holding κ fixed. *Common mistake:* attributing the whole difference between the two trajectories to persistence; part of it is that the $\rho = 0.95$ dynasty is heading to a higher steady state.

Exercise 9.2 Heterogeneous steady states

Example 9.10: $\rho = 0.8$, $\mu = 0.3$ ($\kappa = 1.5$), $\bar{\theta} = 1$, $r^* = 0.1$, $g_M(\theta) = 0.05 + 0.1\theta$ and $g_T(\theta) = 0.03 + 0.02\theta$. With $\mathbf{g} = \mathbf{a} + G\boldsymbol{\theta}$, $\mathbf{a} = (0.05, 0.03)'$ and $G = \mathrm{diag}(0.1, 0.02)$, case (b) of Proposition 9.9 gives the unique steady state

$$\boldsymbol{\theta}^* = \left(I - \kappa W^I G\right)^{-1}\left(\bar{\theta}\mathbf{1} + \kappa(W^I\mathbf{a} - r^*\mathbf{1})\right).$$

(a) With $W^I = I$ the system decouples: $\theta_M = 1 + 1.5(0.05 + 0.1\theta_M - 0.1)$, so $0.85\,\theta_M = 0.925$ and $\theta_M^* = 1.0882$; and $0.97\,\theta_T = 0.895$, so $\theta_T^* = 0.9227$. The gap is

0.1656. Both maps have slope below one (0.83 and 0.806), so each steady state is unique and globally stable.

(b) With $W^I = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix}$, the linear system gives $\boldsymbol{\theta}^* = (1.0332, 0.9709)$. The returns are then $(0.1533, 0.0494)$ and the signals $W^I\mathbf{r} = (0.1222, 0.0806)$. The gap shrinks to 0.0623, matching the example. The local rate is $\lambda = \rho(\rho I + \mu W^I G) = 0.822$ (Proposition 9.11).

(c) *The gap vanishes exactly at* $w = 0.5$. Subtract the two steady-state equations with $W^I = \begin{pmatrix} 1-w & w \\ w & 1-w \end{pmatrix}$:

$$\theta_M^* - \theta_T^* = \kappa\big[((1-w)r_M + wr_T) - (wr_M + (1-w)r_T)\big] = \kappa\,(1-2w)\,(r_M^* - r_T^*).$$

The anchor and the reference cancel; only the difference in *signals* remains. At $w = 0.5$ both rows of W^I are equal, both agents observe the same signal, and since everything else in (9.2) is common, $\theta_M^* = \theta_T^*$. Here both equal 1.000, with returns $(0.15, 0.05)$. Conversely, if the gap is zero then $\theta_M = \theta_T = \theta$ and $r_M - r_T = 0.02 + 0.08\theta > 0$ for $\theta > -0.25$, so $1 - 2w = 0$. The formula, verified numerically:

w	0.3	0.4	0.5	0.6	1.0
$\theta_M^* - \theta_T^*$	$+0.0623$	$+0.0306$	0.0000	-0.0294	-0.1371

The gap does *not* vanish because returns equalize: they remain 0.15 vs 0.05. Beyond $w = 0.5$ it reverses, and each agent is calibrated mostly on the other sector's returns. The answer is not "as large as possible".

(d) Raising W_{12}^I means the Modern agent observes more traditional-sector outcomes; raising W_{21}^I, the reverse. Real-world counterparts include school integration and scholarships that put children of both sectors in the same reference groups, mentoring and job networks across sectors, media representation of the other sector's lives, migration links, and, in the other direction, extension workers and public officials embedded in rural communities.

Two cautions from the chapter apply. First, what changes is *information*, not the return functions: a traditional agent calibrated on modern returns may overaspire relative to what g_T rewards (the Intuition after Proposition 9.15). Second, connecting the agents here *lowers* average ambition: the mean steady-state θ falls from 1.0055 ($W^I = I$) to 1.0021 ($w = 0.3$) and 1.000 ($w = 0.5$). Whether that is good is a welfare question (Chapter 12), not a property of the formation equation.

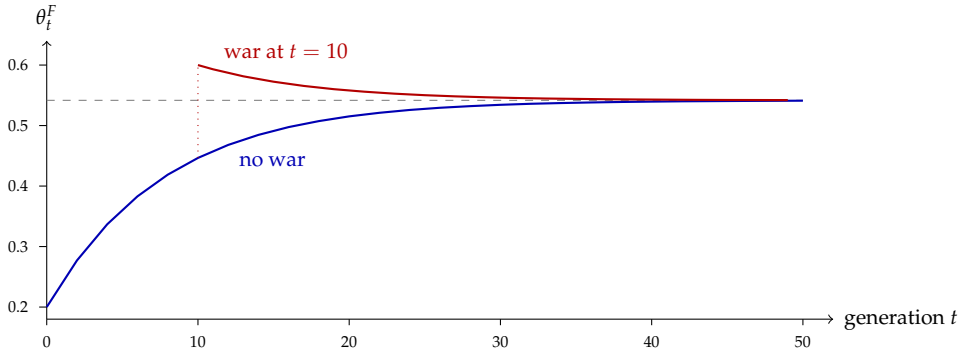
Exercise 9.3 Persistence in female labor supply

(a) Substituting $\bar{r}_t = 0.3\,\theta_t^F$:

$$\theta_{t+1}^F = 0.85\,\theta_t^F + 0.075 + 0.1(0.3\,\theta_t^F - 0.1) = 0.88\,\theta_t^F + 0.065.$$

The steady state is $\theta^{F*} = 0.065/0.12 = 0.5417$. It is unique because the map is affine with slope $0.88 \neq 1$: $\theta = 0.88\theta + 0.065$ has exactly one solution. It is globally stable because $0.88 < 1$. The network effect raises the slope from $\rho = 0.85$ to $\lambda = 0.88$ and slows adjustment (half-life 5.42 generations instead of 4.27), as Proposition 9.11 predicts for $D\mathbf{g} > 0$.

(b, c) From $\theta_0^F = 0.2$ without the war, $\theta_t^F = 0.5417 - 0.3417(0.88)^t$: $\theta_{10}^F = 0.4465$, $\theta_{20}^F = 0.5152$ and $\theta_{50}^F = 0.5411$. With the war, θ_{10}^F jumps to 0.6 and the post-war path is $\theta_t^F = 0.5417 + 0.0583(0.88)^{t-10}$: 0.5930 at $t = 11$, 0.5579 at $t = 20$, 0.5462 at $t = 30$ and 0.5420 at $t = 50$.



θ^F does *not* return to the pre-war level 0.4465. The post-war minimum is 0.542, above it. But the pre-war level was not a steady state: the economy was already converging to 0.5417. The right comparison is with the no-war path.

(d) *No permanent effect.* Let $e_t = \theta_t^{F,\text{war}} - \theta_t^{F,\text{no war}}$. Both paths obey the same affine map from $t = 10$ on, so $e_{t+1} = 0.88\,e_t$ and $e_t = (0.6 - 0.4465)(0.88)^{t-10} = 0.1535(0.88)^{t-10} \to 0$. The difference is 0.0810 at $t = 15$, 0.0428 at $t = 20$, 0.0119 at $t = 30$ and 0.0009 at $t = 50$. Any temporary shock is eventually forgotten, at rate 0.88 per generation, because the map has a unique, globally stable fixed point.

What would make it permanent. Hysteresis needs at least two stable steady states, with the shock moving θ across the boundary of their basins. That requires the composite map to cross the 45° line three times, with slope above one somewhere. An S-shaped return function does this: low returns at low participation (no networks, no childcare, hostile norms), steep returns in a middle range, saturation at high participation (Chapter 10, Example 10.11). A large enough wartime

jump past the unstable middle steady state would then move society to the high-participation steady state for good. A smaller one would be reversed. In this linear model, $0.88 < 1$ everywhere rules this out.

Evidence. Fernández et al. (2004) use cross-state variation in the WWII mobilization rate of men as a shock to the share of men raised by working mothers. They find that wives of such men are more likely to work, a channel of intergenerational preference formation. This fits a model in which a temporary shock persists through θ for at least a generation. The historical evidence on the war's *direct* effect is more mixed. Goldin (1991, *AER*) argues that the war's direct effect on women's employment was modest, since many wartime entrants had left the labor force by 1950; Acemoglu, Autor and Lyle (2004, *JPE*) find higher female labor supply in 1950 in high-mobilization states. A careful answer distinguishes the short-run reversal (consistent with the linear model) from a longer-run effect transmitted through sons' preferences. The latter is persistence through ρ, and it is not by itself evidence of multiple steady states. *Common mistake:* treating any long-lasting effect as hysteresis. With ρ close to one, a single steady state can produce effects that last a century (Example 9.12).

Exercise 9.4 Aspiration and the smartphone

Set-up. The sectors sit at the steady states of Example 9.10: $\theta_M^* = 1.0882$, $\theta_T^* = 0.9227$, with returns $r_M = 0.15882$ and $r_T = 0.04845$. The agent is one of many in her sector, so her own θ does not move the sector returns she observes (the individual case). Before $t = 5$ her signal is r_T; from $t = 5$ on it is $0.7\,r_T + 0.3\,r_M$.

(a) *Old steady state:* $\theta^{\text{old}} = 1 + 1.5(0.04845 - 0.1) = 0.9227$, the sector's value. *New steady state:* $\theta^{\text{new}} = 1 + 1.5(0.7(0.04845) + 0.3(0.15882) - 0.1) = 0.9724$, a gain of 0.0497. Since $\theta_s^* - \bar{\theta} = \kappa(r_s - r^*)$ in each sector, the gain is exactly $w\kappa(r_M - r_T) = w(\theta_M^* - \theta_T^*)$. In the long run, the phone closes the share $w = 0.3$ of the Modern–Traditional aspiration gap.

(b) From $\theta_0 = 0.8$ she first converges toward 0.9227 ($\theta_5 = 0.8825$). After the phone, $\theta_t = 0.9724 - 0.0899(0.8)^{t-5}$:

t	4	5	6	7	8	9	10	11	12
θ_t	0.8724	0.8825	0.9005	0.9148	0.9263	0.9355	0.9429	0.9488	0.9535

Half of the gap to the new steady state (0.0899 at $t = 5$) closes in $\ln 2 / \ln(1/0.8) = 3.11$ generations, so it has closed by $t = 9$, four generations (≈ 100 years) after the phone. The individual's rate is $\rho = 0.8$, because her own aspiration does not feed

back into the returns she observes. If "the aspiration gap" means the gap to the Modern sector, half of it is *never* closed at $w = 0.3$; that would take $w = 0.5$.

(c) *No: it is close to, but not exactly, 100 times*, for two opposite reasons.

1. *Spillover through the sector's return (a social multiplier).* If the sector's return depends on the sector's average aspiration, $r_T = 0.03 + 0.02\,\bar{\theta}_T$, then treated agents raise $\bar{\theta}_T$. That raises r_T, which raises everyone's signal, treated or not. Solving the linear system for a treated share s, the aggregate effect per treated agent is 1.031 times the individual effect for small s, 1.026 at $s = 0.5$, and $1.0215 = 1/(1 - \kappa(0.7)(0.02))$ when the whole sector is treated. With 100 phones in a large sector, the total is about 103 individual effects. The feedback also slows adjustment slightly ($\lambda = 0.804$ instead of 0.8).

2. *General equilibrium in the observed returns.* If many traditional agents aim at modern-sector outcomes, the modern returns *they* can attain may fall: entry, crowding, credential inflation. As an illustration, suppose the observed modern return falls by $0.05s$. The aggregate effect per treated agent is then 0.983, 0.794 and 0.559 times the individual effect at $s = 0.1$, 0.5 and 1.

The individual effect is a partial-equilibrium object, and scaling it up assumes that the return functions do not move. That assumption is what the hint questions, and it is the same caveat as for Ψ (Remark 9.14): the answer depends on the perturbation's scale.

Exercise 9.5 Bernard et al. (2014) as a W^I perturbation

(a) Add a node f (the film's successful farmer, with return r_{film}). For each treated farmer $i \in \mathcal{T}$, the documentary moves weight $w \in (0, 1)$ to f and scales the rest of the row down proportionally:

$$\Delta W^I_{i\cdot} = w\big(\mathbf{e}_f - W^I_{i\cdot}\big)' \quad (i \in \mathcal{T}), \qquad \Delta W^I_{i\cdot} = 0 \quad (i \notin \mathcal{T}).$$

Each row of ΔW^I sums to zero, so $W^I + \Delta W^I$ stays row-stochastic, and nonnegativity holds for $w \leq 1$. The film subject's own row can be anything, since nobody's dynamics depend on θ_f; take $W^I_{f\cdot} = \mathbf{e}'_f$. This states the perturbation as Definition 9.13 requires: layer W^I, rows of the treated, compensation by proportional scaling.

(b) For a treated farmer whose own return does not respond to her own θ (the individual case of Exercise 9.4), the signal changes by $w(r_{\text{film}} - \bar{r}_i)$, where $\bar{r}_i =$

$\sum_j W^I_{ij} r_j$ is her old signal. Then, by (9.2),

$$\Delta\theta^*_i = \frac{\mu}{1-\rho}\, w\,(r_{\text{film}} - \bar{r}_i).$$

If her own return responds ($r_i = g_i(\theta_i)$, $g_i' > 0$, $W^I_{ii} > 0$), subtracting the old and new steady-state equations gives

$$\Delta\theta^*_i = \frac{\kappa\, w\,(r_{\text{film}} - \bar{r}_i)}{1 - \kappa(1-w) W^I_{ii}\, g_i'}, \qquad \kappa = \frac{\mu}{1-\rho},$$

and in general $\Delta\boldsymbol{\theta}^* = (I - \kappa(W^I + \Delta W^I)G)^{-1}\kappa\,\Delta W^I \mathbf{r}^0$ for affine $\mathbf{g}$ (case (b) of Proposition 9.9). The effect is proportional to the formation multiplier κ, the weight w of the new edge, and the *gap* between the film's return and what she observed before. A film about a farmer like her has no effect. The path to $\Delta\theta^*_i$ is $\Delta\theta_{i,T} = \Delta\theta^*_i(1-\lambda^T)$, with $\lambda = \rho$ in the individual case.

(c) *Implied μ.* If the six-month effect is read as the impact effect of one period of the formation equation, $\Delta\theta_{i,1} = \mu\, w\,(r_{\text{film}} - \bar{r}_i)$. A 0.1 SD effect [ILLUSTRATIVE] then implies

$$\mu = \frac{0.1\,\sigma_\theta}{w\,(r_{\text{film}} - \bar{r}_i)}.$$

For example, $\mu = 1.0$ if $w = 0.1$ and the return gap is 1 SD of θ; $\mu = 2.0$ if $w = 0.05$; $\mu = 0.5$ if $w = 0.1$ and the gap is 2 SD. The number is only as good as the assumed w and gap, neither of which the experiment measures.

The time-scale problem. The formation equation is per *generation* (about 25 years), and six months is 1/50 of a generation. Let the six-month process have parameters (ρ_6, μ_6). Fifty sub-periods with a constant signal aggregate to one generation with

$$\rho_{\text{gen}} = \rho_6^{50}, \qquad \mu_{\text{gen}} = \mu_6\,\frac{1-\rho_6^{50}}{1-\rho_6}, \qquad \frac{\mu_{\text{gen}}}{1-\rho_{\text{gen}}} = \frac{\mu_6}{1-\rho_6}.$$

The ratio μ_{gen}/μ_6 is 50 if $\rho_6 = 1$ (no decay), 39.5 if $\rho_6 = 0.99$, 18.5 if $\rho_6 = 0.95$, and 9.95 if $\rho_6 = 0.9$. A six-month impact identifies μ_6 at best. Mapping it to μ_{gen} requires the decay rate ρ_6, which one follow-up round cannot reveal. Only the long-run multiplier κ is invariant to the period length, and it requires the long-run effect, not the impact. There are further problems:

- a six-month change may be a revision of *beliefs* (the fast DeGroot channel of Chapter 8), not of preferences (Section 9.1.4);
- survey measures of aspirations may respond to experimenter demand;

- SD units depend on the sample's dispersion, not on the model's θ.

Common mistake: plugging the six-month coefficient directly into a per-generation μ, which can be off by a factor between 1 and 50.

(d) Yes, if savings respond more than proportionally to θ. Let $s = f(\theta)$ with f convex and $f(0) \leq 0$ (no saving without aspiration). By convexity, the tangent at $\theta > 0$ lies below f at 0: $f(0) \geq f(\theta) - \theta f'(\theta)$. Hence $\theta f'(\theta) \geq f(\theta) - f(0) \geq f(\theta)$, i.e. the elasticity $\theta f'(\theta)/f(\theta) \geq 1$. A given percentage change in θ then produces at least the same percentage change in savings. A 10% rise in θ raises $s = \theta^2$ by 21.0% and $s = e^\theta - 1$ (at $\theta = 1$) by 16.6%, against 10% for a linear f. Convexity is natural with a fixed cost of saving or investing: an account, a minimum lumpy investment, or a threshold aspiration above which forward-looking plans begin.

Two caveats. First, in standardized (SD) units the comparison also depends on the dispersion of s relative to θ, so "larger" must be stated in comparable units. Second, savings also respond to the information in the film directly, through beliefs about returns, which is not formation. Distinguishing the two needs a design in the spirit of Bursztyn et al. (2014).

Exercise 9.6 The Ψ ratio for education

The outcome is $x = \theta w/p$; the perturbation is a permanent 10% cut in p; the horizon is $T = 1$ (Definition 9.13). Write $x_0 = \theta_0 w/p_0$.

(a) *Enforcement* (fixed θ): $E = \theta_0 w/(0.9p_0) - x_0$, so $E/x_0 = 1/0.9 - 1 = 0.1111$, i.e. +11.1%. (The first-order approximation $-d\ln p = 10\%$ understates it, because demand is unit-elastic in p.)

(b) *Formation* (new θ at the new price): $F = 1.05\,\theta_0 w/(0.9p_0) - \theta_0 w/(0.9p_0)$, so $F/x_0 = 0.05/0.9 = 0.0556$, i.e. +5.6% of the original quantity. The total change is $(E+F)/x_0 = 1.05/0.9 - 1 = 0.1667$, which confirms the decomposition $E_i + F_i = y_i(\boldsymbol{\theta}_T, W + \Delta W) - y_i(\boldsymbol{\theta}^0, W)$.

(c) $\Psi = F/E = 0.0556/0.1111 = 0.500$. The enforcement effect is twice as large. An evaluation that ignores formation captures $1/(1+\Psi) = 2/3$ of the effect at $T = 1$ and misses one third (Remark 9.14, item 4).

(d) With a price cut of fraction c and a preference change of fraction g by horizon T,

$$\frac{E}{x_0} = \frac{c}{1-c}, \qquad \frac{F}{x_0} = \frac{g}{1-c}, \qquad \Psi = \frac{g}{c}.$$

Hence $\Psi > 1$ iff the induced proportional change in preferences exceeds the proportional price cut, $g > c$. The common factor $1/(1-c)$ cancels because x is

multiplicative in θ and $1/p$. Inside the formation model g is not free. If the economy starts at a steady state and the policy shifts the observed signal by Δs, then $g(T) = \frac{\kappa \Delta s}{\theta_0}(1 - \lambda^T)$. So $\Psi > 1$ requires:

1. a large formation multiplier $\kappa = \mu/(1-\rho)$;
2. a policy that changes what the reference group visibly earns (a large Δs, e.g. returns to educated women that girls can observe);
3. a long enough horizon relative to the adjustment rate λ;
4. a small direct price effect.

Illustration. If the 5% at $T = 1$ is the first generation of an adjustment with $\lambda = 0.8$, the long-run change is $g_\infty = 0.05/0.2 = 0.25$, and Ψ rises with the horizon: 0.90 at $T = 2$, 1.22 at $T = 3$, 1.68 at $T = 5$, 2.23 at $T = 10$, and 2.5 in the long run. $\Psi > 1$ from $T > 2.29$ generations. The same policy is "mostly enforcement" at one generation and "mostly formation" at three: Ψ is a property of the perturbation, the outcome and the horizon, not of the economy (Remark 9.14).

Exercise 9.7 Furtado's demonstration effect

(a) The elite's W^I row places 0.4 on the Center and 0.6 on the Periphery: travel, media and education abroad give Center consumption a weight comparable to the elite's own society. The row sums to one.

(b) The observed "returns" are the consumption ratios 0.85 (Center) and 0.70 (Periphery), so the signal is $0.4(0.85) + 0.6(0.70) = 0.76$. With $\bar{\theta} = r^* = 0.70$, $\rho = 0.9$ and $\mu = 0.4$ ($\kappa = 4$),

$$\theta^* = 0.70 + 4\,(0.76 - 0.70) = 0.94.$$

The elite consumes 94% of its income, against the sustainable 70%. The map has slope $\rho = 0.9$ (the observed ratios do not depend on θ), so the steady state is unique and stable, with half-life 6.6 generations. Starting from 0.70, θ_t is 0.724, 0.798, 0.856, 0.911 and 0.940 at $t = 1, 5, 10, 20, 60$.

(c) With weight w on the Center, $\theta^*(w) = 0.70 + 4w(0.85 - 0.70) = 0.70 + 0.6w$. It exceeds 1 iff $w > 0.5$: once the Center carries more weight than the elite's own society, the steady state is chronic dissaving. The multiplier $\kappa = 4$ amplifies a modest consumption gap (0.15) into a large preference deviation. The model is linear and $\theta > 1$ is outside its meaningful range (Definition 9.7); read the threshold

as the point where the local approximation predicts unsustainable consumption, i.e. debt, rents or capital inflows.

Variant. If the "Periphery" signal were the elite's *own* consumption ratio rather than a fixed 0.70, the map would have slope $\rho + 0.6\mu = 1.14 > 1$. Its fixed point (0.529) would then be unstable, and consumption would run away from it: a self-reinforcing emulation race.

(d) Each structural lever corresponds to a term of the steady-state formula $\theta^* = \bar{\theta} + \kappa(w\, c_C + (1-w)c_P - r^*)$:

- *Lower w*: reduce the weight of Center consumption in the elite's reference set, for instance through domestic media and cultural production, or taxes on imported luxury consumption that make Center patterns less visible and less attainable.
- *Lower $c_C - c_P$ in what is observed*: make the Periphery's own consumption the relevant comparison, which is Furtado's argument for an autonomous cultural project.
- *Change the anchor or reference* through institutions that stabilize saving norms: pension systems, development banks, savings mandates.
- *Address the distributional root.* For Furtado, income concentration is what lets a minority emulate the Center. Redistribution shrinks the group whose W^I is Center-oriented, and raises the attainable consumption of the majority, so the observed returns become attainable.

The formation equation alone cannot say whether a cross-border edge helps or harms. Here it harms because the returns it carries are not attainable at peripheral incomes (the Prebisch box of Section 9.5). *Common mistake:* prescribing "less information" as such. The problem is the mismatch between observed and attainable returns, and the book's remedies act on that mismatch: access, income, and what the edge carries.

Chapter 10

The Formal Formation Model

All numbers below are printed by `python3 code/solutions/sol10.py`. Throughout, T is the formation map of equation (10.4), $J = DT$ its Jacobian, and $\mathcal{M} = \mu/(1-\rho)$ the formation multiplier (Definition 10.3).

Exercise 10.1 The formation map by hand

(a) With reduced-form returns $r(\theta) = 0.05 + 0.1\theta$, the scalar version of (10.3) is

$$\begin{aligned} T(\theta) &= \rho\theta + (1-\rho)\theta^{\circ} + \mu\big(r(\theta) - r^{*}\big) \\ &= 0.8\theta + 0.2 + 0.3\,(0.05 + 0.1\theta - 0.1) = 0.83\,\theta + 0.185. \end{aligned}$$

The slope is $\rho + \mu \cdot 0.1 = 0.83$ and the intercept $(1-\rho)\theta^{\circ} + \mu(0.05 - r^{*}) = 0.2 - 0.015 = 0.185$.

(b) T is affine with slope $0.83 \neq 1$, so by Theorem 10.4(1) there is exactly one fixed point:

$$\theta^{*} = \frac{0.185}{1 - 0.83} = \frac{0.185}{0.17} = 1.0882.$$

Check: $r(\theta^{*}) = 0.1588$, so the signal gap is $r - r^{*} = 0.0588$ and $\theta^{\circ} + \mathcal{M} \cdot 0.0588 = 1 + 1.5 \times 0.0588 = 1.0882$, as (10.7) requires. The steady state lies above the anchor because the observed return exceeds the reference return.

(c) $J = T'(\theta^{*}) = 0.83$. Since $|J| < 1$ the steady state is globally asymptotically stable (Theorem 10.4(1); for an affine scalar map local and global stability coincide). Deviations shrink by 17% per generation, with half-life $\ln 2/(-\ln 0.83) = 3.72$ generations.

(d) The deviation obeys $\theta_t - \theta^{*} = 0.83^{t}(\theta_0 - \theta^{*})$, so from $\theta_0 = 0.5$ (initial gap 0.5882):

t	0	1	2	3	5	10	15	20
θ_t	0.5000	0.6000	0.6830	0.7519	0.8565	0.9970	1.0523	1.0741
$\theta^* - \theta_t$	0.5882	0.4882	0.4052	0.3363	0.2317	0.0913	0.0360	0.0142

The script iterates the map and checks it against the closed form. After 20 generations $0.83^{20} = 2.4\%$ of the initial gap remains. The path is monotone because $J > 0$. *Common mistake:* using $\rho = 0.8$ as the convergence rate. The loop adds $\mu\beta = 0.03$ to persistence, so convergence is slower than ρ alone suggests.

Exercise 10.2 Why convex returns are not enough

(a) Substituting $r(\theta) = 0.05 + 0.1\theta + 0.05\theta^2$,

$$T(\theta) = 0.85\theta + 0.15 + 0.25\,(0.05\theta^2 + 0.1\theta - 0.05) = 0.0125\,\theta^2 + 0.875\,\theta + 0.1375.$$

Hence $T(\theta) - \theta = 0.0125\,\theta^2 - 0.125\,\theta + 0.1375 = 0.0125\,(\theta^2 - 10\theta + 11)$. This is a quadratic with positive leading coefficient, so it has at most two real roots. More generally, $T - \mathrm{id}$ is convex whenever g is convex, and a convex function that is not identically zero on an interval vanishes at most twice.

(b) The roots are $\theta = 5 \mp \sqrt{14}$:

$$\theta_L = 1.2583, \qquad \theta_H = 8.7417.$$

$T'(\theta) = 0.875 + 0.025\theta$, so $T'(\theta_L) = 0.9065 < 1$ (stable) and $T'(\theta_H) = 1.0935 > 1$ (unstable), by Theorem 10.7.

(c) $T(\theta) - \theta$ is negative strictly between the roots and positive outside them.

- $\theta_0 = 0.9$: $\theta_1 = 0.9351$, $\theta_5 = 1.0435$, increasing to 1.2583.
- $\theta_0 = 1.1$: $\theta_1 = 1.1151$, increasing to 1.2583.
- $\theta_0 = 5$: $\theta_1 = 4.8250$, $\theta_5 = 4.1364$, decreasing to 1.2583.
- $\theta_0 = 9$: $\theta_1 = 9.0250$, $\theta_5 = 9.1529$, and θ_t diverges (it exceeds 10^6 after 44 periods).

There is *no* poverty trap. A trap needs two stable steady states; here there is only one. Every start in $(-78.74, 8.7417)$ converges to θ_L. (The lower edge is the other solution of $T(\theta) = \theta_H$, since T is decreasing below its vertex at $\theta = -35$; it is irrelevant economically.) θ_H is an *explosion threshold*: above it the convex feedback

has nothing to stop it, and aspirations and returns grow without bound. An agent below θ_H, including one far above θ_L such as $\theta_0 = 5$, is pulled *down* to the unique steady state.

(d) *Claim.* Let g be C^1, $\mu > 0$, $T(\theta) = \rho\theta + (1-\rho)\theta^\circ + \mu(g(\theta) - r^*)$, and suppose T has two fixed points $\theta_L < \theta_H$ with $T'(\theta_L) < 1$ and $T'(\theta_H) < 1$ (the derivative condition behind local stability). Then g' is not monotone on $[\theta_L, \theta_H]$: it rises and then falls.

Proof. Let $f = T - \mathrm{id}$, so $f' = \rho - 1 + \mu g'$. Since $f(\theta_L) = 0$ and $f'(\theta_L) < 0$, there is $\varepsilon > 0$ with $f(\theta_L + \varepsilon) < 0$. Since $f(\theta_H) = 0$ and $f'(\theta_H) < 0$, $f(\theta_H - \varepsilon) > 0$. By the mean value theorem there is $b \in (\theta_L + \varepsilon, \theta_H - \varepsilon)$ with $f'(b) > 0$. With $a := \theta_L$ and $c := \theta_H$ we have $a < b < c$ and

$$g'(a) < \frac{1-\rho}{\mu}, \qquad g'(b) > \frac{1-\rho}{\mu}, \qquad g'(c) < \frac{1-\rho}{\mu}.$$

So g' increases somewhere on (a, b) and decreases somewhere on (b, c). If g is C^2, g'' takes both signs: returns are convex and then concave, an S-shape. A convex g (g' nondecreasing) contradicts $g'(b) > g'(c)$; a concave g contradicts $g'(a) < g'(b)$. By the intermediate value theorem there is also a third fixed point $\theta_U \in (\theta_L, \theta_H)$, the basin boundary of Proposition 10.10. □

Interpretation. A trap needs the loop gain $\mu g'$ to be below the damping $1 - \rho$ at low aspirations (so the low state is stable), above it in the middle (so there is a threshold), and below it again at high aspirations (so the high state is stable rather than explosive). The middle condition needs increasing returns; the last one needs *saturation*. Convex returns give increasing returns without saturation, which produces explosion instead of a second steady state. This is the logic of the logistic return in Example 10.11.

Exercise 10.3 The Jacobian

(a) With reduced-form returns $r_i = 0.1 + 0.1\theta_i$, the map is $T(\boldsymbol{\theta}) = \rho\boldsymbol{\theta} + (1-\rho)\theta^\circ\mathbf{1} + \mu(W^I\mathbf{r}(\boldsymbol{\theta}) - r^*\mathbf{1})$, so $J = \rho I + 0.1\,\mu W^I$:

$$J = 0.8\,I + 0.02\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.81 & 0.01 & 0 \\ 0.01 & 0.81 & 0 \\ 0 & 0 & 0.82 \end{pmatrix}.$$

The anchor θ° and r^* only shift the intercept, so they do not enter J; stability does not depend on them.

(b) W^I is block-diagonal. Its eigenvalues are 1 (eigenvector $(1,1,0)'$), 0 (eigenvector $(1,-1,0)'$) and 1 (eigenvector $(0,0,1)'$). Hence the eigenvalues of $J = 0.8I + 0.02W^I$ are

$$0.82, \qquad 0.80, \qquad 0.82,$$

so $\rho(J) = 0.82 < 1$ and the unique steady state is globally stable. The two unit eigenvalues of W^I (one per block, because W^I is reducible) carry the slowest modes: a common shift of each block's preferences decays at 0.82. The within-block difference $\theta_1 - \theta_2$ is averaged away by the signal and decays at $\rho = 0.80$.

(c) Doubling μ to 0.4 gives $J = 0.8I + 0.04W^I$, with eigenvalues $0.84, 0.80, 0.84$. Only the eigenvalues on the unit modes of W^I move. Stability is lost when $\rho + 0.1\,\mu \cdot 1 = 1$, i.e. at

$$\mu = \frac{1-\rho}{0.1} = 2.$$

Caveat (reduced form vs. full model). With returns that depend on the *level* θ_i, a uniform rise in preferences raises everyone's return, so $\mathbf{1}$ is no longer a pure ρ-mode. In the full model the allocation depends on $\theta_i - \bar{\theta}$, and Proposition 10.2(2) gives $J\mathbf{1} = \rho\mathbf{1}$. Computing the full-model Jacobian $\rho I + \mu\phi W^I C P$ with $C = 0.1I$ and $\phi = 1$ gives eigenvalues $0.82, 0.80, 0.80$. The relative mode $(1,1,-2)'$ still has eigenvalue 0.82 (it is an eigenvector of $W^I P$ with eigenvalue 1), so the instability threshold is again $\mu = 2$. The uniform mode, however, now decays at ρ. *Common mistake:* applying the norm bound (10.12) and reporting its threshold as the instability point; that bound is only sufficient.

Exercise 10.4 Three policies for the stylized Bolivia

(a) With the parameters of Example 10.6, $J = \rho I + \mu\phi W^I C P$ has $\rho(J) = 0.8664$. Solving $(I - J)\boldsymbol{\theta}^* = \mathbf{c}$ gives, by block (gas, urban, rural):

$$\boldsymbol{\theta}^*_b = (1.1245,\ 1.0411,\ 0.9871),$$
$$\bar{x}^*_b = (1.2486,\ 0.9852,\ 0.7662), \qquad \bar{r}^*_b = (0.2094,\ 0.1287,\ 0.0764).$$

Checks: $\sum_i x^*_i = 9 = \Omega$; $J\mathbf{1} = \rho\mathbf{1}$; and the fixed point satisfies (10.7), $\theta^*_i = \theta^\circ + \mathcal{M}(\sum_j W^I_{ij} r^*_j - r^*)$ with $\mathcal{M} = 1.333$ (maximum error 6×10^{-16}). The preference gap is $\theta^*_1 - \theta^*_3 = 0.1374$, which equals the bound (10.10) with $\tau_1(W^I) = 0.775$. The allocation gap is $\bar{x}^*_1 - \bar{x}^*_3 = 0.4824$.

(b) Following Corollary 10.13, for each policy we evaluate $E = \mathbf{r}(\boldsymbol{\theta}^0, \text{new}) - \mathbf{r}(\boldsymbol{\theta}^0, \text{old})$. We then solve for the new steady state and set $F(\infty) = \mathbf{r}(\boldsymbol{\theta}^*_{\text{new}}, \text{new}) -$

$\mathbf{r}(\boldsymbol{\theta}^0, \text{new})$. Block values (gas, urban, rural):

	$\mu\Delta\mathbf{s}$ (signal)	E (returns)	$F(\infty)$ (returns)	$\bar{\Psi}(\infty)$	$\bar{\Psi}(1)$	$\rho(J_{\text{new}})$
A	$(-0.0019, -0.0001, 0.0008)$	$(-0.0118, 0, 0.0058)$	$(-0.0013, 0.0002, 0.0005)$	0.117	0.016	0.8664
B	$(-0.0003, 0.0000, 0.0003)$	$(-0.0020, 0, 0.0020)$	$(-0.0003, 0.0000, 0.0001)$	0.091	0.012	0.8658
C	$(0, 0, 0.0040)$	$(0, 0, 0)$	$(-0.0011, -0.0008, 0.0011)$	undef.	undef.	0.8659

The aggregate ratios use $\sum_i |E_i| = 0.0529$ (A) and 0.0122 (B). For C, $\sum_i |E_i| = 0$, so $\bar{\Psi}$ is undefined (Remark 9.14(5)); report $\sum_i |F_i(\infty)| = 0.0091$ and $\sum_i |F_i(1)| = 0.0013$ instead. The pattern matches Proposition 10.15:

- The transfer's signal is $\mu W^I C \Delta\boldsymbol{\omega}$.
- The structural policy's is $\mu W^I \operatorname{diag}(\gamma) \Delta W^P \mathbf{x}^0$, small because $\gamma = 0.02$.
- The information policy's is $\mu \Delta W^I \mathbf{r}^0$. It is nonzero only in the rural rows, which now weight the gas block's higher returns more.

For every policy the script checks $\Delta\boldsymbol{\theta}^* = (I - J_{\text{new}})^{-1} \mu\Delta\mathbf{s}$ (Theorem 10.12).

(c) Long-run changes:

	$\Delta\boldsymbol{\theta}^*_b$	$\Delta(\theta_1 - \theta_3)$	$\Delta(\bar{x}_1 - \bar{x}_3)$
A: transfer	$(-0.0143, -0.0004, 0.0059)$	-0.0201 (14.7%)	-0.2201 (45.6%)
B: structural	$(-0.0024, -0.0000, 0.0022)$	-0.0046 (3.3%)	-0.0046 (0.9%)
C: information	$(-0.0013, -0.0009, 0.0272)$	-0.0285 (20.7%)	-0.0285 (5.9%)

The information policy narrows the *preference* gap most. The transfer narrows the *allocation* gap most, by far: it moves 0.1 from each gas agent to each rural agent, which closes the gap by 0.2 directly, plus 0.0201 through preferences. For B and C the allocation gap moves only through $\phi(\theta_i - \bar{\theta})$, with $\phi = 1$, so $\Delta(\bar{x}_1 - \bar{x}_3) = \Delta(\theta_1 - \theta_3)$. The ranking therefore depends on the outcome chosen. This is why Ψ and policy rankings are always stated for a specific outcome (Remark 9.14(1)).

(d) Combined policy: apply A, B and C together (W^I_C, W^P_B, $\boldsymbol{\omega}_A$). Then $\rho(J_{\text{new}}) = 0.8656$, $\Delta\boldsymbol{\theta}^*_b = (-0.0165, -0.0012, 0.0297)$, and

$$\Delta(\theta_1 - \theta_3) = -0.0462 \ (33.6\%),$$

versus -0.0532 for the sum of the three individual effects. The combined effect is *smaller* than the sum; the interaction term is $+0.0069$ (identical for the allocation gap, -0.2462 versus -0.2532). The model is affine in $\boldsymbol{\theta}$ but not in the policies. The signal contains the product $\Delta W^I \cdot \Delta\mathbf{r}$: policy C makes rural agents watch

the gas block more, while policy A lowers the gas block's returns. Rural agents thus observe more of a return that the transfer has reduced. The policies are mild *substitutes* on this outcome. *Common mistake:* adding the three rows of the table. Additivity holds only to first order in the size of the perturbations.

Exercise 10.5 The formation Leontief inverse

(a) $\operatorname{tr} J = 1.75$ and $\det J = 0.765 - 0.0015 = 0.7635$, so

$$\lambda_{1,2} = \tfrac{1}{2}\big(1.75 \pm \sqrt{1.75^2 - 4 \times 0.7635}\big) = \tfrac{1}{2}\big(1.75 \pm \sqrt{0.0085}\big) = 0.9211,\ 0.8289.$$

Both are real and in $(0,1)$, so $\rho(J) = 0.9211 < 1$. (As a check, $\|J\|_\infty = 0.93 < 1$ already guarantees it.)

(b) $I - J = \begin{pmatrix} 0.15 & -0.05 \\ -0.03 & 0.10 \end{pmatrix}$ has determinant 0.0135, so

$$(I-J)^{-1} = \frac{1}{0.0135}\begin{pmatrix} 0.10 & 0.05 \\ 0.03 & 0.15 \end{pmatrix} = \begin{pmatrix} 200/27 & 100/27 \\ 20/9 & 100/9 \end{pmatrix} = \begin{pmatrix} 7.4074 & 3.7037 \\ 2.2222 & 11.1111 \end{pmatrix}.$$

All entries are positive, as they must be for a nonnegative J with $\rho(J) < 1$ (Neumann series).

(c) $S_{10} := \sum_{k=0}^{10} J^k = \begin{pmatrix} 5.6610 & 1.3229 \\ 0.7937 & 6.9839 \end{pmatrix}$. The remainder is exactly $(I-J)^{-1} - S_{10} = J^{11}(I-J)^{-1} = \begin{pmatrix} 1.7464 & 2.3808 \\ 1.4285 & 4.1272 \end{pmatrix}$. The partial sum captures only 76% and 63% of the diagonal entries and 36% of the off-diagonal ones. The truncation error decays like $\rho(J)^{11} = 0.405$. The off-diagonal entries converge most slowly, because cross-agent feedback needs at least one round to start. Pushing the dominant mode below 1% needs $\lceil \ln 0.01 / \ln 0.9211 \rceil = 57$ terms, i.e. 57 generations. This is the "large and slow" property of the formation Leontief inverse near the stability boundary.

(d) $\mu \Delta \mathbf{s} = 0.2 \times (0.01, 0)' = (0.002, 0)'$, so

$$\Delta \boldsymbol{\theta}^* = (I-J)^{-1} \mu \Delta \mathbf{s} = (0.002 \times 7.4074,\ 0.002 \times 2.2222)' = (0.0148,\ 0.0044)'.$$

Agent 2's long-run preference rises by 0.0044 although her own signal did not change. In round 1 agent 1's higher θ enters agent 2's update through $J_{21} = 0.03$, and the feedback continues in later rounds. Agent 1's own long-run change is

7.4 times the impact 0.002: the formation Leontief inverse amplifies as well as propagates.

Exercise 10.6 Basin boundaries and policy design

Fixed points of Example 10.11: $\theta_L = 0.8814$ (stable), $\theta_U = 0.9801$ (unstable), $\theta_H = 1.1615$ (stable).

(a) T is strictly increasing, so by Proposition 10.10(1) the basin of θ_H is (θ_U, ∞). A one-time push from θ_L escapes iff $\theta_L + \delta > \theta_U$:

$$\delta > \theta_U - \theta_L = 0.0987.$$

Numerically, a push of 0.0967 returns to 0.8814 and a push of 0.1007 converges to 1.1615. At exactly $\delta = 0.0987$ the economy sits on the unstable point, so the minimum is an infimum, not attained. *Common mistake:* using $\hat{\theta} - \theta^\circ$, or the distance to θ_H.

(b) A baseline-return subsidy enters as $T(\theta; \Delta\alpha) = T(\theta) + \mu\Delta\alpha$, so $\partial T/\partial\alpha = \mu = 0.25 > 0$. By Proposition 10.10(2),

$$\frac{d\theta_U}{d\alpha} = -\frac{\mu}{T'(\theta_U) - 1} = -\frac{0.25}{0.1145} = -2.183, \qquad \frac{d\theta_L}{d\alpha} = \frac{\mu}{1 - T'(\theta_L)} = 2.562.$$

The threshold falls and the trap rises, so the required push shrinks at rate 4.745 per unit of α. Finite changes: with $\Delta\alpha = 0.005$ the fixed points are $(0.8955, 0.9680, 1.1709)$, the threshold moves by -0.0121, and the gap falls to 0.0725. With $\Delta\alpha = 0.01$ the gap falls to 0.0333. For $\Delta\alpha > 0.0114$ the trap disappears altogether: θ_L and θ_U merge in a saddle-node (Appendix 10.7), and every trajectory converges to the high state.

(c) The instruments are in different units, so the comparison needs costs: c_θ per unit of θ pushed once, and c_α per unit of return subsidized per generation. In the model a return subsidy paid for one generation is itself a one-time push of size $\mu\Delta\alpha$. A temporary subsidy for n generations escapes if θ_n exceeds θ_U once it is withdrawn. The minimum totals are:

n	1	2	3	5	10	20	40	80
min $\Delta\alpha$	0.3949	0.2032	0.1384	0.0863	0.0472	0.0277	0.0183	0.0140
total $n\Delta\alpha$	0.3949	0.4063	0.4151	0.4314	0.4716	0.5546	0.7323	1.1205

Undiscounted, the cheapest return-based escape is a single large payment, because a push spread over time partly decays at $T'(\theta_L) = 0.90$ before it accumulates. A permanent subsidy just above 0.0114 removes the trap; its present value is $0.0114/(1-\delta_d)$, which is 0.114 at discount factor $\delta_d = 0.9$ and 0.228 at 0.95. However, near the saddle-node the passage takes very long: even 80 generations need $\Delta\alpha = 0.0140$. Summary:

- The *one-time push* dominates when

$$c_\theta \times 0.0987 < \min\{c_\alpha \times 0.3949,\ \text{PV of the cheapest subsidy path}\},$$

  that is, when there is a cheap technology for moving aspirations directly, when the planner is impatient (the push works within one generation), and when θ_U is known precisely.

- The *subsidy* dominates when aspiration technologies are expensive or unreliable, when the planner is patient, or when the threshold is uncertain or heterogeneous. A subsidy above 0.0114 removes the low steady state for *everyone*, whatever their θ_U. A push must be sized to each agent's distance to the threshold, and it fails completely if it falls short.

(d) Bernard et al. (2014) randomize a one-time screening of documentaries about people from similar communities who improved their lives. In the model this is a one-time signal, so a one-time $\Delta\theta_i = \delta_i$ for viewer i. The prediction is a threshold rule. The effect is *permanent* for viewers with $\theta_{i,0} + \delta_i > \theta_U$: they move to θ_H, and their long-run change (0.28 here) exceeds the impact. It is *transient* for the others, decaying back to θ_L at rate $T'(\theta_L) = 0.90$ per period. The long-run average treatment effect is (share escaping)$\times(\theta_H - \theta_L)$. It can exceed the short-run effect, and it implies a bimodal long-run distribution of treated outcomes. A short follow-up cannot tell the two cases apart: in both, the effect at a short horizon is roughly the impact δ_i. Testing requires long horizons, a baseline measure of $\theta_{i,0}$ (to estimate heterogeneity by distance to the threshold), and dose variation in δ. The model's period is a generation, so the experiment's within-generation horizon maps onto it only under the timing assumptions noted in Chapter 11 (Stylized Fact 11.13).

Exercise 10.7 Cardoso and Faletto's "situations"

This is an open modelling exercise; we give one concrete calibration. Agents are ordered (F, B, M, W) = (foreign capital, domestic bourgeoisie, middle class, workers). Row i of W^P gives i's input dependence and row i of W^I whose returns i

observes. Power is the Perron vector of the bargaining layer, which we *assume* to be the composite $\bar{W} = \frac{1}{2}W^P + \frac{1}{2}W^I$. This is a modelling assumption (the node sets coincide, so the composite is admissible; `docs/conventions.md` §1), not a result. All other parameters are common to the three situations, so differences come only from (W^P, W^I):

- returns $\alpha = (0.06, 0.05, 0.04, 0.03)$, $\beta = (0.10, 0.08, 0.06, 0.04)$, $\gamma_i = 0.02$;
- allocation $\omega_i = 0.5$, $\Phi = 2$ (so $\Omega = 4$), $\phi = 1$;
- formation $\rho = 0.85$, $\mu = 0.2$, $\theta^\circ = 1$, $r^* = 0.1$.

Every result below is [ILLUSTRATIVE].

(a) Enclave. Every domestic class depends on foreign capital for inputs. Information is split: F and B observe each other, while M and W observe mainly each other.

$$W^P = \begin{pmatrix} 0.85 & 0.05 & 0.05 & 0.05 \\ 0.60 & 0.30 & 0.05 & 0.05 \\ 0.50 & 0.10 & 0.30 & 0.10 \\ 0.60 & 0.05 & 0.05 & 0.30 \end{pmatrix}, \qquad W^I = \begin{pmatrix} 0.70 & 0.20 & 0.05 & 0.05 \\ 0.40 & 0.50 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.60 & 0.30 \\ 0.02 & 0.03 & 0.30 & 0.65 \end{pmatrix}.$$

(b) National populism. There is a dense domestic production and information cluster (B, M, W), and weak links to F.

$$W^P = \begin{pmatrix} 0.60 & 0.20 & 0.10 & 0.10 \\ 0.05 & 0.45 & 0.20 & 0.30 \\ 0.05 & 0.30 & 0.40 & 0.25 \\ 0.05 & 0.35 & 0.20 & 0.40 \end{pmatrix}, \qquad W^I = \begin{pmatrix} 0.70 & 0.10 & 0.10 & 0.10 \\ 0.05 & 0.35 & 0.30 & 0.30 \\ 0.05 & 0.30 & 0.35 & 0.30 \\ 0.05 & 0.30 & 0.30 & 0.35 \end{pmatrix}.$$

(c) Associated dependent. There is a cross-border elite cluster $\{F, B\}$ in both layers. Workers depend partly on the elite for inputs, but they observe almost only their own class.

$$W^P = \begin{pmatrix} 0.50 & 0.40 & 0.05 & 0.05 \\ 0.40 & 0.45 & 0.10 & 0.05 \\ 0.20 & 0.30 & 0.40 & 0.10 \\ 0.15 & 0.30 & 0.15 & 0.40 \end{pmatrix}, \qquad W^I = \begin{pmatrix} 0.60 & 0.35 & 0.025 & 0.025 \\ 0.40 & 0.50 & 0.05 & 0.05 \\ 0.20 & 0.30 & 0.40 & 0.10 \\ 0.02 & 0.03 & 0.15 & 0.80 \end{pmatrix}.$$

All six matrices are row-stochastic (checked in the script).

(d) Solving $(I - J)\boldsymbol{\theta}^* = \mathbf{c}$ for each situation ($\rho(J) = 0.863, 0.863, 0.861$, all stable):

	$\boldsymbol{\pi}$ (N_{eff})	$\boldsymbol{\theta}^*$
Enclave	$(0.626, 0.152, 0.107, 0.115)$ (2.27)	$(1.179, 1.126, 1.012, 0.994)$
National populism	$(0.125, 0.321, 0.264, 0.290)$ (3.67)	$(1.062, 1.040, 1.037, 1.035)$
Associated dependent	$(0.406, 0.382, 0.101, 0.111)$ (3.00)	$(1.134, 1.117, 1.062, 0.979)$
	$\mathbf{x}^*$	$x_{\max}/x_{\min}$; $\text{Gini}(\mathbf{x}^*)$
Enclave	$(1.854, 0.851, 0.649, 0.646)$	2.87; 0.239
National populism	$(0.768, 1.137, 1.022, 1.072)$	1.48; 0.072
Associated dependent	$(1.373, 1.308, 0.691, 0.628)$	2.19; 0.178

Each $\mathbf{x}^*$ sums to $\Omega = 4$. National populism produces the most equal allocation and the enclave the most unequal; associated-dependent development lies in between. Two observations matter more than the ranking.

- *Power does most of the work.* The allocation decomposes as $\mathbf{x}^* = \boldsymbol{\omega} + \Phi(\boldsymbol{\pi} - \frac{1}{4}\mathbf{1}) + \phi P\boldsymbol{\theta}^*$. In the enclave the power term is $(0.752, -0.197, -0.285, -0.270)$ and the preference term $(0.101, 0.048, -0.066, -0.084)$. Formation *reinforces* power, because the agents who observe high returns are the powerful ones. Under national populism the preference term is almost flat (range 0.027), because the domestic W^I rows are nearly identical. This is Proposition 10.5(1): a small $\tau_1(W^I)$ compresses preferences.

- *The associated-dependent situation shows the worker disconnect.* Workers have the lowest θ^* (0.979) although their power (0.111) is above the middle class's (0.101). They observe only their own low returns, while the middle class observes the elite. This is the channel the case study emphasizes: what each group can *see* differs.

Common mistakes: (i) reading the ranking as a result about Latin America; it follows from the chosen matrices. (ii) Attaching Ψ to a situation. Ψ belongs to a perturbation (Definition 9.13); the comparable statement is that the same policy would have different formation effects in different situations.

Chapter 11

The Ψ Ratio: Measuring Formation

All numbers below are printed by `python3 code/solutions/sol11.py`. Variances of fixed effects are population variances (divisor n), as in `code/ch11.py`. Throughout, $\hat{\Psi}_{\text{var}}$ is the descriptive variance ratio of Definition 11.4, *not* the Ψ of Definition 9.13.

Exercise 11.1 Cohort–year decomposition by hand

The grand mean of the nine cells is 0.5389.

(a) Year means (average over cohorts) are 0.4933 (1980), 0.5433 (1990) and 0.5800 (2000). Centred, the year effects are $\hat{\alpha}_t = (-0.0456, 0.0044, 0.0411)$.

(b) Cohort means minus the grand mean give $\hat{\delta}_c = (-0.1256, 0.0044, 0.1211)$ for the 1940s, 1950s and 1960s cohorts. In a balanced table these row and column means are the OLS estimates of the two-way model (11.2) (with no covariates). The residuals are small but not zero; the largest are 0.0189 and -0.0144, in the corners, so the additive model does not fit exactly. That curvature is what an age profile would pick up.

(c) $\text{Var}(\hat{\alpha}_t) = 0.001262$ and $\text{Var}(\hat{\delta}_c) = 0.010151$, so

$$\hat{\Psi}_{\text{var}} = \frac{0.010151}{0.001262} = 8.045.$$

(d) Under the two-way specification (no age effects), cohort effects are far more dispersed than year effects: their standard deviation is $\sqrt{8.045} = 2.8$ times larger. But see (e).

(e) Ages (cohort mid-years 1945, 1955, 1965) are:

	1980	1990	2000
Born 1940s	35	45	55
Born 1950s	25	35	45
Born 1960s	15	25	35

Index cohorts and years by decade, $c, t \in \{0, 1, 2\}$, so the age index is $a = t - c$. By Proposition 11.5, $(\hat{\alpha}_t - kt,\ \hat{\delta}_c + kc,\ A_a = ka)$ gives fitted values identical to the two-way fit for every k; the script checks this cell by cell. The variance ratios are:

k (per decade)	cohort effects $\hat{\delta}_c + kc$	year effects $\hat{\alpha}_t - kt$	$\hat{\Psi}_{\text{var}}$
-0.10	$(-0.026, 0.004, 0.021)$	$(-0.146, 0.004, 0.141)$	0.027
-0.05	$(-0.076, 0.004, 0.071)$	$(-0.096, 0.004, 0.091)$	0.618
0	$(-0.126, 0.004, 0.121)$	$(-0.046, 0.004, 0.041)$	8.045
$+0.05$	$(-0.176, 0.004, 0.171)$	$(0.004, 0.004, -0.009)$	507.250

Effects centred to mean zero.

Since $\text{Var}(c) = \text{Var}(t)$, the k^2 terms cancel in $\text{Var}(\hat{\delta}_c + kc) - \text{Var}(\hat{\alpha}_t - kt)$. The ratio therefore equals one at a single value,

$$k^* = \frac{\text{Var}(\hat{\alpha}) - \text{Var}(\hat{\delta})}{2\,[\text{Cov}(\hat{\delta}, c) + \text{Cov}(\hat{\alpha}, t)]} = \frac{0.001262 - 0.010151}{2\,(0.08222 + 0.02889)} = -0.0400.$$

The extremes of Proposition 11.5 are reached at the least-squares slopes. At $k_\delta = -0.1233$, which detrends the nearly linear cohort effects, $\hat{\Psi}_{\text{var}} = 0.0005$. At $k_\alpha = 0.0433$, which detrends the year effects, it is 1876. As $|k| \to \infty$ the ratio tends to $\text{Var}(c)/\,\text{Var}(t) = 1$.

What (d) depends on. The answer depends entirely on the unidentified linear trend k, i.e. on the APC restriction. Reporting "cohort effects dominate" means choosing $k > -0.04$; reporting "year effects dominate" means choosing $k < -0.04$. The data identify only the *curvature* (second differences) of the three sets of effects. Here each set has only three points, so a single second difference each. *Common mistake:* treating the two-way fit without age effects as neutral. Omitting age effects is itself a restriction ($A_a \equiv 0$), and it loads the life-cycle profile onto cohorts and years.

Exercise 11.2 Epidemiological design

(a) For daughter i of parents from origin $A \in \{A, \dots, J\}$, living in Argentina (Definition 11.7):

$$\text{LFP}_i = \alpha + \beta\, \text{LFP}_{A(i),t_0} + \gamma' Z_i + \varepsilon_i,$$

where $\text{LFP}_i \in \{0,1\}$ and LFP_{A,t_0} is origin female LFP around the parents' emigration. Z_i contains age, survey year and region of residence. It should *not* include the daughter's education, marital status or fertility, which are bad controls (Proposition 11.8, caveats). Parental education is a separate case: it is a pre-determined control. The regressor varies only across ten origins, so standard errors must be clustered by origin. With ten clusters, use a wild-cluster bootstrap.

(b) $\hat{\beta} = 0.35$: a daughter whose parents come from a country with 10 pp higher female LFP is 3.5 pp more likely to participate, facing the same Argentine labour market. Across the full range of origins (0.20 to 0.75) the predicted difference is $0.35 \times 0.55 = 0.1925$, i.e. 19 pp. Under Proposition 11.8, $\beta = \rho\lambda$ with reliability $\lambda \in [0,1]$, so 0.35 is a *lower bound* on the per-generation persistence ρ. It is not ρ itself, because origin LFP also contains origin enforcement (e_A).

(c) Ten evenly spaced origin rates from 0.20 to 0.75 have spacing 0.0611 and population variance $\text{Var}(\text{LFP}_A) = 0.0611^2\,(10^2 - 1)/12 = 0.030810$. The variance explained by origin is $\hat{\beta}^2 \,\text{Var}(\text{LFP}_A) = 0.1225 \times 0.030810 = 0.003774$. Hence

$$\text{share explained} = \frac{0.003774}{0.003774 + 0.04} = 8.6\%.$$

(With the $n-1$ divisor the variance is 0.034234 and the share 9.5%.) This is not Ψ, for three reasons.

- Ψ is a ratio of the formation effect to the enforcement effect of *one perturbation* (Definition 9.13). The R^2 share is a ratio of variances in a cross-section.
- With a single host country, there is no enforcement variation in the regression at all. The residual variance 0.04 is individual heterogeneity, not enforcement (Appendix 11.9).
- A small explained share is fully compatible with a large Ψ for a given policy, and vice versa.

(d) Generalize assumption 2 of Proposition 11.8 to $\theta_m = \theta_A + \sigma_A$, where σ_A is the selection of migrants on θ relative to their origin, and let the daughter's residual

u_d possibly correlate with origin. Repeating the proof,

$$\beta = \frac{\text{Cov}(\text{LFP}_d, \text{LFP}_A)}{\text{Var}(\text{LFP}_A)} = \rho\lambda + \frac{\rho b \text{ Cov}(\sigma_A, \text{LFP}_A) + \text{Cov}(u_d, \text{LFP}_A)}{\text{Var}(\text{LFP}_A)}.$$

Selection inflates $\hat{\beta}$ (given $b, \rho > 0$) when either term is positive.

- $\text{Cov}(\sigma_A, \text{LFP}_A) > 0$: emigrants from high-LFP origins are *more* work-oriented than their origin average, relative to emigrants from low-LFP origins. Equivalently, selection on θ differs by origin in the same direction as origin culture.
- $\text{Cov}(u_d, \text{LFP}_A) > 0$: migrants from high-LFP origins bring more of something else that raises daughters' work, such as parental education, wealth, or ethnic job networks in Argentina.

Uniform selection (the same σ_A for every origin) does not bias β; it is absorbed in the constant. *Remedies:* control for parental education and occupation; compare origins with similar emigrant selection; use third generations (the ratio $\beta_3/\beta_2 = \rho$ removes λ, though not differential selection).

Exercise 11.3 What adoption designs can and cannot tell you

(a) The design randomizes adoptees across *adoptive* families, so it identifies the effect of the adoptive family's environment. The adoptees' biological parents (in Korea) are not observed in the data, and their characteristics do not vary in a way the design controls. Even if they were observed, their effect would bundle genes with the prenatal and pre-adoption environment. The design therefore has no variation that isolates the biological parents. What it delivers is the contrast between the adoptive-mother gradient for adoptees (environment) and the gradient for biological children (environment plus genes).

(b) Under the additive model, gradient = environment + genes. The adoptee gradient 0.10 estimates the environment component, and $0.30 - 0.10 = 0.20$ the genetic component. The environment share of the biological-child gradient is $0.10/0.30 = 1/3$. Two caveats: additivity rules out gene–environment interaction, and adoptees may respond differently to the same environment (age at adoption, discrimination).

(c) Let schooling be $S = a + b\theta + u$ for both mother and child, with u independent of θ and across generations. Let transmission be environmental only,

$\theta_{\text{child}} = \rho\theta_{\text{mother}} + \text{const} + \text{noise}$, as in Proposition 11.16(1). Then the adoptee gradient is

$$\frac{\operatorname{Cov}(S_{\text{child}}, S_{\text{mother}})}{\operatorname{Var}(S_{\text{mother}})} = \rho\,\frac{b^2 \operatorname{Var}(\theta)}{\operatorname{Var}(S)} = \rho R_\theta.$$

So $0.10 = \rho R_\theta$, and $\rho = 0.10/R_\theta \geq 0.10$. With $R_\theta = 1$, 0.5 or 0.25, the implied ρ is 0.10, 0.20 or 0.40. The additional assumptions needed are:

- the share R_θ of schooling variance due to θ (or b and $\operatorname{Var}\theta$);
- that the mapping $S = a + b\theta + u$ is linear and the same across generations;
- that the mother's schooling affects the child *only* through θ, and not through income or resources. Otherwise the gradient overstates ρ.

(d) Yes, it is consistent, but only for the *non-shared* part of W^I. "Common family environment" in a twin design is everything twins raised together share. That includes much non-family W^I: the same neighbourhood, often the same school, the same local media. So little common-environment variance in risk attitudes (Cesarini et al., 2009) also limits formation through shared peers and media. It leaves room for formation through individual-specific links (one's own friends, workplace) and own experience, the diagonal of W^I (Stylized Fact 11.14). *A test.* Randomly assigned college roommates, following the design of Sacerdote (2001):

- *Treatment:* the baseline risk attitude of one's randomly assigned roommate, elicited with an incentivized lottery task before assignment.
- *Outcome:* one's own risk attitude after one year, with own baseline and dorm fixed effects as controls.
- *Identification:* institutional random assignment removes homophily (item 3 of Section 11.4). Using pre-assignment roommate traits avoids the reflection problem. Dorm fixed effects absorb common shocks.
- *Power:* for an effect of 0.10 SD per SD of roommate risk tolerance, with residual SD 0.8, $\alpha = 0.05$ (two-sided) and power 0.80: $n = [(1.960 + 0.842) \times 0.8/0.10]^2 = 502.3$, i.e. 503 students. Without the baseline control (residual SD 1.0) the requirement is 785; for a 0.05 SD effect it is 2010. Pairs share shocks, so cluster by room; the requirement then rises by the design effect.

A positive coefficient is a formation effect of a specific ΔW^I, which is what Definition 9.13 asks for.

Exercise 11.4 Bolivia's self-employment variance ratio

Data status. The *Encuesta de Hogares* microdata are *not* in `data/processed`. They are on INE's ANDA portal and were not fetched (`data/README.md`, §9). The cohort–year results below are therefore method only; no cohort or departmental value is reported. The one real input available is the *national* self-employment share by survey year, from ILOSTAT (dataflow `DF_EMP_TEMP_SEX_STE_NB`, retrieved 2026-11-05; file `data/processed/ilostat_bolivia_emp_status.csv`). Self-employed (`STE_AGGREGATE_SLF`: employers, own-account and contributing family workers) over total employment, both sexes:

year	2000	2002	2003†	2005	2007	2009†	2013†	2015†	2016	2019
self-employed, %	68.4	68.5	64.0	66.0	62.7	58.2	60.9	61.6	70.0	66.6

† ILOSTAT source "HIES – Household Survey"; the other years are "LFS – Encuesta Continua de Empleo". Over the 19 survey years 2000–2019 (none in 2010) the mean is 64.6%, the range 58.2% (2009) to 70.0% (2016), and the SD of the yearly shares 3.52 pp. The LFS years average 67.1% and the HIES years 61.2%, which suggests a survey-type break. The year effects must therefore include survey-instrument dummies, or else instrument changes are read as "labour-market conditions". The WDI modelled 2019 value is 68.1%.

(a) *Cells.* Pool the EH rounds 2000–2019 (omitting 2010, when there was no survey). Keep employed persons aged 15–64 and define self-employment as above (or own-account only, as a robustness check). Assign five-year birth cohorts, 1935–39 to 1995–99. Compute weighted cell means $\bar{x}_{c,t}$ using the expansion factors, together with cell sizes and sampling variances, and drop cells with fewer than about 100 observations. Restrict to ages 25–64 for the main specification, so that schooling exit does not drive the youngest cells.

(b) *Estimation.* Regress $\bar{x}_{c,t}$ by WLS (weights = inverse sampling variance) on year dummies, cohort dummies, and composition controls (education shares, urban share, sector shares by cell). Year dummies capture labour-market conditions and survey-instrument changes; cohort dummies capture norm persistence. An alternative is to estimate at the individual level with the same dummies and cluster by cohort × year.

(c) *Restriction and sensitivity.* Self-employment has a strong life-cycle profile, so age effects cannot be omitted, and one linear restriction is needed (Proposition 11.5).

- *Main specification:* Deaton's normalization (year effects orthogonal to a linear trend and summing to zero; Deaton 1997, §2.7).
- *Alternative:* equal effects for the two oldest cohorts, or an external age profile (for example from a single cross-section, with year effects set to zero for that year).

Report $\hat{\Psi}_{\text{var}}$ under both and the whole curve $\hat{\Psi}_{\text{var}}(k)$, as in Exercise 11.1. Bias-correct both variances by subtracting the mean sampling variance of the estimated effects (Appendix 11.9), and F-test $\delta_c = 0$. *Expected qualitative result* (not an estimate): Deaton's normalization assigns all trends to age and cohort, so it tends to produce a large $\hat{\Psi}_{\text{var}}$. The national year shares above move by only a few points, apart from the survey-type break. The ranking of cohort vs. year dispersion is likely to reverse under alternative restrictions, as in Exercise 11.1.

(d) Repeat (a)–(c) separately for La Paz (urban) and the rural Altiplano, defined from the department and urban/rural identifiers (rural La Paz, Oruro and Potosí). The comparison *cannot* by itself show that formation is stronger where W^I is more block-diagonal.

1. $\hat{\Psi}_{\text{var}}$ is a ratio: a higher value in one region may reflect smaller year-effect variance (more stable local labour demand), not more cohort dispersion.
2. Rural self-employment is mostly own-account farming. Its persistence across cohorts reflects land tenure and the absence of wage jobs, which is enforcement, not norms.
3. Internal migration (Altiplano to El Alto and La Paz) changes the composition of regional cohorts over time. Selective out-migration creates spurious cohort effects.
4. The APC restriction may be valid in one region and not in the other.

One would need: a measure of W^I by region (network modules in the EH, or proxies such as language, media access and migration ties); an internal epidemiological design (migrants from high- vs. low-self-employment departments now in the same city; Example 11.15(3)); and ideally variation in exposure that is plausibly exogenous, such as rollout of roads, radio or mobile coverage.

Exercise 11.5 The WWII natural experiment

Data status. The figures (∼28%, ∼36%, ∼33%) are the exercise's illustrative values. The Census/BLS series were not fetched (`data/README.md`, row 22), so no

other value is used.

(a) Two observations cannot separate two effects plus a trend without assumptions, so we state them. (i) The wartime rise 28 → 36 (8 pp) is enforcement: mobilization raised labour demand at fixed preferences. (ii) Demobilization removed that demand by 1950. (iii) There is no pre-war secular trend. Under (i)–(iii), the 1950 level decomposes as

$$\underbrace{33-28}_{5\text{ pp}} = \underbrace{E_{1950}}_{\approx 0} + \underbrace{F_{1950} + \text{persistence among wartime workers}}_{5\text{ pp}}.$$

If instead LFP had a pre-war trend of τ pp per year, the residual is $5 - 10\tau$: 3 pp at $\tau = 0.2$, 2 at 0.3, 0 at 0.5. The "formation" share is therefore only as good as the counterfactual trend.

(b) Yes, partly. Between 1945 and 1950 LFP fell by 3 pp, i.e. 37.5% of the 8 pp wartime gain was reversed and 62.5% was not. Under (ii), all of the reversal is the enforcement effect switching off. That enforcement reverses is consistent with the evidence the book cites that much of the wartime entry was reversed by 1950 (Goldin, 1991).

(c) The 5 pp that persisted cannot be cross-generational formation yet. Daughters of wartime workers were mostly children in 1950. Formation through them (Chapter 9: preferences formed in childhood through W^I) would show up as *cohort* effects for women born roughly 1930–1945, as they entered the labour market in the 1950s and 1960s. The 1950 persistence comes instead from the wartime cohorts themselves. That can be own-experience formation (the diagonal of W^I, as in Stylized Fact 11.14), or human capital and job attachment acquired during the war. The latter is state dependence at fixed preferences, i.e. enforcement. A related cross-generational channel runs through sons: men whose mothers worked are more likely to have working wives (Fernández et al., 2004). The test is a cohort comparison in later censuses: do women who were children in high-mobilization states work more as adults, after year and age effects?

(d) Definition 9.13 needs a *permanent* perturbation, an outcome, an agent group and a horizon. The war was a *temporary* ΔW^P (1941–45). Measured after it ended, its enforcement effect is $E \approx 0$, so a Ψ for "the war" is undefined or meaningless. Well-posed alternatives:

- The perturbation is the permanent ΔW^I the war left behind (daughters' exposure to working mothers and to women in market jobs); the outcome is

daughters' adult LFP; the horizon is one generation. A pure ΔW^I has $E = 0$ (Proposition 10.15(3)), so again one reports F, not Ψ.

- The perturbation is a permanent rise in female labour demand (a ΔW^P that stays), with outcome women's LFP and horizon $T = 1, 2$ generations. Then $E > 0$ persists and $\Psi(\Delta W^P, T) = F/E$ is defined.

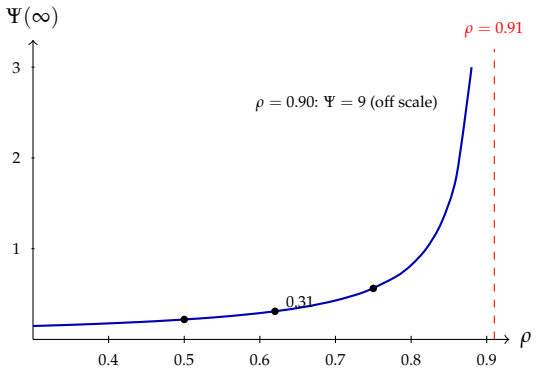
The war is the temporary version of the second case. It identifies hysteresis (persistence after the shock is removed), which in the affine formation model can come only from a lasting change in W^I or in human capital. A transient ΔW^P with an unchanged W^I decays at rate ρ (Proposition 9.11).

Exercise 11.6 Sensitivity analysis

With $\kappa = \mu\beta\eta$ and $\lambda = \rho + \kappa$, Proposition 10.14 gives $\Psi(\infty) = \kappa/(1-\lambda)$. At $\mu = 0.18$, $\beta\eta = 0.5$, $\kappa = 0.09$.

(a) Varying ρ (at $\mu = 0.18$):

ρ	0.50	0.62	0.75	0.90
$\Psi(\infty)$	0.220	0.310	0.562	9.000



The curve is convex and explodes as $\rho \to 1 - \kappa = 0.91$.

(b) Varying μ (at $\rho = 0.62$):

μ	0.10	0.18	0.30	0.50
$\lambda = 0.62 + 0.5\mu$	0.670	0.710	0.770	0.870
$\Psi(\infty)$	0.152	0.310	0.652	1.923

All four values are stable ($\lambda < 1$). Instability starts at $0.62 + 0.5\mu = 1$, i.e. $\mu = 0.76$.

(c) Differentiating, $\partial\Psi/\partial\rho = \kappa/(1-\rho-\kappa)^2 = 1.070$ and $\partial\Psi/\partial\mu = \beta\eta(1-\rho)/(1-\rho-\kappa)^2 = 2.259$. Per unit, Ψ is more sensitive to μ. But the parameters live on different scales, so elasticities are the better comparison:

$$\varepsilon_\rho = \frac{\rho}{1-\rho-\kappa} = 2.138, \qquad \varepsilon_\mu = \frac{1-\rho}{1-\rho-\kappa} = 1.310.$$

A 1% error in ρ moves Ψ by 2.1%, and a 1% error in μ by 1.3%. In general $\varepsilon_\rho/\varepsilon_\mu = \rho/(1-\rho)$, so persistence is the more sensitive parameter whenever $\rho > 1/2$. It is also the parameter estimated least precisely (Exercise 11.1; Proposition 11.16).

(d) $\Psi(\infty)$ diverges as $\lambda \uparrow 1$, i.e. at $\rho = 1 - \mu\beta\eta = 0.91$. This is the stability boundary of Proposition 10.14(3). For the finite-horizon bound, with $|\lambda| \le 1$ and $\kappa > 0$,

$$\Psi(T) = \kappa\,\frac{1-\lambda^T}{1-\lambda} = \kappa \sum_{k=0}^{T-1} \lambda^k \le \kappa \sum_{k=0}^{T-1} |\lambda|^k \le \kappa T.$$

Numerically:

	$T=1$	$T=2$	$T=5$	$T=10$	$T=\infty$
$\rho = 0.50$	0.090	0.143	0.204	0.218	0.220
$\rho = 0.62$	0.090	0.154	0.254	0.300	0.310
$\rho = 0.90$	0.090	0.179	0.441	0.861	9.000

Empirical meaning. At $T = 1$, $\Psi(1) = \kappa = 0.09$ whatever ρ is. A one-generation study therefore cannot distinguish an economy with $\Psi(\infty) = 0.22$ from one with $\Psi(\infty) = 9$; even at $T = 10$ the two differ by a factor of four, not forty. Large long-run formation effects are necessarily slow. Claims about $\Psi(\infty)$ are extrapolations whose size is governed by $\hat\rho$ near one, which is the least precisely estimated parameter.

Exercise 11.7 Designing an aspiration intervention

(a) $\Psi(\infty) = 2.1$ for the childcare subsidy means its long-run total effect is $E + F = (1 + 2.1)E = 3.1E$, where E is the effect at fixed preferences (Remark 9.14(4)). The exercise does not give the budget's enforcement yield. So the answer is *3.1 times whatever direct effect the budget buys*. Reaching the target +25 pp in the long run requires a direct effect of $25/3.1 = 8.06$ pp. A fixed-preference evaluation would record only $1/3.1 = 32\%$ of the long-run effect. Within the first generation the total is $E(1 + \Psi(1))$, much closer to E.

(b) The 2.1 cannot be used. Ψ belongs to the perturbation (Remark 9.14(1)), and a pure formation programme (mentors, media, workshops) changes W^I with $E = 0$. Its Ψ is undefined (Proposition 10.15(3)). Its effect is F itself, given by Theorem 10.12, $\Delta\boldsymbol{\theta}^* = (I - J)^{-1}\mu\,\Delta W^I \mathbf{r}^0$, times $Y = D_\theta(\text{LFP})$. Its size depends on how much the programme shifts whose returns women observe, and on the return gap they then see. Neither is given. *Common mistake:* computing "2.1×" the enforcement effect for a programme that has no enforcement effect.

(c) Let $e(b_E)$ and $f(b_F)$ be the long-run LFP effects of spending b_E on enforcement and b_F on formation, with $b_E + b_F = B$. The long-run total is $(1 + \Psi_E)\, e(b_E) + f(b_F)$, plus an interaction if formation changes the response to enforcement (Exercise 10.4(d)). With concave e, f, an interior optimum equates marginal *total* effects per unit of cost:

$$(1 + \Psi_E)\, e'(b_E) = f'(b_F).$$

With linear technologies the solution is a corner: everything goes to the instrument with the larger total effect per unit of cost. Ψ enters only as the factor $(1 + \Psi_E)$ on one side. It does not determine the split without e and f (Key Insight of Section 11.7). Illustration, with $e = a_E\sqrt{b}$ and $f = a_F\sqrt{b}$: the optimal enforcement share is $((1 + \Psi_E)a_E)^2/[((1 + \Psi_E)a_E)^2 + a_F^2]$. This is 0.906 when $a_F = a_E$, 0.516 when $a_F = 3a_E$, and 0.375 when $a_F = 4a_E$. Two further considerations:

- *Timing:* with discounting, the formation component is worth $\sum_T \delta^T[\Psi(T) - \Psi(T-1)]$, not $\Psi(\infty)$, which tilts the budget toward enforcement.
- *The LFP ceiling:* effects cannot add beyond 100%, so e and f are concave and become complements near the target.

(d) A share $1 - \lambda^T$ of the long-run formation effect is realized by generation T (Theorem 10.12, scalar case). Setting $1 - 0.62^T \geq 0.9$ gives $T \geq \ln 0.1 / \ln 0.62 = 4.82$, so *5 generations* ($0.62^4 = 0.148$ remains after four, $0.62^5 = 0.092$ after five). At 25 years per generation that is more than a century. A target within one generation is unreachable through formation alone.

A consistency note. If $\lambda = 0.62$ were the loop root of the childcare subsidy in the one-loop model, then $\Psi(\infty) = \kappa/(1 - \lambda) = 2.1$ would need $\kappa = 0.798$ and $\rho = \lambda - \kappa = -0.178 < 0$, which is impossible. The λ of (d) must therefore belong to a different perturbation (the formation programme), or to a model with more than one loop. This is another instance of Ψ and convergence rates being perturbation-specific.

Chapter 12

Welfare Under Endogenous Preferences

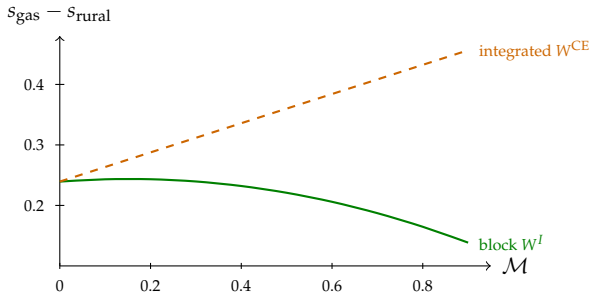
All numbers below are printed by `python3 code/solutions/sol12.py`. Throughout, $v(z) = z - \frac{\kappa}{2}z^2$ (equation (12.1)) and aspirations are $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$ (equation (12.2)). The stylized Bolivia allocation of Example 10.6 is $\bar{x}_b = (1.2486, 0.9852, 0.7662)$, with three agents per block.

Exercise 12.1 The satisfaction paradox

Computation ($\kappa = 0.5$, the near-block-diagonal W^I of Example 12.3; every $\kappa z < 1$):

$\mathcal{M}$	a_b (block W^I)	s_b (block W^I)	gap	gap with W^{CE}	ratio
0	$(0, 0, 0)$	$(0.8588, 0.7425, 0.6194)$	0.2394	0.2394	1.000
0.4	$(0.4771, 0.3954, 0.3275)$	$(0.6227, 0.5028, 0.3906)$	0.2321	0.3359	0.691
0.8	$(0.9541, 0.7908, 0.6550)$	$(0.2728, 0.1849, 0.1081)$	0.1647	0.4324	0.381

The gas-minus-rural gap as a function of $\mathcal{M}$ (on a grid of 0.1):



With integrated aspirations, everyone has the same aspiration $\mathcal{M}\bar{x}$. The gap is then $(x_1 - x_3)[1 - \kappa((x_1 + x_3)/2 - \mathcal{M}\bar{x})]$, which is linear and increasing in $\mathcal{M}$, from 0.2394 to 0.4565 at $\mathcal{M} = 0.9$. With the near-block W^I, the gap is a concave

quadratic in $\mathcal{M}$ (constant second differences -0.0038). It rises slightly up to $\mathcal{M} \approx 0.15$ and then falls, to 0.1647 at $\mathcal{M} = 0.8$. The slight initial rise comes from the cross-block weights (0.025): each block's aspiration includes a little of the others' allocations, and the concavity of v interacts with it. For large $\mathcal{M}$ the compression dominates. The ratio to the integrated gap (0.381 at $\mathcal{M} = 0.8$) is above $1 - \mathcal{M} = 0.2$, because W^I is not exactly block-diagonal and there are three blocks.

Verification of Proposition 12.2 (two equal blocks, pure block-diagonal W^I). We give a route different from the book's. For quadratic v, the midpoint rule is exact:

$$v(z_1) - v(z_2) = \int_{z_2}^{z_1} (1 - \kappa z)\, dz = (z_1 - z_2)\, v'\Big(\frac{z_1 + z_2}{2}\Big).$$

- *Block-diagonal:* $z_b = x_b - \mathcal{M}x_b = (1 - \mathcal{M})x_b$. So $z_M - z_T = (1 - \mathcal{M})(x_M - x_T)$, with midpoint $(1 - \mathcal{M})\bar{x}$, where $\bar{x} = (x_M + x_T)/2$.

- *Integrated:* $z_b = x_b - \mathcal{M}\bar{x}$. So $z_M - z_T = x_M - x_T$, with midpoint $\bar{x} - \mathcal{M}\bar{x} = (1 - \mathcal{M})\bar{x}$.

The midpoints coincide, so the v' factors are equal. They are positive because every $\kappa z < 1$ implies $\kappa \cdot$midpoint < 1. Hence the ratio of the gaps is $(1 - \mathcal{M})(x_M - x_T)/(x_M - x_T) = 1 - \mathcal{M}$. □ Numerically, at $(x_M, x_T) = (1.2, 0.8)$ and $\kappa = 0.5$: for $\mathcal{M} = 0.4$ the gaps are 0.168 and 0.280 (ratio 0.600); for $\mathcal{M} = 0.8$ they are 0.072 and 0.360 (ratio 0.200). The exactness is special to quadratic v. With $v(z) = \ln(1 + z)$ and $\mathcal{M} = 0.8$, the ratio is 0.198 instead of 0.2.

Exercise 12.2 Laundered aspirations

With $W^I_{\text{just}} = \frac{1}{9}\mathbf{1}\mathbf{1}' = W^{\text{CE}}$ and $\mathbf{x}^{\text{ref}}$ the current allocation, Proposition 12.5(1) gives a common laundered aspiration $\hat{a} = \mathcal{M}\bar{x} = 0.8 \times 1.0000 = 0.8$ ($\mathcal{M} = 0.8$, $\kappa = 0.5$). Laundered satisfaction by block is

$$v(\bar{x}_b - 0.8) = (0.3983,\ 0.1766,\ -0.0341).$$

The *rural* block has negative laundered satisfaction: its allocation 0.766 is below the aspiration 0.8 it would hold if it observed the whole economy.

Ranking the current allocation against $\bar{x}_b = 1$, with $W_1 = \sum_i v(x_i - a_i^*(\mathbf{x}))$ and $W_2 = \sum_i v(x_i - 0.8)$:

	endogenous W_1	laundered W_2
current $\mathbf{x}$	1.6974	1.6225
equal $\mathbf{x} = \mathbf{1}$	1.7100	1.7100
difference (equal − current)	0.0126	0.0875

The ranking does *not* change: both criteria prefer the equal allocation. The size of the gain does change. The endogenous criterion records a gain seven times smaller (0.0126 vs. 0.0875), because under W^I the rural block's aspirations are low (0.655), so the rural shortfall that equalization cures is mostly invisible to it. The ranking survives here for a structural reason. This W^I is symmetric, hence doubly stochastic (all column sums equal 1). By Proposition 12.13, the linear part of W_1 is then $(1 - \mathcal{M})\Omega = 1.8$ for *every* allocation. Only the concave part differs, and that part always favours equality. With a W^I that is not doubly stochastic, the endogenous criterion depends on who is watched. With the "media hub" of Chapter 12 (30% of every row moved to the gas block), the current allocation scores $W_1 = 1.1907$ and the equal one 1.7100. Watching the rich makes the unequal allocation look much worse at given aspirations. A ranking reversal needs a W^I under which equalization raises the watched agents' allocations less than it raises aspirations, as in the instance of Proposition 12.5(2).

Exercise 12.3 Capability measurement

Entitlements (Definition 12.6):

$$e_i = \omega_i + \pi_i \Phi : \qquad e_1 = 5 + 0.7 \times 2 = 6.4, \qquad e_2 = 5 + 0.3 \times 2 = 5.6,$$

summing to $\Omega = 12$. With $c_1 = c_2 = 1$, $\mathcal{O}_1 = [0, 6.4]$ and $\mathcal{O}_2 = [0, 5.6]$; agent 1's set is larger by 0.8, a ratio of 1.143. With $c_2 = 0.8$, $\mathcal{O}_2 = [0, 0.8 \times 5.6] = [0, 4.48]$. Agent 1's set is now larger by 1.92 (ratio 1.429). The information deficit adds 1.12 to the 0.8 owed to the power difference. In this example it is the larger of the two capability constraints, although it arises from what agent 2 *knows*, not from what she is *entitled to*.

Varying a_2 does not change the answer: $\mathcal{O}_i = [0, c_i e_i]$ depends on $(\omega_i, \pi_i, \Phi, c_i)$ and not on a_i (Proposition 12.7(1)). An aspiration is a reference point that shapes which point of $\mathcal{O}_2$ she chooses, and how satisfied she reports being. It does not shape which points are available. The conversion factor c_2 is different: it depends on her row of W^I (knowing which options exist), which is why information policy can enlarge $\mathcal{O}_2$ (Proposition 12.7(4)). *Common mistake:* lowering c_2 "because she

aspires to less". That conflates the epistemic constraint (what she can see) with the evaluative one (what she wants).

Exercise 12.4 Γ is not a welfare criterion

Construction. By Proposition 6.16, $W(\gamma, \boldsymbol{\pi}) = \gamma \mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ is row-stochastic, irreducible and aperiodic, with Perron vector $\boldsymbol{\pi}$ and spectral gap γ. Take

$$W_A = W\big(0.9,\ (0.55, 0.25, 0.15, 0.05)\big), \qquad W_B = W\big(0.3,\ (0.30, 0.25, 0.25, 0.20)\big).$$

Then $\Gamma(W_A) = 0.9 > \Gamma(W_B) = 0.3$, while $\mathrm{Var}(\boldsymbol{\pi}_A) = 0.035000 > \mathrm{Var}(\boldsymbol{\pi}_B) = 0.001250$ ($N_{\mathrm{eff}} = 2.564$ vs. 3.922). For instance, W_A written out is

$$W_A = \begin{pmatrix} 0.595 & 0.225 & 0.135 & 0.045 \\ 0.495 & 0.325 & 0.135 & 0.045 \\ 0.495 & 0.225 & 0.235 & 0.045 \\ 0.495 & 0.225 & 0.135 & 0.145 \end{pmatrix}.$$

Fixed-evaluation loss. Take $\Omega = 4$, $\Phi = 2$, $\kappa = 0.5$ and common $\hat{a} = 0$ (W_3). Then $x_i = 1 + 2(\pi_i - 0.25)$, giving $\mathbf{x}_A = (1.6, 1.0, 0.8, 0.6)$ and $\mathbf{x}_B = (1.1, 1.0, 1.0, 0.9)$. All $\kappa x_i \le 0.8 < 1$. By Proposition 12.11,

$$\sum_i v(x_i) = N v(\Omega/N) - \tfrac{\kappa N}{2}\Phi^2 \,\mathrm{Var}(\boldsymbol{\pi}) = 3 - 4\,\mathrm{Var}(\boldsymbol{\pi}),$$

so the loss is 0.1400 under W_A and 0.0050 under W_B. Welfare is 2.8600 under W_A and 2.9950 under W_B; the script checks the identity directly. The economy with the larger spectral gap is ranked lower. To see that Γ plays no role at all, take $W_C = W(0.3, \boldsymbol{\pi}_A)$. It has W_B's gap and W_A's power vector, and its welfare is 2.8600, exactly that of W_A. The ranking follows $N_{\mathrm{eff}}(\boldsymbol{\pi})$ (equivalently $\mathrm{Var}(\boldsymbol{\pi})$), because the allocation depends on W only through $\boldsymbol{\pi}$. Γ measures mixing speed, which the static allocation rule never uses. *Common mistake:* "higher integration means more equal power". Proposition 6.16 shows that the two coordinates are independent.

Exercise 12.5 The Easterlin pattern

Design (illustrative). There are 10 countries with mean incomes $\bar{y}_c$ evenly spaced from 0.5 to 2.0. Each country has five agents, so $N = 50$, with incomes

$$x = \bar{y}_c(1 + d), \qquad d \in \{-0.4, -0.2, 0, 0.2, 0.4\}.$$

Aspirations are $\mathbf{a} = \mathcal{M}W^I\mathbf{x}$ with $\mathcal{M} = 0.8$. The three information structures are:

- *national* (block-diagonal W^I, uniform within a country);
- *global* (W^{CE});
- *partial*: $\frac{1}{2}$ national + $\frac{1}{2}$ global.

The cross-country slope is the OLS slope of country-mean satisfaction on country-mean income. The within-country slope is the pooled slope of satisfaction on income after demeaning both by country.

Analytics (linear v). Agent i in country c has $s_i = x_i - \mathcal{M}[(1-\varepsilon)\bar{x}_c + \varepsilon\bar{x}]$, where ε is the global weight. So the within-country slope is exactly 1, and country means are $\bar{s}_c = [1 - \mathcal{M}(1-\varepsilon)]\bar{x}_c - \mathcal{M}\varepsilon\bar{x}$. The cross-country slope is therefore $1 - \mathcal{M}(1-\varepsilon)$.

Results:

	linear v ($\kappa = 0$)		quadratic v ($\kappa = 0.2$)	
W^I	(a) cross	(b) within	(a) cross	(b) within
national	0.200	1.000	0.170	0.937
partial ($\varepsilon = 0.5$)	0.600	1.000	0.550	0.912
global	1.000	1.000	0.930	0.886

Easterlin's pattern (Easterlin, 1974), a steep within-country gradient and a flat cross-country one, appears under *national* reference groups. There the cross-country slope is compressed by $1 - \mathcal{M}$ (Corollary 12.15), and it tends to zero as $\mathcal{M} \to 1$. Under global reference groups the two slopes coincide (with linear v), or the cross-country slope even exceeds the within-country one (with concave v, which flattens the within-country slope more at high incomes). Stevenson and Wolfers (2008) find similar within- and cross-country gradients in log income. In this model that implies reference groups that are largely *not* national: a large global weight ε in W^I, or a small $\mathcal{M}$. Two caveats. (i) Their evidence is in log income, so the comparison is about *relative* slopes, not levels. (ii) Easterlin's time-series claim (rising income without rising happiness) is a different dimension. In the model

it is the vintage problem of Proposition 12.16, with aspirations catching up with income over time, not a cross-sectional pattern.

Exercise 12.6 Evaluation and preference vintage

Reproduction. Under Policy C the rural rows of W^I put 0.075 on each gas agent and 0.70/3 on each rural agent. At the initial allocation, with $\mathcal{M} = 0.8$, the rural aspiration rises from 0.6550 to 0.7129, and rural satisfaction ($\kappa = 0.5$) falls from 0.1081 to 0.0526, as in Example 12.17.

Gradual adjustment. Rural allocations rise by 0.02 per generation, $x_{\text{rural},t} = 0.7662 + 0.02t$, and all other allocations stay fixed. Aspirations follow $\mathbf{a}_{t+1} = \rho\mathbf{a}_t + \mu W^I_{\text{new}}\mathbf{x}_t$, with $\mu = 0.8(1-\rho)$ and $\mathbf{a}_0 = \mathbf{a}_{\text{old}}$. The three evaluations of rural satisfaction are:

(i) old vintage: $v(x_{\text{rural},t} - 0.6550)$;

(ii) current vintage: $v(x_{\text{rural},t} - a_{\text{rural},t})$;

(iii) new steady state at the current allocation: $v(x_{\text{rural},t} - [0.8\, W^I_{\text{new}}\mathbf{x}_t]_{\text{rural}})$.

For $\rho = 0.75$ ($\mu = 0.2$):

t	0	1	2	3	5	7	10	15	20
(i) old	0.1081	0.1269	0.1454	0.1638	0.2000	0.2354	0.2870	0.3689	0.4458
(ii) current	0.1081	0.1133	0.1192	0.1258	0.1399	0.1549	0.1781	0.2170	0.2553
(iii) new s.s.	0.0526	0.0611	0.0696	0.0781	0.0949	0.1116	0.1363	0.1767	0.2161

When does the policy first look welfare-improving, i.e. when does rural satisfaction first exceed its pre-policy level 0.1081?

- *(i) Old vintage:* at $t = 1$, for every ρ. Aspirations never move, so any allocation gain counts in full.
- *(iii) New steady state:* at $t = 7$, for every ρ. In closed form, $z_t = 0.0533 + 0.02t(1 - 0.8 \times 0.70) = 0.0533 + 0.0088t$ must exceed 0.1112. Each extra unit of the rural block's own allocation raises its aspiration by $0.8 \times 0.70 = 0.56$, so only 44% of the gain shows up.
- *(ii) Current vintage:* at $t = 1$ for $\rho = 0.75$ and $\rho = 0.9$, but only at $t = 4$ for $\rho = 0.5$. With $\rho = 0.5$ aspirations jump toward the new target, and satisfaction first *dips* below the status quo (to 0.0995 at $t = 2$).

Dependence on ρ. The current-vintage path lies between the old-vintage path and the new-steady-state path. Low ρ (fast adjustment) pushes it toward (iii): the policy looks bad for longer, and early cost–benefit studies record a loss. High ρ keeps it near (i): the policy looks good early, and the reckoning comes later. The ranking at any date therefore depends on the vintage and on the speed of formation. This is Proposition 12.16, together with the vintage problem of von Weizsäcker (1971) and Hammond (1976).

Exercise 12.7 Sen vs. Bentham

Data: $\Omega = 10$, $\Phi = 4$, so $\omega_i = 3$. Then $x_i = 5 + 4(\pi_i - 0.5)$, $v(z) = z - 0.05z^2$ (so $\kappa = 0.1$), and $\mathbf{a} = (4, 1.5)$, taken as given.

(a) $\mathbf{x} = (5 + 1.2,\ 5 - 1.2) = (6.2, 3.8)$, so $\mathbf{x} - \mathbf{a} = (2.2, 2.3)$ and $\kappa(x_i - a_i) = (0.22, 0.23) < 1$. Under equal power, $\mathbf{x}' = (5, 5)$, $\mathbf{x}' - \mathbf{a} = (1, 3.5)$, and $\kappa z = (0.10, 0.35)$. With laundered $\hat{a} = 3$ the arguments are $(3.2, 0.8)$ and $(2, 2)$. All are below satiation.

(b) and

(c) Criteria under both structures:

Criterion	$\boldsymbol{\pi} = (0.8, 0.2)$	$\boldsymbol{\pi} = (0.5, 0.5)$	equalizing is
(i) Bentham, current **a**	3.9935	3.8375	worse
(ii) Laundered, $\hat{a} = 3$	3.4560	3.6000	better
(iii) Rawls, current **a**	1.9580	0.9500	worse
(iv) Capability, $\min_i e_i$	3.80	5.00	better

The individual terms are $v(\mathbf{x} - \mathbf{a}) = (1.9580, 2.0355)$ and $(0.9500, 2.8875)$, $v(\mathbf{x} - 3) = (2.6880, 0.7680)$ and $(1.8, 1.8)$, and $\mathbf{e} = (6.2, 3.8)$ and $(5, 5)$. These match Example 12.14. The criteria that use *current* aspirations, (i) and (iii), declare equalization a deterioration. The reason is that the landlord's high aspiration (4) turns his allocation cut into a large satisfaction loss ($1.958 \to 0.95$), which outweighs the worker's gain ($2.036 \to 2.888$). Rawls is worse still: after the change the *landlord* becomes the worst-off agent. The criteria that do not let the status-quo aspirations judge, laundered welfare and capability, declare it an improvement. Both depend only on $\mathrm{Var}(\boldsymbol{\pi})$ or on $\min_i \pi_i$ (Propositions 12.11 and 12.7(3)).

(d) Under every criterion *applied to the landlord himself*, the policy makes him worse off, so his opposition is "valid" as a statement about his own position under all four:

- satisfaction with current aspirations falls by 1.008;
- laundered satisfaction falls by 0.888;
- his entitlement falls by 1.2;
- and he becomes the minimum under Rawls.

It is "valid" as a statement about *social* welfare only under the criteria that use his current aspirations, (i) and (iii). There his high aspiration, itself formed under the power structure being evaluated, gives his loss extra weight.

If aspirations adjust. Suppose each agent keeps the ratio a_i/x_i (landlord 0.645, worker 0.395). Then at $\mathbf{x}' = (5, 5)$ the aspirations become $(3.226, 1.974)$ and satisfaction $(1.617, 2.568)$. Bentham rises to $4.185 > 3.994$: once aspirations adapt, even the current-preference utilitarian prefers equal power. The landlord's loss shrinks from 1.008 to 0.341. Rawls with adapted aspirations is $1.617 < 1.958$, so it still prefers the status quo, because the landlord's adapted satisfaction remains the minimum. The lesson is Proposition 12.16: the verdict depends on the vintage of aspirations. Opposition that rests on aspirations formed under the old structure weakens once those aspirations re-form. *Common mistake:* treating $a_L = 4$ as a fixed "need" rather than as a product of the landlord's past power.

Chapter 13

How Networks Evolve

All numbers below are printed by `python3 code/solutions/sol13.py`. The script takes about a minute, because Exercise 13.7 runs 1,300 simulations. $\Gamma = 1 - |\lambda_2|$ throughout (Definition 6.14).

Exercise 13.1 Edge creation by hand

W_t is symmetric and doubly stochastic, so $\boldsymbol{\pi} = \frac{1}{4}\mathbf{1}$. Its eigenvalues are 1, 0.96 (the block-contrast mode $(1, 1, -1, -1)'$, eigenvalue $0.6 + 0.38 - 0.02$) and 0.22 (twice, the within-block contrasts, eigenvalue $0.6 - 0.38$). Hence $\Gamma = 0.04$. In this family $\Gamma = 4c$, where c is the weight of each cross-block entry.

(a) We use the weighted geodesic distance of Appendix 13.8, with edge length $1/W_{ij}$:

$$d = \begin{pmatrix} 0 & 2.632 & 100 & 100 \\ 2.632 & 0 & 100 & 100 \\ 100 & 100 & 0 & 2.632 \\ 100 & 100 & 2.632 & 0 \end{pmatrix}.$$

Within-block pairs (1,2) and (3,4) are closest ($1/0.38 = 2.632$). Every cross-block pair is at distance 100. An indirect path (e.g. $1 \to 2 \to 3$, length 102.6) never beats the direct one. The effective-resistance distance, treating the off-diagonal W as conductances, gives the same ranking: 2.564 within and 26.282 across. *Common mistake:* using the unweighted hop count. In this complete graph it puts every pair at distance 1 and makes all edges equally likely.

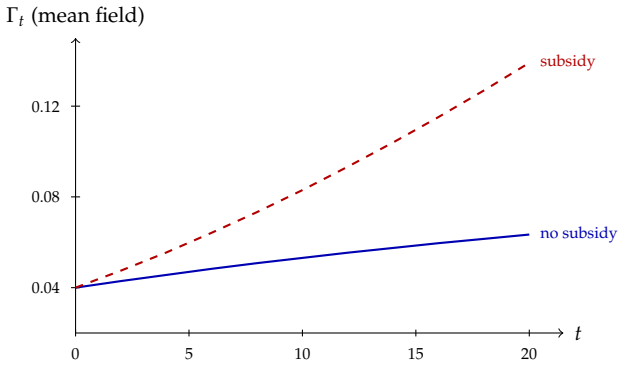
(b) With $\Pr(i, j) \propto 1/d(i, j)$ over the six unordered pairs, the weights are 0.38 (twice) and 0.01 (four times), summing to 0.80:

$$\Pr(1, 2) = \Pr(3, 4) = 0.4750, \qquad \Pr(\text{each cross pair}) = 0.0125.$$

The most likely new edge is within-block, (1,2) or (3,4), with 95% of the probability. A cross-block edge forms with probability 0.05. Note that $1/d_{ij} = W_{ij}$ here, so this creation rule is preferential attachment to existing weight.

(c) Doubling the cross-block weights gives a total of 0.84, so $\Pr(1,2) = \Pr(3,4) = 0.4524$ and each cross pair has probability 0.0238 (cross total 0.0952). The most likely new edge is *still* within-block. The subsidy roughly doubles the chance of integration, but it does not overturn proximity, which favours within-block pairs by a factor of 38.

(d) *Rule.* Each period one edge of weight $\delta = 0.05$ is added to a pair drawn with the probabilities above (recomputed from the current W_t), symmetrically to (i,j) and (j,i). The rows are then renormalized (the operator $\mathcal{N}$ of Definition 13.5). We report the mean-field path (expected increments) and 2,000 stochastic replications.



	mean field Γ_t					stochastic Γ_{20}
	$t=0$	5	10	15	20	mean; median; p_0
no subsidy	0.0400	0.0470	0.0531	0.0586	0.0634	0.0886; 0.0256; 0.510
subsidy	0.0400	0.0598	0.0830	0.1095	0.1391	0.1952; 0.1796; 0.264

Here p_0 is the probability that no cross-block edge forms in 20 periods. Two lessons.

1. In the mean field, Γ rises slowly even without the subsidy. Because $\Pr \propto W_{ij}$, new weight is added in proportion to existing links and the self-weights shrink (W_{ii}: 0.60 $\to$ 0.37). The cross-block share of each row therefore rises (0.01 $\to$ 0.016), while the within/cross ratio stays fixed.

2. The realized process is very different from the mean field. Without the subsidy, in 51% of replications *no* cross edge forms in 20 periods. Within-block edges then dilute the cross weights, and $\Gamma_{20} = 0.025$, *below* its start. The median path fragments. The subsidy halves that probability (26%) and raises the median Γ_{20} to 0.180.

Proximity-driven creation makes integration a low-probability event (Conjecture 13.11(3)), and a cross-block subsidy changes the odds without changing which edge is most likely.

Exercise 13.2 Structural equilibria

(a) *Power and gap along the family.* Let $W(c) = (1-c)B + c\mathbf{1}\mathbf{s}'$ (equation (13.7)). Since $\mathbf{s}$ is constant within blocks, $(\mathbf{s}'B)_j = \sum_{i\in b(j)} \frac{1}{2}s_i = s_j$, so $\mathbf{s}'B = \mathbf{s}'$ (the script prints it). Then

$$\mathbf{s}'W(c) = (1-c)\mathbf{s}'B + c(\mathbf{s}'\mathbf{1})\mathbf{s}' = (1-c)\mathbf{s}' + c\,\mathbf{s}' = \mathbf{s}'.$$

For $c > 0$, $W(c)$ is strictly positive, so the Perron vector is unique, and $\boldsymbol{\pi}(W(c)) = \mathbf{s}$ for every c. For the spectrum, the span of the three block indicators $\mathbf{e}_b$ is invariant: $W(c)\mathbf{e}_b = (1-c)\mathbf{e}_b + c\,s_b\mathbf{1}$, where $s_b = \mathbf{s}'\mathbf{e}_b$. On that span $W(c)$ acts as $(1-c)I_3 + c\mathbf{1}\mathbf{s}'_{\text{blk}}$, with eigenvalues 1 and $1-c$ (twice). On the complement (vectors summing to zero within each block) $B\mathbf{v} = 0$ and $\mathbf{s}'\mathbf{v} = 0$, so $W(c)\mathbf{v} = 0$. Hence $\Gamma(W(c)) = 1-(1-c) = c$. The script confirms the spectrum $\{1, 1-c, 1-c, 0, 0, 0\}$ at $c = 0.05, 0.3, 0.65$.

The three conditions of Definition 13.7.

	c	$\lvert F(c)-c\rvert$	$\max\lvert\boldsymbol{\theta}^* - T(\boldsymbol{\theta}^*)\rvert$	$\max\lvert\mathbf{x}^* - h\rvert$	$g'(c) = 1-\eta+\eta F'(c)$
W_A^*	0.0718	1×10^{-17}	0	0	0.964 (stable)
$\hat{c}$	0.2816	6×10^{-17}	0	0	1.045 (unstable)
W_B^*	0.6425	1×10^{-16}	2×10^{-16}	0	0.955 (stable)

Condition 1 holds because $\mathbf{x}^*$ is computed from $h(\boldsymbol{\pi}(W), \boldsymbol{\theta}^*, \boldsymbol{\omega})$; $\sum_i x_i^* = 6$ in each case. Condition 2 holds because $\boldsymbol{\theta}^* = \theta^\circ\mathbf{1} + \mathcal{M}(W(c)\mathbf{r} - r^*\mathbf{1})$ is the fixed point of the affine formation map with $W^I = W(c)$. Condition 3 holds because $c \mapsto (1-\eta)c + \eta F(c)$ leaves c unchanged exactly when $F(c) = c$. Stability follows from Proposition 13.14: $F'(c_A) = 0.282$, $F'(\hat{c}) = 1.907$, $F'(c_B) = 0.099$. The equilibrium values are $\boldsymbol{\theta}_A^* = (1.128, 1.128, 1.029, 1.029, 0.955, 0.955)$, $\mathbf{x}_A^* = (1.174, \ldots, 0.851)$, $\boldsymbol{\theta}_B^* = (1.089, \ldots, 1.022)$ and $\mathbf{x}_B^* = (1.118, \ldots, 0.902)$, as in Example 13.10.

(b) Bisection on $F(c) - c$ over the bracket containing the middle sign change gives $\hat{c} = 0.2816$. Within the family, $\|W(c) - W(c_A)\|_F = |c - c_A|\,\|\mathbf{1}\mathbf{s}' - B\|_F$, and the basin of B is $\{W(c) : c > \hat{c}\}$. Hence

$$\varepsilon^* = \inf_{c>\hat{c}} |c - c_A|\,\|\mathbf{1}\mathbf{s}' - B\|_F = (0.2816 - 0.0718) \times 1.4629 = 0.2098 \times 1.4629 = 0.307.$$

The infimum is not attained: a push of exactly ε^* lands on the unstable point. *Caveat:* the evolution map is defined only on the one-parameter family, so ε^* is the Frobenius distance to the basin *within that family*. Definition 13.16 over all of $\mathcal{W}_\varepsilon$ would need $\mathcal{E}$ off the family.

(c) Lowering the proximity weight in F from 12 to 8:

proximity weight	c_A	$\hat{c}$	c_B	ε^*
12	0.0718	0.2816	0.6425	0.307
8	0.1434	0.2473	0.6132	0.152

The fragmented equilibrium becomes less fragmented (c_A doubles), the threshold falls, and the integrated equilibrium is slightly less integrated. The escape cost halves. The slopes are $F' = 0.742, 1.246, 0.325$; c_A is still stable but closer to losing stability. Scanning the weight: the trap exists for weights 8–12 and has disappeared at 7, leaving a single integrated equilibrium ($c = 0.592$ at 7, 0.555 at 6, 0.484 at 5). This supports item 2 after Definition 13.16: lock-in is costly when proximity dominates link creation. Proximity is what makes F steep in c, and steepness, i.e. complementarity, is what creates multiplicity.

(d) Welfare at A and B, using Chapter 12's four approaches with $\kappa = 0.5$, aspiration multiplier 0.8 and $W^I = W(c)$:

Criterion	A ($c = 0.072$)	B ($c = 0.643$)	$B - A$
W_1: endogenous, $\mathbf{a} = 0.8\,W(c)\mathbf{x}$	1.1228	1.0404	-0.0824
W_2: laundered, $\hat{a} = 0.8\bar{x} = 0.8$	1.1134	1.1280	$+0.0146$
W_3: anchor, $\mathbf{a} = 0$	4.4734	4.4880	$+0.0146$
Capability, $\min_i e_i$	0.9333	0.9333	0
Process $(N_{\text{eff}}(\boldsymbol{\pi}), N_{\text{eff}}(\boldsymbol{\pi}^I), \Gamma(W^I))$	$(5.263, 5.263, 0.072)$	$(5.263, 5.263, 0.643)$	Γ up

- *Fixed-aspiration criteria* (W_2, W_3; $W_4 = W_2$ with W^{CE}) rank B higher. They depend only on $\text{Var}(\mathbf{x})$, which falls from 0.01773 to 0.00803 because rural preferences, and hence allocations, rise (x_{rural}: 0.851 $\to$ 0.902). The two gains are identical because $\sum_i v(x_i - \hat{a}) = Nv(1 - \hat{a}) - \frac{\kappa N}{2}\text{Var}(\mathbf{x})$ for any common $\hat{a}$ (Proposition 12.11).

- *Endogenous welfare* ranks B *lower*. Integration makes the rural block watch the gas block, so rural aspirations rise faster than rural allocations (rural s: 0.152 $\to$ 0.109). This is the satisfaction paradox in reverse (Proposition 12.2).

- *Capability* records no change: $e_i = \omega_i + \pi_i \Phi$ and $\boldsymbol{\pi} = \mathbf{s}$ in both equilibria. Any capability gain would have to come through conversion factors c_i, which this model does not specify.
- *Process:* B is process-superior, equal on both N_{eff} coordinates and strictly higher on $\Gamma(W^I)$.

The transition is therefore *not* robustly ranked in the multi-criteria sense (Proposition 12.9), because W_1 disagrees. It is a gain under every criterion that does not let the new aspirations judge the new structure. *Common mistake:* expecting integration to redistribute power. In this family $\boldsymbol{\pi}$ is fixed, and the gains come from preference formation alone.

Exercise 13.3 Bolivia's gas nationalization

(a) This is an [ILLUSTRATIVE] four-node model with nodes (MNC, Gas company, Government, Traditional sector); row i gives i's input/revenue dependence. Following Example 13.3, the fiscal linkage is modelled as an edge, which requires the government to be a node.

$$W^P_{\text{before}} = \begin{pmatrix} 0.70 & 0.25 & 0.03 & 0.02 \\ 0.55 & 0.30 & 0.10 & 0.05 \\ 0.05 & 0.15 & 0.50 & 0.30 \\ 0.02 & 0.03 & 0.15 & 0.80 \end{pmatrix}, \qquad W^P_{\text{after}} = \begin{pmatrix} 0.85 & 0.10 & 0.03 & 0.02 \\ 0.10 & 0.40 & 0.40 & 0.10 \\ 0.02 & 0.35 & 0.40 & 0.23 \\ 0.02 & 0.03 & 0.15 & 0.80 \end{pmatrix}.$$

Destruction: the gas company's dependence on the MNC falls from 0.55 to 0.10. *Creation:* gas $\to$ treasury rises from 0.10 to 0.40, and the government's weight on gas revenue from 0.15 to 0.35. The MNC reorients its own row away from Bolivia. The traditional sector's row is unchanged. All rows sum to one.

(b) The eigenvalues before are $(1, 0.844, 0.394, 0.062)$ and after $(1, 0.831, 0.607, 0.012)$, so

$$\Gamma_{\text{before}} = 0.156, \qquad \Gamma_{\text{after}} = 0.169.$$

In this toy Γ rises slightly. The slowest mode (the MNC-and-gas block vs. the domestic block) loosens a little, because gas now links to the government. The direction depends on the matrices chosen, and the measured economy went the other way. On INE's 12-sector domestic tables (base 1990, input shares), $\Gamma(W^P)$ was 0.518 in 2006 and 0.484 in 2013, with 0.514 (2003), 0.502 (2005) and 0.504 (2009)

around it (Instituto Nacional de Estadística (INE), Bolivia, 2026). The toy's rise comes entirely from counting the treasury link as a production edge. INE's industry tables contain no such edge, because the IDH is a fiscal flow, not an intermediate purchase. That is the point of the book's remark that the gas-to-treasury channel is a *fiscal* linkage in Hirschman's sense.

(c) Perron vectors:

	MNC	Gas	Government	Traditional
π_{before}	0.3695	0.1769	0.1489	0.3046
π_{after}	0.2056	0.1870	0.2294	0.3779
$\Delta\pi$	-0.1639	$+0.0101$	$+0.0805$	$+0.0733$

N_{eff} rises from 3.54 to 3.67. The MNC loses most of its power, and the government gains most. The gas company gains little ($+0.010$): its incoming weight from the government rises, but it loses the MNC's weight ($0.25 \to 0.10$). The traditional sector gains indirectly, through the government's continued weight on it. The first-order formula $\Delta\pi' \approx \pi' EZ$ (Lemma 6.19) gives $(-0.149, -0.041, +0.090, +0.101)$. The perturbation is large: first order gets three of the four signs right, but not the gas company's, so use the exact recomputation. The data agree on the direction for hydrocarbons: $\pi_{\text{hydrocarbons}}$ in the measured W^P rose from 0.064 (2006) to 0.085 (2013) (`data/processed/bolivia_io_pi.csv`).

(d) Probably yes, but this is a hypothesis; no data on W^I are used here. Hydrocarbon revenue was channelled through the treasury into visible programmes (the IDH-financed transfers discussed in Chapter 22). That raises the weight the traditional and urban-popular blocks place on the state's returns, i.e. the government's column of W^I, and makes the state a reference point for aspirations. In the formation model this is a ΔW^I toward the government node. By Proposition 10.15(3) its effect works entirely through formation: no enforcement at fixed preferences, and preference effects over generations. The direction for the *MNC* column is the opposite: foreign firms became less observed as a source of returns. Testing it would need media-audience or survey trust data by group, which are not available here (`data/README.md`, §9).

Exercise 13.4 The principle of relatedness

(a) Relatedness density (Hidalgo et al., 2007) is $\omega_{cp} = \sum_q \phi_{pq} M_{cq} / \sum_q \phi_{pq}$. On the ring every row of ϕ sums to $2(0.3) + 2(0.05) = 0.7$. With $M = \{1, 2\}$:

$$\begin{aligned}\omega &= \left(\tfrac{0.3}{0.7},\ \tfrac{0.3}{0.7},\ \tfrac{0.3+0.05}{0.7},\ \tfrac{0.05}{0.7},\ \tfrac{0.05}{0.7},\ \tfrac{0.3+0.05}{0.7}\right) \\ &= (0.4286,\ 0.4286,\ 0.5000,\ 0.0714,\ 0.0714,\ 0.5000).\end{aligned}$$

Product 4 is opposite 1 (distance 3, $\phi = 0$) and at distance 2 from product 2; product 5 is the mirror image of product 4.

(b) Among the products not yet produced, 3 and 6 tie at $\omega = 0.50$, and 4 and 5 trail at 0.071. The next entry is 3 or 6, each seven times as dense as 4 or 5.

(c) Product 5 is at ring distance 2 from product 1 (via 6) and 3 from product 2. Moving one adjacent step per generation, the shortest path is $\{1, 2\} \to +6 \to +5$: two steps, i.e. two generations. After entering 6, $\omega = (0.857, 0.5, 0.5, 0.143, 0.5, 0.5)$, so 5 ties for the top density with 2 and 3. If the tie in (b) is broken toward 3, the path $\{1, 2, 3\}$ gives $\omega_5 = 0.143$ and $\omega_4 = 0.5$, and reaching 5 takes three steps ($3 \to 4 \to 5$). The expected time therefore exceeds two generations unless the first step is steered.

(d) The product-space distance is the discrete analogue of the escape threshold ε^* (Definition 13.16). Both measure how far the current structure is from the basin in which the target becomes self-sustaining. There are two ways to cover the distance:

- *Stepping stones:* a sequence of steps, each small and probable, which costs time (two generations here) but little push per step.
- *A jump:* enter product 5 directly. With a logistic entry rule $\sigma(p_0 + p_1\omega)$ (equation (13.1)), making 5 as likely as 6 requires a subsidy worth $p_1(0.5 - 0.0714) = 0.4286\, p_1$ logit units: 0.857 at $p_1 = 2$ and 2.143 at $p_1 = 5$.

The stronger proximity is (p_1), the costlier the jump relative to the path. This mirrors Exercise 13.2(c), where the escape cost rises with the proximity weight. Strategic "long jumps" are costly for the same reason that ε^* is large.

(e) Data status: not fetched. The country $\times$ product RCA matrix and the product-space proximities are not in `data/processed`. The Growth Lab file needed (about 451 MB) is listed as fetchable but not done (`data/README.md`, §9). No relatedness value is reported here. *Method:*

1. Download the HS92 (4-digit) country $\times$ product export values from the Growth Lab Dataverse (doi:10.7910/DVN/T4CHWJ) for a recent year, e.g. 2019, and record the release.

2. Compute $\mathrm{RCA}_{cp} = (X_{cp}/\sum_{p} X_{cp})/(\sum_{c} X_{cp}/\sum_{c,p} X_{cp})$ and $M_{cp} = \mathbf{1}[\mathrm{RCA}_{cp} \geq 1]$, using a three-year average to reduce noise.

3. Compute proximities $\phi_{pq} = \min\{\Pr(M_{cp} = 1 \mid M_{cq} = 1), \Pr(M_{cq} = 1 \mid M_{cp} = 1)\}$ over all countries (Appendix 13.8), or download the Atlas proximity file.

4. Compute $\omega_{\mathrm{BOL},p} = \sum_{q} \phi_{pq} M_{\mathrm{BOL},q} / \sum_{q} \phi_{pq}$ for all p with $M_{\mathrm{BOL},p} = 0$, and list the top five.

Expected qualitative result: Bolivia's basket is concentrated in natural gas, ores and metals, gold, and soy and other agricultural products (WDI: ores and metals 28.7% and fuels 32.1% of merchandise exports in 2019; `data/processed/bolivia_facts.csv`). The highest-density unexported products should therefore lie near the primary-products periphery: other ores, concentrates and basic metal products, and processed agricultural goods. Densities should be low overall compared with Korea's, as Section 13.4 anticipates. The ranking itself must come from the data.

Exercise 13.5 Sunkel's integration/disintegration

(a)

$$W^{P}_{\text{after}} = \begin{pmatrix} 0.45 & 0.35 & 0.20 \\ 0.25 & 0.60 & 0.15 \\ 0.20 & 0.20 & 0.60 \end{pmatrix},$$

with every row summing to 1 and every entry nonnegative.

(b) Domestic submatrix (Elite, Mass), rows renormalized:

$$\text{before: } \begin{pmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{pmatrix}, \ \lambda_2 = 0.5, \ \Gamma = 0.500;$$

$$\text{after: } \begin{pmatrix} 0.8 & 0.2 \\ 0.25 & 0.75 \end{pmatrix}, \ \lambda_2 = 0.55, \ \Gamma = 0.450.$$

(c) Globally, the eigenvalues before are 1 and 0.4 (twice), so $\Gamma = 0.600$. After, they are $1, 0.4396, 0.2104$, so $\Gamma = 0.560$.

(d) No. Domestic Γ falls, as Sunkel predicts, but global Γ falls too. *Why.* W_{before} is symmetric with a double eigenvalue 0.4, so first-order perturbation must be done on the whole eigenspace. Restrict $E = W_{\text{after}} - W_{\text{before}}$ to it, $V'EV$. Its eigenvalues are $+0.0396$ and -0.1896, so the leading eigenvalue moves to 0.4396, which matches the exact value. In a symmetric three-node world there is no single binding cut. The reroute strengthens the Center–Elite link, but it weakens the Elite–Mass link, which is part of the slowest remaining mode, so integration falls.

A starting matrix for which the prediction holds. Make the Center the bottleneck:

$$W'_{\text{before}} = \begin{pmatrix} 0.90 & 0.05 & 0.05 \\ 0.05 & 0.60 & 0.35 \\ 0.05 & 0.35 & 0.60 \end{pmatrix},$$

with eigenvalues $1, 0.85, 0.25$. The slowest mode is $v_2 \propto (-0.8165, 0.4082, 0.4082)$, Center against periphery. Apply the same reroute: W'_{12} rises by 0.15, taken from the Center's self-weight, and 0.05 moves from Elite→Mass to Elite→Center. Then

$$\Gamma_{\text{global}} : 0.150 \to 0.317 \text{ (rises)}, \qquad \Gamma_{\text{domestic}} : 0.737 \to 0.702 \text{ (falls)}.$$

First order: $d\lambda_2 = v_2' E v_2 = -0.175$ (predicting $\lambda_2 = 0.675$; exact 0.683). Sunkel's pattern requires two things. (i) The binding (slowest) cut of the global network is Center vs. periphery. (ii) The rerouted weight comes from links that do *not* cross that cut: the Center's self-weight, or the periphery's internal links. The domestic loss then lowers domestic Γ without touching the global bottleneck. The Cheeger bounds of Proposition 13.12 formalize (i): Γ moves with the conductance of the weakest cut. If the weakest cut is domestic, as in the symmetric case, elite rerouting lowers both Γ's. The prediction is about where the bottleneck is, not an identity, which is the Key Insight of Section 13.5.

Exercise 13.6 WWII as structural change

The exercise leaves the model open. We use the block-level formation equation (Chapter 10), with every number [ILLUSTRATIVE].

- *Production.* $k_t := W^P_{F,\text{market}}$ is women's weight on market production. The return to market work observed among women is $r_{F,t} = \bar{r} k_t$, with $\bar{r} = 0.2$; men are fully in the market ($r_M = \bar{r}$).
- *Information.* $\ell_t := W^I_{F,M}$ is the weight girls place on market (men's) returns. Women's signal is $s_t = (1 - \ell_t) r_{F,t} + \ell_t \bar{r}$.

- *Formation.* $\theta_{F,t+1} = \rho\theta_{F,t} + (1-\rho)\theta^{\circ} + \mu(s_t - r^*)$, with $\rho = 0.6$, $\mu = 0.5$ ($\mathcal{M} = 1.25$), $\theta^{\circ} = 0.5$, $r^* = 0.16$.

(a) Pre-war: W^P is block-diagonal ($k = 0$) and W^I too ($\ell = 0$). Then $s = 0$ and $\theta_F^* = 0.5 + 1.25(0 - 0.16) = 0.300$, as required. Men's steady state is $\theta_M^* = 0.5 + 1.25(0.2 - 0.16) = 0.550$.

(b) War: k jumps to 0.5, a large cross-block edge in W^P. The *immediate* effect on θ is zero. Preferences are predetermined within a generation, and the war is an enforcement shock to allocations (women's employment), not to θ. What changes at once is the signal, s: $0 \to 0.1$. The next generation inherits $\theta = 0.6(0.3) + 0.2 + 0.5(0.1 - 0.16) = 0.350$ ($+0.050 = \mu\Delta s$).

(c) Post-war: the production edge partly reverses ($k = 0.2$), but children of wartime workers keep information edges to the market ($\ell = 0.3$). Then $s = 0.7(0.04) + 0.3(0.2) = 0.088$. Iterating from 0.350:

$$\theta_F : 0.300 \text{ (pre-war)} \to 0.350 \to 0.374 \to 0.388 \to 0.397,$$

with new steady state $0.5 + 1.25(0.088 - 0.16) = 0.410$.

(d) θ_F^* does not return to 0.3; it converges to 0.41. In the affine model persistence needs a permanent change in structure, not just a past shock. Counterfactuals:

- full reversal ($k = \ell = 0$): $0.330, 0.318, 0.311 \to 0.300$, i.e. no hysteresis (Proposition 9.11);
- only the W^I change persists ($k = 0$, $\ell = 0.3$): steady state 0.375;
- only the partial W^P change persists ($k = 0.2$, $\ell = 0$): steady state 0.350.

Lasting multiplicity would need nonlinear returns, as in Example 10.11, or the network complementarity of Example 13.10. Fernández (2013) models the twentieth-century rise in female participation as *learning* about the cost of working from observing working women. In this model that is the ℓ channel: what daughters see. Her mechanism adds an S-shaped dynamic absent from the affine model: the more women work, the more informative the signal. That would turn a temporary shock into a permanent one if it pushes beliefs past a threshold. Empirically, Fernández et al. (2004) document an intergenerational channel through sons: men raised by working mothers are more likely to have working wives, a cross-gender W^I edge. The reversal of much wartime employment by 1950 (Goldin, 1991) is the k channel switching off.

Exercise 13.7 Network evolution simulation

Program (in `code/solutions/sol13.py`; an open design, with every parameter [ILLUSTRATIVE]). $N = 8$. Every row keeps a self-weight 0.2, and links carry the other 0.8. Each period:

1. *Bargaining:* $\boldsymbol{\pi} = \boldsymbol{\pi}(W_t)$ and $x_i = 1 + 4(\pi_i - 1/8)$.
2. *Surplus:* $s_{ij} = 0.2 - |x_i - x_j|$; links between very unequal agents are unprofitable.
3. *Creation* (equation (13.1)): $C_{ij} = \sigma\big(-2 + p_1(a_{ij} - 1) + 2s_{ij}\big)$. Proximity $a_{ij} = (W_t^2)_{ij}/\text{mean}$ is the two-step reach (friends of friends), normalized to mean one, so p_1 changes the contrast and not the average rate.
4. *Destruction* (equation (13.2)): $D_{ij} = 0.05 + \max(0, -s_{ij})$.
5. *Reweighting:* $+0.02\, s_{ij} W_{ij}$.
6. *Update:* $W_{t+1} = \mathcal{N}\big((\mathbf{1}\mathbf{1}' - D) \circ W_t + 0.05\, C + 0.02\, S \circ W_t\big)$ on the links, then restore the self-weight.

Initial networks: 100 draws with Dirichlet(0.5) rows (median $\Gamma_0 = 0.631$, range 0.446–0.740). The same 100 draws are used for every p_1 (common random numbers). We run $T = 400$ periods.

(a) –

(c) Results ($\Gamma > 0.3$ is integrated, $\Gamma < 0.1$ fragmented):

p_1	0	0.25	0.5	0.75	1.0	1.25	1.5	1.75	2.0	3.0
integrated	1.00	1.00	1.00	0.88	0.29	0.02	0.00	0.00	0.00	0.00
fragmented	0.00	0.00	0.00	0.01	0.08	0.76	0.71	0.99	1.00	1.00
median Γ_T	0.808	0.898	0.803	0.762	0.200	0.085	0.062	0.073	0.068	0.025
settled	1.00	0.86	0.05	0.01	0.00	0.00	0.00	0.03	0.69	0.69

("Settled": last-period change in W below 10^{-4}.) With weak proximity, every starting network converges to a well-mixed structure. For $p_1 = 0$ the off-diagonal entries end between 0.008 and 0.132 and $\Gamma = 0.81$. With strong proximity, every network fragments into small cliques. In a typical $p_1 = 3$ outcome, agents split into groups $\{1, 4, 7\}$, $\{2, 5, 8\}$ and the pair $\{3, 6\}$, with cross weights of 0.01 or less and $\Gamma = 0.0255$. The fraction depends on p_1 monotonically up to sampling noise

(the dip at 1.5 is within it). Between the regimes there is a band ($p_1 \approx 0.75$–1.5) where many runs are "neither" and have not settled after 400 periods. This is critical slowing down near the transition: the system lingers between configurations.

(d) The fragmented share first exceeds 50% between $p_1 = 1.0$ (8%) and 1.25 (76%); linear interpolation gives $p_1^* \approx 1.15$. With $T = 800$ the shares at $p_1 = 1.0, 1.25, 1.5$ are 9%, 89% and 66% fragmented, so the threshold is robust to the horizon. The number itself depends on every design choice (the logistic intercept, the surplus weight, the self-weight); the qualitative result does not. Once p_1 exceeds about 1.15, proximity-driven creation locks in whatever clustering the initial network contains. This is Conjecture 13.11(3) in a simulation, not a proof. *Common mistakes:* classifying after too few periods near the threshold, and comparing p_1 values with different random starting networks.

Chapter 14

Coalitions and Class Alliances

All numbers below are printed by `python3 code/solutions/sol14.py`. Agents are ordered (F, B, M, W): foreign capital, domestic bourgeoisie, middle class, workers. The $\Delta\boldsymbol{\pi}$ vectors of the four alliances are those of Example 14.9; they are [ILLUSTRATIVE].

Exercise 14.1 Coalition payoffs

(a) By Proposition 7.3, $x_i = \Omega/N + (\pi_i - 1/N)\Phi = 1 + 2(\pi_i - 0.25)$:

$$\mathbf{x}^* = (1 + 0.2,\ 1 + 0.1,\ 1 - 0.1,\ 1 - 0.2) = (1.2,\ 1.1,\ 0.9,\ 0.8),$$

which sums to $\Omega = 4$. Then $\kappa x_i = (0.60, 0.55, 0.45, 0.40) < 1$: every allocation is below satiation, as `docs/conventions.md` §6 requires.

(b) With $\Delta\Omega = 0$ and $\Psi = 0$, Proposition 14.4 gives $v_i = \Phi\,\Delta\pi_i$. Moving to $\boldsymbol{\pi} = \frac{1}{4}\mathbf{1}$ gives $\Delta\boldsymbol{\pi} = (-0.10, -0.05, +0.05, +0.10)$, so

$$v_3(\{3,4\}) = 2 \times 0.05 = 0.10, \qquad v_4(\{3,4\}) = 2 \times 0.10 = 0.20.$$

Both are positive; agents 1 and 2 lose 0.20 and 0.10, and the payoffs sum to zero (power is constant-sum). In utility terms with $U = x - \frac{\kappa}{2}x^2$, the changes are $(-0.0900, -0.0475, +0.0525, +0.1100)$, summing to $+0.025$. That sum is exactly the distributional loss $L_{\text{power}} = \frac{\kappa N}{2}\Phi^2\,\mathrm{Var}(\boldsymbol{\pi}) = 0.025$ removed by equalizing (equation (7.9)).

(c) As stated, feasibility cannot be checked: the exercise specifies neither W nor $\phi(\cdot)$, and $\|\Delta W\|_F$ depends on both. We therefore solve the closest well-posed version: find the cheapest ΔW that delivers the target and compare it with $\phi(0.35)$.

Minimum-norm ΔW. Let $\mathbf{u} = \frac{1}{4}\mathbf{1}$. We need $\mathbf{u}'(W + \Delta W) = \mathbf{u}'$ and $\Delta W \mathbf{1} = 0$. The first condition is linear, $\mathbf{u}'\Delta W = \mathbf{r}' := \mathbf{u}'(I - W)$. Its minimum-Frobenius-norm solution is $\Delta W^\star = \mathbf{u}\mathbf{r}'/\|\mathbf{u}\|^2$. This also satisfies the second condition, since $\mathbf{r}'\mathbf{1} = \mathbf{u}'(I - W)\mathbf{1} = 0$. Its norm is $\|\mathbf{r}\|/\|\mathbf{u}\| = 2\|\mathbf{u}'(I - W)\|$.

A concrete W. Take $W = \gamma \mathbf{1}\boldsymbol{\pi}' + (1 - \gamma)I$, which has Perron vector $\boldsymbol{\pi}$ (Proposition 6.16). Then $\mathbf{r} = \gamma(\mathbf{u} - \boldsymbol{\pi})$, and

$$\|\Delta W^\star\|_F = 2\gamma \, \|\mathbf{u} - \boldsymbol{\pi}\|_2 = 2\gamma \times 0.1581 = 0.3162\,\gamma,$$

i.e. 0.3162, 0.1581, 0.0632 for $\gamma = 1, 0.5, 0.2$. Here $\Delta W^\star = \gamma \mathbf{1}(\mathbf{u} - \boldsymbol{\pi})'$, and $W + \Delta W^\star \geq 0$ (checked).

Feasibility. With the linear $\phi(p) = p$ the bound is 0.35, and the change is feasible for every γ. With the convex $\phi(p) = p^2$ the bound is 0.1225, and the change is feasible only if $\gamma \leq 0.387$. That is, only when the economy is weakly integrated, so that $\boldsymbol{\pi}$ is cheap to move (Appendix 14.7). The answer depends on ϕ and on the integration of W, which is the content of Definition 14.2.

(d) Agent 2 is B. In $\{1,2\}$ (Alliance 1) $\Delta\pi_2 = +0.03$; in $\{2,3,4\}$ (Alliance 2) $\Delta\pi_2 = +0.02$. With a common positive factor $(1 + \Psi)\Phi$, she prefers $\{1,2\}$ and stays (condition (14.13)).

Exercise 14.2 Alliance stability

All members gain. Alliance 1: F+0.05, B+0.03. Alliance 2: B+0.02, M+0.05, W+0.03. Alliance 3: F+0.03, B+0.04, M+0.02. Alliance 4: M+0.05, W+0.07. Each vector sums to zero.

Internal stability against the other three alliances. We read (14.13) with each rival alliance's program as the payoff that member i gets by joining it, $\Delta\pi_i(S' \cup \{i\}) := \Delta\pi_i(S')$. For (14.12) we assume that a coalition missing a member cannot implement its program, so $\Delta\pi_i(S \setminus \{i\}) = 0$; the first condition then reduces to $\Delta\pi_i(S) \geq 0$, which holds for every member.

Alliance	Defection test	Internally stable?
1: $\{F, B\}$	B: 0.04 in A3 $>$ 0.03	No (B leaves)
2: $\{B, M, W\}$	B: 0.04 in A3 $>$ 0.02; W: 0.07 in A4 $>$ 0.03; M: tie	No (B, W leave)
3: $\{F, B, M\}$	F: 0.05 in A1 $>$ 0.03; M: 0.05 in A2/A4 $>$ 0.02	No (F and M leave)
4: $\{M, W\}$	M: tie (0.05 in A2); W: 0.07 is her best	Yes (weakly)

Only Alliance 4 is internally stable, and only weakly, because M is indifferent with Alliance 2. This matches Conjecture 14.15, which places the baseline $\boldsymbol{\pi}$ in the

$\{M, W\}$ region. It is also externally stable: admitting B would turn the program into Alliance 2's, under which W gets $0.03 < 0.07$, so W vetoes B.

When each breaks or would hold. To deter every defection, each member must get at least her best rival offer. Compare the total required with what the program delivers to members (equal to what outsiders lose):

- A1 needs $F \geq 0.03$ and $B \geq 0.04$: 0.07 in total, and it delivers 0.08. A1 is *viable after an internal transfer*: if F cedes 0.01 of its gain to B (through a side payment or a different ΔW), both stay. It breaks only because the split favours F.

- A2 needs $B \geq 0.04$, $M \geq 0.05$, $W \geq 0.07$: 0.16 in total, against 0.10 delivered. No reallocation within the program works. The program would have to strip F of 0.16 rather than 0.10: a national-populist program holds together only if it is radical enough against foreign capital to pay both its bourgeois and its worker wings, i.e. only while π_F is large.

- A3 needs $F \geq 0.05$, $B \geq 0.03$, $M \geq 0.05$: 0.13 in total, against 0.09 delivered, and it can only be taken from W, who holds 0.15. A3 is stable only with near-complete exclusion of workers, i.e. where π_W is already small (region 3 of Conjecture 14.15).

- A4 needs $M \geq 0.05$ and $W \geq 0.03$: 0.08, against 0.12 delivered. It has slack: W could give up to 0.04 to keep M strictly loyal. It breaks if Alliance 2 can offer M more than W can spare, or if F and B can use the surplus term $\Delta\Omega/N$ to buy M (with $\Delta\Omega \neq 0$ the stability conditions include it).

Why B does not join Alliance 4 without compensation. Under Alliance 4's program $\Delta\pi_B = -0.02 < 0 = v_B(\{B\})$. Joining is individually irrational, by (14.12). B joins only if paid at least $(1+\Psi)\Phi \times 0.02$ (equal to 0.04 with $\Phi = 2$, $\Psi = 0$) through $\Delta\Omega/N$ or a side payment. That payment must be credible, which is Acemoglu's problem when the enforcer is the current W.

Exercise 14.3 The Latin American pendulum

Specification (an open design; every parameter [ILLUSTRATIVE]).

- *Initial structure:* $W_0 = \frac{1}{2}I + \frac{1}{2}\mathbf{1}\boldsymbol{\pi}_0'$, where $\boldsymbol{\pi}_0 = (0.35, 0.30, 0.20, 0.15)$. Then $\boldsymbol{\pi}(W_0) = \boldsymbol{\pi}_0$ and $\Gamma = 0.5$.

- *Feasibility:* $\mathcal{F}(S) = \{\Delta W : \Delta W\mathbf{1} = 0,\ \|\Delta W\|_F \leq 0.3\,\pi(S)\}$, i.e. $\phi(p) = 0.3p$ (Definition 14.2).

- *Surplus:* $\Omega(W) \equiv 4$. The game is then constant-sum, and $v_i = \Phi\Delta\pi_i$ with $\Phi = 2$ and $\Psi = 0$. By Proposition 14.14(2) its core is empty, so *some* instability is guaranteed.

- *Coalition program:* S maximizes $\sum_{i\in S} \Delta\pi_i$. By Lemma 6.19,

$$d \sum_{i\in S} \pi_i = \boldsymbol{\pi}' \, \Delta W \, Z\mathbf{1}_S,$$

  so the steepest feasible direction is the row-centred matrix $\boldsymbol{\pi}(Z\mathbf{1}_S)'$, scaled to the boundary of $\mathcal{F}(S)$. Entries are floored at 0.02, rows renormalized, and the exact new $\boldsymbol{\pi}$ computed.

- *Institutional inertia:* $W_{t+1} = 0.8(W_t + \Delta W_{S_t}) + 0.2\, W_0$.

- *Defection:* at W_{t+1}, each member of S_t compares the $\Delta\pi_i$ she would get from each *viable* alliance containing her (viable: all its members gain under its own program at W_{t+1}) with that of S_t. If some member gains strictly more elsewhere, the alliance offering the largest improvement forms.

Simulation (8 periods, starting with Alliance 2):

t	in power	π_F	π_B	π_M	π_W	Γ	defection
0	A2 $\{B, M, W\}$	0.217	0.344	0.244	0.194	0.500	$B \to$ A1 (+0.048)
1	A1 $\{F, B\}$	0.304	0.411	0.166	0.119	0.477	none
2	A1 $\{F, B\}$	0.387	0.461	0.089	0.063	0.471	$B \to$ A2 (+0.014)
3	A2	0.244	0.462	0.162	0.131	0.501	$B \to$ A1 (+0.066)
4	A1	0.343	0.504	0.090	0.063	0.473	$B \to$ A2 (+0.038)
5	A2	0.201	0.507	0.162	0.130	0.501	$B \to$ A1 (+0.082)
6	A1	0.307	0.536	0.092	0.065	0.461	$B \to$ A2 (+0.044)
7	A2	0.178	0.532	0.161	0.129	0.494	$B \to$ A1 (+0.089)

($\boldsymbol{\pi}$ is the value after the period's program.) Plotted over time, π_F, π_M and π_W oscillate with period two from $t = 2$ on, while π_B ratchets up from 0.30 to about 0.53.

Does it cycle? Yes, but not in the conjectured four phases. Power swings between the oligarchic-export alliance and the national-populist alliance, and the pivot is always B. Each alliance in power raises its members' π until the domestic bourgeoisie gains more by switching partners. Under $\{F, B\}$, F's power grows and B's marginal gain from staying shrinks ($+0.029$ at $t = 2$, against $+0.043$ from A2), so B turns to the domestic classes. Under $\{B, M, W\}$ the reverse happens. Alliances 3 and 4 never form. Under $\{F, B\}$ rule, M and W are outsiders and cannot

trigger a switch. Under $\{B, M, W\}$ rule, W's best option at W_{t+1} is to stay, since Alliance 4 offers less once π_F has been cut. Without inertia (W_0-reversion set to zero) the sequence is similar (A2, A1, A1, A2, A1, A2, A1, A1), but π_B climbs to 0.70. The pendulum then concentrates power in the swing class.

So the model generates the *mechanism* of Conjecture 14.12: an alliance's success changes $\boldsymbol{\pi}$, which changes its members' outside options. But in this calibration the cycle is the two-phase B-pivot pendulum, not ISI $\rightarrow$ liberalization $\rightarrow$ left populism. Obtaining Phase 3 would need the outsiders M and W to be able to *initiate* a coalition (a challenger rule), or a feasibility function that gives $\{M, W\}$ enough reach. *Common mistake:* assuming the cycle visits every alliance. Which alliances recur depends on who holds the pivot.

Exercise 14.4 Bolivia 1952–2019

This is an interpretive mapping onto the stylized typology of Example 14.9, not a test. The book itself flags the fit as loose. The events below are standard in the historiography (e.g. Klein, 2011; Dunkerley, 1984). The model readings are ours, and every $\Delta\boldsymbol{\pi}$ is qualitative.

(a) 1952, MNR revolution: Alliance 2, national-populist $\{B, M, W\}$. The urban middle-class leadership of the MNR allied with organized miners (the COB) and, after 1953, with peasants.

- ΔW^P: the nationalization of the three large tin companies into COMIBOL redirected the mining edges from foreign and oligarchic owners to the state; the 1953 agrarian reform cut landlord $\rightarrow$ peasant edges.
- ΔW^I and the bargaining layer: universal suffrage and the expansion of rural schooling.
- $\Delta\boldsymbol{\pi}$: the tin oligarchy (in the model, close to F and the export bourgeoisie) lost; workers, peasants and the state-linked middle class gained.

(b) 1964–1982, military governments: a switch away from Alliance 2. The core of the period, Barrientos's regime resting on the Military–Peasant Pact and later Banzer's, is closest to Alliance 3 ($\{F, B, M\}$, with peasants co-opted rather than members). It opened to foreign investment, repressed the miners and favoured agro-export interests in the east, so π_W fell. The period was not uniform. The 1969 nationalization of Gulf Oil and the Torres government (1970–71) were short swings back toward Alliance 2. This is itself pendulum behaviour.

(c) 1985, New Economic Policy: Alliance 3, techno-bureaucratic. Decree 21060 ended hyperinflation, liberalized prices and trade, and led to mass dismissals at COMIBOL. In model terms: $W^P_{\text{domestic}} \downarrow$, $W^P_{\text{global}} \uparrow$, and a large fall in π_W (the organized miners). That it was implemented by an MNR government illustrates the model's claim that the alliance, not the party label, determines ΔW (case study).

(d) 2006, Evo Morales: Alliance 4, left-populist $\{M, W\}$ plus peasant and indigenous organizations.

- ΔW^P: the 2006 hydrocarbons nationalization (Supreme Decree 28701) and renegotiated contracts redirected gas revenue from foreign operators to YPFB and the treasury, the edge-destruction/creation event of Example 13.3.
- In the measured domestic W^P (INE, base 1990, 12 sectors, input shares), hydrocarbons' power rose from 0.064 (2006) to 0.085 (2013), while $\Gamma(W^P)$ fell slightly, from 0.518 to 0.484 (`data/processed/bolivia_io_pi.csv` and `bolivia_io_summary.csv`). The alliance shifted rents and power, not production integration, as the Prebisch box of Section 14.3 notes.

(e) 2019, breakdown. The disputed October 2019 election and Morales's resignation in November ended the alliance. In the terms of Proposition 14.10, two conditions failed.

1. *No-defection* (14.13) *for M.* A large part of the urban middle class came to expect more from an alternative coalition than from continued MAS rule. This was visible in the 2016 referendum against a further re-election and in the 2019 urban protests. M is precisely the member who is indifferent in Exercise 14.2, the weak link of Alliance 4.
2. *The surplus that bought compensation shrank.* The proposition assumes $\Delta\Omega = 0$; side payments to members with weak power gains (and to B, as in Exercise 14.2) came from hydrocarbon rents. Natural-gas rents fell from 5.56% of GDP (2013) to 2.07% (2019), and fuels from 54.5% to 32.1% of merchandise exports (WDI, `data/processed/bolivia_facts.csv`). With a falling $\Delta\Omega$, the payments that held the alliance's weaker members together could no longer be made.

Caveat. The mapping is not unique. A historian could place 1964–82 in Alliance 1, or treat 2006 as Alliance 2 with indigenous organizations as the core. The model disciplines the questions (who gained power, through which edges, and what made a member's outside option better), not the answers.

Exercise 14.5 Why cash transfers are universal

Claim 1: $\Delta\pi(\Delta\omega) = 0$ within the period. Within a period W is held fixed, and π is defined (Definition 6.4) as the unique solution of $\pi' W = \pi'$, $\pi'\mathbf{1} = 1$, for irreducible W. This system involves W only. A transfer $\Delta\omega$ changes the allocation $\mathbf{x} = \omega + \Phi\pi + \ldots$, but enters neither the equation nor the normalization. Hence $\pi(W) = \pi(W)$ before and after the transfer, i.e. $\Delta\pi = 0$, and the power component of every payoff is zero: $v_i = \Delta\omega_i$ on impact. □

Claim 2: transfers can change π_{t+1}. In the co-evolutionary system $W_{t+1} = \mathcal{E}(W_t; \mathbf{x}_t^*)$ (Definition 13.6). If $\mathcal{E}$ is differentiable in $\mathbf{x}$, then $\Delta W_{t+1} \approx D_{\mathbf{x}}\mathcal{E}[\Delta\mathbf{x}_t]$ with $\Delta\mathbf{x}_t = \Delta\omega$ (plus formation effects). This perturbation has zero row sums because $\mathcal{E}$ preserves stochasticity. By Lemma 6.19,

$$\Delta\pi'_{t+1} \approx \pi'_{t+1}\left(D_{\mathbf{x}}\mathcal{E}[\Delta\omega]\right) Z_{t+1}, \qquad Z = (I - W + \mathbf{1}\pi')^{-1},$$

which is generally nonzero. *Example.* Take $W_t = \frac{1}{2}I + \frac{1}{2}\mathbf{1}\pi'_0$, with $\Phi = 2$ and $\omega_i = 0.5$, so $\mathbf{x}_t = (1.2, 1.1, 0.9, 0.8)$. Let the evolution map reweight partners by their resources (surplus-driven reweighting, Definition 13.4): $W_{t+1} = \mathcal{N}(W_t \circ (\mathbf{1}\mathbf{1}' + 0.5\,\mathbf{1}\mathbf{x}'_t))$. A transfer of 0.1 from F to W gives:

	F	B	M	W
π_{t+1} without transfer	0.3758	0.3070	0.1853	0.1319
π_{t+1} with transfer	0.3620	0.3103	0.1872	0.1404
difference (exact)	−0.0138	+0.0033	+0.0020	+0.0085
first-order formula	−0.0137	+0.0033	+0.0020	+0.0084

The recipient's future power rises by 0.085 per unit transferred. The change in W induced by the transfer has Frobenius norm 0.020, against 0.158 for the minimum-norm ΔW that would equalize power outright (Exercise 14.1(c), $\gamma = 0.5$).

Least controversial. On impact, a transfer threatens no one's structural position. It moves resources, not π, so the constant-sum power fight of Proposition 14.14(1) is not triggered. If it is financed out of growth ($\Delta\Omega > 0$), it can be Pareto-improving. And it is *reversible*: a flow that the next coalition can stop, unlike a ΔW, which persists and compounds through feasibility (Definition 14.2). Every coalition can therefore agree to it, which is the last row of Definition 14.16.

Least effective for structural change. For the same reasons. It leaves W, and hence π and Γ, unchanged within the period. The bargaining surplus Φ is redivided by the same π every period, so a transfer must be repeated forever to keep its effect (the Transfer Trap of Chapter 2). Its effect on future π is indirect, second-

round and small: in the example, $+0.0085$ for the recipient after a transfer of 0.1 (8% of the donor's allocation). Its formation effect is also small ($\bar{\Psi}(\infty) = 0.12$ for Policy A in Example 10.16). Structural change requires the ΔW that some coalition will resist.

Exercise 14.6 The formation amplifier in politics

A premise to make explicit. If $\boldsymbol{\pi}$ is the Perron vector of W^P (or of a bargaining layer that excludes W^I), a pure ΔW^I leaves $\boldsymbol{\pi}$ unchanged. Then $E = 0$, and Ψ_B is undefined (Proposition 10.15(3); Remark 9.14(5)). For Alliance B to have the same $\Delta\boldsymbol{\pi}$ as Alliance A and a finite Ψ_B, the bargaining layer must include W^I, e.g. a composite $\sum_k \alpha_k W^k$ on a common node set. We assume this.

(a) –

(c) With $\Delta\Omega = 0$, $v_i = (1 + \Psi)\Phi\,\Delta\pi_i$. Take $\Phi = 2$ and a member with $\Delta\pi_i = 0.05$:

$$v_i^A = (1+1) \times 2 \times 0.05 = 0.20, \qquad v_i^B = (1+3) \times 2 \times 0.05 = 0.40.$$

Alliance B is twice as valuable to its members, and its rivals lose twice as much, since $(1 + \Psi)$ multiplies losses too. Two qualifications matter.

- *Horizon.* $\Psi = \Psi(\Delta W, T)$. Formation effects arrive over generations, so with discounting the relevant multiplier is not $1 + \Psi(\infty)$. In the one-loop model $\Psi(t) = \Psi(\infty)(1 - \lambda^t)$, so a member who receives $\Phi\Delta\pi_i(1 + \Psi(t))$ per generation, discounted at δ, has the effective amplifier

$$1 + \Psi(\infty)\,\frac{\delta(1-\lambda)}{1-\delta\lambda}.$$

- With $\lambda_A = \lambda_B = 0.8$: A 1.643, B 2.929 at $\delta = 0.9$; A 1.167, B 1.500 at $\delta = 0.5$.
- If information effects are slower ($\lambda_B = 0.9$ against $\lambda_A = 0.6$): A 1.783, B 2.421 at $\delta = 0.9$, but A 1.286 against B 1.273 at $\delta = 0.5$. Impatient coalitions can prefer the production alliance.

(d) The W^I alliance is harder to form, for four reasons.

1. *Feasibility sequencing.* Reshaping media and education usually requires control of the state. That requires $\pi(S)$ large enough that $\phi(\pi(S))$ covers the needed $\|\Delta W^I\|$ (Definition 14.2), which requires a prior alliance, usually built on W^P or electoral power. The W^I alliance is a second-stage coalition.

2. *Intertemporal free-riding.* Most of the $(1+\Psi)$ premium accrues to the next generation, i.e. to members' descendants and to whoever holds power then. Current members bear the cost now, so commitment problems are worse.

3. *Stronger resistance.* Losers' losses are amplified by the same $(1+\Psi_B)$, so the opposing coalition fights harder; its stakes double too.

4. *Verifiability.* ΔW^I is hard to observe and contract on, so side payments within the alliance are harder to enforce.

This is the regress the hint points to: power over W^I requires power, which requires an alliance, which is easier to build on the faster, enforceable W^P channel. *Common mistake:* treating Ψ as a property of the layer. Here it belongs to each alliance's specific perturbation (Definition 9.13).

Exercise 14.7 Empty core

The game. Take three agents with $\boldsymbol{\pi} = \frac{1}{3}\mathbf{1}$, $W = \frac{1}{2}\cdot\frac{1}{3}\mathbf{1}\mathbf{1}' + \frac{1}{2}I$, and $\Phi = 2$. The policy space is the family $\Delta W = \gamma \mathbf{1}\Delta\boldsymbol{\pi}'$ with $\gamma = 0.5$. Every such ΔW keeps rows stochastic and moves the Perron vector exactly to $\boldsymbol{\pi} + \Delta\boldsymbol{\pi}$ (Proposition 6.16), with $\|\Delta W\|_F = \gamma\sqrt{3}\|\Delta\boldsymbol{\pi}\|_2$. Feasibility is $\phi(p) = 0.6\max(0, p - \frac{1}{2})$: coalitions without a power majority cannot act. The TU value is

$$v(S) = \max\Big\{(1+\Psi)\Phi\sum_{i\in S}\Delta\pi_i \;:\; \mathbf{1}'\Delta\boldsymbol{\pi} = 0,\ \gamma\sqrt{3}\|\Delta\boldsymbol{\pi}\| \le \phi(\pi(S))\Big\}.$$

Maximizing $\mathbf{1}_S'\Delta\boldsymbol{\pi}$ on a ball in the sum-zero subspace gives

$$r_S\left\|\mathbf{1}_S - \tfrac{|S|}{3}\mathbf{1}\right\| = r_S\sqrt{|S|(3-|S|)/3}, \qquad r_S = \frac{\phi(\pi(S))}{\gamma\sqrt{3}}.$$

Hence, with $\Psi = 0$:

- singletons ($\pi = 1/3 < 1/2$): $v(\{i\}) = 0$;
- pairs: $r = 0.1/(0.5\sqrt{3}) = 0.1155$, so $v(\{i,j\}) = 2\times 0.1155\times\sqrt{2/3} = 0.4\sqrt{2}/3 = 0.18856 =: g$;
- the grand coalition: $v(N) = 0$, since $\mathbf{1}_N$ has no sum-zero component (power is constant-sum).

Coalition $\{1,2\}$ raises $\pi_1 + \pi_2$. Its optimal program is

$$\Delta\boldsymbol{\pi} = (0.0471,\ 0.0471,\ -0.0943),$$

which moves the Perron vector to $(0.3805, 0.3805, 0.2391)$ with $\|\Delta W\|_F = 0.1 = \phi(2/3)$.

The core is empty. A core allocation needs $x_1 + x_2 + x_3 = v(N) = 0$ and $x_i + x_j \geq g$ for each pair. Summing the three pair constraints gives $2(x_1 + x_2 + x_3) \geq 3g > 0$, a contradiction. This is Proposition 14.14(2): with $\Delta\Omega \equiv 0$, any coalition that can gain empties the core.

Adding surplus. Suppose the grand coalition, and only it, can also implement an integrating ΔW that raises output by $\Delta\Omega$ while leaving π unchanged. For example, it can raise γ within the same family, which raises Γ but not π. Then $v(N) = (1 + \Psi)\Delta\Omega$, and pairs keep g. By Bondareva–Shapley (Bondareva, 1963; Shapley, 1967), the core is nonempty iff $\sum_S \lambda_S v(S) \leq v(N)$ for every balanced collection. For three players the minimal balanced collections are:

- $\{\{1\}, \{2\}, \{3\}\}$, which requires $v(N) \geq 0$;
- $\{\{i, j\}, \{k\}\}$ with unit weights, which requires $v(N) \geq g$;
- $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$ with weights $\frac{1}{2}$, which requires $v(N) \geq \frac{3}{2}g$.

The binding one is the last. Hence

$$\text{core} \neq \varnothing \iff (1 + \Psi)\Delta\Omega \geq \tfrac{3}{2}g = 0.2\sqrt{2} = 0.28284.$$

A grid search over the simplex confirms this: no core point at $(1 + \Psi)\Delta\Omega = 0, 0.2, 0.28$; the unique point $\mathbf{x} = (0.0943, 0.0943, 0.0943)$ at 0.28284; and, for example, $\mathbf{x} = (0.028, 0.161, 0.161)$ at 0.35. With Φ general and pair power gain G (in π units), the condition is $(1 + \Psi)\Delta\Omega \geq \frac{3}{2}(1 + \Psi)\Phi G$, i.e. $\Delta\Omega \geq \frac{3}{2}\Phi G$. The amplifier cancels. The grand coalition is stable only if the integration surplus exceeds the power that pairs could grab, by a factor of $3/2$. *Common mistake:* checking only the pairwise or singleton constraints. The binding balanced collection is the cover by all three pairs.

Chapter 15

Operator Design

Every number below is printed by `python3 code/solutions/sol15.py`. Throughout, $\Gamma = 1 - |\lambda_2|$, $\boldsymbol{\pi}$ is the left Perron vector, $\mathrm{Var}(\boldsymbol{\pi}) = \frac{1}{N}\sum_i (\pi_i - 1/N)^2$, and G is the gradient of Γ in (15.5), with left and right eigenvectors normalized biorthogonally (rows of V^{-1}).

Exercise 15.1 Optimal edge selection

(a) Set-up. Two blocks $\{1,2\}$ and $\{3,4\}$; each row puts total weight 0.02 across blocks, split equally, and 0.98 inside its block, split equally:

$$W^P = \begin{pmatrix} 0.49 & 0.49 & 0.01 & 0.01 \\ 0.49 & 0.49 & 0.01 & 0.01 \\ 0.01 & 0.01 & 0.49 & 0.49 \\ 0.01 & 0.01 & 0.49 & 0.49 \end{pmatrix}.$$

Rows sum to 1. Rows inside a block are identical, so the partition is equitable (as in Proposition 18.4(a)): the quotient matrix is $\begin{pmatrix} 0.98 & 0.02 \\ 0.02 & 0.98 \end{pmatrix}$, with eigenvalues 1 and 0.96, and every vector that sums to zero inside each block is mapped to 0. Hence the spectrum is $\{1, 0.96, 0, 0\}$, $\lambda_2 = 0.96$ is real and simple, no other eigenvalue has modulus 0.96, and $\Gamma = 0.04$. The hypotheses of Lemma 6.19 and of (15.5) hold. (An exactly block-diagonal matrix would have $\lambda_2 = 1$ twice, and Γ would only have one-sided derivatives.)

Eigenvectors. $\mathbf{v}_2 = \frac{1}{2}(1,1,-1,-1)'$ and $\mathbf{u}_2 = \frac{1}{2}(1,1,-1,-1)'$, so $\mathbf{u}_2'\mathbf{v}_2 = 1$ and $\mathbf{u}_2'\mathbf{v}_k = 0$ for the other eigenvectors (W is symmetric). Because $\lambda_2 > 0$ is real,

$G_{ij} = -u_{2,i}v_{2,j}$:

$$G = \frac{1}{4}\begin{pmatrix} -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix}.$$

$G_{ij} = +\frac{1}{4}$ on every cross-block pair and $-\frac{1}{4}$ on every within-block pair.

(b) The best move. A feasible direction moves weight within a row, $E = \mathbf{e}_i(\mathbf{e}_j - \mathbf{e}_\ell)'$, at rate $G_{ij} - G_{i\ell}$ (Appendix 15A). The rates are

- $\frac{1}{2}$ for moving weight from a within-block entry to a cross-block entry;
- 0 for a move between two within-block entries or between two cross-block entries;
- $-\frac{1}{2}$ for a move from cross to within.

The maximum $\frac{1}{2}$ is attained by 16 tied moves: any row i, any within-block donor ℓ (including the diagonal), any cross-block recipient j. By Proposition 15.3(1), an ℓ_1 budget B buys a move of $B/2$, so $\Delta\Gamma \approx \frac{B}{2} \cdot \frac{1}{2} = \frac{B}{4}$: 0.25 per unit of ℓ_1 budget. All entries of W are positive, so the nonnegativity constraint is slack for small B.

Checks.

- The finite difference for the move $(1,1) \to (1,3)$ gives 0.5000; for the within-block move $(1,1) \to (1,2)$ it gives 0.
- Exact effect of the move with $\varepsilon = 0.01$: $\Delta\Gamma = 0.00497$ (first order 0.00500). With $\varepsilon = 0.05$: 0.02435 (first order 0.02500). The concavity of the exact path shows the first-order rule overstates large moves slightly.
- Spreading the same ℓ_1 norm over all four rows has the same first-order effect (0.5 per unit moved). Its exact effect at $\varepsilon = 0.01$ is 0.00500, slightly larger than the single move's 0.00497, because a symmetric perturbation keeps the partition equitable.
- Breaking the symmetry at random (entries perturbed by up to ± 0.005, rows renormalized): in all 200 draws the optimal recipient j is in the other block.

(c) Within or cross? Cross-block, always. $G_{ij} = -u_{2,i}v_{2,j}$ is positive exactly when $u_{2,i}$ and $v_{2,j}$ have opposite signs, and the slow mode $\mathbf{v}_2$ has opposite signs on the two blocks. A within-block move cannot change λ_2 to first order, since it leaves the quotient matrix unchanged. With only two agents per block there is no "bridge"

agent, so the core-versus-boundary distinction of the intuition box after Proposition 15.3 does not arise: all cross-block pairs tie.

Common mistakes: computing "$\partial\Gamma/\partial W_{ij}$" without a compensating entry (not a feasible direction), and using an exactly block-diagonal W, where λ_2 is not simple.

Exercise 15.2 ISI-A vs. ISI-B

(a–b) Set-up. From the printed A of Example 15.6, the row sums are $(0.6, 0.6, 0.3, 0.3)$, all below 1, so $\rho(A) < 1$. $W^P = D^{-1}A$ is the matrix printed in the example. Its eigenvalues are $1, 0.8867, 0.1333, 0.08$, so $\Gamma = 0.1133$. Its Perron vector is $\boldsymbol{\pi} = (0.3529, 0.3529, 0.1471, 0.1471)$, with $\mathrm{Var}(\boldsymbol{\pi}) = 0.01060$. The Leontief inverse is

$$L = (I - A')^{-1} = \begin{pmatrix} 1.7358 & 0.6489 & 0.0395 & 0.0395 \\ 0.6489 & 1.7358 & 0.0395 & 0.0395 \\ 0.0329 & 0.0329 & 1.2040 & 0.1794 \\ 0.0329 & 0.0329 & 0.1794 & 1.2040 \end{pmatrix},$$

with backward linkages (column sums) $BL = (2.4506, 2.4506, 1.4625, 1.4625)$, forward linkages (row sums) $FL = (2.4638, 2.4638, 1.4493, 1.4493)$, and total output $\mathbf{1}'L\mathbf{1} = 7.8261$.

The two policies. Each raises one entry of row M1 by 0.05. Row M1 is renormalized in W^P; A changes only in that entry.

	Γ	$\Delta\Gamma$	$\mathrm{Var}(\boldsymbol{\pi})$	$\Delta\,\mathrm{Var}(\boldsymbol{\pi})$	$\mathbf{1}'L\mathbf{1}$	Δ output
Base	0.1133		0.01060		7.8261	
(a) ISI-A, $\Delta a_{M1,M2} = 0.05$	0.1121	-0.0012	0.01155	$+0.00095$	8.1381	$+0.3120$
(b) ISI-B, $\Delta a_{M1,T1} = 0.05$	0.1502	$+0.0369$	0.00032	-0.01027	8.0066	$+0.1805$

After ISI-A, $\boldsymbol{\pi} = (0.3397, 0.3739, 0.1432, 0.1432)$. After ISI-B, $\boldsymbol{\pi} = (0.2646, 0.2677, 0.2449, 0.2228)$. These match the table of Example 15.6 to the precision printed there.

Why the output ranking reverses. The change in A is the rank-one update $\Delta A' = \varepsilon\,\mathbf{e}_j\mathbf{e}'_{M1}$. By the Sherman–Morrison formula,

$$\Delta(\mathbf{1}'L\mathbf{1}) = \frac{\varepsilon\,(\mathbf{1}'L\mathbf{e}_j)(\mathbf{e}'_{M1}L\mathbf{1})}{1 - \varepsilon\,L_{M1,j}} = \frac{\varepsilon\,BL_j\,FL_{M1}}{1 - \varepsilon\,L_{M1,j}}.$$

This is exact: it gives 0.3120 and 0.1805, and the first-order terms $\varepsilon BL_j FL_{M1}$ are 0.3019 and 0.1802. So, given the buyer M1, the output gain is governed by the

backward linkage of the new supplier: $BL_{M2} = 2.4506$ against $BL_{T1} = 1.4625$ (and $L_{M1,M2} = 0.6489$ against $L_{M1,T1} = 0.0395$ in the denominator). The modern supplier is input-intensive, so a unit of demand for it triggers more rounds of production.

Who wins what. ISI-B wins on integration (Ω_1) and on power (Ω_2): it nearly equalizes $\boldsymbol{\pi}$. ISI-A wins on output (Ω_3). No single objective ranks them, as the intuition box "Three objectives, three answers" says.

(c) An input intensity at which ISI-B also wins on output. Scale the traditional recipes, rows T1 and T2 of A, by $k > 1$: the traditional sectors buy more domestic inputs per unit of output, in the same proportions. Two facts make this the clean experiment:

- *W^P does not change.* $D^{-1}A$ is invariant to scaling a row of A. So Γ, $\boldsymbol{\pi}$ and both policies' $\Delta\Gamma$ and $\Delta\,\mathrm{Var}(\boldsymbol{\pi})$ are exactly as above. Only the output objective moves. This is the independence of the A-based and W-based objectives in its sharpest form.
- *BL_{T1} rises with k*, while BL_{M2} changes little.

Bisection on k gives $k^* = 2.0409$, a traditional row sum of 0.6123 (from 0.30), with $\rho(A) = 0.6040 < 1$. At k^* both policies raise output by 0.3445. There $BL_{T1} = 2.5705 > BL_{M2} = 2.5034$: ISI-B's supplier now has the larger linkage, but ISI-A still benefits from the larger denominator term ($L_{M1,M2} = 0.6534$ against $L_{M1,T1} = 0.1342$), so the crossover is slightly past $BL_{T1} = BL_{M2}$. Checks: at $k = 1.5$ the gains are 0.3235 (A) and 0.2364 (B); at $k = 2.5$ (row sum 0.75) they are 0.3793 and 0.5399. So ISI-B wins on all three objectives whenever the traditional sectors use at least about 0.61 of domestic inputs per unit of output, about the input intensity of the modern block (0.60).

A wrong route. Raising the traditional sectors' purchases from the modern block instead ($a_{T,M} = 0.10$) does not work: it raises ISI-A's gain more (0.3948 against 0.3235), and it changes W^P ($\Gamma = 0.4535$), so it is no longer the same experiment.

Common mistake: computing the multiplier from W^P. For row-stochastic W, $(I - \gamma W)^{-1}\mathbf{1} = \mathbf{1}/(1-\gamma)$, so every policy has the same "multiplier" (Section 15.1).

Exercise 15.3 The smartphone operator

Note on the exercise. Example 15.11 is tagged illustrative, and its numbers (θ_T^* from 0.92 to 0.96 after one generation and to 1.01 in the long run) are not generated by a

stated model: `code/ch15.py` only prints them. They cannot be "reproduced". The well-posed version is to put the smartphone operator into the formation model of Definition 10.1, calibrated so that the initial steady state is $\theta_M^* = 1.09$, $\theta_T^* = 0.92$, and compute what it implies.

(a) Model. Two blocks, modern M and traditional T, three agents each ($N = 6$). Parameters follow Example 10.6, collapsed to two blocks:

- W^P: within-block weight $0.94/3$, cross-block 0.02 per pair. W^I: within $0.85/3$, cross 0.05 per pair. Both are row-stochastic.
- Returns $r_i = \alpha_i + \beta_i x_i + \gamma \sum_j W^P_{ij} x_j$ with $\beta = 0.10$ (M), 0.04 (T), $\gamma = 0.02$.
- Allocation (10.1): $\omega_i = 0.5$, $\Phi = 3$ ($\Omega/N = 1$), $\phi = 1$, bargaining weights $\pi_i = 0.20$ (M) and 0.1333 (T), summing to 1.
- Formation (10.3): $\rho = 0.85$, $\mu = 0.2$ ($\mathcal{M} = 1.33$), $\theta^\circ = 1$, $r^* = 0.1$.

The steady state is affine in the baselines (α_M, α_T); solving the 2×2 system gives $\alpha_M = 0.0531$, $\alpha_T = -0.0367$, so that $\theta_M^* = 1.0900$ and $\theta_T^* = 0.9200$ exactly. (A negative baseline is a level normalization; the traditional return $r_T = 0.0127$ is positive.) The Jacobian $J = \rho I + \mu\phi W^I C P$ has $\rho(J) = 0.8623 < 1$, so the steady state is unique and globally stable (Theorem 10.4). Base allocations: $x_M = 1.1850$, $x_T = 0.8150$; returns $r_M = 0.1948$, $r_T = 0.0127$.

The operator. Each traditional row of W^I gains $+0.05$ on each of the three modern columns (from 0.05 to 0.10), and its own-block weights fall equally, from $0.85/3$ to $0.70/3$, so rows still sum to 1. Modern rows, W^P, $\boldsymbol{\pi}$ and endowments are unchanged.

(b) Dynamics for five generations (from the old steady state, under the new W^I):

Generation	0	1	2	3	4	5
θ_T	0.9200	0.9255	0.9301	0.9341	0.9374	0.9403
θ_M	1.0900	1.0900	1.0900	1.0899	1.0898	1.0896
Gap $\theta_M - \theta_T$	0.1700	0.1645	0.1598	0.1558	0.1523	0.1493
x_T	0.8150	0.8177	0.8201	0.8221	0.8238	0.8253

Steady state. $\theta_T^* = 0.9566$ ($\Delta\theta_T^* = +0.0366$) and $\theta_M^* = 1.0876$ (-0.0024). The preference gap falls from 0.1700 to 0.1311, closing by 22.9%. After five generations 55.5% of the long-run movement of θ_T has occurred. The half-life of deviations is $\ln 2/(-\ln \rho(J)) = 4.58$ generations, with $\rho(J) = 0.8596$ after the operator. In the

steady state $\Delta x_T = +0.0195$ and $\Delta x_M = -0.0195$. They sum to zero, as they must: the preference term of (10.1) sums to zero, and Ω is unchanged.

Comparison with the example. The calibrated model closes the gap by 23% in the long run, not 53%, and by 3% after one generation, not 24%. With $\rho = 0.85$, no change in W^I alone can move preferences by a quarter of the gap in one generation, because the per-generation response is bounded by μ times the change in the signal. The example's numbers are optimistic for its own parameter.

(c) Two lessons from the model.

1. *The gain is distributional.* By Proposition 10.2(3), only relative preferences matter, and $\sum_i \Delta x_i = 0$. The traditional block gains 0.0195 per agent exactly what the modern block loses. The aggregate return even falls slightly, from 0.62250 to 0.61900, because resources move toward the block with the lower private return β. The example's "investment increases, returns increase" is not a property of this model. It needs returns that depend on own preferences through investment (as in Exercise 18.4), not only through the bargaining position.

2. *Why Ψ is not defined.* By Definition 9.13, the enforcement effect is $E_i = y_i(\boldsymbol{\theta}^0, W + \Delta W) - y_i(\boldsymbol{\theta}^0, W)$. Allocations and returns depend on W^I only through $\boldsymbol{\theta}$, so at fixed $\boldsymbol{\theta}^0$ the operator changes nothing: the script finds $\max|E| = 0$ for both x and r. The ratio F/E has a zero denominator; this is the case of Remark 9.14(5). Report the formation effect itself: $F_T = +0.0195$ in allocation, $+0.0366$ in preference. A cost-benefit analysis at fixed preferences records zero benefit, which is correct for the enforcement channel and incomplete for the policy.

Common mistake: reading $\Psi = \infty$ as "infinitely effective". Ψ is undefined, and the size of the effect is F, which here is modest.

Exercise 15.4 Sequencing

Note on the exercise. As posed, the exercise is underdetermined: it does not say how big each operator is, what "education" does to W^P, or how information affects later operators. And under the chapter's own definitions the information operator leaves W^P, and hence $\Gamma(W^P)$, unchanged (Definition 15.10). Without a complementarity between operators the order cannot help sequence B. The well-posed version below states a model with an explicit complementarity parameter η and reports how the answer depends on it.

(a–b) Model. Six sectors in three blocks: gas (G_1, G_2), urban (U_1, U_2), rural (R_1, R_2). Raw intensities K are 1 inside each block (diagonal included) and 0.04 (G–U), 0.04 (U–R) and 0.01 (G–R) across blocks; $W^P = D^{-1}K$, with $\Gamma_0 = 0.0571$. The gas block is the weakly coupled enclave. Operators act once, at the start of periods 1, 4 and 7, and edges persist:

- *Information*: the rural cross-block weight in W^I rises from 0.10 to 0.30. Rural aspirations then follow the formation equation $\theta_{R,t+1} = \rho\theta_{R,t} + (1-\rho)\theta^\circ + \mu(s_{R,t} - r^*)$, with signal $s_R = (1-w)r_R + w\,r_U$, fixed returns $r_U = 0.20$, $r_R = 0.08$, and $\rho = 0.85$, $\mu = 0.2$, $\theta^\circ = 1$, $r^* = 0.1$; $\theta_{R,0} = 0.92$, $\theta_U = 1.09$. W^P does not change.
- *Education*: adds 0.05 to the U–R and G–R intensities, and raises rural capability c from 0.5 to 1.
- *ISI-B*: the enclave buys domestic inputs: adds $0.30\,u$ to the G–U and G–R intensities, with uptake $u = (c \cdot \min\{1, \theta_R/\theta_U\})^\eta$. $\eta = 0$ means no complementarity; $\eta > 0$ means ISI succeeds only where rural firms are capable and willing to supply.

(c) Results (Γ_t for $t = 0, \ldots, 10$):

- $\eta = 0$. Sequence A: 0.057, then 0.296 from $t = 1$ (after ISI), 0.368 from $t = 4$. Sequence B: 0.057 to $t = 3$, 0.132 from $t = 4$ (education), 0.368 from $t = 7$. A reaches $\Gamma = 0.30$ at $t = 4$, B at $t = 7$. Both end at the same matrix (additive operators commute) and the same $\Gamma = 0.368$.
- $\eta = 1$. ISI applied first has uptake $0.92/1.09 \times 0.5 = 0.4220$. A reaches only 0.192 after ISI and 0.280 after education, never 0.30. Under B, information raises θ_R to 0.9831 by $t = 7$ (uptake 0.9019), and $\Gamma_7 = 0.354$: B reaches the target at $t = 7$, A never.
- $\eta = 2$. A peaks at 0.231; B reaches 0.342 at $t = 7$.
- On a grid of step 0.25, the smallest η at which B reaches 0.30 no later than A is $\eta = 0.75$.

Without information, θ_R would rise only to 0.9757 by $t = 10$; with it, to 1.0014.

Model answer. The ordering of Conjecture 15.14 is right if and only if the later, "hard" operators are complementary to the earlier, "soft" ones: here, if ISI's uptake rises steeply enough with capability and aspirations ($\eta \geq 0.75$). Without such complementarity the "hard" operator acts on W^P immediately and wins on

Γ, since information never touches $\Gamma(W^P)$. So the conjecture is a claim about the size of a complementarity, which is an empirical parameter (the role Ψ plays in the chapter's keyinsight box). Two side observations: the binding constraint on Γ is the weakest cut (the gas enclave), so operators must target it; and education alone moves Γ only to 0.132, because it does not touch the G–U link.

Common mistakes: plotting $\Gamma(W^I)$ and $\Gamma(W^P)$ on the same axis, and treating the conjecture's ordering as a result rather than a hypothesis about η.

Exercise 15.5 Bolivia in the product space

Data status. The country×product RCA data and product-space proximities needed here were **not fetched** (the Growth Lab file is 451 MB; `data/README.md`, §9 and row 34 of its status table). What *is* in `data/processed` (retrieved 2026-11-05) is used below: Bolivia's ECI and the PCI of selected products (The Growth Lab at Harvard University, 2026).

(a) Top 10 exports and position.

1. Download the Growth Lab country×product×year file (HS92, 4 digits; Harvard Dataverse, doi:10.7910/DVN/T4CHWJ) or the equivalent OEC table.
2. Keep Bolivia, the latest year without a revision flag (e.g. 2019, to match the book), and sort by export value. Report each product's share of the basket.
3. "Core vs. periphery": compute each product's eigenvector or degree centrality in the product-space proximity network ϕ (Definition 16.6), or use the Atlas's published product-space coordinates and cluster labels. Classify as core if above the median of the centrality of all products.

Expected result. Natural gas, gold, zinc and silver ores, soy products, quinoa, Brazil nuts: almost all peripheral. Consistent with this, the PCIs available here are low: natural gas (HS 2711) -2.23, zinc ores (2608) -1.67, soya beans (1201) -1.81 in 2019, and Bolivia's ECI was -1.30 (Example 15.16).

(b) Relatedness density. Build $M_{cp} = \mathbf{1}[R_{cp} \geq 1]$ (Definition 16.5) for all countries; compute $\phi_{pp'}$ by (16.3); then $\omega_{\text{BOL},p} = \sum_{p'} M_{\text{BOL},p'}\phi_{pp'} / \sum_{p'} \phi_{pp'}$ (Definition 16.14). Target codes: lithium-ion batteries are inside HS 8507 (electric accumulators; PCI 0.549 in 2019); food-processing machinery is HS 8438; pharmaceutical intermediates are in HS 2933–2942. Use a 3-year average of exports for R_{cp} to reduce noise, and report the percentile of each ω among all products Bolivia does not export. *Expected:* low densities, well below the median for the country; the book's $\omega \approx 0.08$ is a placeholder.

(c) Edges needed.

- *Lithium batteries.* ΔW^P: links from lithium carbonate (HS 2836, PCI -0.573) to cathode materials, cell assembly and electrical equipment; chemical-engineering inputs and imported capital goods. ΔW^I: engineers trained in electrochemistry; links to foreign battery producers (joint ventures, licensing).

- *Food-processing machinery.* ΔW^P: metalworking, machining and repair shops supplying agro-industry (quinoa, soy, Brazil nuts), which are already domestic buyers. ΔW^I: technical schools and supplier-development programmes that link agro-industry buyers to local metalworkers.

- *Pharmaceutical intermediates.* ΔW^P: organic chemistry and quality-control laboratories. ΔW^I: regulatory know-how (GMP certification), university chemistry, links with regional pharmaceutical firms.

(d) Sequencing cost, order of magnitude. Count the missing capabilities: the number of products p' with high $\phi_{pp'}$ that Bolivia lacks, weighted by ϕ. Then multiply by a unit cost from comparable programmes. The ranking that follows from the method: food-processing machinery is the cheapest, because it sits nearest Bolivia's existing agro-food basket; lithium batteries are intermediate (the raw material is present, the chemistry and assembly are not); pharmaceutical intermediates are the most expensive. No numbers are given, because they would have to be invented.

Exercise 15.6 Hirschman vs. spectral gap

Data. INE supply-use tables 2019, base 2017, converted to a domestic industry×industry table at basic prices and aggregated to the book's 12 sectors (`data/processed/bolivia_Z/Z_2017_2019_book12.csv`, rows = sellers), with gross output from `bolivia_io_linkages.csv` (Instituto Nacional de Estadística (INE), Bolivia, 2026); retrieved 2026-11-05.

(a) Backward linkages from A. With the Chapter 1 convention, $a_{ss'} = Z_{s's}/X_s$ (Appendix 17A). Row sums of A, the domestic intermediate input per unit of output, range from 0 (domestic services) and 0.162 (public administration) to 0.558 (manufacturing); hydrocarbons have 0.225. With $L = (I - A')^{-1}$:

Sector	BL	FL	Sector	BL	FL
Hydrocarbons	1.340	1.195	Transport	1.542	1.522
Mining	1.364	1.296	Financial services	1.433	1.570
Public administration	1.245	1.023	Agriculture	1.549	1.704
Manufacturing	1.855	2.207	Other services	1.356	2.018
Construction	1.574	1.128	Personal services	1.661	1.097
Commerce	1.416	1.576	Domestic services	1.000	1.000

Checks: all linkages are at least 1; mean BL = mean FL = 1.4447; the values agree with `bolivia_io_linkages.csv` to 10^{-6} and with Example 17.2.

The same computation on W^P is uninformative. On the 11 sectors with intermediate flows (domestic services dropped), $W^P = D^{-1}A$ is row-stochastic, so $(I - W^{P\prime})$ is singular and the Leontief inverse does not exist. With attenuation $\gamma < 1$:

- the column sums of $(I - \gamma W^{P\prime})^{-1}$ are $\left((I - \gamma W^P)^{-1}\mathbf{1}\right)' = \mathbf{1}'/(1-\gamma)$: all equal 2.0000 at $\gamma = 0.5$ and 10.0000 at $\gamma = 0.9$;
- the row sums of $(I - \gamma W^P)^{-1}$ are the same constants.

Every sector gets the same "linkage". What $(I - \gamma W^P)^{-1}$ does carry is the Acemoglu et al. influence vector, $\frac{1-\gamma}{N}\mathbf{1}'(I - \gamma W^P)^{-1}$, which tends to $\boldsymbol{\pi}$ as $\gamma \to 1$ (at $\gamma = 0.99$ the largest difference is 0.0027). That is a power measure, not a multiplier.

(b) Gradient of $\Gamma(W^P)$. $\lambda_2 = 0.5326$ (real, simple), $|\lambda_3| = 0.3127$, $\Gamma = 0.4674$. Since $\lambda_2 > 0$, $G_{ij} = -u_{2,i}v_{2,j}$. Scaled to a maximum modulus of 1:

- $\mathbf{v}_2$ is 1 on financial services and -0.48 to -0.67 on personal services, manufacturing and construction;
- $\mathbf{u}_2$ is 1 on financial services, 0.39 on other services, and -0.53 on agriculture.

The slow mode is "financial services against the rest". Financial services buy 56.7% of their inputs from themselves (the largest diagonal entry of W^P; other services 0.511, agriculture 0.331). The full 11×11 matrix G is printed by the script; a finite difference confirms one entry to 10^{-4}. Its largest entries lie in the financial-services row ($G_{\text{fin,constr}} = 0.455$, $G_{\text{fin,fin}} = -0.677$) and column ($G_{\text{agr,fin}} = 0.361$).

Best feasible moves (donor entry $W_{i\ell} > 0.01$):

1. Financial services shift purchases from themselves to construction: rate 1.132. Moving 0.05 of weight raises Γ by 0.0595 exactly (first order 0.0566).

2. The same row, to manufacturing (1.021), personal services (1.000) or hydrocarbons (0.985).

3. Excluding self-purchases: agriculture shifting purchases from construction to financial services (0.604, but only 0.019 of weight is available to move) or from manufacturing to financial services (0.544, with 0.181 available).

(c) Agreement and disagreement. Hirschman ranks *sectors* by BL: manufacturing (1.855), personal services (1.661), construction (1.574). The spectral criterion ranks *edges*. To compare, score each sector j as a supplier by the best rate of moving weight towards it. The ranks agree moderately: Spearman correlation 0.636. Manufacturing, construction and personal services are near the top on both lists. They disagree on two points:

- On the buyer side, the spectral criterion singles out financial services, ranked only 6th by BL (Spearman 0.209 between BL and the buyer score).

- The spectral criterion's best edge is a change in an existing sector's *input mix*, cutting its self-reliance. It is not an investment in a high-linkage sector.

(d) When Hirschman's criterion misleads. It maximizes output per unit of final demand, which is the right target for Ω_3. It misleads, for integration or for power, when the high-BL sector sits inside an already dense core, so that new demand deepens the core without bridging to the periphery (ISI-A in Exercise 15.2). Three cautions specific to these data:

- $\Gamma = 0.467$ and λ_2 is not close to 1, so Bolivia's 12-sector W^P has no weakly coupled block (Example 17.2). The spectral criterion's gains are therefore modest, and its "slow mode" is an artefact of one large diagonal entry, which depends on the aggregation.

- Neither criterion sees the export enclave. Hydrocarbons export 70% of output, and their linkage runs through the fiscal channel, as Hirschman himself stressed.

- Both criteria are first-order: large interventions change the gradient.

Common mistake: ranking sectors by the row sums of $(I - \gamma W^P)^{-1}$, which are identical.

Exercise 15.7 Political cost

Set-up. Agents: Gas (1), Government (2), Urban (3), Rural (4), with power $\boldsymbol{\pi} = (0.35, 0.30, 0.20, 0.15)$, the vector of Example 14.9, so $\pi_1 + \pi_2 = 0.65$. A stylized W with exactly this Perron vector is the Metropolis chain with uniform proposals, $W_{ij} = \frac{1}{4}\min\{1, \pi_j/\pi_i\}$ for $i \neq j$, with the remainder on the diagonal:

$$W = \begin{pmatrix} 0.5357 & 0.2143 & 0.1429 & 0.1071 \\ 0.2500 & 0.4583 & 0.1667 & 0.1250 \\ 0.2500 & 0.2500 & 0.3125 & 0.1875 \\ 0.2500 & 0.2500 & 0.2500 & 0.2500 \end{pmatrix}.$$

It is reversible ($\pi_i W_{ij} = \pi_j W_{ji}$), so $\boldsymbol{\pi}'W = \boldsymbol{\pi}'$. Its eigenvalue moduli are $1, 0.2857, 0.2083, 0.0625$, so $\Gamma = 0.7143$; $\mathrm{Var}(\boldsymbol{\pi}) = 0.006250$. Its smallest entry is 0.107, so every perturbation with $\|E\|_F \leq 0.1$ keeps $W + E \geq 0$. The budget is Frobenius, as in Definition 14.2.

(a) Integration: zero political cost. Take $E = \mathbf{1}\boldsymbol{\pi}' - W$, with $\|E\|_F = 0.3678$. By Proposition 15.2, along $W + tE = (1-t)W + t\mathbf{1}\boldsymbol{\pi}'$ the Perron vector stays $\boldsymbol{\pi}$, so $\Delta\pi_i = 0$ for every i and the constraint $\Delta\pi_i \geq 0$ holds for any coalition. Meanwhile $\Gamma = 1 - (1-t)\cdot 0.2857$. With budget B, $t = \min\{1, B/0.3678\}$:

- $B = 0.1$: $t = 0.2719$ and $\Gamma = 0.7920$;
- $B \geq 0.3678$: $t = 1$ and $\Gamma = 1$, the maximum possible value of Γ, so the constrained and unconstrained optima coincide and $C_{\text{political}} = 0$.

The script confirms that $\boldsymbol{\pi}$ is unchanged at both points. (For $B < 0.3678$ the proposition gives only a lower bound on the constrained optimum.)

(b) Power: the exact constrained problem. A reduction to a three-dimensional problem makes this exact.

Reachability lemma. A target $\tilde{\boldsymbol{\pi}}$ in the open simplex is the Perron vector of $W + E$ for some E with $E\mathbf{1} = 0$ and $\|E\|_F \leq B$ if and only if $\|b\|_2 \leq B\|\tilde{\boldsymbol{\pi}}\|_2$, where $b := (I - W')\tilde{\boldsymbol{\pi}}$; this ignores nonnegativity, which we check afterwards.

Proof. Necessity: $\tilde{\boldsymbol{\pi}}'(W + E) = \tilde{\boldsymbol{\pi}}'$ means $E'\tilde{\boldsymbol{\pi}} = b$, so $\|b\| \leq \|E'\|_2\|\tilde{\boldsymbol{\pi}}\| \leq \|E\|_F\|\tilde{\boldsymbol{\pi}}\|$. Sufficiency: let $E^* = \tilde{\boldsymbol{\pi}}b'/\|\tilde{\boldsymbol{\pi}}\|^2$. Then $E^*\mathbf{1} = \tilde{\boldsymbol{\pi}}(b'\mathbf{1})/\|\tilde{\boldsymbol{\pi}}\|^2 = 0$, because $\mathbf{1}'b = \mathbf{1}'\tilde{\boldsymbol{\pi}} - (W\mathbf{1})'\tilde{\boldsymbol{\pi}} = 0$. Also $E^{*\prime}\tilde{\boldsymbol{\pi}} = b$, and $\|E^*\|_F = \|b\|/\|\tilde{\boldsymbol{\pi}}\|$. When $W + E^*$ is positive it is irreducible, so $\tilde{\boldsymbol{\pi}}$ is *its* Perron vector. E^* is also the minimum-norm solution: any other solution differs from it by some F with $F\mathbf{1} = 0$ and $F'\tilde{\boldsymbol{\pi}} = 0$, and $\langle E^*, F\rangle_F = b'F'\tilde{\boldsymbol{\pi}}/\|\tilde{\boldsymbol{\pi}}\|^2 = 0$. □

So the problem is: minimize $\mathrm{Var}(\tilde{\boldsymbol{\pi}})$ over the simplex, subject to $\|(I - W')\tilde{\boldsymbol{\pi}}\| \leq 0.1\,\|\tilde{\boldsymbol{\pi}}\|$ and, in the constrained case, $\tilde{\pi}_1 \geq 0.35$, $\tilde{\pi}_2 \geq 0.30$. The script solves it on a grid of step 0.001 and checks that $W + E^* \geq 0$ and that $\boldsymbol{\pi}(W + E^*) = \tilde{\boldsymbol{\pi}}$.

	$\tilde{\boldsymbol{\pi}}$	min Var	Ω_2^*	$\|E^*\|_F$
Unconstrained	$(0.304, 0.283, 0.224, 0.189)$	0.002101	0.004149	0.0999
Constrained, $S^* = \{\text{Gas, Gov}\}$	$(0.350, 0.300, 0.175, 0.175)$	0.005938	0.000312	0.0634

Political cost: $C_{\text{political}} = 0.004149 - 0.000312 = 0.003837$, which is 92.5% of the unconstrained gain. Under the constraint the only redistribution left is between the two outsiders, and the best it can do is equalize them at 0.175. That uses only 0.0634 of the budget: the constraint, not the budget, binds. Even the unconstrained planner cannot reach uniform power, which would need $\|E\|_F \geq 0.2434$. As a cross-check, the perturbation lemma's $d\,\mathrm{Var}/dt = \frac{2}{N}\boldsymbol{\pi}' EZ(\boldsymbol{\pi} - \mathbf{1}/N)$ along E^* (-0.005398) matches the finite difference. A linearized solution overshoots in this problem: the first-order objective is linear in E, so it always spends the full budget.

Which coalition change reduces the cost? The cost depends on whether the protected agents are the ones a variance-reducing reform must take power from:

Coalition	$\{G\}$	$\{G, Gov\}$	$\{G, Gov, U\}$	$\{Gov, U\}$	$\{U, R\}$
Political cost	0.001734	0.003837	0.004150	0.000275	0

- Dropping Gas from the coalition, so that Government governs with Urban, cuts the cost by 93%.
- A coalition of the two least powerful agents faces no cost at all: its members want their own power to rise, which is what lowers $\mathrm{Var}(\boldsymbol{\pi})$.
- Adding Urban to $\{G, Gov\}$ makes the cost total: with three members protected, no redistribution is possible.

This is the content of the chapter's conjecture that the cost is large when power is concentrated in the coalition: the cost depends on *who* is protected, not on how many.

Common mistakes: stating a political cost for Ω_1 (there is none once $B \geq \|\mathbf{1}\boldsymbol{\pi}' - W\|$), and reading the cost as a Pareto loss. It is the forgone reduction in a distributional objective.

Chapter 16

The Global Weight Matrix

Every number below is printed by `python3 code/solutions/sol16.py`. The data are those of `data/processed` and the Growth Lab ranking file tracked in `data/raw/atlas`, all retrieved on 2026-11-05 (`data/README.md`, §§5, 7, 8). Most exercises in this chapter are data exercises. Where the data were not fetched, the manual says so and gives the method only.

Exercise 16.1 Build the global W^P

Data. The exercise names BACI (CEPII), which was **not fetched**. The same construction is carried out on the Growth Lab's HS92 bilateral goods-trade totals (The Growth Lab at Harvard University, 2026) (`atlas_bilateral_selected_years.csv.gz`), which contain 2018. BACI reconciles mirror flows differently, so its values will differ somewhat; the steps are identical.

(a) Method.

1. Keep the 2018 rows (exporter, importer, value in current US$).
2. Pivot to $K_{cc'}$ = imports of c from c', and set the diagonal to 0 (imports only, Definition 16.1).
3. Drop, iteratively, countries with zero imports or zero exports inside the set, so that every row and column of K is nonzero. This leaves $N = 230$ economies.
4. For the 50-node version, keep the 49 largest traders (imports plus exports), which account for 91.4% of world trade, and merge the other 181 economies into a Rest-of-World node. Trade among ROW members is genuine cross-border trade, so it stays on ROW's diagonal ($W_{\mathrm{ROW,ROW}} = 0.1237$); the alternative with that entry set to zero is also reported.

(b) $W^P = D^{-1}K$. Row sums equal 1 to machine precision, the diagonal is zero and all entries are nonnegative.

(c) Column sums (2018). They average exactly 1, because the N rows each sum to 1. Largest: China 28.75, the United States 23.99, Germany 10.98, France 8.86, Italy 7.56. Smallest: three tiny territories (Norfolk Island, Saint Pierre and Miquelon, Heard and McDonald Islands), of order 3×10^{-5}. Bolivia: 0.0961 (2019: 0.0997, against 0.10 in Remark 16.2). W^P is therefore far from doubly stochastic, so $\boldsymbol{\pi}$ is far from uniform (`docs/conventions.md`, §2).

(d) Perron vector. Power iteration $\boldsymbol{\pi}'_{k+1} = \boldsymbol{\pi}'_k W$ from the uniform vector converges in 51 steps to tolerance 10^{-14} (Appendix 6A), and agrees with the eigendecomposition to 10^{-10}. It converges because W is irreducible and aperiodic, and $|\lambda_2| = 0.55$ sets the rate. The top ten in 2018 are China 0.1251, the US 0.0883, Germany 0.0848, Japan 0.0436, Korea 0.0348, France 0.0324, the Netherlands 0.0300, Italy 0.0299, the UK 0.0269 and Russia 0.0253; together they hold 52.1%. $\pi_{\text{BOL}} = 0.00051$.

Correlation with GDP per capita. GDP per capita (WDI, PPP, current international $) is in `data/processed` only for the 13 comparator economies (ARG, BOL, BRA, CHL, CHN, COL, DEU, JPN, KOR, MEX, PER, URY, USA). On those, $\text{corr}(\pi_c, \text{GDP pc}) = 0.530$ and $\text{corr}(\log \pi_c, \log \text{GDP pc}) = 0.622$. The correlation of $\log \pi_c$ with the log of the *level* of GDP is 0.970. Power in the trade matrix tracks economic size, as gravity predicts, much more than income per head. The sample is small and hand-picked, so these are descriptions, not estimates; with all 230 economies the method is the same.

(e) Γ, N_{eff} and aggregation.

	N	Γ	$\lvert\lambda_2\rvert$	N_{eff}	N_{eff}/N
Full, 2018	230	0.4479	0.5521	23.00	0.100
Full, 2019 (check)	230	0.4522	0.5478	22.58	0.098
49 + ROW, 2018	50	0.4844		20.29	0.406
49 + ROW, ROW diagonal = 0	50	0.4844		20.47	0.409

The 2019 row reproduces Remark 16.2 ($\Gamma = 0.452$, $N_{\text{eff}} = 22.6$), which validates the pipeline.

Aggregation raises Γ by 0.037. Merging nodes averages rows, which removes some slow modes (the small-island and regional sub-networks). It barely moves N_{eff} (23.0 $\to$ 20.3): the ROW node holds $\pi_{\text{ROW}} = 0.0765$, but power was already concentrated in the 49 economies kept. N_{eff}/N, by contrast, quadruples. This is

the convention's warning: N_{eff} is not comparable across node sets without normalizing by N, and the normalized measure depends on the aggregation.

Gravity benchmark. With world export shares $\mathbf{s}$ (within the 230), $W = \mathbf{1}\mathbf{s}'$ has $\Gamma = 1$ and $\boldsymbol{\pi} = \mathbf{s}$. Measured, $\text{corr}(\boldsymbol{\pi}, \mathbf{s}) = 0.9969$ and $\frac{1}{2}\|\boldsymbol{\pi} - \mathbf{s}\|_1 = 0.0346$ (China 0.1251 against $s = 0.1389$; the US 0.0883 against 0.0901; Bolivia 0.00051 against 0.00047). Compare $\frac{1}{2}\|\boldsymbol{\pi} - \mathbf{1}/N\|_1 = 0.7165$. So $\boldsymbol{\pi}$ is almost exactly the gravity benchmark, while $\Gamma = 0.448$ is far below its benchmark of 1. Power is a matter of size; the gap measures the frictions (distance, regional blocs) that keep rows from being identical.

Common mistake: concluding from the small Γ that power is concentrated. The two coordinates are independent (Proposition 6.16).

Exercise 16.2 Compute the ECI

Data. Growth Lab, "Growth Projections and Complexity Rankings", v12, HS92 ECI (The Growth Lab at Harvard University, 2026), file `data/raw/atlas/growth_proj_eci_rankings.csv` (tracked in the repository; retrieved 2026-11-05). ECI is re-estimated with each release, so the figures belong to this release.

(a) **2018** (145 ranked economies).

- *Top 10:* Japan 2.044, Korea 1.797, Switzerland 1.752, Taiwan 1.742, Singapore 1.631, Germany 1.543, Czechia 1.484, Austria 1.400, Sweden 1.396, Finland 1.393.
- *Bottom 10*, from 136th to 145th: Iraq −1.613, Guinea −1.708, Libya −1.762, Sudan −1.911, DR Congo −2.067, Yemen −2.102, Chad −2.118, Equatorial Guinea −2.181, Nigeria −2.198, Papua New Guinea −2.398.

Oil and mineral exporters dominate the bottom.

(b) ECI and GDP per capita. On the 13 comparators: $\text{corr}(\text{ECI}, \text{GDP pc}) = 0.699$ and $\text{corr}(\text{ECI}, \log \text{GDP pc}) = 0.753$, against $\text{corr}(\log \pi_c, \log \text{GDP pc}) = 0.622$ for the trade Perron vector on the same countries. So ECI is the better correlate of income per head. That is expected: π_c measures size (part (d) of Exercise 16.1), while ECI is a z-score of the second eigenvector of $\tilde{M}$, a measure of the composition of the basket that is independent of scale (Proposition 16.8). Over all 145 ranked economies, $\text{corr}(\text{ECI}, \log \pi_c) = 0.665$: large traders tend to be complex, but the two objects differ.

(c) ECI and Gini. The WDI Gini (latest year in 2014–2019) is available for the same 13 countries; $\text{corr}(\text{ECI}_{2018}, \text{Gini}) = -0.474$. This is the negative associa-

tion of Empirical Result 16.11, but 13 hand-picked countries cannot say whether the low-ECI, low-Gini quadrant is sparse. Bolivia (-1.46, Gini 41.6 in 2019) and Peru (-0.68 in 2019, 41.5) sit in it, with Ginis below Brazil's (53.5) and Colombia's (52.5), which have higher ECIs. *Method for the full answer:* merge the ranking with WDI `SI.POV.GINI` for all countries, take the closest year within ± 3 years, plot, and count the countries in each quadrant defined by the medians. Do not read the empty quadrant, if there is one, as Fajnzylber's casillero vacío: his box is in growth×equity space.

(d) Bolivia and Korea. 2018: Bolivia -1.464 (rank 131), Korea 1.797 (rank 2); the gap is 3.26, i.e. more than three cross-country standard deviations, since ECI is a z-score. (2019: -1.30 and 1.81, as in Chapter 17.)

Exercise 16.3 Relatedness density

Data status. Country×product RCA data and product-space proximities were **not fetched** (`data/README.md`, §9). The ω values of Example 16.16 are placeholders. Method and expected qualitative result only.

(a) Method.

1. Download the Growth Lab HS92 country×product×year file (4 digits) and average exports over 2017–2019.

2. Compute R_{cp} (Definition 16.5) for all countries and set $M_{cp} = \mathbf{1}[R_{cp} \geq 1]$.

3. Compute $\phi_{pp'} = \sum_c M_{cp} M_{cp'} / \max(M_p, M_{p'})$ (Definition 16.6), or use the Atlas's published proximities.

4. Compute $\omega_{cp} = \sum_{p'} M_{cp'} \phi_{pp'} / \sum_{p'} \phi_{pp'}$ (Definition 16.14) for Bolivia and the targets: HS 8507 (accumulators, including lithium-ion; PCI 0.549 in 2019), the processed-food headings (HS chapters 19–21) and pharmaceutical intermediates (HS 2933–2942).

5. Report each ω also as a percentile within the country's distribution over the products it does not yet export, since raw densities are not comparable across product classifications.

(b) Repeat for Chile, Peru and Colombia. *Expected result:* all four Andean economies have low density to batteries and pharmaceuticals, all with ECI near or below zero (2019 ECI: Chile -0.06, Peru -0.68, Colombia 0.14, Bolivia

-1.30). Bolivia's and Peru's mining-heavy baskets should give the lowest densities; Colombia's more diversified manufacturing should give the highest, and processed food should be the most accessible target for all four.

(c) The country with the most diversified basket in proximity-dense products, plausibly Colombia among these, has the highest average density to core products. The reason is mechanical: ω_{cp} is a ϕ-weighted share of neighbours already exported, and core products are each other's neighbours. Diversity in the core raises density to the rest of the core.

(d) Sequence to reach $\omega > 0.3$ for batteries. The density rises only by adding products p' with high $\phi_{p',8507}$: electrical machinery, chemicals and metal products. Following Conjecture 15.14:

1. $\mathcal{L}_{\text{info}}$: technical exchanges and FDI in assembly, which create W^I links to battery producers.
2. $\mathcal{L}_{\text{inst}}$: stable contracts for lithium, lowering the destruction rate δ_1.
3. $\mathcal{L}_{\text{HC}}$: chemistry and electrical-engineering training.
4. $\mathcal{L}_{\text{infra}}$: power and transport links for the processing zones.
5. $\mathcal{L}_{\text{ISI}}$ of type B: domestic supply of lithium compounds to cathode and cell plants.

Use the computed ϕ to list the smallest set of neighbouring products whose acquisition lifts ω above 0.3 (a greedy knapsack on $\phi_{p',8507}$). The length of that list is the honest measure of distance; it is likely to be long.

Exercise 16.4 Prebisch–Singer

(a) Terms of trade: not fetched. The WDI net barter terms-of-trade index (`TT.PRI.MRCH.XD.WD`) is not in `data/processed`, and CEPAL's series was not downloaded. *Method:* download the series for 1960–2023; rebase to 2000 = 100; plot it in logs; date booms as runs of years above a centred 10-year moving average. *Expected:* a long-run decline punctuated by the 1970s and 2003–2014 commodity booms. Whether the trend is significant depends on the sample and on structural breaks, which is why the literature remains divided (Grilli and Yang, 1988; Harvey et al., 2010; Arezki et al., 2014).

(b) ECI during the boom (data available). Bolivia's HS92 ECI was -0.856 in 2003, -1.207 in 2006, -1.345 in 2009 and -1.342 in 2014: a fall of 0.486 over 2003–2014

(2012 shows a transitory -0.916). Before the boom it was flat (-0.004 over 1995–2003); after it, it recovered slightly ($+0.047$ over 2014–2019). Fuels rose from 12.2% of merchandise exports (2000) to 48.8% (2006) and 51.7% (2014), and fell to 32.1% (2019). The boom concentrated the basket in a low-PCI product (natural gas, PCI -2.23), and the ECI fell. Booms do not raise complexity; they lower it by changing the weights of the basket.

(c) Price vs. quantity in π_{BOL}.

- *Available results.* π_{BOL} was 0.00047 in 2006 (boom start) and 0.00071 in 2014 (peak); it fell to 0.00051 in 2018 and was 0.00055 in 2023. Bolivia's world export share moved in step ($0.00033 \to 0.00071 \to 0.00047$).
- *A decomposition by partners* is possible with values alone. Bolivia's column sum of W rose from 0.0713 to 0.1115 between 2006 and 2014. The intensive margin, larger shares in partners present in both years, accounts for $+0.0451$; the extensive margin, new minus lost partners (28 new, 17 lost), accounts for -0.0050. The largest single increase is Bolivia's share of Argentina's imports ($+0.0322$), which is gas. So the boom raised weights on existing edges; it did not add partners, as Example 16.18 says.
- *Price vs. quantity: not computable here.* It needs quantities or unit values (BACI reports quantities by HS6 product; not fetched). *Method:* for a boom year t_1 (2014) and a bust year t_0 (2018 or 2019), deflate each flow by a product price index with base t_0, i.e. value flows at t_0 unit values; rebuild W and $\boldsymbol{\pi}$. The price effect is $\pi_{\text{BOL}}(t_1) - \pi^{\text{const}}_{\text{BOL}}(t_1)$; the remainder is quantities and partners.
- *Expected:* most of the 2006–2014 rise is price, given that the extensive margin was negative and gas prices moved strongly over the period.

Exercise 16.5 Global W^I

Data status. Citation data were **not fetched**. Scopus requires a licence; OpenAlex is an open alternative with the same structure. Method only.

(a) Construction.

1. Take all journal articles in economics (or all fields) published in one window, e.g. 2015–2019.
2. Assign each paper to countries by author affiliation, using fractional counting: a paper with k affiliations gives weight $1/k$ to each.

3. For each citing paper in country c and each cited reference in country c', add the product of the fractional weights to $K_{cc'}$.

4. Keep the 19 largest producers and merge the rest into one node (20 nodes).

5. Decide the diagonal. Self-citation of a country's own literature is real information flow, so the main specification keeps it; report also the zero-diagonal version.

6. $W^I = D^{-1}K$.

(b–c) Compute $\boldsymbol{\pi}^I$ by power iteration and $N_{\text{eff}} = 1/\sum(\pi_c^I)^2$ (Definition 6.15), with bootstrap intervals obtained by resampling papers. *Expected:* the US first by a wide margin, then the UK, Germany, China and France. With a 20-node set, N_{eff} should lie between the number of dominant producers (about 3) and 20; report it as $N_{\text{eff}}/20$. It is *not* comparable with the N_{eff} of the trade matrix, which has a different node set.

(d) Correlate $\boldsymbol{\pi}^I$ with $\boldsymbol{\pi}^P$ from Exercise 16.1 restricted to the same 20 nodes (renormalized to sum to 1); Spearman rank correlation is the more robust choice. A high correlation is the alignment of Remark 16.19. Because both layers are on the same node set, a composite $W = \alpha_P W^P + \alpha_I W^I$ is defined (`docs/conventions.md`, §1). Alignment means its Perron vector is close to both, so concentration in one layer is not offset by the other. Integration of the composite is a separate question: Γ of the composite need not be small (a hub structure mixes fast).

Exercise 16.6 Integration, power and inequality

Data. `data/processed` holds OECD ICIO 2023 domestic blocks for 22 economies, and INE's table for Bolivia (OECD, 2023; Instituto Nacional de Estadística (INE), Bolivia, 2026); 2019, 12 ISIC-based sectors (`book12_isic`), domestic flows, input shares, diagonal kept. That gives 23 countries, below the 30–50 requested. The extra countries need the full ICIO (fetchable, not done; `data/README.md`, §9).

(a) Domestic Γ and N_{eff} (2019, $N = 11$ sectors with intermediate flows):

	Γ	N_{eff}		Γ	N_{eff}		Γ	N_{eff}
VNM	0.333	3.11	ESP	0.494	4.71	DEU	0.544	3.89
MYS	0.381	5.18	BRA	0.496	4.35	IND	0.548	5.76
PER	0.407	5.15	JPN	0.510	3.77	POL	0.552	5.44
COL	0.460	4.78	KOR	0.511	3.62	CHL	0.557	4.22
USA	0.463	3.89	NOR	0.515	5.38	CHN	0.582	3.44
CRI	0.469	4.14	ARG	0.522	5.40	THA	0.604	4.28
BOL	0.475	6.67	ZAF	0.531	5.44	MEX	0.649	4.30
CZE	0.486	4.63				IDN	0.659	6.07

Γ ranges over 0.333–0.659 and N_{eff} over 3.11–6.67; $\text{corr}(\Gamma, N_{\text{eff}}) = 0.154$. Bolivia has the *highest* N_{eff}, the least concentrated sectoral power.

(b) No function of Γ bounds $\text{Var}(\boldsymbol{\pi})$. *Claim.* For every $\gamma \in (0, 1]$ and every $\boldsymbol{\pi}$ in the open simplex, $W_\gamma = \gamma \mathbf{1}\boldsymbol{\pi}' + (1-\gamma)I$ is row-stochastic, irreducible and aperiodic, with $\boldsymbol{\pi}(W_\gamma) = \boldsymbol{\pi}$ and $\Gamma(W_\gamma) = \gamma$. Hence $\{\text{Var}(\boldsymbol{\pi}(W)) : \Gamma(W) = \gamma\} = [0, (N-1)/N^2)$ for every γ, and no function f with $\text{Var}(\boldsymbol{\pi}) \le f(\Gamma)$ can be tighter than the trivial bound.

Proof. $W_\gamma \ge 0$ and $W_\gamma \mathbf{1} = \gamma \mathbf{1} + (1-\gamma)\mathbf{1} = \mathbf{1}$. Every off-diagonal entry is $\gamma \pi_j > 0$, so W_γ is irreducible, and the positive diagonal makes it aperiodic. $\boldsymbol{\pi}' W_\gamma = \gamma(\boldsymbol{\pi}'\mathbf{1})\boldsymbol{\pi}' + (1-\gamma)\boldsymbol{\pi}' = \boldsymbol{\pi}'$, and by uniqueness of the stationary vector of an irreducible matrix it is *the* Perron vector. For the spectrum, $\mathbf{1}\boldsymbol{\pi}'$ has rank one, with eigenvalue 1 on $\mathbf{1}$ and 0 on the $(N-1)$-dimensional space $\{\mathbf{v} : \boldsymbol{\pi}'\mathbf{v} = 0\}$. On that space $W_\gamma = (1-\gamma)I$. So the eigenvalues are 1 and $1-\gamma$ (multiplicity $N-1$), and $\Gamma = 1 - |1-\gamma| = \gamma$. Finally, $\text{Var}(\boldsymbol{\pi})$ ranges over $[0, (N-1)/N^2)$ as $\boldsymbol{\pi}$ ranges over the open simplex: it is 0 at $\mathbf{1}/N$ and approaches the maximum as $\boldsymbol{\pi} \to \mathbf{e}_1$. □

Numerically ($N = 11$, $\pi_1 = p_1$, the rest equal): for $\gamma = 0.05$, 0.5 and 1, $\Gamma = \gamma$ exactly, and $\text{Var}(\boldsymbol{\pi}) = 0$, 0.01674, 0.07380 at $p_1 = 1/11$, 0.5, 0.95, whatever γ. The maximum is $(N-1)/N^2 = 0.08264$.

(c) Regressions (n = 12). Countries with WDI Gini (latest 2014–2019), ECI 2019 and a domestic Γ: ARG, BOL, BRA, CHL, CHN, COL, DEU, JPN, KOR, MEX, PER, USA. OLS, standard errors in parentheses:

	(1)	(2)	(3)
ECI	-3.57 (1.83)	-1.07 (2.92)	-4.72 (5.46)
log GDP pc		-5.49 (5.02)	-4.66 (5.76)
Γ			$+1.61$ (36.36)
N_{eff}			-4.08 (4.81)
R^2	0.277	0.362	0.422

The negative ECI–Gini association appears in (1). With income controlled it shrinks to -1.07, and adding Γ and N_{eff} *enlarges* it to -4.72. A mediator would shrink it, so neither Γ nor N_{eff} mediates in this sample. Γ's coefficient has a standard error of 36 Gini points per unit of Γ. With 12 observations and 5 parameters nothing is identified; the exercise's 30–50 countries are the minimum for a meaningful test. Mediation should be tested properly: estimate the ECI coefficient with and without the mediators on the same sample, bootstrap the difference, and add regional fixed effects.

(d) Lower inequality than predicted. Residuals from specification (2), actual minus predicted Gini: Bolivia -6.9, Peru -5.2, China -4.9, Japan -4.3, Germany -4.0, Korea -4.0. The largest positive residuals are Brazil ($+9.0$), Colombia ($+7.7$) and the United States ($+6.0$). In this sample Bolivia is the strongest over-performer. That is consistent with Stylized Fact 16.13: equity gains through labour income at the bottom and transfers, not through complexity. What the over-performers have in common is a policy and labour-market story (Bolivia, Peru: commodity-boom wage growth; Germany, Japan, Korea: compressed wage structures), not a common Γ. Their Γs range from 0.407 to 0.582.

Exercise 16.7 Bolivia's structural position

(a) Position (sourced values).

- *Global power*: $\pi_{\text{BOL}} = 0.00053$ in 2019, which is 0.121 times the uniform share $1/230$.
- *Complexity*: ECI -1.30 in 2019, rank 131 of 145. It was -1.21 in 2006, and fell by 0.58 over 1999–2019.
- *Domestic integration*: $\Gamma(W^P) = 0.467$ (Chapter 17).
- *Terms of trade*: a boom-and-bust in shares rather than a structural change (Exercise 16.4).
- *Relatedness to core products* and *position in* W^I: not measured. Relatedness needs Exercise 16.3; for W^I, Bolivia is a peripheral receiver, with the expected position of Exercise 16.5.

(b) A 20-year plan (model answer). The binding constraint is complexity, not integration. Bolivia's Γ is already in the comparator range, and its sectoral power is the least concentrated of the 23 economies in Exercise 16.6. So the plan targets

new products near the existing basket, not Γ. Following the sequence of Conjecture 15.14:

1. Years 1–3, $\mathcal{L}_{\text{info}}$ and $\mathcal{L}_{\text{inst}}$: technical exchanges, FDI in assembly, stable rules for lithium and agro-industry contracts.
2. Years 2–10, $\mathcal{L}_{\text{HC}}$: chemical, electrical and food-engineering training tied to target firms.
3. Years 5–15, $\mathcal{L}_{\text{infra}}$: power, logistics and testing laboratories in the processing zones.
4. Years 8–20, $\mathcal{L}_{\text{ISI}}$ of type B: supplier-development contracts linking agro-industry and lithium to domestic metalworking and chemicals.

Target products, in order of expected relatedness: processed food and food machinery (HS 8438), lithium compounds and battery precursors (HS 2836 to 8507), and pharmaceutical intermediates last.

(c) Orders of magnitude.

- *Cost.* A programme of 1% of GDP a year for 20 years is 9.8 bn US$ at the 2019 GDP of 49.1 bn. That is an assumption about scale, not an estimate.
- ΔECI. Over 1999–2019, across the 140 economies ranked in both years, the median 20-year change was $+0.02$, the 75th percentile $+0.28$ and the 90th $+0.53$; only 11.4% gained 0.5 or more. A plan that delivered $+0.5$ would be a top-decile performance, and would still leave Bolivia near -0.8.
- $\Delta\Gamma$. About 0, and not a target. Across comparators Γ moves by a few hundredths per decade in either direction.
- $\Delta Gini$. No structural prediction. The 2005–2019 fall (from 0.600 to 0.426) came from labour income and transfers; the plan's contribution is to make it durable.

(d) Political feasibility. By Proposition 15.18, the integration and complexity components need not cost any coalition power. The constraint bites only where the plan redistributes power, e.g. moving fiscal rents from the hydrocarbons–state coalition to new private suppliers. Under the coalition logic of Chapter 14 (Proposition 14.4), the plan is feasible if its surplus component $\Delta\Omega/N$ is large enough to compensate the losers of $\Delta\pi$. The plan should therefore be designed to raise Ω while leaving π of the current coalition roughly unchanged at first, and so it should be front-loaded with the operators that do not move π.

Chapter 17

Domestic Structural Diagnostics

Every number below is printed by `python3 code/solutions/sol17.py`. Input-output results use INE's supply-use tables (Instituto Nacional de Estadística (INE), Bolivia, 2026) as processed in `data/processed/bolivia_Z` and `bolivia_io_*.csv`; comparators use OECD ICIO 2023 (OECD, 2023); employment uses ILOSTAT (International Labour Organization, 2026). All were retrieved on 2026-11-05. Household-survey microdata (EH) and audience data were **not fetched** (`data/README.md`, §9). Exercises that need them get the method only.

Exercise 17.1 Compute Bolivia's Γ

Method.

1. INE's 2019 supply-use tables (base 2017) are converted to a domestic industry×industry table at basic prices: import proportionality, product-sales-structure model D (`data/README.md`, §2).

2. Aggregate to the book's 12 sectors, rows = sellers.

3. $a_{ss'} = Z_{s's}/X_s$ (Appendix 17A); $W^P = D^{-1}A$, so row s is $Z_{\cdot s}/\sum_{s''} Z_{s''s}$.

4. Domestic services has no intermediate purchases or sales, so its row is undefined and it is dropped; $N = 11$.

Results, input shares (book12). The eigenvalues by modulus are 1; 0.5326; 0.3127; 0.2233; $-0.1951 \pm 0.0986i$; $0.1003 \pm 0.1035i$; -0.0192; 0.0171; -0.0137. Hence $\lambda_2 = 0.5326$ (real, simple) and $\Gamma = 0.4674$. The Perron vector is

Hydrocarbons	0.039	Construction	0.029	Agriculture	0.091
Mining	0.037	Commerce	0.095	Other services	0.270
Public administration	0.005	Transport	0.092	Personal services	0.021
Manufacturing	0.159	Financial services	0.162		

so $G(\boldsymbol{\pi}) = 0.4446$ and $N_{\text{eff}} = 6.48$. Checks: $\sum \pi = 1$; all values agree with Examples 17.2 and 17.7 and with `bolivia_io_summary.csv` to 10^{-9}.

Normalizations and aggregations (2019):

Aggregation	Normalization	Γ	$G(\boldsymbol{\pi})$	N_{eff}	$\arg\max \pi$
book12	input shares	0.467	0.445	6.48	Other services (0.270)
book12	input, zero diagonal	0.614	0.363	7.79	Other services (0.190)
book12	sales shares	0.694	0.319	7.88	Manufacturing (0.241)
book12	sales, zero diagonal	0.653	0.268	8.55	Manufacturing (0.221)
book12_isic	input shares	0.475	0.435	6.67	Other services (0.258)
8 sectors (here)	input shares	0.589	0.517		Finance & other serv. (0.416)
8 sectors (here)	sales shares	0.676	0.237		
detail39	input shares	0.410	0.670		(CSV, $N = 38$)
detail39	sales shares	0.331	0.428		(CSV)

The 8-sector grouping merges hydrocarbons with mining, commerce with transport, finance with other services, and personal with domestic services.

How much do they change? Across book12 and detail39, input or sales shares, with or without diagonal, Γ ranges over 0.331–0.694 (the book's 0.33–0.69) and $G(\boldsymbol{\pi})$ over 0.268–0.670. The spread across normalizations, about 0.36 in Γ, is wider than the spread across the 23 economies of Exercise 16.6 (0.333–0.659). Three regularities:

- Zeroing the diagonal raises Γ, because large self-purchases (finance 0.567, other services 0.511) are the slow modes.
- Sales shares raise Γ and lower $G(\boldsymbol{\pi})$: the hub becomes manufacturing as a buyer rather than other services as a supplier.
- Aggregating to 8 sectors raises Γ, but merging finance with other services creates a single large hub, which raises $G(\boldsymbol{\pi})$.

Model answer. Report Γ with its normalization and aggregation, and compare countries only like-for-like (book12_isic, domestic, input shares). Under any variant, no eigenvalue of Bolivia's W^P is close to 1, so there are no weakly coupled blocks at this resolution.

Exercise 17.2 Between-block variance

Data status. EH 2019 microdata were **not fetched** (INE ANDA portal; row 41 of the status table in `data/README.md`).

(a–b) Method.

1. From the EH 2019 person file, keep employed adults with positive labour income. Use the survey weights throughout; monthly labour income, or per-capita household income for the household version.

2. Assign blocks: (1) workers in ISIC B (mining and hydrocarbons) or O (public administration); (2) formal urban, i.e. urban workers contributing to social security or employed by firms with 5 or more workers; (3) everyone else (informal or rural).

3. With weights w_i, compute $\bar{y} = \sum_i w_i y_i / \sum_i w_i$, block shares $N_b/N = \sum_{i\in b} w_i / \sum_i w_i$, block means $\bar{y}_b$, V_B by (17.2) and $V_T = \sum_i w_i (y_i - \bar{y})^2 / \sum_i w_i$. Then $\sigma_B = V_B / V_T$.

4. Bootstrap σ_B by resampling primary sampling units.

With the chapter's illustrative inputs (Example 17.5): $\bar{y} = 4{,}326$, $V_B = 19.98$M and $\sigma_B = 0.624$ for $V_T = 32$M.

(c) Two and four blocks. Merging blocks 1 and 2 of the illustration into "formal" gives a formal mean of 7,674.4, $V_B = 8.46$M and $\sigma_B = 0.264$: the coarser partition loses 11.52M of between-block variance. That loss is exactly the variance of the two fine means inside the merged block, as the proof below shows. A four-block partition (gas, formal manufacturing, urban services, rural) can only raise σ_B relative to any three-block partition it refines. *Expected with real data:* σ_B rises with the number of blocks, at a decreasing rate.

(d) Refinement never lowers σ_B. Let $\mathcal{P}'$ refine $\mathcal{P}$: every block B of $\mathcal{P}$ is a union of blocks b of $\mathcal{P}'$. With n_b the size of fine block b and $n_B = \sum_{b\subseteq B} n_b$, the coarse mean is $\bar{y}_B = \sum_{b\subseteq B} (n_b/n_B)\bar{y}_b$. Then

$$V_B(\mathcal{P}') = \sum_B \sum_{b\subseteq B} \tfrac{n_b}{N} (\bar{y}_b - \bar{y})^2 = V_B(\mathcal{P}) + \sum_B \tfrac{n_B}{N} \sum_{b\subseteq B} \tfrac{n_b}{n_B} (\bar{y}_b - \bar{y}_B)^2 \;\geq\; V_B(\mathcal{P}).$$

The middle equality writes $\bar{y}_b - \bar{y} = (\bar{y}_b - \bar{y}_B) + (\bar{y}_B - \bar{y})$ and notes that the cross term vanishes within each B (the law of total variance applied to block means). V_T does not depend on the partition, so σ_B weakly rises under refinement, with equality iff all fine means inside every coarse block are equal. The highest σ_B is

reached by the finest partition: with singletons, $V_W = 0$ and $\sigma_B = 1$. The script checks both facts on 500 random samples.

Implications. σ_B is comparable across countries only for partitions of the same number and type of blocks. There is no "natural" partition in the data unless one is fixed in advance on substantive grounds, or read off W^P by spectral clustering (Appendix 17B). A partition chosen by maximizing σ_B is meaningless: the maximum is always the singleton partition.

Exercise 17.3 Power-employment ratio

(a) π is that of Exercise 17.1 (input shares, book12, 2019).

(b) Ratios. Employment is from ILOSTAT 2019 (ISIC Rev. 4 sections, mapped as in Example 17.7). VA per worker is INE value added divided by ILOSTAT employment, in thousand Bs per worker.

Sector	Emp. share	π_s	π_s/share_s	VA per worker
Hydrocarbons and mining	0.013	0.077	5.74	464.6
Public administration	0.026	0.005	0.19	313.0
Manufacturing	0.112	0.159	1.42	60.4
Construction	0.079	0.029	0.37	54.6
Commerce	0.171	0.095	0.55	29.3
Transport	0.071	0.092	1.30	39.1
Financial services	0.007	0.162	23.36	378.1
Agriculture	0.274	0.091	0.33	16.8
Other services	0.121	0.270	2.23	81.7
Personal services	0.107	0.021	0.20	41.3
Domestic services	0.019	—	—	19.0

(c) Ranking and capital intensity. The highest ratio is financial services (23.4); the lowest is public administration (0.19), then personal services (0.20) and agriculture (0.33). Regressing $\log(\pi/\text{share})$ on $\log(\text{VA per worker})$ over the 10 sectors gives a slope of 0.770 and $R^2 = 0.328$ (Spearman 0.515). So value added per worker, a proxy for capital intensity and labour productivity, explains about a third of the variation.

The rest has a clean decomposition:

$$\frac{\pi_s}{\text{emp}_s} = \frac{\pi_s}{\text{VA}_s/\text{VA}} \cdot \frac{\text{VA}_s/\text{emp}_s}{\text{VA}/\text{emp}}.$$

The first factor, power per unit of value added, is 3.41 for finance, 1.83 for transport and 0.03 for public administration. Public administration has the third-highest VA per worker but sells almost nothing to other sectors, so its residual is -2.72 log points. Finance has both a high VA per worker and a high power per unit of VA (residual $+1.93$). As the intuition box after Example 17.7 says, the ratio mixes supplier centrality with productivity; it is not a measure of individual privilege.

(d) Moving from agriculture to manufacturing. Let $\varphi := \Phi/\Omega \in [0,1]$ and $N = 11$ sectors. Sector s receives $x_s = \Omega[1/N + (\pi_s - 1/N)\varphi]$ (Proposition 7.3), shared equally by its $\text{emp}_s \cdot E$ workers. Relative to the national mean per worker Ω/E, a worker in s gets $[1/N + (\pi_s - 1/N)\varphi]/\text{emp}_s$:

φ	0	0.25	0.5	1
Agriculture	0.332	0.332	0.332	0.332
Manufacturing	0.810	0.962	1.113	1.415
Ratio	2.44	2.90	3.35	4.26

At $\varphi = 1$ the relative allocation is exactly the power-employment ratio, so she gains a factor $1.42/0.33 = 4.26$. Agriculture's row does not move with φ because $\pi_{\text{agr}} = 0.091 \approx 1/11$.

What the calculation assumes.

1. Sectoral $\boldsymbol{\pi}(W^P)$ is a bargaining weight (Assumption 7.A). That is a modelling assumption, and this layer is the least natural one for it.

2. Sectors are the bargaining units and divide their share equally among workers.

3. One worker's move changes neither W^P nor $\boldsymbol{\pi}$ (it is marginal).

4. Disagreement payoffs are equal across sectors.

With a different N (e.g. 39 sectors) the numbers change. *Common mistake:* reading the factor 4.26 as a wage premium; it is a division rule, not an estimate.

Exercise 17.4 Information democracy

Data status. Audience data were **not fetched** (Kantar IBOPE is paywalled; `data/README.md`, §9). Method only, plus a formal result on node sets.

(a–b) Method.

1. List outlets (TV channels, radio networks, newspapers, major digital platforms) with audience measures from a ratings service or a representative survey ("where did you get news yesterday?").

2. Convert each outlet's reach to shares of total attention (minutes, or mentions as main source) that sum to 1.

3. Group outlets into ecosystems for the four-node comparison.

Using audience shares $\mathbf{s}$ as $\boldsymbol{\pi}^I$ is exact only if $W^I = \mathbf{1}\mathbf{s}'$, i.e. everyone attends to the same mix, which has $\Gamma(W^I) = 1$ (Example 17.9). With survey data on *who* watches what, build W^I by groups and take its Perron vector instead.

(c) $N_{\text{eff}} = 1/\sum_j s_j^2$. For the chapter's four ecosystems, $N_{\text{eff}} = 3.381$.

(d) Node sets. *Claim:* refining the node set never lowers N_{eff}. If node e splits into outlets $o \in e$ with shares s_o summing to s_e, then $\sum_{o\in e} s_o^2 \le (\sum_{o\in e} s_o)^2 = s_e^2$, since the cross terms are nonnegative. Summing over e, $\sum_o s_o^2 \le \sum_e s_e^2$, with equality iff no node is actually split. Examples:

- an equal split of every ecosystem into k outlets multiplies N_{eff} by k: 6.761 for $k = 2$ and 16.903 for $k = 5$;

- an unequal split (0.8/0.2 for state media, 0.5/0.5 for private, none for community, 0.6/0.4 for social) gives 5.666.

So "Bolivia 3.4 vs. the US 15–20" compares different node sets and says nothing. Compare on the same four-ecosystem partition, or report N_{eff}/N on comparable outlet-level sets. *Expected:* on four ecosystems most countries lie between 2 and 4; on outlet sets the numbers scale with the number of outlets measured.

Exercise 17.5 Bolivia's structural trajectory

(a) Every available year (book12, domestic, input shares; from `bolivia_io_summary.csv`, reproduced by the script from the flow matrices for the years in Example 17.15):

Year	N	Γ	$G(\boldsymbol{\pi})$	π_{hyd}	Year	N	Γ	$G(\boldsymbol{\pi})$	π_{hyd}
1988	10	0.563	0.442	0.025	2004	10	0.524	0.385	0.057
1989	10	0.565	0.429	0.029	2005	10	0.502	0.366	0.074
1990	10	0.567	0.420	0.037	2006	11	0.518	0.424	0.064
1991	10	0.578	0.431	0.030	2007	10	0.489	0.350	0.087
1992	10	0.571	0.417	0.030	2008	10	0.486	0.341	0.081
1993	10	0.573	0.419	0.026	2009	10	0.504	0.353	0.077
1994	10	0.567	0.413	0.022	2011	11	0.478	0.369	0.088
1995	10	0.578	0.411	0.025	2012	11	0.506	0.379	0.079
1996	10	0.566	0.407	0.026	2013	11	0.484	0.371	0.085
1997	10	0.553	0.409	0.027	2014	11	0.361	0.352	0.117
1998	10	0.541	0.449	0.017	2017	11	0.506	0.446	0.045
1999	10	0.642	0.425	0.035	2018	11	0.493	0.444	0.042
2000	10	0.494	0.420	0.036	2019	11	0.467	0.445	0.039
2001	10	0.493	0.410	0.038	2020	11	0.418	0.463	0.035
2002	11	0.501	0.447	0.039	2022	11	0.400	0.475	0.026
2003	11	0.514	0.444	0.047	2024	11	0.419	0.470	0.021

(2021: 0.432, 0.462, 0.028; 2023: 0.381, 0.470, 0.030. The rule separates base 1990 from base 2017.)

Outliers. On base 1990 the z-scores of Γ exceed 1.5 in absolute value in two years: 1999 ($\Gamma = 0.642$, $z = 2.15$) and 2014 (0.361, $z = -3.14$). In 2014 the slow mode $\mathbf{v}_2$ is concentrated on hydrocarbons (1.0, against at most 0.45 in absolute value elsewhere). The hydrocarbons self-purchase share jumps from 0.478 (2013) to 0.658 (2014), the largest diagonal change of any sector. A near-absorbing diagonal entry creates an eigenvalue near the diagonal value, so the 2014 Γ reflects one cell of the table, not a disintegration of the economy. It is most likely a compilation artefact of the last base-1990 table.

Why not compare across the change of base. The 2017 base has a new classification (42 products, 39 industries against 35), a new SNA vintage and new valuation conventions. Γ, $G(\boldsymbol{\pi})$ and π_{hyd} depend on aggregation and conventions (Exercise 17.1), so a jump at the break mixes measurement with structure. A further caveat: on base 1990, N alternates between 10 and 11, because public administration has no intermediate sales in diagonal-make years. Changes between, say, 2005 ($N = 10$) and 2006 ($N = 11$) are partly a change of node set.

(b) σ_B **by year: not computable here** (EH microdata not fetched). Method: Exercise 17.2 applied to EH 2000, 2005, 2011 (there is no 2010 survey), 2015 and 2019, with a fixed partition and fixed definitions of formality; then plot. INE's published Gini fell from 0.600 (2005) to 0.426 (2019) (Chapter 16), so the total variance fell. Whether σ_B fell too depends on whether the fall was between or within blocks; decompositions attributing it to labour income at the bottom (Vargas and Garriga, 2015) suggest a fall in V_B.

(c) Structural breaks in Γ**.** Chow tests on the annual base-1990 series, 1988–2013

(2014 excluded), with a model of level plus linear trend and a break in both:

	All years ($n = 25$)		Excluding 1999 ($n = 24$)	
Break at	$F(2, n-4)$	p	F	p
2006 (nationalization)	0.108	0.898	0.940	0.407
2009 (constitution)	0.170	0.845	1.193	0.324

No break is detected at either date, and the OLS trend is -0.042 per decade (-0.041 without 1999). The p-values come from the F distribution, integrated numerically; the integration is checked against the 5% critical value of $F(2, 20)$, which is 3.49. Consistent with Example 17.15: the nationalization changed revenue flows, not W^P. π_{hyd} shows the gas boom (from 0.047 in 2003 to 0.088 in 2011), but that rise starts before 2006.

(d) Projection of σ_B. No data. With the book's *illustrative* trajectory, 0.68 (1990) to 0.62 (2019), the trend is -0.021 per decade, and σ_B would reach 0.40 around 2125. That number illustrates the arithmetic only; a projection needs the EH series of part (b) and a model of why σ_B moves.

Exercise 17.6 Dashboard comparison

(a) Measured instruments (2019, book12_isic, domestic, input shares; INE for Bolivia, OECD ICIO 2023 for Chile and Costa Rica):

	Γ	$G(\boldsymbol{\pi})$	$N_{\text{eff}}(W^P)$ of 11	ECI 2019	Gini (WDI)
Bolivia	0.475	0.435	6.67	-1.295	41.6 (2019)
Chile	0.557	0.568	4.22	-0.059	45.3 (2017)
Costa Rica	0.469	0.611	4.14	0.257	not in `data/processed`

Costa Rica's ECI is from the Growth Lab rankings file in `data/raw/atlas`. The largest power weights are other services in Chile (0.429) and in Costa Rica (0.414). σ_B needs household microdata (CASEN for Chile, ENAHO for Costa Rica, EH for Bolivia); not fetched.

(b) $N_{\text{eff}}(W^I)$: not computable (audience data not fetched). Method of Exercise 17.4 on a common four-ecosystem partition: state or public, private commercial, community, social media.

(c) Dashboard and "healthiest". Rescaled bars (benchmark = 1):

- Γ: Bolivia 0.475, Chile 0.557, Costa Rica 0.469;

- $1 - G(\boldsymbol{\pi})$: 0.565, 0.432, 0.389;
- N_{eff}/N: 0.606, 0.384, 0.377.

Chile is the most integrated, Bolivia the least concentrated. On both production-network coordinates Costa Rica, the most complex of the three, is the furthest from the benchmark. "Healthiest" is therefore undefined without weights on independent coordinates (`docs/conventions.md`, §4), and the measured instruments rank the countries opposite to complexity. As in Chapter 17, a concentrated $\boldsymbol{\pi}(W^P)$ marks a service-hub structure, not exclusion.

(d) Recommendation. The largest deficit depends on the coordinate:

- Bolivia: complexity (ECI -1.30). The recommended operators are $\mathcal{L}_{\text{info}}$ and $\mathcal{L}_{\text{HC}}$ aimed at related products, not Γ-raising operators.
- Costa Rica, on power concentration: an operator that diversifies suppliers away from the services hub. Proposition 15.18 says integration is free politically, but redistribution is not.

Choose the operator by the instrument, then sequence by Conjecture 15.14.

Exercise 17.7 Bolivia's escape plan

(a) Current position. $\Gamma = 0.475$ (like-for-like) and $G(\boldsymbol{\pi}) = 0.435$, both measured. ECI -1.30 in 2019, down from -1.21 in 2006. σ_B, $N_{\text{eff}}(W^I)$ and Ψ remain illustrative (0.62, 3.4, 1.56).

(b) Binding constraint. Among the *measured* instruments, none is far from comparators: Γ sits inside the range, and $G(\boldsymbol{\pi})$ is the lowest of 23 economies. The binding constraint is complexity, which is not a dashboard instrument but is the memo row of Example 17.13. Among the illustrative rows, σ_B and $N_{\text{eff}}(W^I)$ would be the candidates if the microdata confirmed them. Because Γ and $G(\boldsymbol{\pi})$ are independent coordinates, an operator chosen to raise Γ does not address either.

(c) Sequence: information, then institutions, education, infrastructure and ISI-B (Conjecture 15.14), aimed at products related to the existing basket (Exercise 16.7(b)). Exercise 15.4 shows when this order beats the reverse: only if later operators are complementary to earlier ones.

(d) Political feasibility. Integration and complexity operators need not cost any coalition power (Proposition 15.18), so a coalition of the middle class and workers ($\{M, W\}$), or of the bourgeoisie, middle class and workers ($\{B, M, W\}$), can

support them without internal conflict. The redistributive part, shifting rents or power from the hydrocarbons–state nexus, is where the coalition's own members' $\Delta\pi_i \geq 0$ constraints bind (Exercise 15.7). A $\{B, M, W\}$ coalition is compatible if the plan's surplus share $\Delta\Omega/N$ compensates B for any power loss (Proposition 14.4).

(e) Simulation. Use the three-block model of Example 10.6 with $\rho = 0.85$ and $\mu = 0.2$. The script reproduces its steady state ($\theta^* = 1.1245, 1.0411, 0.9871$; $\rho(J) = 0.866$). Apply an information operator: cross-block W^I weight per pair from 0.025 to 0.05, with own-block weights cut equally.

- *Long run*: $\theta^* = (1.1009, 1.0439, 1.0062)$; the gas–rural preference gap falls from 0.1374 to 0.0947. The variance of allocations falls from 0.03890 to 0.03229; its power part (0.01985) is untouched, because $\boldsymbol{\pi}$ is.
- $\Gamma(W^I)$ rises from 0.225 to 0.450. $\Gamma(W^P) = 0.090$ and $N_{\text{eff}}(\boldsymbol{\pi}) = 8.34$ do not move.
- *30 periods, generational* ($\rho = 0.85$; one period is one generation, Chapter 10): 14.0% of the adjustment occurs after 1 period, 77.9% after 10, 98.9% after 30.
- *30 years, annual* ($\rho_a = 0.85^{1/25}$, with $\mu_a = 0.00864$ chosen to keep $\mathcal{M}$ fixed; $\rho(J) = 0.9940$): only 0.6% after one year, 5.9% after ten and 16.6% after thirty. The gap falls from 0.1374 to 0.1303.

Answer. Target the information structure ($N_{\text{eff}}(W^I)$, the cross-block weight of W^I) and complexity, not $\Gamma(W^P)$, which is already in the comparator range. With generational persistence $\rho = 0.85$, thirty *years* buy about a sixth of the long-run preference convergence; most of the effect takes several generations. Plans should be judged against that horizon.

Common mistake: simulating 30 periods of the generational model and calling them years, which overstates the speed of formation by a factor of about six here (98.9% against 16.6%).

Chapter 18

The Individual Scale

Every number below is printed by `python3 code/solutions/sol18.py`. Throughout, $K = \phi \circ K^{\max}$ are raw intensities and $W = D^{-1}K$. When K is symmetric, $\pi \propto K\mathbf{1}$ (Proposition 18.6(i)).

Exercise 18.1 Assortative mixing and Γ

Set-up. Three men and three women, $K^{\max}_{ij} = 1$ for every pair including $i = j$ (so $a = c = 1$ in Proposition 18.4(a)), $\phi = 1$ within gender and λ_g across. Each row of K sums to $3(1+\lambda_g)$, so W has $1/(3(1+\lambda_g))$ within and $\lambda_g/(3(1+\lambda_g))$ across.

Spectrum. The partition is equitable, with quotient $\frac{1}{1+\lambda_g}\begin{pmatrix} 1 & \lambda_g \\ \lambda_g & 1 \end{pmatrix}$, so the eigenvalues are 1 and $\frac{1-\lambda_g}{1+\lambda_g}$. Every vector summing to zero within each gender is mapped to 0, because rows are constant within blocks. Hence

$$\Gamma(\lambda_g) = 1 - \frac{1-\lambda_g}{1+\lambda_g} = \frac{2\lambda_g}{1+\lambda_g}.$$

λ_g	1	0.7	0.4	0.1	0
$\lvert\lambda_2\rvert$	0	0.1765	0.4286	0.8182	1
Γ	1.0000	0.8235	0.5714	0.1818	0
Γ, no self-loops	0.8000	0.7561	0.6875	0.2609	0

At $\lambda_g = 0$, W is block-diagonal and reducible; 1 is a double eigenvalue and $\Gamma = 0$ (`docs/conventions.md`, §3). The script confirms the closed form to 10^{-12} at every λ_g.

Not linear. Γ is strictly increasing and strictly concave on $[0,1]$: $\Gamma' = 2/(1+\lambda_g)^2$ falls from 2 at $\lambda_g = 0$ to 0.5 at $\lambda_g = 1$, and $\Gamma'' = -4/(1+\lambda_g)^3 < 0$. At

$\lambda_g = 0.5$, $\Gamma = 0.6667$, against 0.5 on the chord. Integration is most sensitive to openness when groups are nearly segregated: the first cross-group edges buy the most mixing.

Robustness: self-loops matter. Without self-loops ($K_{ii} = 0$), each row has within weight $1/(2+3\lambda_g)$ on each of the other two members of its group. The within-group zero-sum vectors then have eigenvalue $-1/(2+3\lambda_g)$, and the group eigenvalue is $(2-3\lambda_g)/(2+3\lambda_g)$. Hence $\Gamma = 6\lambda_g/(2+3\lambda_g)$ for $\lambda_g \leq 1/3$ and $\Gamma = (1+3\lambda_g)/(2+3\lambda_g)$ above, where the negative eigenvalue binds. Γ is still increasing, with a kink at $\lambda_g = 1/3$, and reaches only 0.8 at $\lambda_g = 1$. The qualitative claim, segregation lowers Γ, survives; the exact formula depends on the diagonal, which must be stated.

Exercise 18.2 Bolivia's gender power gap

Data. EH 2019 microdata were not fetched. The sector-by-sex tabulation available in `data/processed` is ILOSTAT's 2019 series of employment by sex and economic activity (source survey: Encuesta Continua de Empleo, processed by the ILO from the microdata), which is what the chapter uses (International Labour Organization, 2026). Combined with INE 2019 (base 2017, book12, domestic) for $\boldsymbol{\pi}$ and BL (Instituto Nacional de Estadística (INE), Bolivia, 2026). Retrieved 2026-11-05. Sector mapping as in Example 17.7.

(a) Classification (female share of employment):

Sector	Female share	Class	BL	π_s
Hydrocarbons and mining	0.065	male	1.352	0.077
Public administration	0.343	male	1.245	0.005
Construction	0.031	male	1.574	0.029
Transport	0.039	male	1.542	0.092
Manufacturing	0.441	mixed	1.855	0.159
Financial services	0.469	mixed	1.433	0.162
Agriculture	0.460	mixed	1.549	0.091
Other services	0.539	mixed	1.356	0.270
Commerce	0.614	female	1.416	0.095
Personal services	0.689	female	1.661	0.021
Domestic services	0.939	female	1.000	—

BL for hydrocarbons and mining is the output-weighted average of the two sectors' backward linkages.

(b) Average Leontief multiplier (BL, column sums of $(I - A')^{-1}$):

- male-dominated 1.428 (employment-weighted 1.501);
- mixed 1.548 (1.569);
- female-dominated 1.359 (1.478).

The mixed sectors have the highest multiplier, because they include manufacturing (1.855). The female group is pulled down by domestic services, which has no intermediate flows ($BL = 1$). The differences are modest, because Bolivia's linkages are compressed (all between 1.0 and 1.86).

(c) Per-capita power ratio. Give each sector's π_s equally to its workers (none to domestic service, which is outside W^P). Then

$$\frac{\pi^F_{\text{avg}}}{\pi^M_{\text{avg}}} = \frac{\sum_s \pi_s f_s / F}{\sum_s \pi_s (1 - f_s) / M} = 0.8555,$$

where f_s is the female share of sector s and F, M are total female and male employment. This reproduces the 0.86 of Example 18.7.

(d) Women at 40% in male-dominated sectors. By the $>$ 60%-male rule, the male-dominated sectors are hydrocarbons and mining, public administration, construction and transport, with a combined female share of 8.0%. (The chapter's "about 10%" excludes transport.) *Counterfactual:* set each of these sectors' female share to 40% at fixed sector employment. That moves 339.4 thousand women, 13.25% of female employment. Two assumptions about where they come from:

- From all other sectors in proportion to employment, with displaced men going the other way: the ratio rises to 0.8923, closing 25.4% of the gap ($1 - r$ falls from 0.1445 to 0.1077).
- From the female-dominated sectors only: the ratio becomes 1.0315, overshooting parity, because it empties the low-π commerce and personal-service jobs.

Which sector matters? Leaving out one sector's change at a time (main counterfactual), the ratio would be:

- 0.8173 without hydrocarbons and mining, the largest contribution, since $\pi/\text{share} = 5.7$;
- 0.8630 without transport;

- 0.8966 without public administration, and 0.9607 without construction. Moving women into these two *lowers* the ratio, because their power per worker (0.19, 0.37) is below women's current average.

Model answer. Entering male-dominated sectors helps only where those sectors are central in W^P; occupational desegregation is not the same as power equalization. The calculation assumes fixed $\boldsymbol{\pi}$ and equal division within sectors (as in Exercise 17.3(d)).

Exercise 18.3 Intersectionality

(a) Separable constraint matrix. Order: 1 non-indigenous man, 2 non-indigenous woman, 3 indigenous man, 4 indigenous woman. $\phi_{ij} = \lambda_g^{s_{ij}} \lambda_r^{e_{ij}}$, where $s_{ij} = 1$ if i and j differ in sex and $e_{ij} = 1$ if they differ in ethnicity (both 0 otherwise):

$$\phi = \begin{pmatrix} 1 & 0.3 & 0.35 & 0.105 \\ 0.3 & 1 & 0.105 & 0.35 \\ 0.35 & 0.105 & 1 & 0.3 \\ 0.105 & 0.35 & 0.3 & 1 \end{pmatrix}.$$

(b) Uniform $\boldsymbol{\pi}$. With $W^{\max} = \frac{1}{4}\mathbf{1}\mathbf{1}'$, $K = \frac{1}{4}\phi$ is symmetric, so $\pi_i \propto \sum_j \phi_{ij}$. Every row of ϕ sums to $(1+\lambda_g)(1+\lambda_r) = 1.755$: each agent has one partner identical on both attributes, one differing in sex only, one in ethnicity only, and one in both. Hence $\boldsymbol{\pi} = (0.25, 0.25, 0.25, 0.25)$. Edge suppression is symmetric, so with equal opportunities it cannot create a power gap (Proposition 18.6(i)–(ii) with $p = \frac{1}{2}$).

(c) Population-weighted opportunities. $K_{ij} = \phi_{ij} n_j$ is not symmetric, but W is reversible with respect to $\pi_i \propto n_i d_i$, where $d_i = \sum_j \phi_{ij} n_j$: indeed $n_i d_i W_{ij} = n_i n_j \phi_{ij}$ is symmetric. So the per-capita power is $\pi_i / n_i \propto d_i$, and $d_i = (1+\lambda_g)(n^{\text{own eth.}} + \lambda_r n^{\text{other eth.}})$, where $n^{\text{own eth.}}$ and $n^{\text{other eth.}}$ are the population shares of a single sex in the agent's own and in the other ethnic group (0.295 and 0.205 for a non-indigenous agent). Numerically:

- $\boldsymbol{\pi} = (0.3156, 0.3156, 0.1844, 0.1844)$;
- $\pi_i / n_i = (1.070, 1.070, 0.899, 0.899)$.

The indigenous groups have the lowest per-capita power, and men and women within an ethnic group are identical: gender makes no difference, because the two sexes are of equal size in each ethnic group and ϕ treats them symmetrically. $\Gamma(W) = 0.4615$, with eigenvalues $1, 0.5385, 0.4694, 0.2528$.

(d) The ratio.

$$\frac{\pi_4/n_4}{\pi_1/n_1} = \frac{0.205 + 0.35 \times 0.295}{0.295 + 0.35 \times 0.205} = 0.8405,$$

which equals formula (18.4) with $p = 0.41$ and $\lambda = \lambda_r$, to 10^{-12}. Compare $\lambda_g \lambda_r = 0.105$. The indigenous woman's *edge* to a non-indigenous man is suppressed to 10.5% of its unconstrained weight. Her *power* is 84% of his per capita, and all of that gap comes from group size; gender contributes nothing. Edge suppression and power are different objects: power depends on the whole degree, and symmetric suppression of cross edges hurts the larger group too. This is the first caution of Section 18.3.2.

Exercise 18.4 The formation trap

(a) Set-up. With own weight $w = 1/(1+\lambda)$,

$$\theta^g_{t+1} = \rho\theta^g_t + (1-\rho)\bar{\theta} + \mu\big(w\, r^g + (1-w)\, r^{-g} - r^*\big), \qquad r^A = 0.05 + 0.1\theta^A,\ r^D = 0.03 + 0.1\theta^D,$$

with $\rho = 0.85$, $\mu = 0.3$, $\bar{\theta} = 1$, $r^* = 0.1$. The system is affine, $\boldsymbol{\theta}_{t+1} = J\boldsymbol{\theta}_t + \mathbf{c}$, with $J = \rho I + 0.1\mu \begin{pmatrix} w & 1-w \\ 1-w & w \end{pmatrix}$. Its eigenvalues are $\rho + 0.1\mu$ and $\rho + 0.1\mu(2w-1)$, so $\rho(J) = 0.88 < 1$ for every λ, and the steady state is unique and globally stable (Theorem 10.4).

(b) Steady states, $\lambda = 0.25$ ($w = 0.8$). Solving $(I - J)\boldsymbol{\theta} = \mathbf{c}$: $\theta^{*A} = 1.11364$, $\theta^{*D} = 1.08636$, so $\Delta\theta^* = 0.02727$; returns are $r^A = 0.16136$ and $r^D = 0.13864$.

Closed form. Subtracting the two steady-state equations, with $q := 2w - 1 = \frac{1-\lambda}{1+\lambda}$ and $\Delta r = 0.02 + 0.1\Delta\theta$:

$$(1-\rho)\Delta\theta^* = \mu q(0.02 + 0.1\Delta\theta^*) \quad \Longrightarrow \quad \Delta\theta^* = \frac{0.02\,\mu q}{(1-\rho) - 0.1\,\mu q}.$$

At $q = 0.6$ this gives $0.0036/0.132 = 0.02727$. With fixed returns, Proposition 18.13 would give $\mathcal{M}q \times 0.02 = 2 \times 0.6 \times 0.02 = 0.024$. Endogenous returns amplify it by $1/(1 - 0.1\mathcal{M}q) = 1.1364$: the self-confirming feedback of item 3 of that proposition.

(c) $\lambda = 0.5$ ($w = 2/3$, $q = 1/3$): $\theta^{*A} = 1.10714$, $\theta^{*D} = 1.09286$, and $\Delta\theta^* = 0.002/0.14 = 0.01429$. The gap shrinks by 0.01299, i.e. by 47.6%.

(d) Threshold. $\Delta\theta^* < 0.01$ iff $0.006q < 0.01(0.15 - 0.03q)$, iff $q < 0.0015/0.0063 = 0.238095$, iff $\lambda > (1-q)/(1+q) = 0.6154$. Above this openness the aspiration gap is below 0.01, even though the opportunity gap of 0.02 is unchanged.

(e) Identical opportunity structures. If $r^g = 0.05 + 0.1\theta^g$ for both groups, then $\Delta r = 0.1\Delta\theta$ and the difference equation becomes $\big[(1-\rho) - 0.1\mu q\big]\Delta\theta^* = 0$. The bracket is at least $0.15 - 0.03 = 0.12 > 0$ for every $q \in [0,1]$, so $\Delta\theta^* = 0$ for every λ, including complete segregation ($\lambda = 0$, $q = 1$). Numerically, both groups converge to $\theta^* = 1.12500$, which the script checks for $\lambda \in \{0, 0.25, 0.5, 0.9\}$. *Interpretation:* assortative information does not create a gap; it propagates and amplifies an opportunity gap that exists elsewhere, here in the baseline returns. Removing the opportunity gap closes the aspiration gap without any change in W^I; raising λ shrinks it without any change in opportunities. These are the two levers of Chapter 10.

Common mistake: solving each group's equation with the other group's θ held at the initial value. The two equations must be solved jointly.

Exercise 18.5 The cost of discrimination

Inputs (Example 18.16): two equal groups, $\kappa = 0.5$, $\Omega/N = \Phi/N = 1$. The power ratio is $r = (a_F + \lambda_g c)/(a_M + \lambda_g c)$ with $a_F = 0.454 a_M$ and $c = a_M$: $r = 0.5800$ at $\lambda_g = 0.3$ and 0.7270 at $\lambda_g = 1$. (The chapter rounds the latter to 0.73; with 0.73 exactly the loss would be 0.0061.) With $N^2 \operatorname{Var}(\boldsymbol{\pi}) = ((1-r)/(1+r))^2$:

- $L_{\text{power}}/N = \frac{\kappa}{2} N^2 \operatorname{Var}(\boldsymbol{\pi}) = 0.01767$ at $\lambda_g = 0.3$, with $x_M = 1.2658$, $x_F = 0.7342$ and $\kappa x < 1$;
- 0.00625 at $\lambda_g = 1$;
- cost of discrimination $\Delta L/N = 0.01142$.

Under the illustrative assumption $L_{\text{formation}} = \Psi L_{\text{power}}$ with $\Psi = 2.3$, the total is $3.3 \times 0.01142 = 0.03768$. The script checks the formula against $U(1) - \frac{1}{2}[U(x_M) + U(x_F)]$.

From utility to GDP. The losses are in utils, so a conversion is needed. Two choices, stated as assumptions:

1. *Linear money metric*: divide by the marginal utility at the mean, $U'(1) = 1 - \kappa = 0.5$. The power channel then costs 0.0228 of per-capita GDP, and the total 0.0754.
2. *Equally distributed equivalent income* (Atkinson): find x^{ede} with $U(x^{\text{ede}}) = U(1) - L/N$. Then the cost is $x^{\text{ede}}(\lambda_g = 1) - x^{\text{ede}}(\lambda_g = 0.3)$, which is $0.98758 - 0.96527 = 0.02231$ for the power channel and $0.95959 - 0.88951 = 0.07007$ in total.

So discrimination costs about 2–2$\frac{1}{4}$% of GDP through the power channel, and 7–7$\frac{1}{2}$% once formation is included under $\Psi = 2.3$.

Comparison with the literature. The two numbers measure different things.

- Here Ω is fixed: the bargaining allocation is Pareto efficient, and L_{power} is a *distributional* loss (`docs/conventions.md`, §5). The "share of GDP" is an equivalent-income measure of inequality, not output lost.
- Cuberes and Teignier (2016) estimate *output* losses from the misallocation of women's labour and entrepreneurial talent in an occupational-choice model. Their losses are of the order of several percent to well over ten percent of income per capita, depending on the region. (Check the paper's tables for the exact figures; they are not reproduced here.)

A complete account adds the two: the misallocation loss, which needs Ω to depend on λ_g (outside this chapter's model), and the distributional loss computed here. *Common mistake:* calling L_{power} a GDP loss.

Exercise 18.6 Optimal intervention design

This is an open design question. The numbers below follow from stated illustrative assumptions and the chapter's formulas; they are not estimates.

(a) Three interventions.

1. *Quota* (ΔW^P): a 30% female share in new hires in public administration, construction and transport contracts. Assumed effect: λ_g^P from 0.30 to 0.45.
2. *Mentoring* (ΔW^I): pairing young women with female and male professionals in male-dominated sectors, one year per cohort. Assumed effect: λ_g^I from 0.30 to 0.40.
3. *Media campaign* (ΔW^I): national broadcast and social-media role models. Cheaper, but a weaker change in who observes whom. Assumed effect: λ_g^I from 0.30 to 0.35.

(b) $\Delta\lambda_g$ **and** $\Delta\Gamma$ for each layer, using $\Gamma = 2\lambda/(1+\lambda)$ ($a = c$, Proposition 18.4(a)) from a base $\Gamma = 0.4615$:

- quota: $\Delta\Gamma(W^P) = +0.1592$;
- mentoring: $\Delta\Gamma(W^I) = +0.1099$;

- media: $\Delta\Gamma(W^I) = +0.0570$.

(c) Total effects.

- *Quota.* The enforcement effect on the power ratio is $(a_F + 0.45c)/(a_M + 0.45c) - 0.58 = +0.0434$ (Proposition 18.6(iii)). With $\Psi = 2.3$, the total is $(1+\Psi)E = +0.1434$. That would lift the ratio to 0.72, close to its $\lambda_g = 1$ value of 0.727. This shows how strong the assumption "formation $= \Psi\times$ enforcement" is.

- *Mentoring and media.* At fixed preferences they change no allocation, so $E = 0$ and Ψ is undefined (Remark 9.14(5)). Their effect is pure formation. By Proposition 18.8, the steady-state aspiration gap is proportional to $(1-\lambda_g)/(1+\lambda_g)$, which falls by 20.4% (mentoring) and 10.6% (media).

(d) Sequencing. Following Conjecture 15.14 and the remark "Policy Instruments for Attribute Constraints": mentoring first, then media (cheap, and it shifts the next cohort's aspirations), then the quota. The quota's take-up and the retention of hires depend on aspirations, which is the complementarity of Exercise 15.4. If there is no such complementarity, the quota first is better, because it acts on W^P at once.

Evaluation design, which turns the assumed $\Delta\lambda_g$ into estimates:

- *Design*: a cluster-randomized trial of the mentoring programme across municipalities.

- *Primary outcome*: female business ownership or employment in male-dominated sectors after two years. *Secondary*: network composition, the measured share of cross-gender contacts, which gives a direct estimate of λ_g^I.

- *Power*: baseline $p_0 = 0.12$ (an assumed value), minimum detectable effect 4 points, $\alpha = 0.05$ two-sided, power 0.8. Individual randomization needs $n = (z_{0.975} + z_{0.8})^2 \cdot 2p_0(1-p_0)/\text{MDE}^2 = 2.8016^2 \times 0.2112/0.0016 = 1{,}036$ per arm. With $m = 20$ women per cluster and intra-cluster correlation 0.05, the design effect is $1 + 19 \times 0.05 = 1.95$, giving 2,020 women per arm, i.e. 102 municipalities per arm.

- *Identification threats*: spillovers to control municipalities through regional media (use distant controls or a buffer), and differential attrition of women who migrate for work (track them).

Exercise 18.7 Aspirations data exercise

Data status. No survey with an aspirations module was fetched. Method and expected qualitative result only.

(a) Aspiration gap.

1. Choose a survey with the question "how far in school do you want your child to go?" and with ethnicity (self-identification or mother tongue), household income, location and the child's age. Candidates are a national household survey round with an education module, or a multi-country survey of the MICS or Young Lives type in a country where it exists; check availability before committing.

2. Code aspirations as desired years of schooling, with a binary "university" outcome as a robustness check.

3. Estimate by survey-weighted OLS

   $$\text{asp}_i = \beta_0 + \beta_1 \text{Indig}_i + \beta_2 \log y_i + \beta_3 \text{Urban}_i + \beta_4 \text{Indig}_i \times \text{Urban}_i + X_i'\gamma + \varepsilon_i,$$

   with X containing the child's age and sex, the mother's education and region fixed effects; cluster standard errors by PSU. The conditional gap is β_1 (rural) and $\beta_1 + \beta_4$ (urban).

(b) Consistency with the model. Proposition 18.13 predicts $\Delta\theta^* = \mathcal{M}\frac{1-\lambda_r}{1+\lambda_r}\Delta\bar{r}$. Two predictions:

- the conditional gap is larger where W^I is more assortative (lower λ_r);
- the gap is zero where opportunity structures are equal (Exercise 18.4(e)).

Measure λ_r directly where possible: the share of the mother's reported contacts or information sources from the other group, or the local index of residential or school segregation. Then interact the indigenous indicator with that measure instead of urban/rural.

(c) Urban La Paz vs. rural Altiplano. The model predicts a smaller conditional gap in urban La Paz (higher λ_r), i.e. $\beta_4 < 0$ if $\beta_1 < 0$. The return gap $\Delta\bar{r}$ may also differ between the two places, so a smaller urban gap is not by itself evidence for the W^I channel. Control for local returns to schooling by group, e.g. the Mincer returns estimated from the same survey. Identification threats:

- selective migration of high-aspiration indigenous families to the city (use the mother's birthplace, or restrict to non-migrants);
- self-identification changing with urban residence (use mother tongue as an alternative classification).

Expected result: a negative conditional indigenous gap that is smaller in urban areas. Without data, no magnitude is given.

Chapter 19

Scale Invariance

Every number below is printed by `python3 code/solutions/sol19.py`. Sourced inputs come from `data/processed` (retrieval date 2026-11-05; `data/README.md`). Numbers without a source are illustrative and are labelled as such.

Exercise 19.1 Three-scale comparison

(Global column) *Data and method.* Growth Lab bilateral goods trade, HS92, 2019 (`data/processed/atlas_bilateral_selected_years.csv.gz`). Set K_{ij} to the imports of i from j with a zero diagonal, drop countries with zero imports or exports inside the set until none is left, and put $W_G = D^{-1}K$ (Definition 16.1). The script recomputes the matrix from the raw flows and matches `global_wp_summary.csv` to 10^{-9}.

	Global ($n = 230$)	National ($n = 11$)
λ_2 (real in both)	0.5478	0.5326
Γ	0.452	0.467
$\pi_{\max}$	0.1302 (CHN)	0.2702 (Other services)
$\pi_{\min}$	1.38×10^{-8} (HMD)	0.0051 (Public administration)
$\max \pi / \min \pi$	9.4×10^{6}	53.2
N_{eff} (N_{eff}/n)	22.58 (0.098)	6.48 (0.589)
$G(\boldsymbol{\pi})$	0.854	0.445

The US/Laos ratio used in Example 19.2 is 193.1, but the true global minimum is the Heard and McDonald Islands (HMD), an uninhabited territory with recorded imports. So $\max \pi / \min \pi$ is 9.4×10^6, and even the ratio of the 90th to the 10th percentile is 7986.5. A max/min ratio is driven by the smallest node in the data set. Concentration should be reported as N_{eff}/n or $G(\boldsymbol{\pi})$, as the conventions require.

(National column) INE supply–use tables 2019 (base 2017), domestic flows at basic prices, 12-sector book aggregation, input shares $W^P = D^{-1}A$ (`bolivia_Z/Z_2017_2019_book12.csv`). Domestic services has no intermediate flows and drops out, so $n = 11$. The values are in the table; $|\lambda_3| = 0.313$. The national gap depends on how it is built. Across the 12 variants in `bolivia_io_summary.csv` (book12, book12_isic and detail39; input or sales shares; diagonal kept or zeroed), Γ_N ranges from 0.331 to 0.694. The two measured gaps are almost equal: $\Gamma_N/\Gamma_G = 1.034$.

(σ_B) Not computed: the household microdata were not fetched. *Method.*

1. National: take the INE Encuesta de Hogares 2019 (ANDA portal) and per-capita household income y_h with expansion factors w_h. Assign each household to a block (for instance the sector of the head, crossed with urban/rural). Compute the between-block variance $V_B = \sum_b s_b(\bar{y}_b - \bar{y})^2$ and the total variance V, and report $\sigma_B = V_B/V$ (Definition 17.3).

2. Global: apply the same formula to harmonized microdata for many countries (LIS, or the PIP distribution files), with countries as blocks.

Expected result: σ_B rises with the number of blocks, since refining a partition never lowers V_B (the refinement check in `code/ch17.py`). So σ_B values at the two scales are comparable only for a stated, fixed partition. The book's 0.55 and 0.62 remain illustrative.

(Individual scale) No data set observes W_I for 10^6–10^8 persons. A feasible design has two parts.

1. Aggregated relational data (ARD) in a national household survey: "How many people do you know who are [indigenous / women / work in mining / live in El Alto]?" A latent-space model recovers group-level ties and centralities from such answers (Breza et al., 2020).

2. Complete-network censuses in a stratified random sample of communities, in the manner of the village networks of Example 21.6. These validate the ARD estimates and supply within-community structure.

Sample size: to estimate a cross-group tie share near 0.2 within ± 0.02 at 95%, simple random sampling needs $1.96^2(0.2)(0.8)/0.02^2 = 1537$ respondents per group, and 3073 with a design effect of 2 for clustering. The sampling, censoring and misreporting problems of Section 21.4 apply.

(Why a lower fine-scale Γ is not evidence of fragmentation) Under exact lumpability, Proposition 19.5 gives $WV = V\bar{W}$. Every eigenvector x of $\bar{W}$ then lifts to the eigenvector Vx of W, so $\sigma(\bar{W}) \subseteq \sigma(W)$ and $\Gamma(\bar{W}) \geq \Gamma(W)$ for every lumpable W. The inequality holds whatever the economics, so observing it tells us nothing. Without lumpability no ordering holds in either direction. Take

$$W_a = \begin{pmatrix} 1-a & a & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \det(W_a - \lambda I) = -(\lambda - 1)(\lambda^2 + a\lambda + a),$$

so the non-unit eigenvalues are complex with $|\lambda_2| = \sqrt{a}$, and $\Gamma(W_a) = 1 - \sqrt{a}$. The stationary vector is $\pi = (1, a, a)/(1 + 2a)$. With blocks $\{1\}$ and $\{2, 3\}$, the π-weighted aggregate (which here equals the unweighted one) is $\bar{W} = \begin{pmatrix} 1-a & a \\ 1/2 & 1/2 \end{pmatrix}$, with $\lambda_2 = \frac{1}{2} - a$. Rows 2 and 3 place different mass on block $\{2, 3\}$ (1 and 0), so W_a is not lumpable. For $a = 0.25$, $\Gamma_{\text{fine}} = 0.500 < \Gamma_{\text{coarse}} = 0.750$. For $a = 0.01$, $\Gamma_{\text{fine}} = 0.900 > \Gamma_{\text{coarse}} = 0.510$: aggregation *lowers* the gap. The real tables show the same thing.

- Bolivia's 39-industry matrix has $\Gamma = 0.410$ against 0.467 for its 12-sector aggregate.
- In ICIO 2019, the 45-industry and 12-sector values are KOR 0.455 $\to$ 0.511, DEU 0.372 $\to$ 0.544, MEX 0.560 $\to$ 0.649 and BRA 0.436 $\to$ 0.496, but USA 0.465 $\to$ 0.463.

A comparison of Γ across scales is therefore uninformative unless it is made on a common node set or after an explicit aggregation. *Common mistake:* reading $\Gamma_I \approx 0.10 < \Gamma_N = 0.467$ as a "more fragmented" individual scale.

Exercise 19.2 Power nesting: three Bolivian agents

(Set-up) By Proposition 19.7, $\pi_{I,i} = \pi_{G,c}\, \pi_{N,s|c}\, \pi_{\text{within},i|s,c}$. Each factor is a share that sums to one over its level. With L_c workers in c, of whom a fraction $e_{s|c}$ work in s, the within-sector share of person i is $w_i/(e_{s|c}L_c)$, where the per-capita weights w_i have mean one in the sector. Hence

$$\pi_{I,i} = \frac{\pi_{G,c}}{L_c} \cdot \frac{\pi_{N,s|c}}{e_{s|c}} \cdot w_i.$$

All three agents are Bolivian, so the country factor $\pi_{G,\mathrm{BOL}}/L_{\mathrm{BOL}}$ cancels from every ratio.

(Sector factors (sourced)) π_N comes from the 2019 INE W^P (Exercise 19.1) and employment shares from ILOSTAT 2019. Hydrocarbons are pooled with mining (ISIC B) because ILOSTAT does not separate them.

Sector	π_N	e	π_N/e	female share
Hydrocarbons and mining (B)	0.0767	0.0134	5.737	0.065
Agriculture (A)	0.0911	0.2741	0.332	0.460
Personal services (I, R, S)	0.0211	0.1067	0.198	0.689

(Within factors (illustrative)) Chapter 18's calibration is relative to a non-indigenous man (Example 18.11): an indigenous person has 0.57 of his per-capita power and a woman 0.58. If the two factors are separable, the reference weights are $\rho_a = 1$, $\rho_b = 0.57$ and $\rho_c = 0.57 \times 0.58 = 0.3306$.

(Ratios) The per-capita indices are 5.7369, 0.1895 and 0.0654. The pairwise ratios, with the share of the log gap due to the sector factor, are:

Pair	ratio	sector factor	within factor	sector share of log gap
(a)/(b) gas man / farmer	30.27	17.26	1.754	0.835
(a)/(c) gas man / vendor	87.67	28.99	3.025	0.753
(b)/(c) farmer / vendor	2.90	1.68	1.724	0.488

The sector factor dominates the two comparisons that involve the gas sector. Between the farmer and the vendor the two factors contribute about equally. The ratio $a/c = 87.67$ is the chapter's "about 88" (Example 19.9). The chapter rounds the within factor to 0.33, but the conclusion does not depend on that rounding.

(Two refinements a strong answer adds)

1. *Pooling.* Hydrocarbons alone have $\pi = 0.0392$. Their employment is unknown but below the pooled 0.0134, so their own factor is at least $0.0392/0.0134$, that is 2.93, and a/c is at least 44.8. The pooled 5.7 is a property of "hydrocarbons and mining", not of gas.

2. *Mean-one normalisation.* The formula needs $w_i = \rho_i/m_s$, where m_s is the mean of ρ in sector s. Using the reference ratios directly treats every sector as if all its workers were non-indigenous men. Correcting for gender composition alone (ILOSTAT female shares; ethnic composition by sector was not fetched)

gives $m_s = 1 - f_s + 0.58 f_s = 0.973$, 0.807 and 0.710. The ratios become $a/b = 25.11$, $a/c = 64.05$ and $b/c = 2.55$. The ordering survives and the sector factor still dominates, but the gas/vendor gap shrinks by about a quarter.

Common mistake: comparing π_N itself across sectors (gas 0.039 against personal services 0.021) without dividing by employment. The sector factor is high in gas because the sector employs very few people, not because it is central in W^P.

Exercise 19.3 The global comparison

(Country factor per worker (sourced)) The country shares come from the 2019 trade matrix and the labour forces from WDI (SL.TLF.TOTL.IN, 2019): Germany $\pi_G = 0.082407$ with 43.98 M workers, Brazil 0.013642 with 106.26 M, Bolivia 0.000526 with 5.95 M. Per worker and relative to Bolivia, the country factor is 21.19 for Germany and 1.452 for Brazil.

(Ratios) The sector and within factors of the two foreign workers are set to one: a sector whose power share equals its employment share, and the reference worker. This is an assumption, because the repository has sector employment only for Bolivia. The vendor keeps 0.198 and 0.3306 from Exercise 19.2.

	ratio	country share	sector share (of the log gap)	within share
German engineer / Potosí vendor	323.9	0.528	0.280	0.191
Brazilian factory worker / vendor	22.2	0.120	0.523	0.357
German engineer / Brazilian worker	14.6		(country factor only)	

Against the German engineer the global scale contributes most, about half of the log gap. Against the Brazilian worker the national (sector) scale contributes most: Brazil's country factor per worker is only 1.45 times Bolivia's. The chapter's 325 uses $1/0.33 = 3.03$ in place of $1/0.3306 = 3.025$. If Chapter 18's illustrative gender ratio 0.58 is applied to the German engineer as well, her ratio falls to 188.

(Forgetting the workforce) Using π_G in place of π_G/L multiplies each ratio by L_c/L_{BOL}. The German ratio becomes 2396 (a factor 7.40 too large) and the Brazilian one 397 (a factor 17.87 too large). The error is larger for Brazil, whose workforce is larger, and it reverses the ranking of the two foreign workers' advantages: Germany/Brazil becomes 6.04 instead of 14.6. Shares at the country level are totals, and per-capita comparisons must divide by group size at every level.

Exercise 19.4 Five processes as matrix properties

(Construction ([ILLUSTRATIVE])) Nodes C1, C2 (core), S1, S2 (semi-periphery) and P1, P2 (periphery: an enclave and a subsistence node):

$$W = \begin{pmatrix} 0.50 & 0.46 & 0.03 & 0 & 0.01 & 0 \\ 0.46 & 0.50 & 0 & 0.03 & 0.01 & 0 \\ 0.04 & 0 & 0.60 & 0.36 & 0 & 0 \\ 0 & 0.04 & 0.36 & 0.60 & 0 & 0 \\ 0.30 & 0.20 & 0 & 0 & 0.45 & 0.05 \\ 0.15 & 0.10 & 0.05 & 0 & 0.20 & 0.50 \end{pmatrix}.$$

W is row-stochastic and irreducible (the script checks $(I + W)^6 > 0$). Its eigenvalues are 1, 0.9306, 0.5718, 0.3676, 0.2458 and 0.0342, all real. The Perron vector is $\boldsymbol{\pi} = (0.2826, 0.2814, 0.2122, 0.2121, 0.0106, 0.0011)$, with $N_{\text{eff}} = 4.01$ of 6 (sanity check: $\sum \pi_i = 1$).

(The five phenomena)

1. *Underdevelopment* (low Γ from weak coupling). $\Gamma(W) = 1 - 0.9306 = 0.0694$. The bottleneck is the core–semi link. The semi block has stationary outflow 0.01697 and conductance $\Phi = 0.0400$, and Cheeger's bound holds: $1 - \lambda_2 = 0.0694 \leq 2\Phi = 0.0800$.

2. *Development* (high and rising Γ). The core's own renormalized block has $\Gamma_{\text{core}} = 0.958$. Moving weight t from the diagonal to the core–semi edges raises the whole economy's Γ to 0.168, 0.264 and 0.407 at $t = 0.05$, 0.10 and 0.20, while N_{eff} barely moves (4.01 $\rightarrow$ 4.08). Integration rises without deconcentration (Proposition 6.16).

3. *Dependence* (asymmetric cross-block rows). Periphery rows place mean weight 0.375 on the core, and core rows place 0.010 on the periphery. The periphery's power is $\pi_P = 0.0117$, against its population share $2/N = 0.333$.

4. *Marginality* ($\pi_i \approx 0$). $\pi_{P2} = 0.0011$, against $1/N = 0.167$. Its column sum is 0.550, but 0.50 of that is its own self-loop: other nodes place only 0.050 on it, all of it from P1.

5. *Spatial inequality in integration.* The renormalized regional blocks have $\Gamma_r =$ 0.958 (core), 0.750 (semi) and 0.386 (periphery), so $\text{Var}_r(\Gamma_r) = 0.0560$.

(Are all five visible in the same matrix?) Yes, but only because the dictionary of Definition 19.10 is applied to *different submatrices*. "Development = high Γ" and "underdevelopment = low Γ" cannot both describe the whole matrix: the first is a property of the core block, the second of the full W. Two further points come out of the computation. First, the small Γ is produced by the weak core–semi coupling, not by the periphery: the periphery's rows are strongly coupled to the core, but it carries almost no stationary mass, so it does not slow mixing. Second, dependence and marginality are column (in-weight) properties, while Γ is a spectral property. That is why a policy that raises Γ (item 2) leaves the periphery's π essentially unchanged. *Common mistake:* building the periphery as a weakly coupled block and then reading its small π off the small Γ. The two are independent coordinates.

Exercise 19.5 Cross-scale feedback simulation

(Model ([ILLUSTRATIVE])) There are 3 countries, 3 sectors per country and 2 agents per sector, an advantaged type A and a disadvantaged type D: 18 nodes. Country c has three state variables: national integration e_c, individual openness λ_c and global attractiveness (complexity) a_c. The raw intensities are:

- 1 on the self-loop;
- λ_c to the other type in the same sector;
- e_c to the same type in another domestic sector, and $\lambda_c e_c$ to the other type there;
- $g\, a_{c'}$ to a foreign agent of type A in country c', and $g\, a_{c'}\lambda_{c'}$ to one of type D, with $g = 0.05$.

$W = D^{-1}K$ has exactly the nested form of Definition 19.6. Each period the script computes $\boldsymbol{\pi}$, the country shares $\pi_{G,c}$, and $\Gamma_{N,c}$, the gap of country c's stochastic complement (Proposition 19.7). The four links of Conjecture 19.11 are then implemented as targets with logistic response, $\sigma(z) = 1/(1+e^{-z})$:

$$
\begin{aligned}
e_c^* &= 0.02 + 0.98\,\sigma\big(\beta[(3\pi_{G,c} - 1) + (\lambda_c - 0.5)]\big),\\
\lambda_c^* &= 0.05 + 0.95\,\sigma\big(\beta[\Gamma_{N,c} - 0.5]\big),\\
a_c^* &= 0.05 + 0.95\,\sigma\big(\beta[\Gamma_{N,c} - 0.5]\big).
\end{aligned}
$$

The first line carries the global→national and individual→national links, the second national→individual and the third national→global. Each variable then moves

part of the way to its target, by partial adjustment $z_{t+1} = z_t + 0.2(z^* - z_t)$ for $z \in \{e, \lambda, a\}$. The parameter β is the loop gain. The initial states are core $(0.9, 0.9, 0.9)$, semi-periphery $(0.5, 0.5, 0.5)$ and periphery $(e, \lambda, a) = (0.05, 0.1, 0.1)$; the run lasts 400 periods.

(Results)

- $\beta = 1$ and $\beta = 2$: all three countries converge to the same state (at $\beta = 2$, $e = 0.594$, $\lambda = a = 0.672$, $\Gamma_N = 0.821$, $3\pi_G = 1$). No circle.
- $\beta = 3$: the semi-periphery joins the core, and the periphery is trapped at $e = 0.114$, $\lambda = 0.462$, $3\pi_G = 0.289$ and $\Gamma_N = 0.412$, against 0.877 in the core.
- $\beta = 4$: periphery $e = 0.039$, $3\pi_G = 0.153$, $\Gamma_N = 0.326$.
- $\beta = 6$: periphery $e = 0.021$, $\lambda = a = 0.305$, $3\pi_G = 0.097$. Its type-D agents hold 0.603 of type A's per-capita power, against 0.997 in the core.

At the trapped state the script checks the nested factorisation $\pi_{I,i} = \pi_{G,c}\pi_{N,s|c}\pi_{\text{within},i}$ exactly.

(When does the circle emerge and break?) Bisection on β gives a critical loop gain $\beta^* = 2.704$. Below it, all three countries converge to the same state from these initial conditions. Above it, the periphery's low state is a second stable equilibrium: the vicious circle. At $\beta = 6$ the script applies a one-shot push to the periphery's state and asks whether it escapes:

- a push on e alone, on λ alone or on a alone fails even when it raises the variable to its maximum of 1, because the other two scales pull it back;
- a simultaneous push of the same size on all three succeeds once it reaches 0.497.

This is the content of Conjecture 19.11: with strong loops, piecemeal interventions fail and a coordinated one succeeds. *What is general and what is not.* Bistability above a threshold loop gain is generic for loops with sigmoidal response (the same mechanism as Proposition 13.9). The values $\beta^* = 2.70$ and 0.497 are properties of this calibration only.

Exercise 19.6 The binding scale

Ill-posed as stated. Γ_G is a property of the whole global matrix, so there is one value, 0.452 in 2019, not one per country. The country-specific global object is the

position $\pi_{G,c}$, best expressed per worker. We use it in place of a country-specific Γ_G.

(Global position (sourced)) Per worker and relative to the world (WDI labour force: world 3.509 bn), $\pi_{G,c}/(L_c/L_W)$ is 0.310 for Bolivia ($\pi_G = 0.00053$, 5.95 M workers), 3.972 for Korea (0.03233, 28.56 M) and 1.120 for Uruguay (0.00055, 1.72 M). Uruguay is small in total but not peripheral per worker.

(National (sourced)) Γ_N is 0.475 for Bolivia (ISIC-based 12 sectors; 0.467 on the book's sectors) and 0.511 for Korea (OECD ICIO 2023). Uruguay is not in ICIO, so its Γ_N is not available. A national I-O table from the Banco Central del Uruguay would be needed; it was not fetched.

(Individual (partly sourced)) *Method.* Occupational segregation gives a group-level exposure matrix. If workers interact uniformly within their ISIC section, the probability that a woman's co-worker is a woman is $Q_{FF} = \sum_s (f_s/F)(f_s/n_s)$, and likewise Q_{MM}. The 2×2 matrix Q is row-stochastic, with Perron vector equal to the sex shares of employment and $\lambda_2 = Q_{FF} + Q_{MM} - 1$. For Bolivia, ILOSTAT 2019 (21 ISIC sections) gives a Duncan index $D = 0.312$, $Q_{FF} = 0.564$, $Q_{MM} = 0.635$, $\lambda_2 = 0.198$ and $\Gamma_I^{\text{gender}} = 0.802$. In the two-equal-group model of Proposition 18.4(a) with $a = c$, $\lambda_2 = (1 - \lambda_g)/(1 + \lambda_g)$, so this corresponds to $\lambda_g = 0.669$, far above Chapter 18's illustrative 0.3. Ethnicity needs EH microdata (not fetched). Korea and Uruguay need the same ILOSTAT tables, which were not fetched.

(Which scale binds?) Definition 19.13 ranks scales by $-c_s^{-1}\partial L_{\text{total}}/\partial \Gamma_s$. None of these data identifies a marginal loss or a cost, and the Γ values at different scales are not comparable (Exercise 19.1). The data can only rule some answers out.

- *Bolivia.* Neither Γ_N (0.475) nor the gender-exposure Γ_I (0.802) is low. What is low is the global position per worker (0.310) and export complexity (ECI -1.30 in 2019). On these data the case for a binding *national aggregate* is weak. The defensible reading is the chapter's own caveat: the constraint runs through the export enclave and complexity.

- *Korea.* Its global position per worker (3.97) rules out the global scale. The data cannot discriminate between the national and individual scales.

- *Uruguay.* Per worker it sits above the world average (1.12), which weakens the chapter's suggestion that the global scale binds.

Whether these readings match the policy history (Korea's export discipline, Bolivia's rent capture in 2006) can be argued only qualitatively. The test the exercise

has in mind needs time series of all three scales.

Exercise 19.7 Multi-scale structural plan for Bolivia

An open design question. Sourced inputs are marked; the operator magnitudes are design choices.

(a) *Global: three products.* The current basket sits at low product complexity. The 2019 PCI (Growth Lab HS92) is -2.226 for natural gas (2711), -1.670 for zinc ores (2608), -2.809 for gold (7108) and -1.808 for soya beans (1201). One value chain offers a ladder of increasing PCI that starts from a national resource:

- lithium carbonate (2836), PCI -0.573;
- lithium oxides and hydroxides (2825), -0.276;
- electric accumulators, including Li-ion (8507), $+0.549$.

Feasibility should be judged by relatedness density (Stylized Fact 15.15), which needs the country $\times$ product RCA matrix. That matrix was not computed here (`data/README.md`, §9). *Effect on π_G:* across the 230 economies π_G tracks world export shares almost exactly (correlation 0.9974). If every economy's imports from Bolivia doubled, π_{BOL} would go from 0.000526 to 0.001046 (a factor 1.991), while Γ_G would move from 0.4522 to 0.4520. Bolivia can change its own position; it cannot change global integration.

(b) *National: the operator sequence of Conjecture 15.14, quantified on the real 2019 W^P* (11 sectors; baseline $\Gamma = 0.4674$, $\pi_{\text{hydro}} = 0.0392$, $N_{\text{eff}} = 6.480$).

1. *Information.* It changes W^I, not W^P, so $\Delta\Gamma(W^P) = 0$ by construction. Its effect is on formation.
2. *Education and infrastructure.* These enable the next two steps; their direct ΔW^P is not identified by any data here.
3. *Forward linkage (ISI).* Every sector replaces 10% of its imported intermediate inputs by domestic gas-based inputs: 4249 million Bs, or 0.164 of hydrocarbons output. Then $\Gamma = 0.4670$ (-0.0004) and $\pi_{\text{hydro}} = 0.0582$ ($+0.0190$).
4. *Backward linkage.* Hydrocarbons double their domestic purchases ($0.225 \to 0.449$ per unit of output), with the new half bought from agriculture and manufacturing. Then $\Gamma = 0.4648$ (-0.0026) and $\pi_{\text{hydro}} = 0.0378$ (-0.0014).
5. *Both together:* $\Gamma = 0.4625$ (-0.0049), $\pi_{\text{hydro}} = 0.0558$ ($+0.0166$), $N_{\text{eff}} = 6.788$.

On Bolivia's measured matrix, linkage-building operators *lower* Γ slightly, because they concentrate flows on gas, agriculture and manufacturing. The targets should therefore be the enclave's linkages and π_{hydro}, not Γ_N.

(c) *Individual.* The instruments are those of Remark 18.17: mentoring and anti-discrimination for λ_g, bilingual media and schooling for λ_r. Using Chapter 18's formulas (Proposition 18.6(ii) with $p = 0.41$, (iii) with $a_F = 0.454$ and $c = a_M = 1$; Proposition 18.13):

(λ_g, λ_r)	$(0.30, 0.35)$	$(0.50, 0.50)$	$(0.70, 0.70)$
women/men per-capita π	0.580	0.636	0.679
indigenous/non-indigenous (homophily only)	0.840	0.887	0.938
aspiration-gap factor $(1-\lambda_g)/(1+\lambda_g)$	0.538	0.333	0.176
two-group $\Gamma = 2\lambda_g/(1+\lambda_g)$	0.462	0.667	0.824

(d) *Combined* $\Delta\Gamma$ *and timeline.* Global: $\Delta\Gamma_G \approx 0$ (-0.0001 even when the world's imports from Bolivia double). National: -0.0049, the wrong yardstick, as shown in (b). Individual: $+0.205$ at the group level for $\lambda_g = 0.3 \to 0.5$. The scales move on different clocks.

- At $\rho = 0.85$ the formation half-life is $\ln 2 / \ln(1/0.85) = 4.27$ generations, so aspiration gaps take generations to halve.
- Measured structural velocities are small: Bolivia -0.036 per decade over 1990–2013.
- A credible horizon is therefore 30 years or more, with intermediate targets set on π_{hydro}, domestic input intensity, ECI and λ_g rather than on Γ (compare the plan of Example 26.7).

(e) *Political feasibility.* Under the combined operators of (b), items 3 and 4, the gainers in π are agriculture ($+0.0208$), hydrocarbons ($+0.0166$), manufacturing ($+0.0065$) and mining ($+0.0003$). The losers are other services (-0.0182), financial services (-0.0098), transport (-0.0081), commerce (-0.0039), and smaller amounts in construction, personal services and public administration. In the class terms of Chapter 14 this is the ISI alliance $\{B, M, W\}$ (domestic bourgeoisie, middle class, workers) of Definition 14.16, here drawn from the gas, agricultural and manufacturing sectors, with the individual-scale reforms bringing in women and indigenous workers. Under power-bounded feasibility (Definition 14.2) the coalition is weak. The gainers hold $\sum \pi = 0.327$ today, against 0.673 for the losers. The plan is feasible only if the formation steps come first and shift preferences, which is the logic of the sequencing conjecture, or if the state compensates the service sectors.

Chapter 20

Social Accounting Matrices as Weight Matrices

Every number below is printed by `python3 code/solutions/sol20.py`. Exercise 20.1 asks for a Bolivian SAM built from INE, EH and fiscal data. The production, value-added and import blocks can be built from INE's 2019 supply–use tables, which are in the repository (retrieved 2026-11-05). The household and fiscal blocks need EH microdata and MEFP tables, which were not fetched. Exercises 20.2–20.6 refer to "your SAM". They are solved on the chapter's illustrative SAM (Example 20.7), and the real production block is reported alongside wherever it answers part of the question.

Exercise 20.1 Build a SAM

(Method, step by step)

1. *Production × production.* Start from the INE supply–use tables for 2019 (base 2017; `ine_sut_long.csv`). Remove net product taxes and trade margins, allocate imports proportionally, and form industry-by-industry domestic flows with the fixed product-sales assumption (`data/compute/compute_bolivia_io.py`). Then aggregate the 12 book sectors to the template's four accounts: Gas/Mining (hydrocarbons and mining), Agriculture, Manufacturing, and Services (everything else, including construction and public administration).

2. *Factors ← production.* The use table's value-added rows split gross value added (B1b) into compensation of employees (D1) → Labour, operating surplus (B2) → Capital, mixed income (B3), and taxes on production net of subsidies → Government.

3. *Government and RoW ← production.* Net product taxes on intermediates (output minus domestic inputs, imported inputs and value added) go to Government, and imported intermediates go to RoW.

4. *Production ← final demand.* Exports are a RoW payment. Domestic final demand (household and government consumption, investment) is the remainder of output after intermediate sales and exports.

5. *Households, Government, RoW.* Factor income by household type and household consumption come from the EH 2019 income and expenditure modules. Taxes and transfers come from MEFP. Remittances come from the balance of payments; WDI gives the total, 9.30 bn Bs in 2019.

6. *Balance.* Reconcile the independent sources with RAS or cross-entropy (Appendix 20.6).

(Results for the blocks that can be built (INE 2019, million Bs))

Paying column →	Gas/Mining	Agriculture	Manufacturing	Services
Gas/Mining (domestic input)	319	278	16917	4593
Agriculture	128	5444	22006	3195
Manufacturing	1756	2978	12494	29019
Services	9554	7753	12433	59493
Labour (D1)	5762	4457	12933	104361
Capital (B2)	28779	3104	18719	60946
Mixed income (B3)	111	18258	6089	44861
Gov: net production taxes	271	55	355	905
Gov: net product taxes on inputs	639	734	1780	5954
RoW: imported inputs	3552	3412	10779	25529
Column total = gross output	50872	46473	114506	338857

On the receipt side, intermediate sales are 22107, 30773, 46247 and 89234; exports are 27905, 1122, 24783 and 10700; and domestic final demand is 861, 14578, 43476 and 238923.

Balance checks.

- The value-added identity (D1 + D29 − D39 + B2 + B3 = B1b) holds to 2.8×10^{-16}.
- INE's own supply = use check has a maximum gap of 7.3×10^{-11}, and the output identity (intermediate plus final sales equal output) holds to 7.8×10^{-16} (`bolivia_io_checks.csv`).

- Every production account therefore balances, row against column.

The household, government and capital rows and columns cannot be balanced until the EH and MEFP blocks are added.

(What the real blocks already say)

- *Gas/Mining.* Per unit of output it pays 0.566 to capital and 0.113 to labour, buys 0.231 of domestic inputs, imports 0.070, and exports 0.549 of its output. Hydrocarbons alone export 70.1% and buy 0.225 of domestic inputs.
- *Manufacturing* pays 0.113 of output to labour and buys 0.558 of domestic inputs.
- *Agriculture.* Mixed income is 0.393 of its output. Its "labour" is self-employment, so whether B3 is booked to Labour or to Capital changes the agricultural column completely.

(Two design points the template hides) The template has no savings–investment account. In real data investment demand is large and has to be paid by some account, so a real SAM needs an eleventh account (Capital account), or the RoW column has to carry net saving, as in the toy. And the household split needs EH microdata: poverty status is a household attribute, so the Labour $\rightarrow$ HH-P cell cannot be read off any national-accounts table.

Exercise 20.2 Row-normalize and compute Γ

(Spectrum) For $W = \mathrm{diag}(y)^{-1}T$ on the illustrative SAM, the eigenvalues ordered by modulus are 1, $0.013 \pm 0.598i$, -0.486, 0.400, $-0.064 \pm 0.232i$, $0.202 \pm 0.017i$ and -0.041. So $\Gamma_{\mathrm{SAM}} = 1 - 0.598 = 0.402$.

(Production block) The toy production block has $\Gamma = 0.391$ for receipt shares (row-normalized T_{PP}) and 0.248 for input-cost shares $W^P = D^{-1}A$. The smallest gap in this SAM is therefore that of W^P, and the full SAM mixes faster than its own production block (Proposition 20.9(3): no general ordering). The real INE 2019 production block, aggregated to the same four accounts, has $\Gamma = 0.763$ for input shares ($\lambda_2 = 0.237$, real) and 0.801 for receipt shares. At 11 sectors the input-share gap is 0.467, so aggregation raises it, as in Exercise 19.1.

(Complex eigenvalues and the circular flow) The leading pair has argument 88.8°, so it rotates by about a quarter turn per step, a period of 4.06 steps. The phases of its right eigenvector, measured relative to Gas, fall into blocks:

- production accounts: $0°$, $-8°$, $30°$, $82°$;
- factors: $-42°$, $-74°$;
- households: $-135°$, $-152°$;
- Government and RoW: $119°$, $117°$.

The rotation runs around the four-stage circuit production $\to$ factors $\to$ households $\to$ Government/RoW $\to$ production. A pure three-stage cycle would have eigenvalues 1 and $-0.5 \pm 0.866i$, all of modulus 1, and $\Gamma = 0$. Intra-block purchases and leakages damp the rotation to modulus 0.598. A small Γ_{SAM} can therefore signal a strong circular flow rather than fragmentation. Non-real eigenvalues also certify asymmetry: $T \neq T'$ (the largest asymmetry is the Cap $\to$ HH-NP cell, 6.3 against 0). If $T = T'$, W would be similar to the symmetric matrix $Y^{-1/2} T Y^{-1/2}$ and its spectrum would be real.

(Perron vector = account totals) *Proof.* With $y = T\mathbf{1} = T'\mathbf{1}$, we have $y' \operatorname{diag}(y)^{-1} T = \mathbf{1}'T = y'$. So $y/\mathbf{1}'y$ is stationary. W has the zero pattern of T and is irreducible, so the stationary vector is unique. Numerically, for (Gas, …, RoW),

$$\boldsymbol{\pi} = (0.0861, 0.0614, 0.1238, 0.1518, 0.1119, 0.0764, 0.0732, 0.1485, 0.0635, 0.1033),$$

which equals $y/\mathbf{1}'y$ with $\mathbf{1}'y = 92.9$ to machine precision (sum = 1), and $G(\boldsymbol{\pi}) = 0.179$. *Common mistake:* reading this π as power. It is turnover (Proposition 20.4).

Exercise 20.3 SAM multipliers

(Computation) On the illustrative SAM, with endogenous accounts production, factors and households and exogenous Government and RoW, $\rho(A_n) = 0.801$. $M = (I - A_n)^{-1}$ exists, and $M = M_3 M_2 M_1$ holds to machine precision (Proposition 20.11). Stone's additive components, as column sums by injected account:

	Gas	Agr	Mfg	Svc	Lab	Cap	HH-P	HH-NP
initial	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
own-block	0.53	0.99	1.16	1.03	0.00	0.00	0.00	0.00
open-loop	1.62	1.73	1.52	1.59	2.06	1.82	2.14	1.01
closed-loop	2.02	2.48	2.13	2.23	1.96	1.52	2.28	1.06
total	5.18	6.20	5.82	5.85	5.02	4.34	5.42	3.08

Sanity check. The leakage coefficients $\ell_j = 1 - \sum_i (A_n)_{ij}$ are Gas 0.125, Agr 0.053, Mfg 0.122, Svc 0.099, Lab and Cap 0, HH-P 0.25 and HH-NP 0.645, and $\ell' M = \mathbf{1}'$ exactly: every injected unit eventually leaks.

(Answers) *Highest total multiplier:* Agriculture, 6.20, both among the production sectors and over all eight accounts. *Largest open-loop component:* Agriculture among the production sectors (1.73), and HH-P over all accounts (2.14). A unit given to poor households reaches production and factors more strongly than a unit injected into any sector, because poor households leak less (0.25 against 0.645). For a gas injection the 5.18 splits into 2.68 in production, 1.25 in factors and 1.25 in households.

(Real comparison and a warning) The INE 2019 own-block (Type I output) multipliers, the column sums of $(I - A)^{-1}$ for the real four-account production block, are Gas/Mining 1.348, Agriculture 1.550, Manufacturing 1.857 and Services 1.446. The toy's own-block multipliers $\mathbf{1}' M_1$ are 1.534, 1.989, 2.161 and 2.026. The ranking of gas at the bottom survives in the real data, but the real own-block effects are smaller. Multiplying the column sums of M_1, M_2, M_3 is not a decomposition: for gas it gives 5.69, not 5.18.

Exercise 20.4 The gas enclave in the SAM

(The path) Per unit of gas output (Bs per Bs), the first stage pays:

- domestic intermediates: 0.2625 to gas itself, 0.0375 to manufacturing and 0.025 to services;
- labour 0.10 and capital 0.45;
- government 0.125, and RoW 0.

Following the dominant branch of a 1B Bs injection:

Stage	mass entering	leakage rate (Gov + RoW)	passed on along the path
Gas	1.000	0.125	0.450 to Capital
Capital	0.450	0.000	0.887 to HH-NP
HH-NP	0.399	0.645	0.355 to domestic goods
back in production	0.142		

Households' leakage of 0.645 splits into 0.174 to Government and 0.471 to RoW. The capital stage leaks nothing in this SAM because it has no capital-to-RoW cell.

Foreign profit repatriation is absent by construction; in real data it would be a Cap $\to$ RoW cell taken from the balance of payments (primary income), which was not fetched. For comparison, labour passes 0.404 of its income to poor households and leaks nothing, and poor households leak 0.250.

(Round by round) Let $x_k = A_n^k e_{\text{gas}}$ be the income created in round k. The leakage in round k is $\ell' x_k$.

round k	in circulation	production	factors	households	cumulative leakage
0	1.000	1.000	0.000	0.000	0.125
1	0.875	0.325	0.550	0.000	0.165
2	0.835	0.119	0.166	0.550	0.498
3	0.502	0.280	0.057	0.166	0.621
4	0.379	0.214	0.108	0.057	0.675
5	0.325	0.134	0.083	0.108	0.745
6	0.255	0.120	0.052	0.083	0.801

Where the flow dies. The decisive leak is at the household stage. In round 2, when the injection reaches households, 0.333 leaks in a single round, and cumulative leakage jumps from 0.165 to 0.498. Over the whole circuit, HH-NP accounts for 0.626 of all leakage, gas's own taxes for 0.178, HH-P for 0.070, services for 0.062, manufacturing for 0.054 and agriculture for 0.010; 0.564 goes to RoW and 0.436 to Government. Cumulative leakage passes 90% after round 10 and 99% after round 20. The total multiplier is 5.18, and the leakages sum to one (sanity check).

(Real first stage (INE 2019, hydrocarbons alone)) Per unit of output, hydrocarbons buy 0.225 of domestic inputs, import 0.030 and generate value added of 0.736, of which compensation is 0.044 of output (6.0% of VA). The real enclave is therefore starker at stage 1 than the toy's. Very little of the injection goes to domestic suppliers or to labour, and most becomes surplus and taxes. Their destination (treasury, foreign owners, domestic capital) is exactly the information a real SAM must add.

Exercise 20.5 Counterfactual: land reform

(Set-up) "10% of capital income" is 0.71 of 7.1, so the Cap $\to$ HH-P coefficient rises by 0.1000 and the Cap $\to$ HH-NP coefficient falls by the same amount. Following Definition 20.15, we change A_n, hold the injections x (Government and RoW columns) fixed, solve $y'_e = (I - A'_n)^{-1} x$, and rebuild the SAM with the leakages at their coefficients. In the rebuilt SAM every endogenous account balances

to 3.6×10^{-15}. The other reading, 10% of the Cap $\to$ HH-NP cell (0.63, a coefficient shift of 0.0887), is reported in brackets.

(Results)

	before	after [other reading]	change
Γ_{SAM}	0.4018	0.4050 [0.4047]	+0.0032
gas total multiplier	5.180	5.365 [5.344]	+3.56%
capital-injection multiplier	4.342	4.619 [4.587]	+6.39%
other multipliers			+1.00% to +1.72%
poor households' income share	0.3301	0.3642 [0.3604]	+3.41 points
total endogenous income	77.400	79.090 [78.897]	

(Which moves most, and why) In relative terms Γ rises by 0.81% and the poor's share by 10.33%. The poor's income share moves most. It responds directly, because every Bs of capital income is now redistributed. The multipliers move next, because the redistribution moves income from a household that leaks 0.645 to one that leaks 0.25: the capital column's multiplier rises most, and gas, whose output goes mostly to capital, gains more than the other sectors. Γ barely moves. It measures how fast the row-normalized receipt matrix mixes, and the reform changes only the composition of two households' receipts. As in Example 20.16, distribution should be read off the multipliers and the household accounts, not off Γ. *Caveat:* this is a fixed-price multiplier closure. A real land reform also changes production (the Capital and Agriculture columns), which the exercise does not model.

Exercise 20.6 Counterfactual: education spending

(Set-up) Government spending on labour (account 9 $\to$ 5) is exogenous, so the injection vector gains 0.5 in the Lab entry. The tax on gas is a new leakage in the gas column: the Gas $\to$ Gov coefficient rises by t, and the tax is paid out of gas's capital income (Gas $\to$ Cap falls by t). We choose t so that the tax raises exactly 0.5 in the new equilibrium, $t\,y'_{\text{gas}} = 0.5$, which keeps the operation budget-neutral. Bisection gives $t = 0.06243$ (the first-pass rate $0.5/8$ is 0.06250).

(Recomputed multipliers) Only the gas column of A_n changes, but every multiplier moves, because every circuit passes through gas. The gas multiplier falls from 5.180 to 4.797 (-7.40%). The others fall by 0.39% (Cap) to 0.98% (Mfg), and none changes by more than 0.057. Measured by the multiplier matrix, the economy is slightly *worse*: a tax is a leakage.

(Is the reallocation multiplier-improving?) Yes, in total income. The 0.5 that used to flow Gas → Cap now flows Gov → Lab. Total endogenous income rises from 77.400 to 77.741 (+0.341), which equals $0.5\,(m_{\text{Lab}} - m_{\text{Cap}}) = 0.5\,(5.024 - 4.342)$ to three decimals. The reason is again distributional: labour pays 40% of its income to poor households and capital 11%, and poor households leak less. Gas output barely changes (8.000 → 8.009), and the poor's share of household income rises from 0.3301 to 0.3371. *What this omits:* the multiplier model counts education only as wage income. Its structural effect, new capabilities and edges in W^P and W^I (Definition 15.9), is invisible in a fixed-coefficient SAM and would have to be modelled as a change in A_n.

Exercise 20.7 Cross-country SAM comparison

(Data status) No published SAM for Bolivia, Chile or South Korea is in the repository, and INE does not publish one (Example 20.6). Parts (a)–(c) are therefore answered by method, with the production-block analogues that the repository does support.

(a) *Method.* Map each country's SAM to the same account classification (the 10-account template plus a savings–investment account), with the same treatment of mixed income and the same exogenous accounts. Row-normalize, and report Γ_{SAM} together with the argument of λ_2: a near-periodic circular flow can produce a small Γ by itself. *Available analogue*, $\Gamma(W^P)$ for 2019 on the ISIC-based 12 sectors: Bolivia 0.475 (INE), Chile 0.557 and Korea 0.511 (ICIO 2023).

(b) *Method.* Apply Proposition 20.11 to each SAM and report the manufacturing column of Stone's open-loop and closed-loop terms. *Available analogue:* the own-block effect, the manufacturing Type I output multiplier from domestic flows. It is 1.851 in Bolivia, 1.898 in Chile and 2.013 in Korea (forward linkages 2.224, 2.289 and 4.688).

(c) *Expected result.* The circular flow is strongest where the closed-loop multipliers are largest. That requires low household leakage (import and saving propensities) and factor income that reaches households who spend domestically. The production block points to Korea, which has the deepest manufacturing linkages, especially forward. But the closed loop is determined by the factor and household blocks, which these data do not contain. The answer cannot be inferred from $\Gamma(W^P)$: Chile has the highest Γ of the three and a smaller manufacturing multiplier than Korea's.

(d) The SAM supplies two of the dashboard's coordinates (Definition 17.12): integration (and the multipliers that locate thin parts of the circular flow), and the size distribution of accounts. It does not supply σ_B, $N_{\text{eff}}(W^I)$ or Ψ (Remark 20.14). Its $G(\boldsymbol{\pi})$ is the Gini of account sizes, not of power. A complete comparison therefore pairs the SAM diagnostics with household microdata and audience data.

Chapter 21

Measuring the Information Network

Every number below is printed by `python3 code/solutions/sol21.py`. The audience table S and the population shares p are the chapter's illustrative inputs (Example 21.2). Internet-use figures are from WDI (IT.NET.USER.ZS, ITU estimates; retrieved 2026-11-05). No audience survey, Facebook or Latinobarómetro data were fetched.

Exercise 21.1 Media overlap

(The raw overlap) $O = SS'$, with groups ordered as urban non-indigenous (NI) men, urban NI women, urban indigenous, rural NI, rural indigenous men and rural indigenous women:

$$O = \begin{pmatrix} 0.3218 & 0.3004 & 0.2100 & 0.2135 & 0.1290 & 0.1116 \\ 0.3004 & 0.2838 & 0.2110 & 0.2140 & 0.1416 & 0.1286 \\ 0.2100 & 0.2110 & 0.2150 & 0.2000 & 0.1920 & 0.1790 \\ 0.2135 & 0.2140 & 0.2000 & 0.2250 & 0.1990 & 0.2205 \\ 0.1290 & 0.1416 & 0.1920 & 0.1990 & 0.2448 & 0.2610 \\ 0.1116 & 0.1286 & 0.1790 & 0.2205 & 0.2610 & 0.3148 \end{pmatrix}.$$

(The population-weighted matrix) With reach

$$m = p'S = (0.2430, 0.2084, 0.1388, 0.1934, 0.2164),$$

the matrix is

$$W^{I}_{\text{media}} = S\,\mathrm{diag}(m)^{-1}S'\,\mathrm{diag}(p) = \begin{pmatrix} 0.2446 & 0.2429 & 0.2434 & 0.0965 & 0.0924 & 0.0802 \\ 0.2294 & 0.2307 & 0.2466 & 0.0976 & 0.1026 & 0.0931 \\ 0.1655 & 0.1776 & 0.2673 & 0.0965 & 0.1527 & 0.1404 \\ 0.1640 & 0.1756 & 0.2412 & 0.1053 & 0.1505 & 0.1634 \\ 0.1047 & 0.1231 & 0.2545 & 0.1003 & 0.2062 & 0.2112 \\ 0.0909 & 0.1117 & 0.2339 & 0.1089 & 0.2112 & 0.2434 \end{pmatrix}.$$

Its rows sum to one (sanity check). The eigenvalues are 1, 0.2584, 0.0323, 0.0062, 0.0007 and 0, all real. So $\Gamma = 0.7416$, $\boldsymbol{\pi}^{I} = p = (0.17, 0.18, 0.25, 0.10, 0.15, 0.15)$ and $N_{\text{eff}} = 5.593$ of 6. Row-normalizing O directly gives $\Gamma = 0.7398$, with $\boldsymbol{\pi} \propto$ the row sums of O, $(0.1732, 0.1722, 0.1625, 0.1713, 0.1572, 0.1636)$, and $N_{\text{eff}} = 5.992$. Both normalizations round to the chapter's $\Gamma = 0.74$.

(Proofs) Let $X = S$ $(G \times K)$ and $Y = \mathrm{diag}(m)^{-1}S'\,\mathrm{diag}(p)$ $(K \times G)$. Then $W_g = XY$ and $W_k = YX$.

1. *Stochasticity.* $Y\mathbf{1} = \mathrm{diag}(m)^{-1}S'p = \mathrm{diag}(m)^{-1}m = \mathbf{1}$ and $X\mathbf{1} = \mathbf{1}$, so $W_g\mathbf{1} = XY\mathbf{1} = \mathbf{1}$ and $W_k\mathbf{1} = \mathbf{1}$.

2. $\boldsymbol{\pi} = p$.

$$\begin{aligned} p'W_g &= p'S\,\mathrm{diag}(m)^{-1}S'\,\mathrm{diag}(p) = m'\,\mathrm{diag}(m)^{-1}S'\,\mathrm{diag}(p) \\ &= \mathbf{1}'S'\,\mathrm{diag}(p) = (S\mathbf{1})'\,\mathrm{diag}(p) = p'. \end{aligned}$$

   Here $S > 0$, so $W_g > 0$ is irreducible and its stationary vector is unique.

3. $\boldsymbol{\pi} = m$ *for outlets.*

$$m'W_k = m'\,\mathrm{diag}(m)^{-1}S'\,\mathrm{diag}(p)S = \mathbf{1}'S'\,\mathrm{diag}(p)S = p'S = m'.$$

4. *Same* Γ. For any X $(G \times K)$ and Y $(K \times G)$, $\det(\mu I_G - XY)\,\mu^K = \det(\mu I_K - YX)\,\mu^G$ (Sylvester's determinant identity). So XY and YX have the same nonzero eigenvalues with the same algebraic multiplicities, and differ only in the number of zero eigenvalues (here W_g has one more). The unit eigenvalue has the same multiplicity in both, so $|\lambda_2|$, and hence Γ, coincide. Numerically $\Gamma_g = \Gamma_k = 0.7416$, and outlet reach gives $N_{\text{eff}} = 4.855$ of 5.

5. *Reversibility.* $\mathrm{diag}(p)W_g = \mathrm{diag}(p)S\,\mathrm{diag}(m)^{-1}S'\,\mathrm{diag}(p)$ is symmetric, so $p_g W_{gg'} = p_{g'}W_{g'g}$ (detailed balance; verified). The spectrum is therefore real.

Common mistake: reporting $N_{\text{eff}} = 5.59$ as "information is unconcentrated". It is the concentration of the population, by construction, whatever the audience shares.

Exercise 21.2 Reference groups: a five-question module

(Questions) For each named person, record sex, self-identified indigenous belonging or first language, urban/rural residence (municipality), occupation and sector, and relationship to the respondent.

1. *Degree (uncensored).* "About how many people outside your household do you talk with regularly about work, prices or money?"
2. *Comparison references.* "Think of the people whose economic situation you pay most attention to when you judge how your household is doing. Name up to five."
3. *Advice.* "The last time you made an important economic decision (a job, migration, a purchase on credit, schooling), whom did you ask for advice? Name up to three."
4. *Aggregated relational data.* "How many people do you know who [speak Aymara or Quechua at home / live in a rural community / are women who run a business / work in mining or gas / live in Santa Cruz]?"
5. *Role model and intensity.* "Whom would you like your children to resemble when they are adults?" Then "If people like that earn more, does that make you want to change what you do? (never / sometimes / often)", a proxy for μ.

(Construction of W^{I}_{ref})

1. Assign each named alter to one of the six groups g' from the recorded attributes.
2. With survey weights w_i and $n_{ig'}$ the number of alters of respondent i in group g' (from Q2, or Q2 and Q3 pooled), set

$$W^{I}_{\text{ref},gg'} = \frac{\sum_{i \in g} w_i \, n_{ig'}}{\sum_{i \in g} w_i \, n_i}, \qquad n_i = \sum_{g'} n_{ig'}.$$

   This is row-stochastic by construction (the script checks it on simulated responses). The indicator version of Definition 21.3, $R_{ig'} \in \{0,1\}$, gives the

share of g naming anyone in g'. Its rows need not sum to one, so it must be row-normalized before $\boldsymbol{\pi}$ or Γ are computed.

3. *Censoring correction.* Respondents who report d_i contacts in Q1 but name only $k_i \leq 5$ have their alters weighted by d_i/k_i (Griffith, 2022).

4. *ARD.* Q4 identifies group-level tie propensities through a latent-space model (Breza et al., 2020), which serves as an independent estimate of the cross-group entries.

(Sample size) Two power calculations:

- *Precision.* Estimating a reference share near 0.3 within ± 0.03 at 95% needs 896 respondents per group, or 1345 with design effect 1.5: 8067 for six groups.
- *Test.* Detecting a cross-group naming rate of 0.20 against 0.30 (two-sided $\alpha = 0.05$, power 0.8) needs $290.4 \to 291$ per group, or 436 with the design effect: 2616 in total.

(Identification threats) Reference groups are not influence (Manski's reflection problem; Appendix 21.8). Alter attributes are reported by proxy, so ethnicity may be misclassified. Social desirability can suppress cross-ethnic comparisons, and homophily confounds contagion. The module measures W^I; validating it requires the tests of Definition 21.13.

Exercise 21.3 Digital W^I from Facebook data

Data not fetched. The Marketing API returns estimated audience sizes for targeting specifications (demographics crossed with interests or pages), not attention shares. No such estimates are in the repository.

(Construction) Define K page or interest clusters (national news pages, regional pages, indigenous-language pages, entertainment, and so on). For each group g, obtain the estimated reach a_{gk} of cluster k among g's users, and set $s^{\text{dig}}_{gk} = a_{gk} / \sum_k a_{gk}$. Users follow several clusters, so these are exposure shares, not exclusive choices. With p^{dig} the group composition of the user base, build $W^I_{\text{digital}} = S^{\text{dig}} \operatorname{diag}(m^{\text{dig}})^{-1} S^{\text{dig}\prime} \operatorname{diag}(p^{\text{dig}})$ as in Definition 21.1. Then compute O^{dig}, Γ and the λ_k, and compare cell by cell with W^I_{media}, on the same six groups. A composite exists only on a common node set (Definition 21.9).

(Is N_{eff} higher or lower?) The proof in Exercise 21.1 settles the group-level question *without* any digital data. For a population-weighted overlap matrix,

$\boldsymbol{\pi} = p^{\text{dig}}$ for every S^{dig} (verified on 200 random audience matrices). So $N_{\text{eff}}^{\text{groups}} = 1/\sum_g (p_g^{\text{dig}})^2$ depends only on who is online, not on what they see. If penetration is lower among rural and indigenous groups, p^{dig} shifts mass to the urban groups. Whether N_{eff} then falls below 5.59 depends on the shift: moving mass towards the largest group lowers it, while evening out the shares raises it. Bolivia's internet use rose from 47.5% (2019) to 79.7% (2024), which narrows such differences; their group pattern needs the EH or census. At the cluster level, $N_{\text{eff}} = 1/\sum_k (m_k^{\text{dig}})^2$ measures the concentration of reach across pages. One would expect it to be lower than on traditional media if a few national pages dominate. That is a hypothesis, not a result. *The substantive comparison is* Γ *and the* λ_k, which do depend on S^{dig}.

Exercise 21.4 Validation with Latinobarómetro

Data not fetched (registration required; `data/README.md`, §9). The method follows.

1. *Groups.* In the Bolivia sample of a given wave, build the six groups from sex, urban/rural (size of town) and ethnic self-identification or language. Check each wave's questionnaire for the three outcomes. If an item is not asked (for example a quantitative inflation expectation), that test cannot be run for that wave.

2. *Group means.* Compute weighted means $\bar{\theta}_g$ for (a) expected inflation, (b) trust in government and (c) support for redistribution, with their sampling variances.

3. *Dyadic statistic.* For each of the 15 unordered pairs, let $D_{gg'} = |\bar{\theta}_g - \bar{\theta}_{g'}|$. Compute $r = \text{corr}(O_{gg'}, D_{gg'})$ over the dyads. The prediction is $r < 0$.

4. *QAP.* Permute the group labels jointly on the rows and columns of D and recompute r. With six groups there are 720 permutations, so enumerate them all. For the chapter's O, all 720 permuted dyad vectors are distinct. The exact one-sided p-value is the share of permutations with $r_\pi \leq r$, and the smallest attainable value is $1/720 = 0.0014$.

5. *Noise correction.* Under the null of equal group means, $E\, D_{gg'} > 0$ because of sampling noise. Subtract its expectation, or use $(\bar{\theta}_g - \bar{\theta}_{g'})^2 - \hat{V}_g - \hat{V}_{g'}$.

Expected result and its limits. Fifteen dyads give little power, and the O used is illustrative, so a test on it validates a hypothesis, not Bolivia's W^I. The design should use municipality $\times$ group cells and pooled waves (Definition 21.13, item 2). Given the high illustrative integration ($\Gamma = 0.74$), media overlap should predict

little. Politically salient outcomes (b, c) are more likely to show dyadic structure than inflation expectations. A null result is evidence against media overlap as the channel, not against W^I as such.

Exercise 21.5 λ_k estimation

(a) *Gender.* Only the pairs (urban NI men, urban NI women) and (rural indigenous men, rural indigenous women) differ in gender alone. $\lambda_g = O_{12}/\frac{1}{2}(O_{11} + O_{22}) = 0.992$, and $O_{56}/\frac{1}{2}(O_{55} + O_{66}) = 0.933$.

(b) *Ethnicity* (NI $= \{1, 2, 4\}$, indigenous $= \{3, 5, 6\}$): the mean cross-block overlap divided by the mean of the two within-block means is $\lambda_r = 0.717$. If self-overlaps are excluded from the within means, $\lambda_r = 0.760$.

(c) *Urban/rural* (urban $= \{1, 2, 3\}$): $\lambda_u = 0.685$, or 0.718 without the diagonal.

(d) Geography shows the strongest assortative mixing, then ethnicity, and gender is negligible. The two main dimensions are confounded: rural groups are mostly indigenous in this design, and five pairs differ on both. A log-linear fit on the 21 cells,

$$\log O_{gg'} = c + \mathbf{1}[\text{eth}] \log \lambda_r + \mathbf{1}[\text{urb}] \log \lambda_u + \mathbf{1}[\text{sex}] \log \lambda_g,$$

gives partial values $\lambda_r = 0.748$, $\lambda_u = 0.709$ and $\lambda_g = 1.034$ ($R^2 = 0.767$). The ordering survives conditioning. *Comparison with Chapter 18.* Its illustrative values are $\lambda_r^P \approx 0.35$ and $\lambda_r^I \approx 0.30$ (Example 18.11) and $\lambda_g = 0.3$. Exercise 19.6 found that gender exposure across ISIC sections in Bolivia's 2019 employment data corresponds to $\lambda_g \approx 0.669$. So the audience table implies an information layer far *less* assortative than both. Two cautions. The objects differ: Chapter 18's ϕ scales edge weights, while an overlap ratio compares inner products of attention vectors and stays well above zero unless two groups use disjoint outlets. And audience shares cannot see segregation *within* channels.

Exercise 21.6 The operator from data

(a) The darkest entry is $O_{16} = 0.1116$ (urban NI men and rural indigenous women), 0.351 of the mean of their self-overlaps. It is also the smallest normalized overlap $O_{gg'}/\sqrt{O_{gg}O_{g'g'}}$ (0.351). It crosses all three dimensions at once.

(b) *Simulation.* Add 0.10 to O_{16} and O_{61}, and row-normalize, $\hat{W} = \text{diag}(O\mathbf{1})^{-1}O$. The population-weighted matrix cannot take this perturbation, because it is built from S and p, and an added overlap has no counterpart in S. The results:

- Γ rises from 0.7398 to 0.7914 ($+0.0516$);
- N_{eff} goes from 5.9924 to 5.9814 (-0.0110);
- $\boldsymbol{\pi}$ moves from

$$\begin{aligned} & (0.1732, 0.1722, 0.1625, 0.1713, 0.1572, 0.1636) \\ \text{to} \quad & (0.1817, 0.1677, 0.1582, 0.1668, 0.1530, 0.1725). \end{aligned}$$

Why Γ moves. The eigenvector of $\lambda_2 = 0.2602$ is the urban-NI-versus-rural-indigenous axis, $(0.893, 0.725, 0.069, -0.068, -0.735, -1)$ after normalization. The operator adds weight exactly across that axis. The perturbation lemma (Lemma 6.19, eigenvalue part) predicts $d\lambda_2 = u_2' E v_2 = -0.0587$, against an exact change of -0.0516. *Why N_{eff} does not move.* $\hat{W}$ is reversible with $\pi \propto O\mathbf{1}$, so the operator changes π only through two row sums, and it moves mass towards the two groups it connects. It is an integration operator, not a deconcentration operator (Proposition 6.16).

(c) Partly ill-posed. Under Definition 9.13, $\Psi = F/E$ is defined only if the enforcement effect $E \neq 0$. A pure W^I operator has $E = 0$: at fixed preferences, outcomes do not depend on W^I. So "$\Psi = 2$ for this operator" is undefined. We solve the closest well-posed version in two steps.

1. *Formation model.* By Proposition 18.13, the steady-state aspiration gap is $\mathcal{M}\,\lambda_2^{\text{grp}}\,\Delta\bar{r}$, where λ_2^{grp} is the second eigenvalue of the two-group information matrix. Aggregating $\hat{W}$ to NI versus indigenous (π-weighted, Definition 19.6) gives $\lambda_2^{\text{grp}} = 0.1636$ before and 0.1331 after. At fixed returns the long-run aspiration gap therefore falls by 18.65%. Using instead λ_r from O ($0.7174 \to 0.7636$) and the factor $(1-\lambda)/(1+\lambda)$ ($0.1646 \to 0.1340$) gives 18.55%.

2. *Reading $\Psi = 2$ as a property of the existing ethnic outcome gap.* Interpret it as formation explaining $\Psi/(1+\Psi) = 2/3$ of that gap, the rest being enforcement. The information operator acts only on the formation part, so the outcome gap falls by $\frac{2}{3} \times 18.65\% = 12.43\%$ (or 12.37%).

Both figures are long-run, reached at the rate ρ per generation.

Exercise 21.7 Cross-country comparison

(Sourced inputs) Internet users as a share of the population (WDI): Bolivia 47.5% (2019) and 79.7% (2024); Chile 85.0% and 95.6%. Sweden is not in the repository's

WDI extract. It is fetchable with the same script, but was not fetched. Audience shares and media-ownership concentration were not fetched for any of the three countries (`data/README.md`, §9), so outlet-level N^I_{eff}/n cannot be computed here.

(Method)

1. Fix a common outlet list: the top $n-1$ outlets by audience plus "other", with the same n in every country, because N_{eff}/n depends on n and lies in $[1/n, 1]$.
2. Obtain reach shares m_k from audience measurement, or ownership-group shares from a media-pluralism source, and compute $N_{\text{eff}} = 1/\sum_k m_k^2$.
3. Report the result alongside internet penetration. Penetration changes which outlets enter the list; it does not change the formula.

(Ranking and Ψ) Expected result, stated as a hypothesis: N_{eff}/n measures the concentration of attention, not pluralism or independence. A strong public broadcaster can give a pluralistic system a *low* N_{eff}, so the ranking need not follow press-freedom rankings. On this chapter's illustrative table, Bolivia's outlet-level $N_{\text{eff}}/n = 0.971$. Whether the ranking predicts the Ψ ranking cannot be tested. Ψ has not been estimated for any of the three countries (Chapter 11's values are illustrative). With three countries, even perfect rank agreement would occur by chance with probability $1/3! = 0.1667$. A test needs many countries, or within-country variation across regions.

Chapter 22

Measuring the Utility Network

Every number below is printed by `python3 code/solutions/sol22.py`. The transfer matrix, alignment positions and audience table are the book's illustrative inputs. The EH microdata, MEFP budget tables and Latinobarómetro were not fetched (`data/README.md`, §9). Remittances and Gini coefficients are from WDI (retrieved 2026-11-05).

Exercise 22.1 Transfer-revealed W^U

(Received transfers in the EH: method) EH microdata were not fetched, so this part is method only.

1. From the EH 2019 income module, take each household's transfer income by source: transfers from other households in the country, remittances from abroad, and in-kind transfers valued as reported. Use the variable names and value conventions of the survey's own documentation.

2. Classify each recipient household into the four groups: urban/rural from the survey's area variable, and indigenous from the head's self-identification or language.

3. Report weighted totals and means by group and source. As a check, compare total remittances with the balance of payments: WDI gives US$1.336 bn = 9.30 bn Bs in 2019, or 2.72% of GDP (the peak was 8.04% in 2007). Survey totals typically fall short because of under-reporting.

(Why the sender is not identified) The EH interviews the *recipient* household. It records the type of source (a household in Bolivia, a person abroad), not the identity or group of the sender. Each row of the estimable object is therefore a column

total of τ: we observe $\sum_i \tau_{ij}$ by type of i, not τ_{ij} by group of i. Definition 22.3 needs rows (shares of what i sends), and many different sender matrices share the same column sums. A dyadic module would ask each household, for every transfer sent *and* received, the counterpart's relationship, location, ethnicity and sex. That identifies $\tau_{gg'}$ directly, gives both margins as a consistency check, and supports dyadic regressions with dyad-robust standard errors (Fafchamps and Gubert, 2007).

(The illustrative matrix) Row-normalizing the chapter's raw shares gives (senders in rows):

$$W^U = \begin{pmatrix} 0.7500 & 0.1333 & 0.0833 & 0.0333 \\ 0.1250 & 0.5000 & 0.0625 & 0.3125 \\ 0.1127 & 0.0423 & 0.7042 & 0.1408 \\ 0.0375 & 0.1875 & 0.0875 & 0.6875 \end{pmatrix}.$$

The eigenvalues are 1, 0.6999, 0.6282 and 0.3137, all real, so $\Gamma = 0.3001$. The Perron vector is $\boldsymbol{\pi} = (0.2493, 0.2083, 0.2120, 0.3304)$ (sum $= 1$), and $N_{\text{eff}} = 3.851$ of 4 ($N_{\text{eff}}/n = 0.963$). The ethnic mixing parameter is $\lambda^U_{\text{eth}} = \bar{O}_{\text{cross}}/\bar{O}_{\text{within}}$, taken over the 8 cross-ethnic and 8 same-ethnicity cells, diagonal included: $0.0828/0.4172 = 0.198$. Geography gives $0.1089/0.3911 = 0.279$. Rural indigenous households carry the largest Perron weight, because their column sum is the largest (1.1742): urban indigenous households send them 0.3125 of their transfers.

Exercise 22.2 Experimental design

(Design) A directed dictator game run within subjects. Each dictator receives an endowment E and makes nine allocation decisions:

- one for each of the $2 \times 2 \times 2 = 8$ recipient profiles: (a) same or different ethnicity, (b) same or different community, (c) same or different gender;
- one for an anonymous stranger, as a baseline.

One decision is drawn at random and paid. Further details:

- Recipients are real people drawn from lists in the dictator's own and a paired community, so there is no deception.
- Each recipient is described by the three attributes alone, through standardized cards and in the respondent's language, so that ethnicity is not confounded with signals of class.
- The profile order is randomized.

Number of treatments: 8 profiles plus one baseline. The full factorial is what identifies the separable model and its interaction ι (Definition 18.12). A design with only three "main effect" arms cannot test separability.

(Sample size) The standard deviations used here are design assumptions to be replaced with pilot data, not estimates.

- *Main effects.* To detect a 5-point difference in the share given ($\delta = 0.05$), with an SD of the paired difference of 0.15, $n = ((1.96 + 0.84) \cdot 0.15/0.05)^2 = 70.6 \to 71$ dictators.
- *Two-way interaction* (the ethnicity $\times$ community contrast, SD 0.25): $n = 196.2 \to 197$. With communities of $m = 10$ dictators and an intra-community correlation of 0.05, the design effect is 1.45, so 286 dictators.
- *Between-subject alternative* (SD of shares 0.20): 251 per arm, 2268 in total. The within-subject design is about eight times cheaper.

(From shares to weights) Under log altruism, $W^U = s/(1-s)$ (Definition 22.5). The delta method gives $\mathrm{SE}(W) = \mathrm{SE}(s)/(1-s)^2$. With $n = 197$ and $\mathrm{SE}(s) = 0.0142$, this is 0.0184 at $s = 0.12$ ($W = 0.136$), 0.0253 at $s = 0.25$ ($W = 0.333$) and 0.0371 at $s = 0.38$ ($W = 0.613$). The mapping is a modelling assumption: other utility functions give other mappings.

(Threats) Experimenter demand and norm compliance, which are higher for co-ethnics observed by enumerators; enumerator ethnicity, to be randomized and controlled for; spillovers between dictators in the same community; and the stranger baseline, who may be perceived as a co-villager (Example 22.6).

Exercise 22.3 Fiscal W^U

Data not fetched: the MEFP/UDAPE fiscal tables (`data/README.md`, §9). Method:

1. *Revenue.* IDH collections by year, from the tax administration and MEFP execution reports.
2. *Legal distribution.* The coparticipation rules that split IDH among the central treasury, departmental governments, municipalities, universities and earmarked funds, taken from the legal texts and verified against execution data. The rules are not reproduced here.
3. *Spending.* Each recipient's spending by programme (Renta Dignidad, Bono Juancito Pinto, investment, wages).

4. *Incidence.* Programme benefits allocated to the four household groups using EH benefit receipt (for cash transfers) or a stated rule (for public goods).

5. *Matrix.* Assemble the shares into W^U_{fiscal}.

The chapter's illustrative incidence (12% to poor households, 8% to non-poor, 25% reinvested in the sector) leaves 55% unallocated. A real tracing must account for it: administration, other investment, the treasury.

(Construction) Definition 22.11 gives $W^U_{\text{fiscal},gg'} = (\text{net transfer } g \to g')/\text{capacity}_g$. Its row sums are the share of g's fiscal capacity sent to others, so they are below one. To compute π or Γ, add the retained share on the diagonal, which makes the matrix row-stochastic. One complication is specific to the IDH: it is paid by the gas sector, and ultimately by its export customers, not by a household group. In a household-group matrix it has no sender row. It must be treated as an exogenous inflow, or attributed to the owners of the sector's surplus, or to the state.

(Why it is not anyone's utility weights) W^U_{ij} is a parameter of U_i (Definition 22.1). A fiscal flow is the *outcome* of a political process: $\tau^{\text{fiscal}} = F(W^U, \boldsymbol{\pi}, \text{institutions})$.

1. *Non-identification.* Two polities with different W^U but the same $\boldsymbol{\pi}$ and rules can produce the same flows. A ruler with $W^U = 0$ can redistribute for political survival, and a solidaristic electorate can be blocked by veto players.

2. *Aggregation.* The inverse-optimal-tax approach recovers at most a *vector* of marginal social welfare weights for the planner (Bourguignon and Spadaro, 2012), not a matrix of interpersonal concern. By Arrow's theorem there is in general no coherent aggregate of individual W^U rows to recover.

3. *No utility at the source.* The IDH's "sender", the gas sector, has no utility at all.

W^U_{fiscal} is therefore a *revealed allocation rule of the polity*. It belongs in the composite as an institutional channel, and should not be read as solidarity.

Exercise 22.4 The alignment test

Partly ill-posed with the book's inputs. The Chapter 20 SAM has two household accounts (poor, non-poor), not the four ethnic–geographic groups. Mapping it requires poverty incidence by group from the EH, which was not fetched. And the book's illustrative inputs use inconsistent population shares: Chapter 21's six groups aggregate to $(0.35, 0.25, 0.10, 0.30)$, while Chapter 22 uses $(0.35, 0.25, 0.15, 0.25)$.

We solve the closest well-posed version: the measured-style layers where they exist, with Example 22.16's illustrative r^P in place of the SAM.

(The three layers on four groups)

- W^I: Chapter 21's 6×6 media matrix, aggregated by the π-weighted rule of Definition 19.6. Its Perron vector is the block sums, $\boldsymbol{\pi}^I = (0.35, 0.25, 0.10, 0.30)$, and $\Gamma(W_4^I) = 0.7449$.
- W^U: the transfer matrix, with $\boldsymbol{\pi}^U = (0.2493, 0.2083, 0.2120, 0.3304)$.
- W^P: $\boldsymbol{\pi}^P = r^P \circ p = (0.525, 0.200, 0.160, 0.115)$, from $r^P = (1.5, 0.8, 1.067, 0.46)$.

(Correlations)

- Levels: $\text{corr}(\boldsymbol{\pi}^P, \boldsymbol{\pi}^U) = -0.175$, $\text{corr}(\boldsymbol{\pi}^P, \boldsymbol{\pi}^I) = 0.566$, $\text{corr}(\boldsymbol{\pi}^I, \boldsymbol{\pi}^U) = 0.525$.
- Per capita (with the Chapter 22 shares): $r^U = \boldsymbol{\pi}^U / p = (0.712, 0.833, 1.413, 1.322)$, and $\text{corr}(r^P, r^U) = -0.532$.
- The media layer is degenerate per capita: $r^I = \boldsymbol{\pi}^I / p^{(21)} \equiv 1$ (Exercise 21.1), so its correlation with anything is undefined. Dividing by the other chapter's shares produces the spurious $(1, 1, 0.667, 1.2)$.

(Aligned or compensating?) On the chapter's own transfer matrix, *compensating*. Rural indigenous households are the most cared-about group in W^U ($r^U = 1.32$) and among the weakest in the illustrative W^P ($r^P = 0.46$). This contradicts Example 22.16, which assumes $r^U = 0.30$ for them and reports correlations of 0.96, 0.98 and 0.99. The two examples of the same chapter cannot both describe Bolivia. The test result depends entirely on which illustrative input one adopts, which is the point: alignment is an empirical question (Proposition 22.15). Two further points. Correlations of π *levels* mostly reflect group size, so the per-capita version is the relevant one. And with four groups none of these correlations has meaningful sampling precision.

Exercise 22.5 Misalignment as policy

(Rank-one layers (Proposition 22.15(b))) Equal weights, Example 22.16's positions. *Before:* composite per-capita positions $r = (1.417, 0.900, 1.244, 0.370)$, $\text{Var}(\boldsymbol{\pi}) = 0.02247$, $N_{\text{eff}} = 2.942$. *Policy:* the utility layer moves towards the indigenous groups, $\boldsymbol{\pi}^{U\prime} = (1 - s)\boldsymbol{\pi}^U + s\,q$, where q is the indigenous population share normalized to one.

	r_{RI}^{U}	composite r	$\mathrm{Var}(\boldsymbol{\pi})$	N_{eff}	$\mathrm{corr}(\boldsymbol{\pi}^{P}, \boldsymbol{\pi}^{U})$
before	0.30	$(1.417, 0.900, 1.244, 0.370)$	0.02247	2.942	0.96
$s = 0.25$	0.725	$(1.304, 0.983, 1.132, 0.512)$	0.01600	3.185	$+0.792$
$s = 0.50$	1.15	$(1.192, 1.067, 1.019, 0.653)$	0.01128	3.388	-0.027

With rank-one layers the composite moves by exactly α_U times the change in the utility layer (verified): $\Delta r_{\mathrm{RI}} = \frac{1}{3}(1.15 - 0.30)$ at $s = 0.5$. Compound disadvantage falls. Rural indigenous households go from 0.370 to 0.653 of the average, and concentration falls with them, because the policy lowers the covariance between layers.

(Full (non-rank-one) layers) Composite of the transfer W^{U}, the aggregated media W_4^{I} and a rank-one W^{P}, with equal weights. *Policy:* the two indigenous rows of W^{U} move 0.10 of their weight from non-indigenous to indigenous recipients (mutual aid among indigenous communities).

- $\boldsymbol{\pi}^{U}$ goes from $(0.2493, 0.2083, 0.2120, 0.3304)$ to $(0.1230, 0.2846, 0.0923, 0.5001)$, and $\Gamma(W^{U})$ from 0.3001 to 0.2138.
- The composite $\boldsymbol{\pi}$ and its per-capita version move as follows:

$$\boldsymbol{\pi} : (0.4207, 0.2207, 0.1426, 0.2159) \to (0.4097, 0.2286, 0.1327, 0.2290),$$
$$\boldsymbol{\pi}/p : (1.202, 0.883, 0.951, 0.864) \to (1.170, 0.915, 0.884, 0.916).$$

- Composite Γ falls from 0.6840 to 0.6626, N_{eff} rises from 3.417 to 3.446, and Var falls from 0.01067 to 0.01004.

(Answer) Yes, compound disadvantage of indigenous households decreases: rural indigenous $0.864 \to 0.916$, urban indigenous $0.883 \to 0.915$. Two qualifications. In the full-matrix case the gain is smaller than the rank-one formula suggests, and *rural non-indigenous* households lose ($0.951 \to 0.884$), because the policy withdraws concern from them. And the policy makes W^{U} more assortative ($\Gamma(W^{U})$ falls), so it trades integration for compensation. Proposition 6.16 again: the two coordinates move separately.

Exercise 22.6 Cross-country comparison

Partly ill-posed. Latinobarómetro's standard trust item is *generalized* interpersonal trust, one number per respondent, as Example 22.10 notes. A matrix W^{U} on groups needs trust *in* each group, so it cannot be estimated from generalized

trust. A respondent-group-by-target-group matrix requires a group-specific battery. Where a wave contains trust items towards specific groups, they can be used; this was not verified, because the data were not fetched (registration).

(Method (with a group-specific module))

1. Define the same groups in each country (for instance indigenous or ethnic minority $\times$ urban/rural).
2. Let t_{gh} be the mean trust of respondents in g towards group h on the 1–4 scale. Rescale to $[0,1]$, set $K_{gh} = (t_{gh} - 1)/3$ and $W^U = D^{-1}K$.
3. Compute N_{eff}^U/n, Γ and λ^U on the common node set.

(Inequality (sourced)) WDI Gini: Bolivia 41.6 (2019), Chile 45.3 (2017, the latest survey year up to 2019), Uruguay 39.7 (2019). *Expected result:* no inference is possible. With three countries a correlation has one degree of freedom, and even a perfect rank agreement has chance probability 1/6 (Exercise 21.7). Moreover N_{eff}^U/n and Γ^U are different coordinates. High within-group trust lowers Γ without concentrating π, as in Exercise 22.1, so "more solidarity" can move them in opposite directions.

Exercise 22.7 The complete measurement

Illustrative, and partly ill-posed. Bolivia's welfare loss cannot be decomposed from data in this repository. The SAM's W^P is not on household groups (Exercise 22.4), and the welfare theorem needs Ω, κ, Φ and γ_U, which are not measured. We carry out the computation on the four groups with explicitly illustrative inputs, treating each group as one representative agent. This ignores group sizes. A population-weighted version would replicate agents in proportion to p.

(Composite) The layers are W^U, the transfer matrix (Exercise 22.1); W^I, the aggregated media matrix (Exercise 22.4); and $W^P = \mathbf{1}\boldsymbol{\pi}^{P\prime}$ (illustrative). The weights are Chapter 5's $\alpha_U = 0.2$, $\alpha_P = 0.5$, $\alpha_I = 0.3$ (Example 5.12):

$$W = \begin{pmatrix} 0.5545 & 0.2002 & 0.1258 & 0.1195 \\ 0.3904 & 0.2802 & 0.1214 & 0.2079 \\ 0.3869 & 0.1808 & 0.2524 & 0.1798 \\ 0.3345 & 0.2108 & 0.1289 & 0.3258 \end{pmatrix}.$$

It is row-stochastic, with eigenvalues 1, 0.2148, 0.1279 and 0.0703. So $\Gamma = 0.7852$, $\boldsymbol{\pi} = (0.4541, 0.2167, 0.1436, 0.1856)$ and $N_{\text{eff}} = 3.244$.

(Welfare decomposition (Theorem 7.8)) The calibration is Chapter 2's four-agent one: $N = 4$, $\Omega = 20$, $\kappa = 0.1$, $\bar{d} = 3$, so $\Phi = 8$, and $\gamma_U = 0.3$. The care weights c_j are the column sums of the zero-diagonal, row-normalized W^U: $c = (0.7510, 1.2762, 0.7383, 1.2345)$, summing to 4.

- Bargaining allocation, with $\max \kappa x = 0.663 < 1$:

$$x^B = \Omega/N + (\pi - \mathbf{1}/N)\Phi = (6.6326, 4.7337, 4.1488, 4.4849).$$

- First best, from $(1 + \gamma_U c_j)(1 - \kappa x_j) = \mu$ and $\sum x = \Omega$:

$$x^{FB} = (4.7136, 5.3160, 4.6972, 5.2732).$$

- Social welfare: $\mathcal{S}(x^{FB}) = 19.5226$, $\mathcal{S}(x^E) = 19.5000$, $\mathcal{S}(x^B) = 19.2126$.

Then

$$L = 0.31000 = \underbrace{0.18630}_{L_{\text{power}}} + \underbrace{0.02264}_{L_{\text{externality}}} + \underbrace{0.10105}_{L_{\text{interaction}}},$$

which holds exactly (sanity check). The interaction term also matches its closed form, with $\text{Cov}(c, \pi) = -0.011820$.

(Fractions and reading) Power accounts for 0.601 of the loss, externality for 0.073 and interaction for 0.326. Power dominates because the composite concentrates on urban non-indigenous households ($\pi = 0.454$). The interaction term is large and *adds* to the loss, because $\text{Cov}(c, \pi) < 0$: the powerful group is the least cared-about in W^U, so the bargaining tilt takes resources from the groups that others care about most. Had the powerful also been the most cared-about, the term would offset part of L_{power}. *What would be needed for Bolivia:* all three layers measured on the same groups, and Φ, κ, γ_U calibrated or estimated. The fractions above are a property of the illustrative inputs, not an estimate.

Chapter 23

Tracking Structural Change Over Time

Every number below is printed by `python3 code/solutions/sol23.py`. The sources, all in `data/processed` and retrieved 2026-11-05, are:

- INE's input–output matrices (base 1990) and supply–use tables (base 2017), as domestic flow matrices on the book's 12 sectors (`bolivia_Z/`, `bolivia_io_summary.csv`, `bolivia_io_pi.csv`);
- the OECD ICIO domestic blocks, 2021 and 2023 editions, on ISIC-based 12 sectors (`oecd_io_summary.csv`);
- Growth Lab ECI, HS92 (`eci_facts.csv`).

Unless stated otherwise, Γ is that of $W^P = D^{-1}A$ with domestic flows and the diagonal kept. EH microdata were not fetched, so no σ_B is reported.

Exercise 23.1 Bolivia's trajectory

(Method)

1. For each year, take INE's domestic industry $\times$ industry flow matrix Z_t at basic prices (12 book sectors).
2. Drop sectors without intermediate purchases or sales: domestic services always, and public administration in years without a published make matrix.
3. Form W_t^P and compute Γ_t, $\boldsymbol{\pi}_t$ and $G(\boldsymbol{\pi}_t)$.
4. Keep base 1990 (2000–2014) and base 2017 (2017–2019) as separate series.

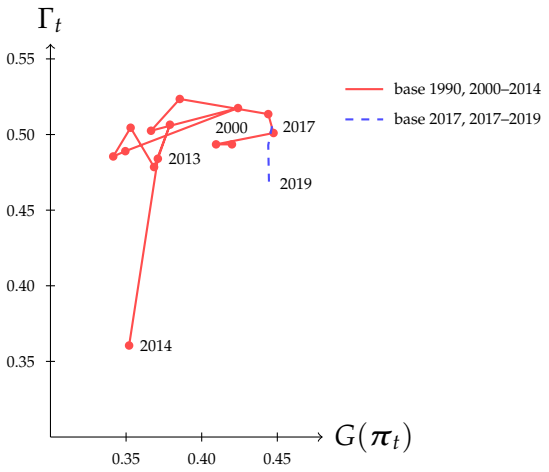
5. Repeat with sales shares ($D^{-1}Z$) and with the diagonal set to zero before normalizing.

$\sigma_{B,t}$ would be computed from each year's EH as in Exercise 19.1; the microdata were not fetched.

(The trajectory (input shares, diagonal kept))

	2000	2002	2003	2004	2005	2006	2007	2009	2013	2014
Γ_t	0.494	0.501	0.514	0.524	0.502	0.518	0.489	0.504	0.484	0.361
$G(\pi_t)$	0.420	0.447	0.444	0.385	0.366	0.424	0.350	0.353	0.371	0.352

The other base-1990 years are 2001 (0.493, 0.410), 2008 (0.486, 0.341), 2011 (0.478, 0.369) and 2012 (0.506, 0.379). On base 2017: 2017 (0.506, 0.446), 2018 (0.493, 0.444), 2019 (0.467, 0.445).



(Which coordinates move?) On base 1990, 2000–2013, Γ barely moves (-0.010) while $G(\pi)$ falls by 0.049. The year-to-year co-movement of the two is weak and positive (correlation $+0.401$ over 2000–2014). Including the outlier year 2014 changes the picture: Γ then falls by 0.133. On base 2017, Γ falls (-0.038, 2017–2019) and $G(\pi)$ is flat (-0.002).

(Robustness to normalization) Changes over the same periods:

	2000–2013 (base 1990)		2017–2019 (base 2017)	
	$\Delta\Gamma$	$\Delta G(\pi)$	$\Delta\Gamma$	$\Delta G(\pi)$
input shares	-0.010	-0.049	-0.038	-0.002
input shares, zero diagonal	$+0.044$	$+0.025$	-0.018	-0.003
sales shares	-0.204	$+0.039$	$+0.005$	$+0.001$
sales shares, zero diagonal	$+0.000$	-0.032	-0.007	$+0.000$

The direction of change does not survive. Over 2000–2013 the sign of $\Delta\Gamma$ is negative for two variants and non-negative for the other two, and $\Delta G(\boldsymbol{\pi})$ has both signs as well. Over 2017–2019 three of the four variants agree on a falling Γ (sales shares give $+0.005$). The magnitudes are of the order of the year-to-year noise (Exercise 23.2), except the sales-share fall of 0.204, which is driven by the composition of sales. The trajectory is not a robust object at this aggregation: report a band across variants, not a single line.

Exercise 23.2 Structural break detection

(Criterion) Definition 23.8 with $\hat{\sigma}_{\Delta\Gamma}$ estimated from the 9 quiet changes of 1989–1997: $\hat{\sigma} = 0.0090$, so $2\hat{\sigma} = 0.0181$. On base 1990 the year-to-year changes include $+0.101$ (1999), -0.148 (2000), -0.021 (2005), -0.029 (2007), $+0.019$ (2009), $+0.028$ (2012), -0.022 (2013) and -0.123 (2014). There is no 2010 table, so 2011 has no change. At $k = 2$ the flagged years are:

- base 1990: 1999, 2000, 2005, 2007, 2009, 2012, 2013 and 2014;
- base 2017, with the same $\hat{\sigma}$: 2019 (-0.026), 2020 (-0.049), 2022 (-0.032), 2023 (-0.019) and 2024 ($+0.038$).

At $k = 2$ the criterion flags a third of all years. A threshold calibrated on a quiet decade is too tight for a series whose construction changes.

(Bootstrap band) The noise model is an assumption, standing in for INE's unpublished revision ranges: every cell of Z_t is multiplied by independent lognormal noise with mean one and coefficient of variation 10%, and Γ_t is recomputed ($B = 300$ per year). The mean bootstrap s.d. of Γ_t is 0.0220. The 95% bootstrap band of $\Delta\Gamma_t$ excludes zero only for 1999 ($[+0.030, +0.211]$), 2000 ($[-0.253, -0.070]$) and 2014 ($[-0.188, -0.076]$). Every other flagged year, including all of base 2017 (for example 2020, $[-0.104, +0.009]$), is within noise. *Breaks surviving both criteria: 1999, 2000 and 2014.* None corresponds to a policy event. 1999 and 2000 lie in a run of diagonal-make years, and 2014 is the last year of the base. These are the construction effects the chapter identifies (Example 23.11). Note that cell noise does not capture construction error, such as a missing make matrix. The bootstrap tests measurement noise, not methodology.

(τ_1 and $\Gamma^{(5)}$) Products require a common node set, so we use the 10 sectors defined in every base-1990 year (dropping domestic services and public administration).

- *Dobrushin coefficient.* $\tau_1(W_t)$ ranges over 0.722–0.847. Every W_t is scrambling, $1 - \tau_1 \geq 0.153$, so Hajnal's condition holds trivially (Proposition 23.7). The bound $\Gamma_t \geq 1 - \tau_1(W_t)$ of Proposition 23.5 holds in every year, but it is loose (0.15–0.28 against $\Gamma_t \approx 0.5$). τ_1 moves little and shows no break.
- *Realized-product gap.* $\Gamma_t^{(5)} = 1 - |\lambda_2(W_{t+4} \cdots W_t)|^{1/5}$, over windows of five consecutive years, ranges over 0.506–0.578. It declines smoothly: 0.572 for the 1990 window, 0.558 for 1996 (which contains 1999–2000), 0.508 for 2001 and 0.506 for 2005. The per-period spike $\Gamma_{1999} = 0.642$ is averaged away. Windows containing 1999 average 0.547, the others 0.542.

Do they tell the same story? No. Γ_t shows spikes. The realized product, which is what a dynamic process driven by the sequence W_t actually experiences, shows a gradual decline of about 0.07 over 15 years with no break. This is the warning of Remark 23.4 in reverse: here the per-period gap *exaggerates* the instability.

Exercise 23.3 Structural velocity

(Results) Velocities per decade (v_Γ by endpoints and by OLS on all years in the window), with ΔECI over the same years:

	window	$\Gamma_{t_0} \to \Gamma_T$	endpoint	OLS	ΔECI
Chile	1995–2014 (ICIO 2021)	0.467 → 0.640	+0.091	+0.079	−0.088
Chile	2011–2020 (ICIO 2023)	0.449 → 0.544	+0.107	+0.090	−0.190
Peru	1995–2014 (ICIO 2021)	0.621 → 0.507	−0.060	−0.100	−0.065
Peru	2011–2020 (ICIO 2023)	0.448 → 0.405	−0.047	−0.053	−0.205
Bolivia	1995–2014 (base 1990)	0.578 → 0.361	−0.115	−0.067	−0.490
Bolivia	1995–2013 (excl. 2014)	0.578 → 0.484	−0.052	−0.050	−0.433
Bolivia	2017–2024 (base 2017)	0.506 → 0.419	−0.124	−0.161	+0.270

The endpoint values reproduce `gamma_velocity.csv` (checked).

(Sign stability) Chile's velocity is positive in both windows and Peru's negative in both, by both estimators. No single base spans 2011–2020 for Bolivia. The partial windows give −0.392 (2011–2014, 4 points) and −0.292 (2017–2020, 4 points, including the pandemic). Four-point slopes are noise. Bolivia's two bases agree on the sign, but the 1995–2014 endpoint value is inflated by the 2014 outlier (−0.115 against −0.052 without it).

(Comparison with ECI and interpretation) The sign of v_Γ is unrelated to the change in complexity. Chile has the fastest-rising Γ in the sample and a falling

ECI in both windows. Peru has a falling Γ and a falling ECI. Bolivia's ECI rose ($+0.270$) over 2017–2024 while its Γ fell fastest. That two countries kept their sign here does not rescue v_Γ: Korea and Argentina flip (Example 23.13). v_Γ measures the change in the mixing speed of an aggregated input-share matrix. It is not a measure of development, which concerns capabilities (complexity, productivity, distribution). A positive v_Γ can come from changes unrelated to capabilities, for example a concentration of input flows on a commodity hub.

Exercise 23.4 Structural-cyclical decomposition

(Data and method) The series is $\pi_{\text{gas},t}$ on base 1990, book12, domestic, input shares, 1988–2014. It rises from 0.025 (1988) to 0.047 (2003), 0.074 (2005), 0.087 (2007), 0.085 (2013) and 0.117 (2014). There is no 2010 table. We use an HP filter with a missing observation: the trend τ_t is defined on every year 1988–2014, and the fit term uses observed years only,

$$\min_{\tau} \sum_{t \text{ observed}} (\pi_t - \tau_t)^2 + \lambda \sum_t (\tau_{t+1} - 2\tau_t + \tau_{t-1})^2.$$

This equals the ordinary HP filter when nothing is missing (checked on 1988–2009). It interpolates 2010, giving $\tau_{2010} = 0.0841$ at $\lambda = 100$.

(Results) The observed rise from 2003 to 2013 is $+0.0380$ ($0.0467 \to 0.0847$). The structural share is $(\tau_{2013} - \tau_{2003})/(\pi_{2013} - \pi_{2003})$:

	$\lambda = 6.25$	$\lambda = 100$	$\lambda = 400$
with 2014	1.179	1.217	1.180
without 2014	0.880	1.007	1.029

At $\lambda = 100$ with 2014, the trend runs from 0.0507 (2003) to 0.0969 (2013). The gain-$\frac{1}{2}$ cutoff periods are 9.8, 19.8 and 28.0 years.

(Interpretation) Taken at face value, the filter calls essentially *all* of the rise structural (88% to 122%). That is an artefact of the window. A rise that lasts a decade and does not reverse *within the sample* has most of its power below the cutoff frequency, so any HP trend absorbs it. Shares above one appear because the 2014 spike pulls the trend's endpoint above the observed 2013 value. Dropping 2014 lowers the share at $\lambda = 6.25$ from 1.179 to 0.880. The persistence criterion of Definition 23.15 can classify the boom only with post-boom data on the same base. These do not exist: the base-1990 series ends in 2014.

The composition of the node set is not the problem. N alternates between 10 and 11 as public administration enters and leaves, but on the common 10-sector set π_{gas} is almost identical (0.047 in 2003, 0.086 in 2013, 0.119 in 2014), and the share at $\lambda = 100$ is 1.216.

(Why the base-2017 years must not be appended) Base 2017 gives $\pi_{\text{gas}} = 0.045$ (2017), 0.039 (2019) and 0.021 (2024), against 0.117 in 2014 on base 1990. The jump between 2014 and 2017 mixes three things: a new classification (39 industries against 35), a new SNA vintage and price base, and a new table construction (supply–use tables against symmetric matrices, with the make treatment changed). A filter run across the junction would read a construction break as a cyclical reversal, making the boom look "cyclical" by fiat. The honest answer is that the data cannot decompose the 2003–2013 rise. The falling base-2017 series is *suggestive* of reversal, but only within its own base.

Exercise 23.5 Commodity boom counterfactual

(Baseline (INE 2019)) Hydrocarbons have a domestic input intensity of 0.225, $\pi_{\text{hydro}} = 0.0392$, $\Gamma = 0.4674$ and backward linkage $BL = 1.340$ (column sum of $L = (I - A)^{-1}$). Output is 25863 million Bs and value added 19033 million Bs. Taking "gas revenue" as the sector's value added, 30% of it is 5710 million Bs. The data do not identify the mapping from Bs invested to new linkages, so the simulation is parametrized by its outcome: domestic input intensity raised to 0.30, or to 0.381, the sector's 2006 level. The extra domestic purchases this requires are 1949 and 4044 million Bs a year, 0.34 and 0.71 times the 30% of value added.

(Simulations on the real 2019 W^P)

Operator	intensity	π_{hydro}	Γ	BL_{hydro}
baseline	0.225	0.0392	0.4674	1.340
(i) proportional scale-up, to 0.30	0.300	0.0392	0.4674	1.456
(i) proportional scale-up, to 0.381	0.381	0.0392	0.4674	1.582
(ii) new purchases from agr. and mfg, to 0.30	0.300	0.0385	0.4660	1.470
(ii) new purchases from agr. and mfg, to 0.381	0.381	0.0381	0.4652	1.611
(iii) forward: others buy gas (10% of imports)	0.225	0.0582	0.4670	—

(Which indicator detects the change, and why Γ might not)

1. *Domestic input intensity* (and BL, computed from A) detects every backward-linkage operator: 0.225 $\to$ 0.381, BL 1.340 $\to$ 1.611.

2. π_{hydro} does *not* detect backward linkages. Under (i) it is exactly unchanged. Under (ii) it even falls slightly, because π measures how much *others* depend on the sector (its column), while the operator changes its own purchases (its row). It does detect forward linkages: $+0.0190$ under (iii).

3. Γ *moves by at most* 0.0022. There are three reasons. First, $W^P = D^{-1}A$ normalizes each row, so a proportional scale-up of the sector's purchases leaves W^P, and hence Γ and $\boldsymbol{\pi}$, *exactly* unchanged: the intensity level is divided out. Second, Γ is the slowest mode of the whole matrix, and hydrocarbons carry only 0.0171 of the ℓ_1 mass of the left eigenvector u_2. By the perturbation lemma (Lemma 6.19), $d\lambda_2 = u_2' E v_2$ is therefore small for any change confined to that sector. Third, linking the enclave to agriculture and manufacturing concentrates flows and slightly *lowers* Γ.

The enclave is measured by intensity, linkages and π, not by Γ (Example 23.16).

Exercise 23.6 SCI computation

(Results) The Latin American economies in ICIO are ARG, BRA, CHL, COL, CRI, MEX and PER. Mexico is not extractable from the 2021 edition. Costa Rica has no ECI in the repository's Growth Lab extract.

	1995–2014 (ICIO 2021)					2011–2020 (ICIO 2023)			
	Γ_{t_0}	Γ_T	SCI	ΔECI		Γ_{t_0}	Γ_T	SCI	ΔECI
CHL	0.467	0.640	5.30	-0.088	CHL	0.449	0.544	1.86	-0.190
BRA	0.489	0.527	3.40	-0.174	PER	0.448	0.405	-0.82	-0.205
CRI	0.413	0.506	1.07	n/a	ARG	0.538	0.492	undef.	-0.405
ARG	0.407	0.476	0.75	-0.205	BRA	0.532	0.472	undef.	-0.326
COL	0.553	0.484	undef.	$+0.002$	COL	0.514	0.456	undef.	-0.101
PER	0.621	0.507	undef.	-0.065	CRI	0.520	0.456	undef.	n/a
					MEX	0.660	0.635	undef.	-0.041

Undefined: 2 of 6 economies in 1995–2014 and 5 of 7 in 2011–2020, because $\Gamma_{t_0} \geq 0.50$.

(Rankings) By SCI, 1995–2014: Chile, Brazil, Costa Rica, Argentina. By ΔECI over the same years: Colombia ($+0.002$), Peru (-0.065), Chile (-0.088), Brazil (-0.174), Argentina (-0.205). The Spearman correlation between $\Delta\Gamma$ and ΔECI is -0.600 over the 5 economies with both. Over 2011–2020 it is $+0.429$ over 6. The rankings are unrelated, and the sign of the association flips between windows.

(Why ECI is not a proxy for Γ)

1. *Different matrices.* ECI is the second eigenvector of a country–product similarity matrix built from the binary RCA export matrix (Definition 16.7). Γ is a spectral property of the domestic input–output matrix. One concerns what a country exports relative to others; the other concerns how its sectors buy from each other.

2. *Relative versus absolute.* ECI is standardized across countries every year. A country's ECI can fall while its capabilities grow, if others grow faster. Γ is an absolute property of one country's matrix.

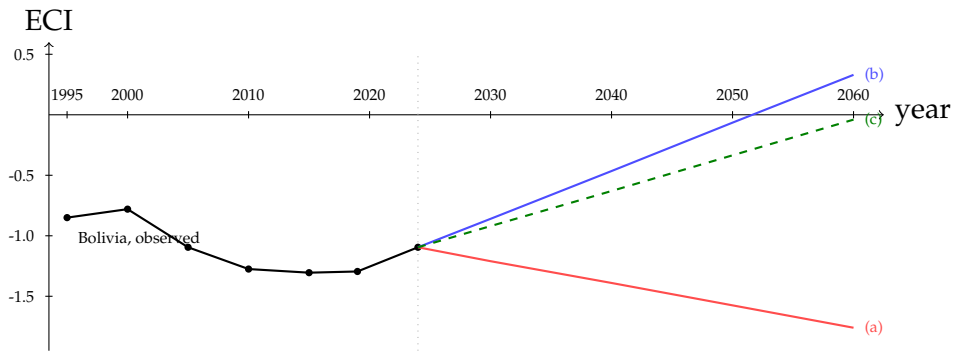
3. *Aggregation.* Γ at 12 sectors is dominated by the normalization and aggregation choices (Exercise 23.1).

The SCI with $\Gamma^{\text{ref}} = 0.50$ is mostly undefined, and where it is defined it is driven by the denominator. Chile's 5.30 comes from a gap of 0.033 (Example 23.18).

Exercise 23.7 Trajectory projection

(Inputs (sourced)) Bolivia's ECI (HS92) is -0.852 (1995), -1.295 (2019) and -1.097 (2024); Korea's is $+1.103$ (1995) and $+1.807$ (2019). Every scenario starts from the 2024 level, with a linear slope:

Scenario	slope/yr	2030	2040	2060	reaches 0
(a) Bolivia 1995–2019 trend ($-0.443/24$)	-0.0185	-1.208	-1.392	-1.761	never
(b) Bolivia 2019–2024 trend ($+0.198/5$)	$+0.0396$	-0.859	-0.463	$+0.329$	2051.7
(c) Korea's 1995–2019 pace ($+0.704/24$)	$+0.0293$	-0.921	-0.628	-0.041	2061.4



(Answers) Under (a) ECI never reaches 0: it keeps falling, to -1.761 by 2060. Under (b) it crosses 0 in 2051.7. Under (c) it crosses 0 only in 2061.4, just after the projection horizon (-0.041 in 2060). Bolivia would need to sustain Korea's 1995–2019 pace for 37 years to reach the cross-country mean.

(What a trend in ECI means) ECI is a z-score: in each release and year it is standardized to mean zero and unit variance across the countries in the sample. Four consequences follow.

1. A "trend" is a change in *relative* position. Bolivia's rise after 2019 can come from others' baskets becoming simpler, as much as from Bolivia's becoming more complex. In this case it partly reflects fuels leaving the export basket (Example 23.11).

2. "ECI $= 0$" means reaching the cross-country mean of that year's sample, not a fixed level of capability.

3. A z-score cannot trend linearly for ever: the cross-country distribution is renormalized every year, so ranks are a cleaner object. Bolivia's rank was 131 in 2019 and 123 in 2024.

4. Values are revised with each release, so a projection must state the release used. Here it is the release documented in `data/README.md`, §7.

Common mistake: treating ΔECI like a growth rate of an absolute quantity and extrapolating it.

Chapter 24

Hegemony and Counter-Hegemony

All numbers below are printed by `python3 code/solutions/sol24.py`. Unless a source is named, every parameter is [ILLUSTRATIVE]: Chapter 24 contains no sourced data except the input–output values of Example 24.10, and the data catalogue (`data/README.md`, row 100) records that the N^I_{eff} values of the chapter remain illustrative because no historical audience data exist.

Exercise 24.1 Hegemony as fixed point

A remark on the definition before the solution. Condition 2 of Definition 24.1 asks that $U_i(x_i^*, \theta_i^*) \geq U_i(x_i', \theta_i^*)$ for all *feasible* x_i'. If "feasible" means resource-feasible ($\sum_i x_i' = \Omega$) and U_i is increasing in x, the condition fails for every dominated agent whenever any transfer is possible, so no hegemony could exist. If it means "attainable under the current W", it holds trivially. The workable reading, and the one used in the last sentence of Definition 24.2 ("none demands a change in W"), is that no agent prefers, *given her formed preferences*, the action that would change W. We formalize that action as mobilization at a cost.

(a) The system. Two groups, E and M. The co-evolutionary system (Definition 13.6) has three blocks.

1. *Bargaining* (Proposition 7.3, equal disagreement points): $x_i = \Omega/N + (\pi_i^P - 1/N)\Phi$, with $\boldsymbol{\pi}^P$ the Perron vector of the production layer.

2. *Formation.* θ_t is the masses' counter-hegemonic preference, as in model (24.3): $\theta_{t+1} = \rho\theta_t + (1-\rho)[g(\theta_t) + s(W)]$, with logistic g of slope β. The counter-signal is what the masses see of the allocation gap through channels the elite does not control:

$$s(W) = \sigma\, W^I_{MM}\,(x_E - x_M) = \sigma(1-q)(x_E - x_M),$$

where $q = W^I_{ME}$ is the weight the masses put on elite-controlled media. This is the chapter's $(1-\rho)s = \mu\|\Delta(W^I\mathbf{r})\|$ with returns $\mathbf{r} = \mathbf{x}$ and the elite-filtered part of the signal removed.

3. *Network evolution* $\mathcal{E}$. A member of M who mobilizes pays c and, with the subjective weight θ she attaches to the alternative, expects the counter-hegemonic allocation x^{ctr}_M. She mobilizes iff $\theta(x^{\text{ctr}}_M - x_M) > c$, i.e. iff $\theta > \bar\theta := c/(x^{\text{ctr}}_M - x_M)$. Then $\mathcal{E}(W;\theta) = W^{\text{heg}}$ if $\theta \le \bar\theta$ and W^{ctr} otherwise.

A fixed point $(W^*, \theta^*, \mathbf{x}^*)$ of these three maps is a structural equilibrium (Definition 13.7); with the three conditions of Definition 24.1 it is a hegemonic equilibrium (Definition 24.2).

(b) Parameters and the hegemonic fixed point. Take

$$W^{P,\text{heg}} = \begin{pmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{pmatrix}, \qquad W^{I,\text{heg}} = \begin{pmatrix} 0.9 & 0.1 \\ q & 1-q \end{pmatrix}, \; q = 0.7,$$

$\Omega = 2$, $\Phi = 1$, $\sigma = 0.5$, $\rho = 0.85$, $\beta = 10$, $c = 0.15$. For a 2×2 row-stochastic matrix with off-diagonal entries a, b, $\boldsymbol{\pi} = (b,a)/(a+b)$ and $\lambda_2 = 1-a-b$. Hence $\boldsymbol{\pi}^P = (0.8, 0.2)$, $\Gamma(W^P) = 0.500$, and $\mathbf{x} = (1+0.3,\ 1-0.3) = (1.3, 0.7)$, which sums to Ω. The information layer has $\boldsymbol{\pi}^I = (0.7, 0.1)/0.8 = (0.875, 0.125)$ and $\Gamma(W^I) = 0.800$. The counter-signal is $s = 0.5 \times 0.3 \times 0.6 = 0.090$, below the saddle-node value $s^* = 0.1810$ of Proposition 24.6 (for $\beta = 10$; the book rounds to 0.181).

At $s = 0.09$ the formation map has three fixed points, $\theta = 0.1098,\ 0.4366,\ 1.0872$, with slopes $F'(\theta) = \rho + (1-\rho)g'(\theta) = 0.8791,\ 1.1897,\ 0.8542$: stable, unstable, stable. The hegemonic one is $\theta_h = 0.1098$. Under W^{ctr} (below) the masses' allocation would be $x^{\text{ctr}}_M = 1$, so $\bar\theta = 0.15/0.3 = 0.50$. At θ_h the perceived gain from mobilizing is $0.1098 \times 0.3 = 0.0329 < c$: nobody mobilizes, $\mathcal{E}$ returns W^{heg}, and $(W^{\text{heg}}, \theta_h, \mathbf{x})$ is a fixed point. Because $\mathcal{E}$ is locally constant at θ_h, the co-evolutionary Jacobian (Appendix 24.6) is block-triangular and its only nonzero eigenvalue is $F'(\theta_h) = 0.879 < 1$: the fixed point is locally stable. A simulation from $\theta_0 = 0$ converges to 0.1098 with no regime switch.

(c) The three conditions.

1. *Power concentration*: $N_{\text{eff}}(\boldsymbol{\pi}^P) = 1/(0.64 + 0.04) = 1.4706$ of $N = 2$; $\text{Var}(\boldsymbol{\pi}^P) = 0.0900$, 36% of its maximum $(N-1)/N^2 = 0.25$; $d_\pi = 0.300$ of a maximum 0.5. With $N = 2$, "well below N" can only mean this kind of fraction of the attainable range.

2. *Consent*: shown in (b). The elite consents trivially ($x_E = 1.3$ exceeds its counter-hegemonic 1.0).

3. *Formation alignment*: $\pi_E^I = 0.875 > \pi_M^I$ and $\pi_E^P = 0.8 > \pi_M^P$; the rank correlation is $+1$ (with two groups, the only possible positive value).

Sanity checks: both Perron vectors sum to one; $\sum_i x_i = \Omega$; $\mathrm{Var}(\boldsymbol{\pi}^P) \leq (N-1)/N^2$.

(d) When is there a counter-hegemonic equilibrium as well? Let the counter-hegemonic structure be $W^{P,\mathrm{ctr}} = \left(\begin{smallmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{smallmatrix}\right)$ (doubly stochastic, so $\boldsymbol{\pi}^P = (0.5, 0.5)$ by the Chapter 6 criterion, Proposition 6.8) and $q = 0.3$. Then $\mathbf{x} = (1, 1)$ and $s = 0$. At $s = 0$ the fixed points are 0.0072, 0.5 and $\theta_c = 0.9928 > \bar{\theta}$, so the masses keep mobilizing and $\mathcal{E}$ returns W^{ctr}: $(W^{\mathrm{ctr}}, \theta_c, (1, 1))$ is a second structural equilibrium. A simulation started at $\theta_0 = 0.6$ under W^{heg} switches at $t = 1$ and settles at 0.9928.

Coexistence holds exactly when both of the following are true.

- *The hegemonic equilibrium exists*: $s(W^{\mathrm{heg}}) < s^*(\beta)$ (Proposition 24.6, part 2; $\theta_h < a < \frac{1}{2} = \bar{\theta}$ then holds automatically). Here this is $\sigma(1-q)(x_E - x_M) < s^*$, i.e. $q > q^* = 1 - s^*/(\sigma \Delta x) = 0.397$: the elite must control at least about 40% of the masses' attention. With $q = 0.35$ the hegemony is not a fixed point: from $\theta_0 = 0$ the masses cross $\bar{\theta}$ at $t = 79$ and the system ends at θ_c.

- *The counter-hegemonic equilibrium exists*: $\theta_c(s^{\mathrm{ctr}}) > \bar{\theta}$. With $s^{\mathrm{ctr}} = 0$ this needs $\beta > 4$: $s^* = 0.0692$ and $\theta_c = 0.9293$ at $\beta = 6$, $s^* = 0.3029$ and $\theta_c \approx 1.000$ at $\beta = 20$. For $\beta \leq 4$, $g' \leq \beta/4 \leq 1$ everywhere (with equality at most at $\theta = \frac{1}{2}$), so $h(\theta) = g(\theta) - \theta$ is strictly decreasing and the formation map has a unique fixed point; at $s = 0$ it is the knife-edge $\theta = \frac{1}{2} = \bar{\theta}$, at which nobody mobilizes. This is Theorem 10.4, part 3, in miniature: multiplicity needs a nonlinearity strong enough to break the uniqueness condition.

So both equilibria coexist iff $\beta > 4$ and $q > q^*(\beta, \sigma, \Delta x)$. Stronger conformity ($\beta$) widens the coexistence region in two ways: it raises s^* and it makes θ_c more extreme.

Common mistakes. Reading consent as "the masses are well off"; it is a statement about their formed preferences, which is why θ_h must be computed. Using Γ for concentration: here $\Gamma(W^I) = 0.800$ although the elite holds 87.5% of $\boldsymbol{\pi}^I$ (Proposition 6.16).

Exercise 24.2 Maintenance cost

Data status. This is a data exercise. The budget data it needs are not in `data/processed` and were not fetched. No values are reported for them here; what

follows is the method and the expected qualitative result.

(a) Method.

1. *Spending on W^I control.* Source: the executed budgets (*Presupuesto General del Estado*, executed) of the Ministry of Economy and Public Finance, 2006–2019. Collect the line items of the state media named in Example 17.9 (Bolivia TV, Radio Patria Nueva, the newspaper *Cambio*), the ministry responsible for government communication, and transfers to community and indigenous radio networks. Add government advertising placed in private media; it is usually reported separately, by ministry. Use current bolivianos, deflated by the GDP deflator.

2. *Denominator*: total executed central-government expenditure for the same years, from the same source.

3. *Ratio*: report the share year by year and the 2006–2019 average. Check robustness to including state-enterprise advertising.

(b) From spending to N^I_{eff}: the model. Definition 24.3 gives $C_H = \mu_H \sum_{j \notin S_H} \pi^I_j |b_T - b^{(0)}_j| (1 - \lambda_2)$. Note that it depends on the *independent share* $\sum_{j \notin S_H} \pi^I_j$, not on N^I_{eff} directly. To get a threshold in N^I_{eff} an assumption is needed. Take n outlets: the coalition controls one, with weight $1 - \iota$; the other $n - 1$ share the independent weight ι equally and sit at a common distance $\delta = |b_T - b^{(0)}_j|$. Then

$$C_H = \mu_H \, \delta \, (1 - \lambda_2) \, \iota, \qquad N^I_{\text{eff}}(\iota) = \frac{1}{(1-\iota)^2 + \iota^2/(n-1)},$$

and N^I_{eff} is strictly increasing in ι on $(0, (n-1)/n)$ (checked in the script), reaching n at $\iota = (n-1)/n$. Let WTP be the coalition's willingness to pay, for instance a share of the rents its control secures, and let $\kappa := \text{WTP}/(\mu_H \delta (1 - \lambda_2))$. The cost exceeds WTP iff $\iota > \kappa$, that is, iff $N^I_{\text{eff}} > N^I_{\text{eff}}(\kappa)$. For $n = 4$ (the four ecosystems of Example 17.9):

κ	0.30	0.50	0.66
Threshold N^I_{eff}	1.923	3.000	3.834

For the record, the book's illustrative state share 0.34 gives $\iota = 0.66$ and, with symmetric independents, $N^I_{\text{eff}} = 3.834$. (The book's 3.38 uses the actual unequal shares. The symmetric version is the largest N^I_{eff} consistent with $\iota = 0.66$.)

(c) Expected qualitative result and interpretation. The direct budget share of state media is likely to be a small fraction of total spending. The model explains

why a small budget can still be binding. What matters is the cost *per unit of independent attention* times the independent share, against the rents the coalition defends. Two features of (24.2) should be stressed. (i) The threshold moves with $1-\lambda_2(W^I)$: when W^I is weakly coupled, both the coalition and its challengers can move the consensus cheaply (Proposition 8.9). (ii) Fragmentation of the private media lowers δ for the coalition only if the new outlets are closer to b_T. Digital channels that carry the opposition's narrative raise ι *and* δ.

Common mistakes: using Γ^I as the concentration measure; using the total communication budget without separating the share spent on independent outlets (advertising placed in private media buys weight in $j \notin S_H$, which is exactly what the formula prices).

Exercise 24.3 Counter-hegemony simulation

Well-posedness. λ^I is not defined in Chapter 24. We read it as the cross-group mixing weight of Chapter 18 (Definition 18.2): the share of the dominated group's attention on counter-hegemonic sources. It must still be mapped into the signal s of (24.3). The exercise is solvable once that map is stated, and the answer to "how does it depend on ρ" turns on it. We give both natural conventions.

(a) Set-up. Model (24.3), $\beta = 10$, start at the hegemonic equilibrium for $s = 0$, $\theta_0 = \theta_h = 0.0072$, with $\lambda_0^I = 0$ and $\lambda_t^I = 0.02\,t$ for $t = 1, \dots, 50$.

- *Convention A*, the signal in preference units: $s_t = 0.5\,\lambda_t^I = 0.01\,t$.
- *Convention B*, the raw signal in return units: $\mu\|\Delta(W^I\mathbf{r})\| = 0.075\,\lambda_t^I$. Then $s_t = 0.075\lambda_t^I/(1-\rho)$, as the chapter's scaling $(1-\rho)s = \mu\|\Delta(W^I\mathbf{r})\|$ prescribes. At $\rho = 0.85$ the two conventions coincide.

Two collapse dates are reported. The *equilibrium* collapses at the first t with $s_t > s^* = 0.1810$ (the saddle-node). The *state* collapses when θ_t first exceeds $\frac{1}{2}$.

(b) Results.

	Convention A		Convention B	
ρ	equilibrium lost	$\theta_t > \frac{1}{2}$	equilibrium lost	$\theta_t > \frac{1}{2}$
0.50	19	27	never	never (0.185)
0.70	19	31	37	never (0.406)
0.85	19	37	19	37
0.95	19	never (0.408)	7	29

In parentheses: θ_{50} when θ never crosses $\frac{1}{2}$.

In the baseline ($\rho = 0.85$) the hegemonic equilibrium ceases to exist at period 19 ($s_{19} = 0.19 > 0.181$). The population crosses the separatrix, and hence "collapses" visibly, only at period 37, eighteen periods later. (Beyond the crossing θ exceeds 1: with $s > 0$ the upper fixed point $\theta_c(s) = g + s$ lies above 1. If θ is read as a share, that is the cost of the linear s term.)

(c) Dependence on ρ. The fold is located by s^*, which depends on β only, through $g'(a) = 1$ and $s^* = a - g(a)$. So in convention A the date at which the equilibrium is lost does not depend on ρ. Persistence acts only on the *transit*: the excess drift is $(1-\rho)[g(\theta) + s - \theta]$, so higher ρ slows the move to θ_c (27, 31, 37 and more than 50 periods). In convention B, ρ also enters the signal, through the factor $1/(1-\rho)$, the formation multiplier of Definition 10.3. Stickier preferences then turn the same flow of counter-hegemonic edges into a larger long-run shift. The equilibrium is lost earlier (period 7 at $\rho = 0.95$, period 37 at $\rho = 0.70$), and for $\rho = 0.5$ it is never lost within 50 periods ($s_{50} = 0.15 < s^*$). The general lesson is that ρ has two opposite effects: it amplifies a persistent signal through $\mathcal{M} = \mu/(1-\rho)$, and it slows adjustment. Which dominates depends on whether policy moves s or the raw signal.

Common mistake: reporting the crossing date as the date the equilibrium disappeared. The fold comes first; the gap between the two dates is the period in which the hegemony survives in practice although it is no longer an equilibrium.

Exercise 24.4 Bolivia's hegemonic transitions

Broken reference. Chapter 24 has no "Prebisch Connection" box (it contains no `prebisch` environment). The five Bolivian hegemonies are those of Example 24.4 (the table without the UDP interregnum and the post-2019 contested period), and that example is used here. The N^I_{eff} values are the example's and are [ILLUSTRATIVE].

(a) The mapping. The coalitions are the four class alliances of Example 14.9 (foreign capital F, domestic bourgeoisie B, middle class M, workers W). The script checks that each illustrative $\Delta\pi$ vector there sums to zero and that every member gains ($\Delta\pi_i > 0$), the first half of the internal-stability condition (14.12).

Hegemony	Coalition (Ch. 14)	W^I control mechanism	N^I_{eff}	Trigger of collapse
Tin oligarchy (to 1952)	$\{F, B\}$ (Alliance 1)	Press owned by or allied with mining interests; literacy-restricted suffrage limits who enters political W^I	1.5	Chaco War defeat (an external shock in Skocpol's sense), the annulled 1951 election, the armed miners in April 1952
MNR (1952–64)	$\{B, M, W\}$ (Alliance 2)	Party, COB and union networks, miners' radios; land reform and universal suffrage widen W^I	2.0	Split between the MNR government and the COB after stabilization; elite defection; 1964 coup
Military (1964–82)	$\{F, B, M\}$ (Alliance 3)	Censorship, state broadcasting; consent largely replaced by coercion	1.8	Economic crisis of the late 1970s and early 1980s; democratization in 1982
Neoliberal (1985–2003)	$\{F, B, M\}$ (Alliance 3)	Private commercial media framing privatization as modernization; "pacted" party system	2.5	Water War 2000, fiscal crisis of 2003, Gas War 2003 (Example 24.10)
MAS (2006–19)	$\{M, W\}$ (Alliance 4)	State media, indigenous-movement and union networks, community radio	3.0–3.4	Disputed 2019 election, urban civic committees, social-media mobilization (Example 24.7)

(b) What the mapping shows. (i) Every transition is an alliance switch in the sense of Example 14.11. The 1964 and 1985 switches are the defection of B (and M) from a domestic alliance to $\{F, B, M\}$, the switch condition (14.14). (ii) The W^I mechanism changes character over time. The early hegemonies controlled *access* to the information network (who could read, vote or broadcast); the later ones compete for *weight* within a network everyone can enter. That is the reading of a rising N^I_{eff} in Definition 24.3. (iii) The military period is the one hegemony whose stability came from coercion rather than consent. Example 24.4 labels it "coercion-dependent", and it shows that condition 2 of Definition 24.1 is the weak link of the formalization (see Exercise 24.7). (iv) Triggers are of two kinds: external or fiscal shocks (1932–35, the early 1980s) and accumulated ΔW^I (2000–03, 2019). The first kind corresponds to conditions 3–4 of Conjecture 24.9, the second to conditions

1–2.

Common mistakes: treating the N^I_{eff} column as data (it is not); treating the 1982–85 UDP government as a hegemony (the book lists it as "no stable hegemony").

Exercise 24.5 Skocpol's conditions

(a) The four conditions, 2000–2006 (Example 24.10).

1. *Counter-signal*: met. Community radio and the movements of the 1990s built alternative W^I channels. The book's N^I_{eff} path ($2.0 \to 2.8$) is illustrative.

2. *Exposing shock*: met. The Water War of 2000 turned the "returns to privatization" signal negative for poor households, a shock to $\mathbf{r}_t$.

3. *State breakdown under external pressure*: met only in weak form. There was fiscal pressure and an IMF-backed tax proposal (February 2003), followed by riots and a police mutiny, and the governing coalition split. But there was no international military pressure, and the coercive apparatus did not disintegrate. The successions of 2003 and 2005 followed constitutional rules, and the transition was completed through elections. In Skocpol's terms this is a crisis of government, not a state breakdown, as the chapter notes after Definition 24.8.

4. *Autonomous organization of the dominated*: met strongly. The El Alto *juntas vecinales*, cocalero and campesino unions and middle-class groups formed the $\{M, W\}$ alliance that forced two presidents to resign.

(b) Which was binding? In model (24.3), conditions 1 and 2 are *substitutes*. Condition 1 raises s; condition 2 is a one-time push $\Delta\theta$. The hegemony falls if $s > s^*$, or if the shock exceeds the distance to the separatrix, $\theta_u(s) - \theta_h(s)$. The script computes that distance:

s	0.00	0.05	0.10	0.15	0.17	0.18
Required shock $\theta_u - \theta_h$	0.4928	0.4037	0.3061	0.1827	0.1072	0.0314

It is strictly decreasing in s and vanishes at s^*. A decade of counter-hegemonic channel building (condition 1) is what made a local tariff conflict (condition 2) sufficient. Conditions 3 and 4 act through a different block of the system: the coalition's feasibility set $\mathcal{F}(S)$ (Definition 14.2). A preference shift changes W only

if a coalition that can implement ΔW forms. The best-supported answer is therefore that **condition 4 was binding** for the change of structure, and **condition 1** for the change of preferences. Condition 3 was present but not Skocpolian.

(c) Counterfactuals, one condition at a time. (i) Without condition 1 (low s), the 2000 shock would have had to be several times larger: about 0.49 preference units at $s = 0$ against 0.03 near s^*. A purely local conflict would probably have been absorbed. (ii) Without condition 2, the hegemony could have persisted while $s < s^*$, as a hegemony that is still an equilibrium but fragile (Exercise 24.3 shows how long that phase can last). (iii) Without condition 3, the governing coalition does not split. Removal of presidents by protest remains possible, but the elite keeps $\mathcal{F}(S)$ and can reconstitute the neoliberal W^P through elections. (iv) Without condition 4, no feasible counter-coalition exists, and the outcome is an interregnum (Appendix 24.6), not a new hegemony. One episode cannot identify these counterfactuals. The model only says which comparisons would be informative: cases with high s but no organization, or the reverse.

Common mistake: calling 2003 a Skocpolian revolution. Skocpol's scope conditions (agrarian-bureaucratic state, international military competition) fit 1952, not 2003.

Exercise 24.6 The acceleration of the cycle

Broken reference. As in 24.4, there is no "Prebisch Connection box". The Latin American epochs are in the intuition box "The Latin American hegemonic cycle" of Section 24.4. That box gives only approximate dates and gives N^I_{eff} for three of the five epochs. All inputs below are therefore [ILLUSTRATIVE], and the exercise cannot be answered with data.

(a) Inputs. Durations, chosen to match the box's "between about 13 and 40 years": oligarchic 1890–1930 (40), national-populist 1930–1964 (34), bureaucratic-authoritarian 1964–1985 (21), neoliberal 1985–2003 (18), left-populist 2006–2019 (13). N^I_{eff}: 1.5 (midpoint of "1–2"), 2.5 (midpoint of "2–3") and 3.0, as in the box. For the two epochs the box leaves blank we borrow Bolivia's values from Example 24.4: 2.0 (MNR) and 1.8 (military).

(b) Correlation. Pearson $r = -0.829$. The exact one-sided permutation p-value is 0.042 (all 120 orderings of the five N^I_{eff} values). The OLS line is duration $= 59.41 - 15.84\, N^I_{\text{eff}}$. As a check, Bolivia's four closed periods (12, 18, 18, 13 years against N^I_{eff} = 2.0, 1.8, 2.5, 3.2) give $r = -0.313$ and $p = 0.417$ (24 orderings). The sign is the same, but there is no pattern to speak of.

(c) Interpretation. The negative correlation in the regional series is not evidence for the acceleration hypothesis, for four reasons. (i) The N^I_{eff} values are invented placeholders (data catalogue row 100), two of them borrowed from another unit. (ii) Both variables trend with calendar time: the correlation of duration with the epoch's end year is -0.979, and that of N^I_{eff} with end year is 0.900. *Any* trending variable would "explain" durations as well, so the design cannot separate N^I_{eff} from the time trend. (iii) Reverse causality: a long hegemony suppresses N^I_{eff} (censorship, ownership concentration). N^I_{eff} measured within the epoch is an outcome of the hegemony, not a predetermined cause. (iv) The last epoch is right-censored in spirit, because recent epochs have had less time to be long, and the end dates are coded by the analyst. The formal object for a real test is a duration (hazard) model for hegemonic spells, $h(t \mid N^I_{\text{eff},0})$, estimated on many countries, with N^I_{eff} measured at the start of each spell from audience data. Remark 24.12 adds that a cycle needs recurring shocks or a slow reversing feedback; a monotone N^I_{eff} trend on its own produces one transition, not a cycle.

(d) Prediction for the current period. Read literally, the fitted line predicts a spell of $59.41 - 15.84 \times 3.5 = 4.0$ years at $N^I_{\text{eff}} = 3.5$ (the book's value for 2019 onward). That is consistent with the book's "no hegemony" label, but it is an extrapolation beyond the largest observed N^I_{eff}, from five illustrative points, and it should be reported only to show how fragile the inference is. Bolivia's own line predicts 13.4 years. The two "predictions" differ by a factor of three.

Exercise 24.7 Gramsci vs. Skocpol

(a) Complementary, in a precise sense. Gramsci's hegemony is "consent armoured by coercion" (Definition 24.1). A regime survives a period if *either* consent holds ($\theta_t \le \bar{\theta}$, the Gramscian W^I loop) *or* coercion suppresses the mobilization that dissent would produce (capacity $K_t \ge \bar{K}$, which Skocpol's international and fiscal pressure erodes). It falls at the first date at which both fail:

$$T_{\text{collapse}} = \max\{T_{\text{consent}},\ T_{\text{coercion}}\}.$$

Each theory explains one of the two necessary conditions for collapse, so they are complements as explanations. A third route runs through the coalition block: an elite split under the switch condition (14.14) changes W directly, even with consent intact. Skocpol's "dominant-class obstruction" belongs there.

(b) When each dominates. The mechanism that fails *last* sets the date, and it

is the one an observer will call "the cause". In model (24.3), with a linearly rising counter-signal $s_t = s_0 + \delta_s t$, consent fails at $T_{\text{consent}} = (s^*(\beta) - s_0)/\delta_s$. With coercive capacity falling linearly, $T_{\text{coercion}} = (K_0 - \bar{K})/\delta_K$. Hence

$$\text{Gramsci dominates} \iff \frac{s^*(\beta) - s_0}{\delta_s} > \frac{K_0 - \bar{K}}{\delta_K}.$$

Gramsci's mechanism dominates when conformity is strong (large β, hence large s^*), when the elite controls W^I (s_0 small, q large in 24.1), when counter-channels grow slowly, and when coercive capacity erodes quickly or cannot be used (democracies with high repression costs). Skocpol's mechanism dominates when consent is already weak (s_0 near s^*) but the coercive apparatus is strong and erodes only under external pressure: military regimes and agrarian-bureaucratic states. With $s_0 = 0.05$, $\delta_s = 0.01$ per year and $s^* = 0.181$, consent fails after 13.1 years. With $K_0 = 1$ and $\bar{K} = 0.4$, coercion fails after 60.0, 30.0 and 12.0 years for $\delta_K = 0.01, 0.02, 0.05$. The regime falls at 60.0 and 30.0 years in the first two cases (Skocpol's timing) and at 13.1 years in the third (Gramsci's timing).

(c) Application. Bolivia 1964–82 ("coercion-dependent", Exercise 24.4) is the Skocpol regime: consent was low, and the timing was set by the economic and fiscal crisis. Bolivia 2003 is the Gramsci regime: a democracy in which coercion was costly to use, whose timing was set by the counter-signal crossing the threshold. The two theories compete only if each is read as a sufficient account of collapse. Read as accounts of the two necessary failures, they are parts of one model. The synthesis is the book's (Conjecture 24.9), not Skocpol's, who gives ideas no causal role.

Common mistake: putting consent and coercion in a sum ("more consent offsets less coercion") instead of in the max / "or" structure that "armoured by" describes.

Chapter 25

Truncated Industrialization

All numbers below are printed by `python3 code/solutions/sol25.py`. Sourced values come from `data/processed` (retrieved 2026-11-05; sources and scripts in `data/README.md`). Every exercise in this chapter needs country × product RCA data and product-space proximities. The data catalogue lists both as "fetchable, not done" (rows 101–102 and 34): the Growth Lab country × product file is 451 MB and was not downloaded. Those parts are therefore answered with the method and the expected result only. The quantities that *are* in the repository (ECI, PCI, diversity, Γ) are reported.

Exercise 25.1 Identify the dense subgraph

Data status: method only. The SITC proximity matrix was not computed (data catalogue row 101). Part (b) also shows that, applied to the weighted proximity matrix, the book's own definition does not return the capital-goods core. The exercise should be run on a thresholded proximity graph.

(a) Method.

1. *Data*: Growth Lab, International Trade Data (SITC rev. 2), doi:10.7910/DVN/H8SFD2, country × product × year at the 4-digit level. Use one year (for comparability with Hidalgo et al., 2007, a year around 2000), or the average of three adjacent years to reduce noise. Apply and state minimum-size filters for countries and products.

2. *RCA and M*: R_{cp} by (16.2); $M_{cp} = \mathbf{1}[R_{cp} \geq 1]$ (Definition 16.5).

3. *Proximity*: $\phi_{pp'} = \sum_c M_{cp} M_{cp'} / \max(M_p, M_{p'})$ (Definition 16.6), with the diagonal excluded as Definition 25.1 requires.

4. *2-digit averages*: for each SITC 2-digit group S, $\bar{\phi}_S = 2e(S)/(|S|(|S|-1))$.

5. *Peeling* (Appendix 25.6): repeatedly remove the product with the smallest remaining weighted degree and record $e(S)/|S|$ along the way. Use a priority queue or incremental degree updates: $O(P^2)$ for a dense ϕ.

6. *Report*: $\mathcal{D}$, $|\mathcal{D}|$, its composition by SITC section, $\bar{\phi}_{\mathcal{D}}/\bar{\phi}_{\text{all}}$ and the mean PCI of $\mathcal{D}$. Check robustness to the year, the HS92 classification and the threshold of part (c).

(b) An obstacle in the definition. For any set S, $e(S)/|S| = (|S|-1)\bar{\phi}_S/2$. On a weighted proximity matrix in which almost every pair has positive proximity, the objective therefore rewards *size* almost as much as density. With the chapter's own illustrative values (Stylized Fact 25.2 and Example 25.7):

$$\frac{e(\mathcal{D})}{|\mathcal{D}|} = \frac{99 \times 0.35}{2} = 17.325 \quad < \quad \frac{e(\text{all})}{775} = \frac{774 \times 0.08}{2} = 30.960.$$

The 100-product core is *not* the Goldberg maximizer: the whole space beats it. A set with mean proximity 0.35 would need more than 178 products to beat the whole space. The synthetic check in the script confirms this. It uses a product space that is *not data*: 775 products with a 100-product core, core proximities Beta$(3.5, 6.5)$ (mean 0.354) and other proximities with mean 0.0755, so that the overall mean is 0.080. Weighted peeling returns all 775 products (objective 31.019, density ratio 1.00).

(c) The fix: threshold first. Binarize the product space at $\phi_{pp'} \geq \phi_0$, as Hidalgo et al. (2007) did to draw it, and run peeling, or exact Goldberg max-flow, which is cheap at 775 nodes, on the 0–1 graph. In the synthetic space with $\phi_0 = 0.25$, 74.0% of core pairs and 5.67% of the other pairs survive. Peeling then returns exactly the 100 core products (Jaccard index 1.000; objective 36.650) with a weighted density ratio of 4.42, the chapter's illustrative 4.4. Report the results over a grid of thresholds. A $\mathcal{D}$ that is stable across thresholds is the object Definition 25.1 intends.

(d) Expected result. On the real data (not computed here), the thresholded $\mathcal{D}$ should be the core Hidalgo et al. (2007) describe: machinery, metal products and chemicals, with high mean PCI (Appendix 25.6). Whether it is "the capital-goods cluster" depends on the threshold and on whether chemicals and plastics enter. The 2-digit averages $\bar{\phi}_S$ will be highest for SITC 7 (machinery and transport equipment) and parts of SITC 5 and 6, and lowest for primary commodities (SITC 0–3).

Common mistakes: including $\phi_{pp} = 1$ (every singleton then has density 1; Section 25.1); reading the peeling output as optimal (it is a 2-approximation); reporting $\mathcal{D}$ from one threshold only.

Exercise 25.2 Truncation index

Data status: method only for τ_c; ECI is sourced.

(a) Method. Fix $\mathcal{D}$ from Exercise 25.1 (one year, one threshold). For each country, compute M_{cp} for the same year and $\tau_c = 1 - |\{p \in \mathcal{D} : M_{cp} = 1\}|/|\mathcal{D}|$ (25.2). Use ECI from the same classification and year.

(b) What the repository has. ECI (Growth Lab, HS92, 2019; The Growth Lab at Harvard University, 2026): Korea $+1.807$ (rank 2), Germany $+1.544$ (6), Brazil $+0.481$ (52), Chile -0.059 (76), Bolivia -1.295 (131). The expected τ ranking is the reverse: Bolivia highest, then Chile, Brazil and the two industrial economies. The chapter's illustrative values are $\tau_{\text{BOL}} = 0.98$ and $\tau_{\text{KOR}} = 0.18$.

(c) Is the correlation strong, and does that mean anything? It will almost certainly be strongly negative, and for a partly mechanical reason. Both τ_c and ECI are functions of the same matrix M. $\mathcal{D}$ is, by Appendix 25.6, rich in high-PCI products. And ECI is, up to an affine standardization, the average PCI of the products a country exports. (From $\tilde{M}\mathbf{v} = \lambda\mathbf{v}$ and $\mathbf{q} = U^{-1}M'\mathbf{v}$ one gets $v_c = \lambda^{-1}k_c^{-1}\sum_p M_{cp}q_p$, with k_c the diversity; Definition 16.7.) A country that holds many $\mathcal{D}$ products has a high average PCI almost by construction. With five countries, the Pearson coefficient is not informative: run the regression on all countries in the Atlas, and report the correlation of τ with ECI *net of* diversity k_c to see whether τ carries information beyond "many complex products".

Exercise 25.3 The truncation-Γ relationship

Data status: Γ results are sourced (12 sectors and detail); τ_c, the product-space proxy and the bridge test are method only.

(a) 12-sector results (2019; book12_isic, domestic flows, input shares, diagonal kept). Twenty-three economies have a like-for-like Γ (Bolivia from INE, base 2017; the rest from OECD ICIO 2023). It ranges from 0.333 (Viet Nam) to 0.659 (Indonesia), with mean 0.511 and standard deviation 0.077. Sixteen of them also have an HS92 ECI in the repository:

	ARG	BOL	BRA	CHL	CHN	COL	CZE	DEU
Γ	0.522	0.475	0.496	0.557	0.582	0.460	0.486	0.544
ECI	+0.063	−1.295	+0.481	−0.059	+1.301	+0.140	+1.478	+1.544
	JPN	KOR	MEX	MYS	NOR	PER	POL	USA
Γ	0.510	0.511	0.649	0.381	0.515	0.407	0.552	0.463
ECI	+2.047	+1.807	+0.953	+1.031	+0.962	−0.680	+1.060	+1.294

Across them $\mathrm{corr}(\Gamma, \mathrm{ECI}) = 0.265$ (Spearman 0.176), and OLS gives $\Gamma = 0.493 + 0.0188\,\mathrm{ECI}$ (s.e. 0.0183). The slope is not distinguishable from zero. ECI is not τ, but if truncation lowered aggregate Γ, the least complex economies should show it, and they do not. This is Result 25.8 on a larger sample.

(b) Does the answer change with resolution? Yes, at least for the ranking. For the 22 ICIO economies with both resolutions, the correlation between 12-sector Γ and detail Γ (44 industries for most, 43–45 for a few) is 0.476 (Spearman 0.296). The mean falls from 0.512 to 0.411. Germany moves from 0.544 to 0.372 and China from 0.582 to 0.377, from the upper half of the distribution to the bottom; the United States barely moves (0.463 and 0.465); Korea goes from 0.511 to 0.455. Aggregation can raise Γ, and does so unevenly across countries (Chapter 19). That is why the 12-sector test cannot see a capital-goods bridge. Bolivia's 39-industry value (0.410) cannot be compared with these until INE's industries are mapped to ICIO's. Method: map INE's 39 industries to ISIC Rev. 4 divisions, aggregate both sources to the finest common partition (about 30 groups), and recompute Γ on that partition.

(c) The product-level test (method).

1. τ_c for 30 countries from Exercise 25.2.

2. The proxy $W_c = D_c^{-1}\phi_c$ on c's products, with $\Gamma(W_c) \le 1 - \lambda_2(W_c) \le 2\Phi_c(S)$ (Proposition 25.4). The proxy's node set differs across countries. By the book's convention on node sets, compare $1 - \lambda_2$ on a common product set, for example the union of all countries' products with c's absent products given zero weight, or on random subsamples of fixed size.

3. *Is $\mathcal{D}$ a bridge?* Compute the betweenness centrality and the Guimerà–Amaral participation coefficient of $\mathcal{D}$'s products relative to a community partition of the periphery, and the conductance of each commodity cluster with and without $\mathcal{D}$. A bridge has high participation, and the conductance falls sharply when $\mathcal{D}$ is removed (Remark 25.5).

4. Regress Γ_c on τ_c, both for the proxy and for ICIO input shares. Estimate f by OLS and by isotonic (monotone) regression, and test $f' < 0$.

Expected result. For the proxy, a negative slope if $\mathcal{D}$ is a bridge. For ICIO Γ at 12 sectors, no relation, as in part (a). At 44 industries the result is open. The finding of part (b) is that the conclusion may well depend on the resolution.

Exercise 25.4 Counterfactual: Bolivia with the dense subgraph

Partly ill-posed. The chapter has no model linking $\mathcal{D}$ to σ_B or to the size of $\Delta\Gamma$, so "how much would Γ increase / σ_B decrease" presupposes both a sign and a magnitude the book does not deliver. We give the computable parts, the sign analysis and the method.

(a) τ**.** With the illustrative counts ($|\mathcal{D}| \approx 100$, Bolivia holds 2): $\tau = 1 - 2/100 = 0.98 \to 1 - 22/100 = 0.78$.

(b) ECI. The exact method: set $M_{\mathrm{BOL},p} = 1$ for the 20 products, recompute ubiquities, $\tilde{M}$ and the second eigenvector for all countries in the same year, and standardize (Definition 16.7). A first-order approximation uses the identity of Exercise 25.2(c): ECI is an affine function, common to all countries in a year, of the mean PCI of a country's products, $\mathrm{ECI}_c = \alpha + \beta\,\overline{\mathrm{PCI}}_c$. Holding the PCI vector fixed,

$$\Delta\mathrm{ECI}_{\mathrm{BOL}} \approx \beta\,\frac{20}{k_{\mathrm{BOL}} + 20}\left(\overline{\mathrm{PCI}}_{\mathrm{new}} - \overline{\mathrm{PCI}}_{\mathrm{BOL}}\right).$$

Bolivia's 2019 diversity is $k_{\mathrm{BOL}} = 75$ (Growth Lab HS92 country-year file), so the 20 products carry weight $20/95 = 0.21$ in the new average, and its diversity rises by 26.7%. β and $\overline{\mathrm{PCI}}_{\mathrm{BOL}}$ need the country × product file (not fetched). The approximation overstates the gain slightly: the 20 products become more ubiquitous, so their PCI falls a little.

(c) Γ**.** The sign is not determined. Adding products changes the product-space proxy W_c in every row that touches them, and Remark 25.5 shows that the effect can go either way. In the chapter's two toy spaces (recomputed in the script), adding the core raises Γ from 0.009 to 0.548 when it is a bridge and *lowers* it from 0.800 to 0.218 when it is a module. For the 12-sector W^P, the 20 products sit inside "Manufacturing". They change Γ only through the change in that sector's domestic input and sales shares, a perturbation E with $E\mathbf{1} = 0$ whose first-order effect $u_2' E v_2$ (Lemma 6.19) has no general sign. Approximate method: append a capital-goods row and column to Bolivia's 2019 domestic table (`data/processed/bolivia_Z`), with input shares taken from a comparator (Korea's

ICIO machinery industries) and output sold to domestic users in proportion to Bolivia's current imports of capital goods; then recompute Γ.

(d) σ_B**.** Between-block inequality falls only if the new activities employ, or buy from, the low-income blocks. Core products are capital- and skill-intensive. Unless accompanied by the human-capital operator of Chapter 15, their direct effect on σ_B is likely to be small in the short run. Measuring it requires the household microdata that are not available here (data catalogue row 105).

Exercise 25.5 The divergence

Data status: method only for τ. ECI (SITC) is sourced from 1995 on, which is too late for 1960 and 1980.

(a) Method. Growth Lab SITC rev. 2 country $\times$ product data begin in 1962. Use 1962 (for "1960"), 1980 and 2000, and a fixed $\mathcal{D}$ (from Exercise 25.1, computed once for a reference year so that τ moves only with countries' baskets). Compute τ_c for Brazil, Mexico, South Korea and Taiwan. *Taiwan*: UN Comtrade has no Taiwanese reporter (it appears as "Other Asia, n.e.s."), so check whether the Atlas reconstructs Taiwan from partner (mirror) data; Taiwan is not in this repository's extract. Also plot the annual series, not only three dates, so that the divergence date is located to within a few years.

(b) What the repository shows (SITC ECI). 1995: Korea +1.231, Mexico +0.982, Brazil +0.819. 2000: +1.389, +1.169, +0.812. 2019: +1.706, +1.132, +0.277. Brazil's complexity fell by 0.54 after 1995 while Korea's rose by 0.48. So a post-1995 divergence between Brazil and Korea is visible in ECI. Mexico, measured by exports, stays close to Korea.

(c) Expected result and the measurement trap. On the chapter's account (Example 25.12), Korea's τ should fall steeply in the 1970s and 1980s, during and after the HCI drive (1973–79), and Brazil's should fall partly in the 1970s (II PND) and then stall. A divergence date in the mid-1970s would support a policy reading, but the exports-based τ cannot separate policy from the timing of world demand. Mexico is the trap. Maquiladora assembly of imported components gives Mexico RCA in machinery and electronics *exports*. Its export-based τ can therefore fall although, on the chapter's reading, it "imported $\mathcal{D}$ as a service". RCA measures what is exported, not what is produced with domestic capabilities. To test Fajnzylber's claim, compute τ also on domestic value added in exports (OECD TiVA) where available, or weight M_{cp} by the domestic content of exports.

Exercise 25.6 Bolivia's incremental path

Data status: method only for (a) and the gross entry rate; net diversity, ECI and v_Γ are sourced.

(a) Top-5 products by relatedness density. For every $p \in \mathcal{D}$ compute $\omega_{\mathrm{BOL},p} = \sum_{p'} M_{\mathrm{BOL},p'}\phi_{pp'} / \sum_{p'} \phi_{pp'}$ (Definition 16.14) and rank. Use relatedness to $\mathcal{D}$ as the target. A useful refinement is to rank products *outside* $\mathcal{D}$ by the product of their $\omega_{\mathrm{BOL},p}$ and their proximity to $\mathcal{D}$: these are the stepping stones. The products are not listed here because the data were not computed. The chapter's Step 1–2 products (gas processing, zinc smelting, lithium compounds, basic chemicals, fertilizers) are illustrative candidates, not results.

(b) Capabilities and operators. For each product: the process technology (hydrometallurgy, chemical synthesis, precision machining), the skills (chemical and process engineers, technicians), the inputs and infrastructure (energy, water, transport to ports), and the standards and certification needed to sell. Map them to Chapter 15's operators: $\mathcal{L}_{\mathrm{HC}}$ for skills (Definition 15.9), $\mathcal{L}_{\mathrm{infra}}$ for energy and transport, $\mathcal{L}_{\mathrm{inst}}$ for contract and standards regimes, $\mathcal{L}_{\mathrm{info}}$ for links to buyers and engineers abroad, and an $\mathcal{L}_{\mathrm{ISI}}$-type operator for domestic supply contracts. Order them with the sequencing heuristic, Conjecture 15.14, which is a conjecture.

(c) Time at the observed rate of entry. The quantity needed is the *gross* entry rate: the number of products per year that move from no RCA to stable RCA (for instance, $R_{cp} < 0.5$ for three years, then $R_{cp} \geq 1$ for three years). That requires the country $\times$ product file. The repository holds only *net* diversity (number of HS92 products with RCA ≥ 1), and net diversity is not a usable proxy. Over 1995–2019 Bolivia's diversity went from 97 to 75 (net -22, -0.92 per year) and Korea's from 282 to 257 (net -25, -1.04 per year), although Korea's ECI rose from $+1.10$ to $+1.81$. Bolivia's diversity also swings widely from year to year, from 68 (2014) to 194 (2004), which reflects commodity-price-driven changes in RCA rather than capability gains. Once gross entry rates e_c are measured, the time to traverse a path of n_s steps, with m_s products to enter at step s, is roughly $\sum_s m_s / e_c(\omega_s)$, with the entry rate itself rising in relatedness density (Stylized Fact 16.15).

Why v_Γ is the wrong clock. (i) *Wrong object*: Γ is the mixing speed of the sectoral input-share network. Entering a product inside "Manufacturing" does not add a node, and its effect on Γ has no definite sign (Exercise 25.4). (ii) *Wrong sign*: Bolivia's velocity is negative (-0.036 per decade over 1990–2013; -0.078 if the outlying 2014 is kept), so a clock based on v_Γ gives a negative or infinite time. Korea's was -0.055 per decade over 1995–2014 while it deepened its core, and

+0.055 over 2011–2020 in the next ICIO vintage. (iii) *Wrong units*: a product-space step is a change in capabilities, measured by relatedness density, not a change in Γ. Example 25.14 makes the same point.

Exercise 25.7 The lithium question

(a) Relatedness density. $\omega_{\text{BOL,LiB}} = \sum_{p'} M_{\text{BOL},p'} \phi_{\text{LiB},p'} / \sum_{p'} \phi_{\text{LiB},p'}$ (Definition 16.14) with LiB = HS 8507, computed from the HS92 country × product file and the Atlas proximities. *Not computed here* (data not fetched). Expected: low, because Bolivia's RCA products (gas, ores, gold, soy) are far from electrical machinery in the product space. Note that HS92 has no separate lithium-ion heading: 8507 pools all electric accumulators, including lead-acid batteries, so ω and PCI refer to the pooled heading. A finer heading for lithium-ion cells exists only in later HS revisions.

(b) Intermediate products and their complexity (PCI, Growth Lab HS92, 2019; The Growth Lab at Harvard University, 2026). Natural gas (2711) −2.226; carbonates, including lithium carbonate (2836), −0.573; the heading that includes lithium hydroxide (2825) −0.276; electric accumulators (8507) +0.549 (+0.811 in 2023). The steps in PCI are +1.653 (gas to carbonate) and +1.122 (carbonate to accumulators). Cathode active materials, the step between hydroxide or carbonate and cells, fall under HS chapters 28 and 38 and need their own concordance. The chain does not start from Bolivia's current gas exports: it starts from a new extractive activity, brine.

(c) Operator sequence. (1) Brine to battery-grade carbonate or hydroxide: process chemistry and quality control ($\mathcal{L}_{\text{HC}}$, $\mathcal{L}_{\text{info}}$ through technology partners), plus energy, water and transport to the salt flats ($\mathcal{L}_{\text{infra}}$). (2) Cathode materials: precursor chemistry, purity standards, and reliable supply of other metals (nickel, manganese, cobalt, or iron phosphate), with offtake contracts ($\mathcal{L}_{\text{inst}}$). (3) Cells: electrode coating, assembly at scale, a large capital commitment, and proximity to vehicle or storage buyers. This is where an industrial-policy operator ($\mathcal{L}_{\text{ISI}}$-type procurement or joint ventures) is needed. Each step needs different capabilities, and in the product space each is a separate move.

(d) A realistic entry point? As an entry to $\mathcal{D}$: probably not directly. Carbonate and hydroxide have negative PCI and are commodities. The first two steps would raise complexity but leave Bolivia outside the core. The cell step is where the product-space distance is largest and scale economies are steepest. Risks: price volatility in lithium compounds; technical difficulty of the brine; competition from

established producers and from new extraction technologies; the capital and market access needed for cells; governance and contract design (Obaya, 2019); and water use in an arid region. As a *path* it is coherent with the incremental strategy of Remark 25.13: it builds chemical and metallurgical capabilities that are closer to the core than gas is.

(e) Is HS 8507 in $\mathcal{D}$? Cannot be answered without Exercise 25.1. Two remarks. (i) Membership in $\mathcal{D}$ is decided by *proximity*, not by PCI, so a product with moderate PCI can be in the dense set if it is co-exported with many core products. (ii) Its PCI ($+0.549$) is 1.251 below the illustrative mean PCI of $\mathcal{D}$ (1.8). If the real $\mathcal{D}$ has a mean of that order, 8507 is at best on its edge. The rise to $+0.811$ by 2023 indicates that the heading is moving toward the core as battery production concentrates in complex economies.

Chapter 26

The Neo-Structuralist Synthesis

All numbers below are printed by `python3 code/solutions/sol26.py`. Sourced values come from `data/processed`, retrieved 2026-11-05: Γ from INE (base 2017) and OECD ICIO 2023, `gamma_comparison.csv`; ECI from Growth Lab HS92, `eci_facts.csv`; GDP, Gini and gas rents from World Bank WDI, `wdi_long.csv`; Bolivian distribution from INE and World Bank PIP. Thresholds, plan costs and plan targets are the chapter's [ILLUSTRATIVE] values.

Exercise 26.1 The three conditions

Data status. Γ and ECI are sourced for 16 economies (not 30). N^I_{eff} "from media concentration data" is not available: audience data are paywalled or need registration (data catalogue, §9). Condition (3) is therefore not evaluated.

(a) Method. $\Gamma_c = 1 - |\lambda_2(W^P_c)|$ for $W^P_c = D^{-1}A_c$ on the ISIC-based 12-sector aggregation, using domestic flows, basic prices, input shares, diagonal kept, and the same year (2019) for every country. ECI from the Atlas (HS92, 2019). The repository has Γ for 23 economies and ECI for 22. They overlap in 16. To reach 30, run `data/compute/compute_oecd_io.py` on all 76 ICIO economies and extend the ECI extract.

(b) Why Γ must not be proxied by ECI. They are spectral statistics of *different matrices*. Γ is the second eigenvalue modulus of a country's domestic input-share matrix and measures how fast shocks mix across its sectors. ECI is the standardized second *eigenvector* of the global country-country matrix $\tilde{M} = D^{-1}MU^{-1}M'$ built from export baskets (Definition 16.7), and it is positional: it is z-scored across countries, so about half of all countries have ECI < 0 by construction. Empirically they are nearly unrelated: across the 16 economies $\text{corr}(\Gamma, \text{ECI}) = 0.265$. Proxying one by the other would make condition (1) a restatement of condition (2).

(c) The $(\Gamma, \mathbf{ECI})$ plane, 2019. Every economy clears the placeholder $\Gamma^* = 0.30$. In the full set of 23, Γ ranges from 0.333 to 0.659. The conditions therefore separate countries by ECI alone:

- *Satisfy (1) and (2)* (13): Argentina (0.522, +0.063), Brazil (0.496, +0.481), China (0.582, +1.301), Colombia (0.460, +0.140), Czechia (0.486, +1.478), Germany (0.544, +1.544), Japan (0.510, +2.047), Korea (0.511, +1.807), Mexico (0.649, +0.953), Malaysia (0.381, +1.031), Norway (0.515, +0.962), Poland (0.552, +1.060), United States (0.463, +1.294).
- *Violate (2)* (3): Bolivia (0.475, −1.295), Chile (0.557, −0.059), Peru (0.407, −0.680).

Not plotted: Costa Rica, Spain, Indonesia, India, Thailand, Viet Nam and South Africa have Γ but no ECI in the extract; Ecuador, Ethiopia, Laos, Paraguay, Uruguay and Venezuela have ECI but no ICIO table. The interpretation matches the remark under the chapter's viz panel. Condition (1) does not discriminate at the 12-sector level, and Argentina, Colombia and Brazil fall in the "growth with equity" zone on two of the three conditions despite well-known inequality. So either condition (3) is doing all the work, or the conditions are not sufficient. The chapter claims only necessity.

Exercise 26.2 Pathology classification

(a) Brazil. ECI +0.481 (2019), $\Gamma = 0.496$ (Korea 0.511), WDI Gini 53.5 (2019; Korea 33.7 in 2018). On measured values Brazil satisfies conditions (1) and (2), so the table's "(1) only: growth without equity" row does not fit: its "Low Γ" is not observed. Either Brazil violates condition (3) (plausible given media concentration, but unmeasured here), and the correct row is "(3) only: fragile growth", which does not describe high inequality; or the table is wrong. Brazil is evidence that conditions (1)–(2) are not sufficient for equity.

(b) Cuba. No value can be computed: Cuba is not in OECD ICIO, publishes no official income Gini, and is not in the repository's ECI extract. To check "high Γ" one would need a national input–output table (from Cuba's statistical office, if published at all) mapped to the 12 sectors. For "low ECI", one needs mirror trade data: partners' imports from Cuba, since its own reporting is incomplete. For equity, survey-based income distribution data are needed. Until then the classification is the table's conventional reading, "(2) and (3): trapped equality", and not a measurement.

(c) Venezuela. ECI -1.021 (2019) is sourced. Γ is unavailable (not a separate ICIO economy), and the last WDI Gini is for 2006. The "all three violated" row rests on one measured coordinate.

(d) Costa Rica. $\Gamma = 0.469$ (2019, OECD ICIO, like-for-like), which is "moderate" only in the sense that it is close to the comparator mean (0.511). Its ECI is not in the extract and must be added. If its ECI is positive and N_{eff}^{I} high, as the exercise posits, it lies in the "none violated" row. That is a prediction to be checked against its income distribution.

(e) Bolivia, 2005–2019. $\Gamma = 0.475 > \Gamma^{*}$, ECI $-1.295 < 0$, and $N_{\text{eff}}^{I} = 3.4 < N_{\text{eff}}^{*} = 8$ (both illustrative). So conditions (2) and (3) are violated: the "trapped equality" row, illustrated in the table by Cuba. Yet Bolivia's Gini fell from 0.600 to 0.426 (INE) while GDP grew 4.52% a year (Remark 26.4): growth *and* falling inequality, the opposite of "trapped". *Implication.* The pathology table is a set of predictions, and Bolivia contradicts the row it falls in. The table survives only under the narrow reading of Remark 26.4: its rows describe what cannot be *sustained*, not what cannot happen for a decade and a half. A correct table would separate "level" outcomes (equity achieved) from "persistence" outcomes (equity maintained through a commodity cycle).

Exercise 26.3 Complementarity

(a) Model. Measure each pillar as progress toward its target: $z_k \in [0,1]$ for $k \in \{\Gamma, \text{ECI}, N_{\text{eff}}\}$, with the target zone $z_k \geq 1$ for all k. Investment $u_k \geq 0$ with $\sum_k u_k = 1$ per unit of time. Dynamics:

$$\dot{z}_k = u_k\Big(\alpha + \gamma \sum_{l \neq k} z_l + \eta\, z_k\Big),$$

increasing in all three levels as the exercise requires. Here $\gamma > 0$ is cross-pillar complementarity, the dynamic analogue of the supermodularity of Appendix 26.6, and $\eta \geq 0$ is own increasing returns (learning by doing). Normalize $\alpha = 1$. "Simultaneous" sets $u_k = 1/3$ for each pillar below target. "Sequential" puts the whole budget on one pillar until it reaches 1, then moves to the next.

(b) Closed forms. Sequential: pillar $j = 0, 1, 2$ grows at $1 + \gamma j + \eta z$, so

$$T_{\text{seq}} = \sum_{j=0}^{2} \frac{1}{\eta} \ln\Big(1 + \frac{\eta}{1+\gamma j}\Big) \quad \Big(\to \textstyle\sum_j \frac{1}{1+\gamma j} \text{ as } \eta \to 0\Big).$$

Simultaneous: by symmetry $z_k = z$ with $\dot{z} = \frac{1}{3}(1 + (2\gamma + \eta)z)$, so $T_{\text{sim}} = 3\ln(1 + 2\gamma + \eta)/(2\gamma + \eta)$. The script checks both against an Euler simulation (step 10^{-4}).

γ	η	T_{seq}	T_{sim}
0	0	3.000	3.000
1	0	1.833	1.648
3	0	1.393	0.973
0	3	1.386	1.386
1	3	0.999	1.075

(c) Which is faster? *Simultaneous iff $\gamma > \eta$; sequential iff $\eta > \gamma$; they tie at $\gamma = \eta$. Proof of the tie*: at $\gamma = \eta$, $T_{\text{seq}} = \eta^{-1}[\ln(1 + \eta) + \ln\frac{1+2\eta}{1+\eta} + \ln\frac{1+3\eta}{1+2\eta}] = \eta^{-1}\ln(1 + 3\eta) = T_{\text{sim}}$ (for $\gamma = \eta = 2$ both equal 0.972955). *Why the sign flips there*: write $Z = \sum_k z_k$. Then $\dot{Z} = \sum_k u_k[1 + \gamma Z + (\eta - \gamma)z_k]$. At $\gamma = \eta$, $\dot{Z} = 1 + \gamma Z$ for *every* allocation, so all non-wasteful policies take the same time. For $\gamma > \eta$ the term $(\eta - \gamma)z_k$ penalizes investing in advanced pillars, which favours balanced investment. For $\eta > \gamma$ it rewards them, which favours concentration. The grid in the script confirms the sign pattern: $T_{\text{seq}} - T_{\text{sim}}$ is positive above the diagonal $\gamma > \eta$ and negative below it at every grid point. A second force favours sequencing: increasing returns to the *intensity* of investment (indivisibilities), $\dot{z}_k \propto u_k^{\varphi}$ with $\varphi > 1$. With $\varphi = 1.5$ and $\eta = 0$, sequential is faster for $\gamma = 0, 1, 3$ (3.000 vs 5.197, 1.834 vs 2.855, 1.393 vs 1.687), and simultaneous wins only for strong complementarity ($\gamma = 10$: 0.793 vs 1.139). For a Bolivia-like start in which condition (1) is already met, $z = (1, 0, 0)$, with $\gamma = 1$, $\eta = 0$: simultaneous 0.811, sequential 0.833.

Interpretation. Complementarity alone (supermodularity) argues for joint reform. That is Conjecture 26.6 in dynamic form. It does *not* follow that joint reform is always faster: own-pillar increasing returns and minimum scales argue for concentration. Chapter 15's sequencing argument (Conjecture 15.14) is a third, separate consideration, cost per unit of Ψ. Appendix 26.6 makes the same point: supermodularity does not say where to start.

Exercise 26.4 Bolivia's plan: cost-benefit

(a) Is there a Γ–GDP relation? Across the 12 economies with both a like-for-like Γ and WDI GDP per capita (PPP, current international \$, 2019), $\text{corr}(\Gamma, \ln y) = 0.129$: none to speak of at the 12-sector level. *ECI and growth.* For the 13 economies with both series, OLS of annual growth of GDP per capita (PPP, constant prices) on

initial ECI and log income gives (standard errors in parentheses):

$$g_{1995\text{–}2019} = 28.01 + \underset{(0.636)}{1.432}\,\mathrm{ECI}_{1995} - \underset{(0.606)}{2.744}\,\ln y_{1995}, \qquad g_{2009\text{–}2019}: \ \hat{b}_{\mathrm{ECI}} = \underset{(0.651)}{0.993}.$$

This regression does not support the use that follows. Dropping China reverses the sign (-0.541 and -0.521), so the positive coefficient is a China effect in a sample of 13 hand-picked economies. Taken at face value, raising ECI by 1.30 would add 1.86 (1995–2019 coefficient) or 1.29 percentage points (2009–2019) of annual growth. The published ECI–growth literature uses many more countries (Empirical Result 16.9), and its coefficient, not this one, is the right input. It was not fetched, and no value is quoted here. We therefore report the NPV as a function of the coefficient b.

(b) NPV at 5%. *Costs* (Example 26.7, spread evenly over each item's period): broadband 1.5 (2025–34); polytechnics 2, corridors 5, lithium 3 and agriculture 1 (2030–44; the agriculture period is our assumption); industrial policy 4 (2040–54). Total \$16.5 bn, which the book rounds to \$17 bn; present value at 5% (base 2024) \$8.45 bn. *Benefits*: baseline GDP \$54.9 bn in 2024, growing at 3.36% (geometric mean 2006–2024, WDI). ECI follows the plan's path: -1.30 to 2035, linear to -0.30 in 2045 and to 0 in 2055. The plan adds $b\,(\mathrm{ECI}_{t-1} + 1.30)$ points of growth. Benefits are the GDP gap against the baseline, counted to 2055 only.

b (pp per ECI unit)	0	0.5	0.993	1.432
NPV at 5% (\$ bn, to 2055)	-8.5	$+13.4$	$+35.9$	$+56.9$
IRR	—	12.4%	17.9%	20.9%

The NPV at 5% is positive iff $b > 0.1961$ percentage points per ECI unit.

(c) Break-even discount rate. It is the IRR in the table: 12.4% to 20.9% for b between 0.5 and 1.43. With $b = 0$ the plan never breaks even. The decisive number is therefore not the discount rate but b, together with the assumption that the plan delivers its ECI path, an illustrative target. The plan's apparent profitability rests on (i) a causal reading of a cross-country correlation, (ii) ECI gains that Bolivia has never achieved (its ECI fell by 0.44 over 1995–2019), and (iii) zero cost overruns. A credible appraisal would use a published, causally interpreted b, a distribution of plan outcomes, and the opportunity cost of public funds.

Exercise 26.5 The neoliberal counterfactual

(a) Method and data. Extrapolate linearly the structural velocities (Definition 23.12) of the neoliberal period. Γ (INE base 1990, `book12`, domestic, input shares) is available from 1988 and ECI from 1995, so "1985–2005" becomes 1988–2005 for Γ and 1995–2005 for ECI. σ_B is not available: it needs household microdata (data catalogue row 105). We use the income Gini (World Bank PIP) as the observable distributional proxy.

(b) Projections.

- Γ: 0.563 (1988) to 0.502 (2005), $v_\Gamma = -0.0356$ per decade. Projected 0.474 (2013), 0.453 (2019) and 0.449 (2020). Actual: 0.484 in 2013, the last year of the comparable base. The 2019 value of 0.467 is on base 2017 and not comparable.
- ECI: -0.852 (1995) to -1.096 (2005), $v = -0.244$ per decade. Projected -1.438 (2019) and -1.462 (2020). Actual 2019: -1.295.
- Distribution: PIP Gini 0.581 (1997), 0.580 (1999), 0.616 (2000), 0.574 (2001), 0.593 (2002), 0.550 (2004), 0.585 (2005). These come from non-comparable survey spells, but there is no downward trend: the mean is 0.583. A flat projection gives about 0.58 in 2019, against an actual 0.416.

(c) Target zone and comparison. Neither path enters the zone. $\Gamma > 0.30$ holds in both, so condition (1) does not discriminate, and ECI stays far below zero in both. The actual trajectory did better than the projection on ECI (-1.295 against -1.438 in 2019) and on Γ (0.484 against 0.474 in 2013), and much better on distribution. *Caveats.* (i) Linear extrapolation of a seven- or ten-year velocity over 15 years. (ii) The commodity boom of 2004–2014 was exogenous and would have occurred under either regime, so part of the actual gains, and part of what a neoliberal regime would also have obtained, is a price effect. (iii) The 1995–2005 ECI decline already includes the gas-discovery epoch (data catalogue: -0.24, 1995–2005). The counterfactual is a benchmark, not an identified estimate. A synthetic-control design with comparable gas exporters would be the next step.

Exercise 26.6 The populist counterfactual

Ill-posed as stated. The formation model (Definition 10.1) maps a *given* W into preferences $\boldsymbol{\theta}^*$. It holds W^P fixed and contains neither ECI nor any map from investment to ΔW, so it cannot deliver Γ or ECI. The closest well-posed version uses the

operator calculus of Chapter 15 and bounds the outcome with observed rates of structural change. The premise also needs a check: the "$50B" is illustrative.

(a) How large was the gas revenue? WDI natural gas rents (NY.GDP.NGAS.RT.ZS × NY.GDP.MKTP.CD) sum to $12.04 bn (current US$) over 2006–2019, and $8.35 bn over 2006–2014. Rents are an economic concept, value minus estimated extraction cost, not fiscal revenue. Government hydrocarbon revenue (royalties, the IDH tax, YPFB transfers) should be taken from the fiscal accounts, which were not fetched. The two can differ, and the 2017 GDP rebasing affects the rent series. The order of magnitude, though, is a quarter of the $50 bn premise.

(b) What could it have bought? Scaling the plan's illustrative ECI effect linearly (1.30 per $17 bn) gives $+0.92$ for $12 bn and $+3.82$ for $50 bn. The second would put Bolivia far above Korea, which shows that linear scaling is wrong: time-to-build and product-space distance (Chapter 25: steps of decades) bind before money does. A data-based bound: the largest 14-year ECI gain among the 22 economies in the repository is $+0.949$ (Laos, 2009–2023), and Korea gained $+0.704$ over the whole of 1995–2019. Even at that record pace from its 2005 level (-1.096), Bolivia would have reached -0.147 in 2019, still below zero. Γ was already inside the comparator range and is not the binding constraint.

(c) Target zone and trade-offs. No: even the most favourable bound leaves ECI negative by 2019, and condition (3) is not modelled. The counterfactual also has a cost the question omits. The transfers it would have cancelled financed part of the fall in poverty and inequality (Gini 0.600 to 0.426). Chapter 2's transfer result (Proposition 2.14) says transfers do not change the power term. It does not say that they have no distributional value. A sound answer compares plans that split the revenue, not "all or nothing".

Exercise 26.7 The 78-year argument

A reading exercise. The manual cannot do the reading, and it does not supply quotations or page numbers, which would have to be checked against the editions. What follows is the model answer: each work's central claim, as it is usually summarized, and the result or conjecture of the book it maps to.

(a) Claims and mappings.

1. Prebisch (1949): the world economy is divided into a center that generates and retains technical progress and a periphery specialized in primary goods.

The gains from trade are distributed asymmetrically through the terms of trade, so the periphery must industrialize. *Book*: the global W^P (Definition 16.1) and the Prebisch–Singer hypothesis restated spectrally, which is a *conjecture* (Conjecture 16.17).

2. Furtado (1961): underdevelopment is not a stage on the road to development but a distinct structure produced by the way the periphery was inserted into the world economy, and it reproduces itself. *Book*: structural equilibrium and the possibility of multiple equilibria (Definition 13.7, Proposition 13.9), and the formation channel (Chapter 10).

3. Pinto (1970): structural heterogeneity, the coexistence within one economy of strata with very different productivity that technical progress does not spread across. *Book*: block structure of W and between-block inequality (Proposition 2.11), stratified steady states (Proposition 10.5). In Bolivia's data the heterogeneity appears in the enclave's π and export orientation rather than in aggregate Γ (Remark 26.4).

4. Fajnzylber (1983): industrialization was truncated, lacking an endogenous nucleus of technical progress (capital goods and engineering capacity). *Book*: the bridge bound (Proposition 25.4, a theorem) and the truncation conjecture (Conjecture 25.6), which the 12-sector data do not support (Result 25.8).

5. CEPAL (2012): equality is the goal and structural change is the path; redistribution without a change in the productive structure is not sustainable. *Book*: the growth-equity conjecture (Conjecture 26.1) and the result that transfers leave the power term unchanged (Proposition 2.14). Bolivia 2005–2019 tests the sustainability clause.

6. Ocampo et al. (2009): growth is driven by the dynamics of the productive structure (innovation and complementarities), macroeconomic policy should serve structural change under a balance-of-payments constraint, and equity is a precondition for growth. *Book*: complementarity of the pillars (Conjecture 26.6, Appendix 26.6). The macroeconomic pillar has no counterpart in the conjecture (case study of Chapter 26).

(b) Synthesis (one paragraph). The single argument is this. An economy's income and its distribution depend on its *position* in a non-anonymous structure: who supplies whom, who learns from whom, who captures the gains from technical progress. That structure is not a transitory stage. It reproduces itself through

specialization, through the strata it creates and through the expectations it forms. It can be changed only by deliberate action on the structure itself, not by prices or transfers alone. In the book's language these are claims about the integration Γ and the power vector π of the layers of W, and about how operators move them. The mapping shows how uneven the formal status is: one theorem (the bridge bound), several conjectures, and one empirical record (Bolivia 2005–2019) that strains the strongest form of the shared conclusion.

Common mistakes: attributing to an author the book's formalization (the mappings are the book's); presenting paraphrases as quotations.

Chapter 27

Algorithmic W^I

All numbers below are printed by `python3 code/solutions/sol27.py`. The only sourced series is internet use (World Bank WDI, IT.NET.USER.ZS, retrieved 2026-11-05). Every network, engagement and policy parameter is [ILLUSTRATIVE]. As in the chapter, λ denotes a user's cross-group share of attention, $\lambda_i = \sum_{j \notin g(i)} W^I_{ij}$.

Exercise 27.1 Algorithmic $\Delta\lambda$

(a) Baseline. Four equal groups: urban men (UM), urban women (UW), rural men (RM), rural women (RW). Organic attention weights are multiplicative in two attribute factors (Definition 18.2): a factor $\lambda_g = 0.5$ when the genders differ and $\lambda_r = 0.3$ when the geographies differ. Row-normalizing $K_{ij} = \lambda_g^{\mathbf{1}[\text{gender differs}]} \lambda_r^{\mathbf{1}[\text{geo differs}]}$, whose row sums are all $(1+\lambda_g)(1+\lambda_r) = 1.95$:

$$W^I_0 = \begin{pmatrix} 0.5128 & 0.2564 & 0.1538 & 0.0769 \\ 0.2564 & 0.5128 & 0.0769 & 0.1538 \\ 0.1538 & 0.0769 & 0.5128 & 0.2564 \\ 0.0769 & 0.1538 & 0.2564 & 0.5128 \end{pmatrix} = B_r \otimes B_g, \quad B_x = \tfrac{1}{1+\lambda_x} \begin{pmatrix} 1 & \lambda_x \\ \lambda_x & 1 \end{pmatrix}.$$

Its eigenvalues are the products of those of the factors, $\{1, \frac{1-\lambda_r}{1+\lambda_r}, \frac{1-\lambda_g}{1+\lambda_g}, \text{their product}\} = \{1, 0.5385, 0.3333, 0.1795\}$, so $\Gamma = 0.4615$. The cross-gender share is $\lambda_g/(1+\lambda_g) = 0.3333$, the cross-geography share $\lambda_r/(1+\lambda_r) = 0.2308$, and $N_{\text{eff}} = 4$.

(b) The algorithm. Multiply the own-group entry (the diagonal cell) by 1.2 and

every other entry by 0.85, then renormalize:

$$W_1^I = \begin{pmatrix} 0.5978 & 0.2117 & 0.1270 & 0.0635 \\ 0.2117 & 0.5978 & 0.0635 & 0.1270 \\ 0.1270 & 0.0635 & 0.5978 & 0.2117 \\ 0.0635 & 0.1270 & 0.2117 & 0.5978 \end{pmatrix}.$$

Cross-gender share 0.2752 ($\Delta\lambda_g = -0.0581$); cross-geography share 0.1905 ($\Delta\lambda_r = -0.0402$). The eigenvalues are now $1, 0.6189, 0.4496, 0.3225$ (the matrix is no longer a Kronecker product), so $\Gamma = 0.3811$ ($\Delta\Gamma = -0.0805$). $N_{\text{eff}} = 4$ ($\Delta N_{\text{eff}} = 0$). Sanity checks: rows sum to one, and both shares fall, as Proposition 27.3(i) predicts with $e_s/e_c = 1.2/0.85$.

(c) Why $\Delta N_{\text{eff}} = 0$. Every row of W_1^I is a permutation of the same four numbers, so all rows have the same sum before normalization. The engagement matrix is symmetric, so the normalized matrix is still symmetric. A symmetric row-stochastic matrix is doubly stochastic, so $\boldsymbol{\pi} = \mathbf{1}/4$ (Proposition 6.8) and $N_{\text{eff}} = 4$ before and after. Both hypotheses matter. With unequal group sizes $p = (0.3, 0.3, 0.2, 0.2)$, attention proportional to source size ($W_{ij} \propto p_j K_{ij}$) gives a reversible chain with $\boldsymbol{\pi} = (0.3253, 0.3253, 0.1747, 0.1747)$. The same algorithm moves it to $(0.329, 0.329, 0.171, 0.171)$ and N_{eff} from 3.6673 to 3.6369: the larger groups gain power because their own-group cells are reinforced. The general point is Remark 27.6: block reinforcement acts on Γ; it reaches N_{eff} only through an asymmetry.

Exercise 27.2 The echo chamber result

(a) User-specific engagement. Fix user i with $0 < \lambda_i < 1$ and define the W^I-weighted mean engagements

$$\bar{e}_s := \frac{\sum_{j \in g(i)} W_{ij}^I e_{ij}}{1 - \lambda_i}, \qquad \bar{e}_c := \frac{\sum_{j \notin g(i)} W_{ij}^I e_{ij}}{\lambda_i}.$$

Under (27.2) the cross-group mass of row i after reweighting is $\lambda_i \bar{e}_c$ and the own-group mass is $(1 - \lambda_i)\bar{e}_s$, so

$$\lambda_i' = \frac{\lambda_i \bar{e}_c}{\lambda_i \bar{e}_c + (1 - \lambda_i)\bar{e}_s},$$

which is (27.3) with group-level engagement replaced by weighted means. Then $\lambda_i' < \lambda_i$ iff $\bar{e}_c < \lambda_i \bar{e}_c + (1 - \lambda_i)\bar{e}_s$ iff $(1 - \lambda_i)(\bar{e}_s - \bar{e}_c) > 0$ iff $\bar{e}_s > \bar{e}_c$. The

weighted-mean condition is therefore *necessary and sufficient*, and equality $\bar{e}_s = \bar{e}_c$ gives $\lambda_i' = \lambda_i$. The min/max condition of the exercise implies it, since $\bar{e}_s \geq \min_{j \in g(i)} e_{ij} > \max_{j \notin g(i)} e_{ij} \geq \bar{e}_c$, but the converse fails: a user can have one cross-group source more engaging than some same-group source and still lose cross-group exposure. If $\lambda_i \in \{0, 1\}$, $\lambda_i' = \lambda_i$. The script checks the formula and the equivalence on 20,000 random users. Part (ii) of the proposition does *not* extend: with user-specific e_{ij} the reweighted matrix leaves the two-parameter family, it need not be doubly stochastic, and Γ and N_{eff} can move either way.

(b) Two attributes. Twenty users: urban with belief L (8), urban R (2), rural L (2), rural R (8). The organic feed weights same-geography sources 4 and other-geography sources 1 (no self-weight). The algorithm triples same-belief sources. For a rural-L user, the organic cross-geography share is $10/(4 \cdot 9 + 10) = 10/46 = 0.217$. After reweighting, the same-geography mass is $4(1 \cdot 3 + 8) = 44$ and the other-geography mass $8 \cdot 3 + 2 = 26$, so the share rises to $26/70 = 0.371$. Urban-R users are symmetric (0.217 to 0.371). Majority users lose: a rural-R user goes from $10/46$ to $14/106 = 0.132$. Cross-*belief* exposure falls on average from 0.426 to 0.218. These are exactly the values of `code/ch27.py`. Sorting on belief sends belief minorities to like-minded sources in the other region, raising their cross-geography exposure. The "for all attributes k" claim fails.

(c) Top-K. Let the feed show only the K sources with the highest e_{ij}, with organic weights renormalized among them. *Group-level engagement* ($e_s > e_c$): every same-group source outranks every cross-group one. If K is at most the number n_s of same-group sources, $\lambda_i' = 0 \leq \lambda_i$. If $K > n_s$, the shown cross-group mass is $c \leq \lambda_i$ and $\lambda_i' = c/(1 - \lambda_i + c) \leq \lambda_i$, with equality iff all cross-group sources are shown. So the weak inequality survives. Integration can collapse completely: with two equal groups of three and $K = 3$, the feed becomes block-diagonal, $|\lambda_2| = 1.000$ and $\Gamma = 0$. The matrix is reducible and consensus fails (Appendix 27.6). *User-specific engagement*: the result fails even under the weighted-mean condition of (a). Take weights $(0.45, 0.45 \mid 0.10)$ and engagement $(2.0, 0.2 \mid 1.0)$, with the bar separating same-group from cross-group sources. Then $\bar{e}_s = 1.100 > \bar{e}_c = 1.000$, so multiplicative ranking lowers λ from 0.100 to 0.0917. Top-2 shows the first same-group source and the cross-group source, and raises λ to $0.10/0.55 = 0.1818$. Truncation responds to the *rank* of the best cross-group source, not to average engagement.

Exercise 27.3 Bolivia's digital divide

Data status. Only the national internet-use series is in the repository. Department-, area-, sex- and age-specific use is in INE's household survey (ICT module) and was not fetched. The N_{eff} estimates of Chapter 21 are illustrative. Method and expected result only.

(a) Sourced national figures. Internet users were 22.4% of the population in 2010, 47.5% in 2019, 77.3% in 2023 and 79.7% in 2024 (World Bank, 2026).

(b) Method. The fraction of W^I that is algorithmically mediated is

$$F = \sum_g p_g \, u_g \, \omega_{d,g} \, a_g,$$

where p_g is group g's population share (department $\times$ area $\times$ sex $\times$ age), u_g its internet-use rate (INE survey), $\omega_{d,g}$ the digital weight in its composite W^I (Definition 21.9), and a_g the share of digital attention spent on *ranked* feeds. WhatsApp is not ranked (Chapter 27, introduction), so $a_g < 1$. u_g comes from the survey; $\omega_{d,g}$ and a_g need a time-use or attention module (Appendix 21B's survey design) or app-usage panels. Report F and the population share with $u_g \omega_{d,g} a_g > 0$. With the illustrative composite weights of Example 21.10 ($\omega_d \leq 0.50$ for every group), $F \leq 0.5 \times 79.7\% = 39.9\%$ of national W^I in 2024, an upper bound that ignores the unranked share.

(c) Correlation with N_{eff}. N_{eff} is a property of a whole network, not of a group (Conjecture 27.2, last paragraph), so the correlation must be computed either across departments, each with its own W^I and its own N_{eff}, or at user level with feed diversity e^{H_i}. *Expected result.* The sign is ambiguous by Remark 27.8. Across channels, digital media add outlets and raise N^I_{eff}. Within ranked platforms, attention concentrates. The likely pattern is a hump: N_{eff} rising with F where digital access mainly adds sources (low-F, rural departments), and flat or falling where ranked platforms dominate (high-F, urban, young). *Common mistake*: correlating group-level internet use with a national N_{eff}.

Exercise 27.4 Objective function design

(a) The problem. Let W^I_0 be the organic network, $e_{ij} > 0$ predicted engagement and $\mathcal{S}$ the set of row-stochastic matrices supported on the organic graph ($W_{ij} = 0$

where $W^I_{0,ij} = 0$). The platform chooses the feed W:

$$\max_{W \in \mathcal{S}} \; N_{\text{eff}}(W) = \Big(\sum_j \pi_j(W)^2\Big)^{-1}$$
$$\text{s.t.} \quad \frac{1}{N}\sum_{i,j} W_{ij} e_{ij} \geq \bar{E}, \qquad \Gamma(W) \geq \Gamma_{\min}, \qquad \lambda_i(W) \geq \lambda_{\min} \; \forall i.$$

Structure. $N_{\text{eff}} \leq N$, with equality iff $\boldsymbol{\pi} = \mathbf{1}/N$ iff W is doubly stochastic (Proposition 6.8). So whenever the engagement constraint allows it, the solution set is $\{W \in \mathcal{S} : W'\mathbf{1} = \mathbf{1}, \text{ constraints}\}$; without the Γ floor (which is not convex) it is a polytope. The non-convex objective becomes the *linear* constraint "columns sum to one".

(b) Why the Γ floor is needed. Without it, maximizing engagement over doubly stochastic matrices is a linear program over the Birkhoff polytope, whose vertices are permutation matrices (Birkhoff–von Neumann). The optimum is therefore a permutation: each user sees exactly one source. $N_{\text{eff}} = N$, but all eigenvalues have modulus one, so $\Gamma = 0$ and there is no learning or consensus. In the six-user example below, the best permutation achieves engagement 2.760. A well-posed design therefore needs $\Gamma_{\min}$, or entropy regularization. The tractable version is

$$\max_{W \in \mathcal{S},\, W'\mathbf{1}=\mathbf{1}} \; \sum_{i,j} W_{ij} \ln e_{ij} - \frac{1}{\tau}\sum_{i,j} W_{ij} \ln \frac{W_{ij}}{W^I_{0,ij}},$$

whose solution is the Sinkhorn scaling $W = \operatorname{diag}(\mathbf{a})(W^I_0 \circ e^{\tau})\operatorname{diag}(\mathbf{c})$: engagement ranking (27.2) with exponent τ, followed by a rescaling of sources that equalizes their total attention. The engagement floor $\bar{E}$ pins down τ.

(c) A computed example. Six users in two groups of three, organic $\lambda = 0.22$. Engagement is 2.07 for same-group sources and 1 otherwise, and source 1 is viral ($\times 3$).

Feed	engagement	N_{eff}	Γ	mean λ_i
Organic	2.446	6.000	0.440	0.220
Engagement ranking (27.2)	3.190	4.002	0.261	0.130
Balanced ranking ($\tau = 1$)	2.589	6.000	0.240	0.120

Varying τ traces the trade-off. At $\tau = 0.5$, 1, 2 and 4, engagement is 2.526, 2.589, 2.672 and 2.738 with $N_{\text{eff}} = 6$ throughout, while Γ falls to 0.328, 0.240, 0.124 and 0.030.

(d) Trade-offs. (i) *Engagement against concentration*: balancing costs engagement, in the example $3.190 \to 2.589$. (ii) *Concentration against fragmentation*: holding $N_{\text{eff}} = N$, more engagement means lower Γ, so a N_{eff} mandate alone produces echo chambers. The objective must carry a Γ or λ floor (Remark 27.6). (iii) *Accuracy*: uniform power is not the epistemic optimum. Under the wisdom-of-crowds result (Theorem 8.5) the efficient weights depend on source precision, so forcing equal weight on low-quality or fringe sources can spread misinformation. (iv) *Feasibility*: balancing requires that the organic support admit a doubly stochastic pattern (total support), and it overrides user choice, which the evidence in Section 27.2 shows matters as much as ranking.

Exercise 27.5 AI mentoring simulation

(a) Design and ΔW^I edges. An assistant on WhatsApp, the dominant channel (introduction to Chapter 27), matches each rural entrepreneur to three urban or modern-sector mentors in the same sector. It provides weekly market prices and buyer contacts, and relays short mentor videos in Spanish, Quechua or Aymara. It is Chapter 15's information operator (Definition 15.10): $\Delta W^I_{ij} > 0$ for rural i and modern-sector j, financed by an equal reduction of within-group weight, so that rows stay stochastic. The design target is to raise the cross-group share of the rural users from $\lambda = 0.22$ to 0.32.

(b) $\Delta\lambda^I$ and ΔN_{eff}. Six-node version (three urban, three rural, organic $\lambda = 0.22$). Mentoring raises rural rows to $\lambda = 0.32$, spread evenly over the three urban mentors: $\Delta\lambda^I = +0.10$, Γ rises from 0.440 to 0.540, and N_{eff} *falls* from 6.000 to 5.801, because the urban mentors' total Perron weight rises to 0.5926. If the extra 0.10 goes to one star mentor, Γ is again 0.540, $N_{\text{eff}} = 5.764$ and the star's weight is 0.2247. Integration and concentration move in opposite directions, as Proposition 6.16 allows. A design that rotates mentors and caps any mentor's audience limits the concentration.

(c) Formation prediction. With the chapter's illustrative parameters ($\rho = 0.85$, $\mu\Delta\bar{r} = 0.09$; Example 27.4), the rural signal rises by $\Delta\lambda\,\Delta\bar{r}$. In the formation equation (10.3), one generation (one period) changes rural preferences by $\mu\,\Delta\lambda\,\Delta\bar{r} = 0.09 \times 0.10 = 0.0090$ preference units. In steady state the change is $\mathcal{M}\mu^{-1}$ times that, $0.0090/0.15 = 0.0600$ (Definition 10.3), the same order as the gap the chapter attributes to algorithmic filtering. That holds if returns do not respond to θ. If they do, the matrix multiplier $(I - J)^{-1}\mu$ applies (Theorem 10.12).

(d) Evaluation design. Cluster-randomize communities: half receive the assistant, half receive a placebo price-only bot, which separates the mentoring edges from market information. *Power*: minimum detectable effect 0.10 SD on an aspirations index, intra-cluster correlation 0.05, 20 entrepreneurs per community, two-sided $\alpha = 0.05$, power 0.80. Then

$$J = \frac{(z_{0.975} + z_{0.80})^2 \left[\rho_{\text{ICC}} + (1 - \rho_{\text{ICC}})/n\right]}{P(1-P)\,\text{MDE}^2} = \frac{2.8016^2 \times 0.0975}{0.25 \times 0.01} = 306.1,$$

so 308 communities (154 per arm) and 6,160 entrepreneurs. Outcomes: λ_i from usage logs (first stage), aspirations and investment (second stage), sales after 24 months. *Identification threats*: spillovers between neighbouring communities (SUTVA), handled with buffer zones or by randomizing at a coarser level; differential attrition of non-users; experimenter demand on survey aspirations, handled with behavioural outcomes; and the fact that the measured effect is one period of a multi-generation process, so only $\mu\Delta\lambda\Delta\bar{r}$, not the steady state, is identified.

Exercise 27.6 Algorithmic center-periphery

Data status: method only. SimilarWeb is commercial and no traffic panel was fetched. No traffic shares are reported.

(a) Method.

1. *Source*: a web and app traffic panel for Bolivia (SimilarWeb, or an app-usage panel). Web-only panels miss in-app use, which dominates on mobile, so use the app panel for Facebook, Instagram, TikTok, YouTube and WhatsApp.

2. *Classification*: for each destination d record its headquarters country, whether its feed is ranked (recommender) or chronological and contact-based (WhatsApp, Telegram groups), and whether it is domestic news, entertainment or other.

3. *Fraction designed abroad*: $f = \sum_d s_d \, \mathbf{1}[\text{ranked}_d] \, \mathbf{1}[\text{foreign}_d]$, with s_d the share of attention (time, not visits).

4. *Share of W^I*: $F_{\text{abroad}} = \sum_g p_g \, \omega_{d,g} \, f_g$, combining f with the digital weight $\omega_{d,g}$ of Exercise 27.3.

(b) Expected result. The ranked destinations with large shares are likely to be US-based platforms (Google search, YouTube, Facebook, Instagram), plus TikTok,

whose parent company is based outside the US. Domestic news sites are likely to have small shares. So f is likely to be close to the ranked share of attention. The fraction of W^I designed abroad is then bounded by the digital weight ω_d. With the book's illustrative composite weights it is at most about half for urban youth and about a tenth for rural and older users. *Caveats*: traffic is not attention, and attention is not influence (W^I weight). Much Bolivian digital information flows through WhatsApp groups, which a foreign firm hosts but does not rank. "Designed abroad" must distinguish infrastructure from curation.

Exercise 27.7 The policy frontier

(a) A common test bed (illustrative). The six-user network of 27.4, organic $\lambda = 0.22$. *(a) No regulation*: engagement ranking with $e_s/e_c = 2.07$ plus a viral source taking a share $a = 0.3$ of everyone's attention. *(b) Transparency*: assumed to move the feed a quarter of the way back to organic, $W_b = 0.75W_a + 0.25W_0$; this assumption carries the result for (b). *(c) Diversity mandate*: every user's cross-group share raised to at least $\lambda^{\min} = 0.3$ by proportional reallocation. *(d) Public-interest algorithm*: the balanced (Sinkhorn) version of the regime-(a) feed, the N_{eff}-maximizing design of 27.4.

Regime	N_{eff}	Γ	mean λ_i	min λ_i	π_1	engagement
Organic	6.000	0.440	0.220	0.220	0.167	2.446
(a) None	3.388	0.468	0.234	0.084	0.474	3.194
(b) Transparency	4.164	0.461	0.230	0.118	0.398	3.007
(c) Mandate $\lambda^{\min} = 0.3$	4.665	0.684	0.342	0.300	0.361	2.831
(d) Public-interest	6.000	0.325	0.163	0.163	0.167	2.474
(c) + (d)	6.000	0.574	0.287	0.287	0.167	2.272

(b) Effects on N_{eff} and Γ. No single regime fixes both dimensions. The unregulated feed concentrates power ($N_{\text{eff}} = 3.388$, the viral source holds 0.474) while *raising* Γ (0.468), the viral-hub pattern of Remark 27.6, and leaves its least-connected users at $\lambda_i = 0.084$. The mandate restores mixing ($\Gamma = 0.684$, every $\lambda_i \geq 0.3$) but leaves the hub at 0.361. Balancing restores $N_{\text{eff}} = 6$ but keeps the engagement sorting, so Γ falls to 0.325 and λ_i to 0.163. Combining (c) and (d) gives $N_{\text{eff}} = 6$, $\Gamma = 0.574$ and $\lambda_i = 0.287$: balancing after the mandate slightly undoes it, so the two must be imposed jointly, not in sequence. Its engagement (2.272) is below the organic feed's: that is the commercial cost.

(c) Feasibility and cost (qualitative). (b) is cheap, politically easy and enforceable through disclosure (the EU's Digital Services Act is the template; Remark 27.10).

Its structural effect depends entirely on whether users and regulators respond. (c) needs audit access to feeds and a regulator able to verify λ_k by group. Bolivia's regulator lacks the mandate and data (case study), and platforms can contest it. Its cost is the platform's engagement loss. (d) needs public funding or a cooperative operator, and it reaches only its own users: with adoption share σ, its effect on national W^I scales with $\sigma\omega_d$. *Which regime best advances the structural trajectory?* If the binding problem is fragmentation, as on the formation channel of Chapters 10 and 18, where cross-group λ drives the aspiration gap, the mandate does most. If it is concentration of epistemic power (hegemony, Chapter 24), the public-interest algorithm does. The defensible recommendation is (b) as a precondition, since without transparency neither (c) nor (d) can be monitored, plus (c) with N_{eff} monitoring (instrument 4), with (d) as a complement where a public broadcaster can run it. All magnitudes here are illustrative; the ranking of regimes, not the numbers, is the result.

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